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PREPARATION Series

NEET UG

Previous 5 Years Question Papers With Solutions 2021-2025

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The NEET Previous Five Years Question Papers with Solutions eBook (2021–2025) is a comprehensive and trusted resource designed to help students thoroughly understand the actual NEET exam pattern and level of difficulty. This eBook brings together the official NEET question papers from 2021, 2022, 2023, 2024, and 2025, along with accurate answer keys and fully solved, expert-verified solutions.

Each question is explained using clear, step-by-step solutions that focus on concept clarity, smart shortcuts, and proven problem-solving techniques commonly required in NEET. These explanations help students strengthen their fundamentals, avoid common mistakes, and improve overall accuracy.

Beyond being a collection of question papers, this eBook offers in-depth chapter-wise analysis, topic-wise trends, and subject-wise difficulty levels across Physics, Chemistry, and Biology. It highlights high-weightage chapters, frequently tested concepts, and recurring question patterns, enabling students to prioritise their preparation effectively.

Whether you are revising important topics, evaluating your performance, or preparing for a fresh or repeat NEET attempt, this eBook serves as a powerful guide to learn directly from past NEET exams and maximise your score potential.

How This eBook Will Benefit You

This eBook helps you gain a realistic understanding of the NEET exam pattern by providing access to five years of actual NEET question papers. The detailed, step-by-step solutions simplify complex concepts and enhance your problem-solving approach. With chapter-wise and topic-wise analysis, you can clearly identify your strengths and areas that need improvement.

Overall, this eBook builds confidence and exam readiness for NEET.

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NEET 2021

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

All questions with detailed solutions of Physics, Chemistry and Biology

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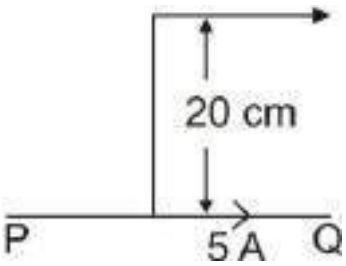
Physics

Q. 1 An infinitely long straight conductor carries a current of 5A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant.
Calculate the magnitude of the force experienced by the electron at that instant.

$\text{Electron} v = 10^5\ m/s$



Option 1:
$4 \times 10^{-20} N$

Option 2:
$8\pi \times 10^{-20} N$

Option 3:
$4\pi \times 10^{-20} N$

Option 4:
$8 \times 10^{-20} N$

Correct Answer:
$8 \times 10^{-20} N$

Solution:
$i = 5A$

$e = 1.6 \times 10^{-19} c\ \ v = 10^5 m/s$

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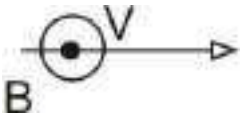
$$F = qvB\ \sin\theta = qvB\sin 90^o$$

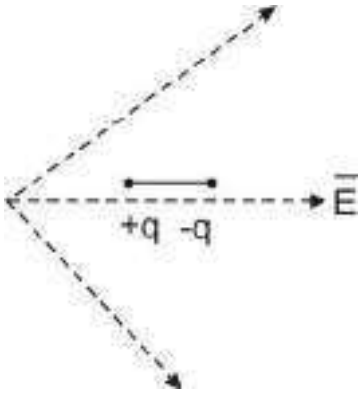


$$B = \frac{\mu_o I}{2\pi r}$$

$$F = 1.6 \times 10^{-19} \times 10^5 \times \frac{2 \times 10^{-7} \times 5}{2 \times 10^{-1}}$$

$$F = 8 \times 10^{-20} N$$

Q. 2 A dipole is placed in an electric field as shown. In which direction will it move?



Option 1:
towards the left as its potential energy will increase.

Option 2:
towards the right as its potential energy will decrease.

Option 3:
towards the left as its potential energy will decrease.

Option 4:
towards the right as its potential energy will increase.

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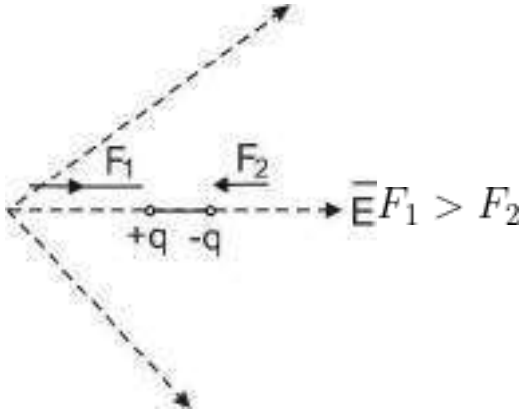
Correct Answer:
towards the right as its potential energy will decrease.

Solution:



F_{net}(towards right)

electric field intensity is more at $+q$ as field lines are closer

$V_2 - V_1 = -PE_2 \cos\theta_2 - (-PE_1 \cos\theta_1)$

$\theta_1 = \theta_2 = 180^o$(along givenline)

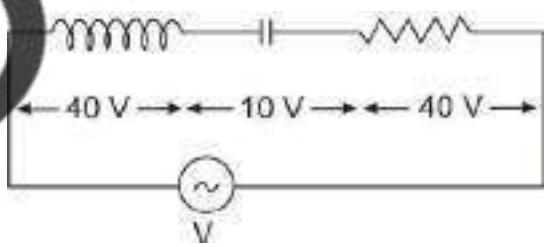
$E_2 < E_1$(as it move towards right)

$= (+PE_2) - (+PE_1)$

$\therefore \Delta U \Rightarrow Decrease$

Q. 3 An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure.

Potential difference across, C and R is 40 V, 10V and 40V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}A$. The impedance of the circuit is:



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Option 1:

$4\sqrt{2}\Omega$

Option 2:

$\frac{5}{\sqrt{2}}\,\Omega$

Option 3:

$4\,\Omega$

Option 4:

$5\,\Omega$

Correct Answer:

$5\,\Omega$

Solution:

$$V_{rms} = \sqrt{V_R^2 + (V_L - V_c)^2}$$

$$= \sqrt{(40)^2 + (30)^2} = 50$$

$$V_{peak} = (V_{rms}) \times \sqrt{2}$$

$$= 50\sqrt{2}$$

$$Z = \frac{V_{peak}}{I_{peak}} = \frac{50\sqrt{2}}{10\sqrt{2}} = 5\Omega$$

Q. 4 For a plane electromagnetic wave propagating it x-direction, which one of the following combination given the correct possible direction for electric field (E) and magnetic field (B) respectively?

Option 1:

$\hat{j} + \hat{k}, \hat{j} + \hat{k}$

Option 2:

$-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$

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Option 3:

$\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

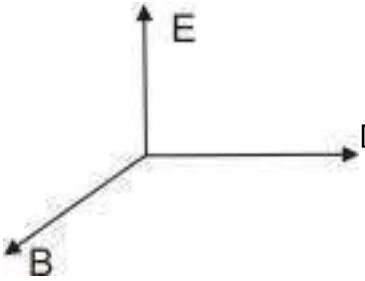
Option 4:

$-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

Correct Answer:

$-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$

Solution:



Direction of proparation $\bar{E}, \bar{B}\&$ dirction of proparation are mutually perpendicular to each other

$$\bar{E} \cdot \bar{B} = EB\cos 90^o = 0$$

Q. 5 A lens of large focal length and large aperture is best suited as an objective of an astronomic telescope since:

Option 1:

A large aperture contributes to the quality and visibility of image.

Option 2:

A large area of the objective ensures better light gathering power.

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Option 3:
a large aperture provides a better resolution

Option 4:
all of the above

Correct Answer:
all of the above

Solution:

Magnification of telescope$= \frac{f_o}{\mu_o} = f_o\left[\frac{1}{fe} + \frac{1}{Ve}\right]$

Large focal length$\longrightarrow$better Magnification

Large aperture$\longrightarrow$better light gathering power

Hence, the answer is the option (4).

Q. 6 **Column - I** gives certain physical terms associated with the flow of current through a metallic conductor.
Column - II gives some mathematical relations involving electrical quantities.
Match Column - I and Column - II with appropriate relation.

	Column I		Column II
(A)	Drift Velocity	(P)	$\frac{m}{ne^2p}$
(B)	Electrical Resistivity	(Q)	nev_d
(C)	Relaction Period	(R)	$\frac{eE}{m}\tau$
(D)	Current Density	(S)	$\frac{E}{J}$

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Option 1:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Option 2:
(A)-(R), (B)-(S), (C)-(Q), (D)-(P)

Option 3:
(A)-(R), (B)-(P), (C)-(S), (D)-(Q)

Option 4:
(A)-(R), (B)-(Q), (C)-(S), (D)-(P)

Correct Answer:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Solution:

$$\frac{E}{J}=\frac{1}{\sigma}=\rho=Electrical Resistivity$$

Driff Velocity $=V_2=\frac{eE\tau}{m}$

Relaxation Period $=\tau=\frac{m}{ne^2\varrho}$

Current density$=J=neV_2$

$A-R, B-S, C-P, D-Q$

Hence, the answer is the option (1).

Q. 7 An electromagnetic wave of wavelength $'\lambda'$ is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then:

Option 1:

$$\lambda=\left(\frac{2m}{hc}\right)\lambda_d{}^2$$

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Option 2:

$$\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$$

Option 3:

$$\lambda = \left(\frac{2mc}{h}\right)\lambda_d{}^2$$

Option 4:

$$\lambda = \left(\frac{2h}{mc}\right)\lambda_d{}^2$$

Correct Answer:

$$\lambda = \left(\frac{2mc}{h}\right)\lambda_d{}^2$$

Solution:

Work Function$\rightarrow$negligible

$$\phi \simeq 0$$

$$\frac{hc}{\lambda} = \phi + kE_{max}$$

$$\frac{hc}{\lambda} = 0 + \frac{p^2}{2m} = \frac{(h/\lambda d)^2}{2m}$$

$$\frac{kc}{\lambda} = \frac{h}{\lambda_d^2 \times 2m}$$

$$\left(\frac{2mc}{h}\right)\lambda_d^2 = \lambda$$

Hence, the answer is the option (3).

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Q. 8 A particle is released from height S from the surface of the earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of the earth and the speed of the earth and the speed of the particle at that instant are respectively:

Option 1:
$\frac{S}{4}, \frac{3gS}{2}$

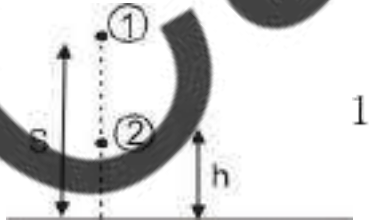
Option 2:
$\frac{S}{4}, \frac{\sqrt{3gS}}{2}$

Option 3:
$\frac{S}{2}, \frac{\sqrt{3gS}}{2}$

Option 4:
$\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

Correct Answer:
$\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

Solution:



$1 \rightarrow Initial$

$2 \rightarrow Final$

$TE_1 = TE_2$

$mgs + 0 = (mgh) + KE$

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$(kE) = 3\,(mgh)\,(Given)$

$mgs = 4mgh$

$h = \frac{s}{4}$

$3mgh = \frac{1}{2}mv^2$

$v = \sqrt{2gh \times 3} = \sqrt{2g\frac{s}{4} \times 3} = \sqrt{\frac{gs}{2} \times 3}$

$v = \sqrt{\frac{3gS}{2}}$

Q. 9 A screw gauge gives the following reading when used to measure the dimeter of a wire

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is:

Option 1:
0.52 cm

Option 2:
0.026 cm

Option 3:
0.26 cm

Option 4:
0.052 cm

Correct Answer:
0.052 cm

Solution:

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$1\ MSD = 100CSD = 1mm$

$CSR = 52Division = 0.52mm$

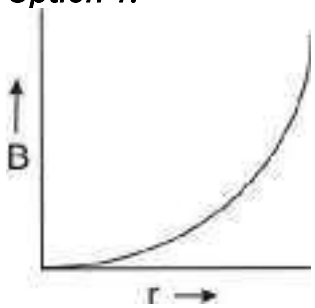
$Reading = MSR + CSR$

$= 0 + 0.52mm$

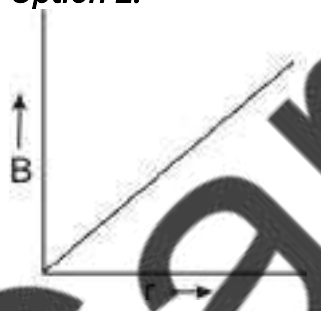
$= 0.052cm$

Q. 10 A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field B(r) due to the cable with the distance 'r' from the axis of the cable is represented by:

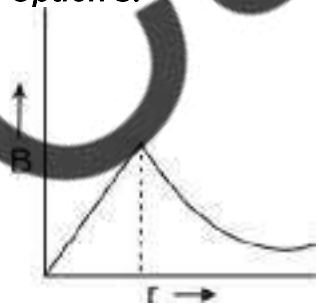
Option 1:



Option 2:



Option 3:



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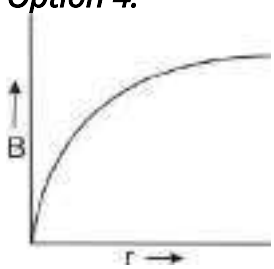
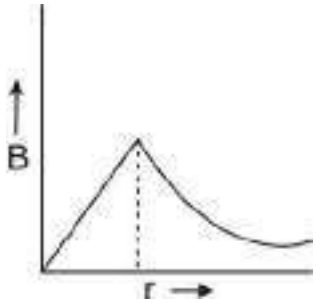
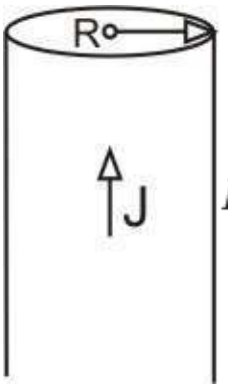
Option 4:



Correct Answer:

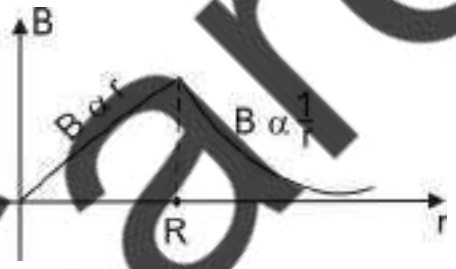


Solution:



$$B = \frac{\mu_o I r}{2\pi R^2} (r < R)$$

$$B = \frac{\mu_o I}{2\pi r} (r > R)$$



Q. 11 Two charged spherical conductors of radius $R_1 \; and \; R_2$ are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :

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Option 1:

$\frac{R_1}{R_2}$

Option 2:

$\frac{R_2}{R_1}$

Option 3:

$\sqrt{\left(\frac{R_1}{R_2}\right)}$

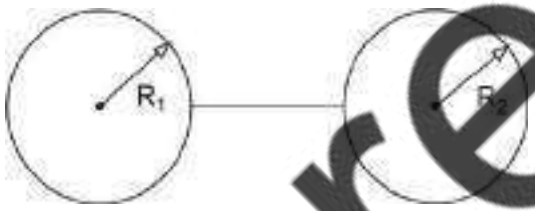
Option 4:

$\frac{{R_1}^2}{{R_2}^2}$

Correct Answer:

$\frac{R_2}{R_1}$

Solution:



$V_1 = V_2$

$\frac{kQ_1}{R_1} = \frac{kQ_2}{R_2}$

$\frac{Q_1}{Q_2} = \frac{R_1}{R_2}$

$\frac{\sigma_1}{\sigma_2} = \frac{Q_1/4\pi R_1^2}{Q_2/4\pi R_2^2} = \frac{R_2^2}{R_1^2} \times \frac{R_1}{R_2}$

$\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$

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Q. 12 The number of photons per second on average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of $3.3 \times 10^{-3} watt$ will be :$\left(h = 6.6 \times 10^{-34} Js\right)$

Option 1:
10^{18}

Option 2:
10^{17}

Option 3:
10^{16}

Option 4:
10^{15}

Correct Answer:
10^{16}

Solution:
$\lambda = 600nm$

$P = 3.3 \times 10^{-3} w$

$h = 6.6 \times 10^{-34} Js$

$$\text{No. of photons per second} = \frac{p}{\frac{hc}{\lambda}} = \frac{3.3 \times 10^{-3}}{\frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{6 \times 10^{-7}}} = 1 \times 10^{16}$$

Hence, the answer is the option (3).

Q. 13 A cup of coffee cools from $90°C \; to \; 80°C$ in t minutes, when the room temperature is $20°C$. The time taken by a similar cup of coffee to cool from $80°C \; to \; 60°C$ at a room temperature same at $20°C$ is:

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Option 1:

$\frac{13}{10}t$

Option 2:

$\frac{13}{5}t$

Option 3:

$\frac{10}{13}t$

Option 4:

$\frac{5}{13}t$

Correct Answer:

$\frac{13}{5}t$

Solution:

$$\frac{\Delta T}{\Delta t} = -k\left(T_{ang} - T_0\right)$$

$$\frac{\frac{90-80}{t}}{\frac{(80-60)}{\Delta t}} = \frac{\left(\frac{90+80}{2} - 20\right)}{\left(\frac{80+60}{2} - 20\right)}$$

$$\frac{\frac{10}{t}}{\frac{20}{\Delta t}} = \frac{65}{50} = \frac{13}{10}$$

$$\frac{\Delta t}{2t} = \frac{13}{10}$$

$$\Delta t = \frac{13\,t}{5}$$

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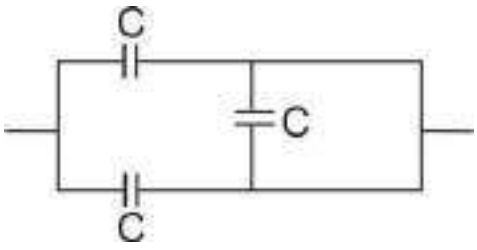
Q. 14 The equivalent capacitance of the combination shown in the figure is:



Option 1:
$3C$

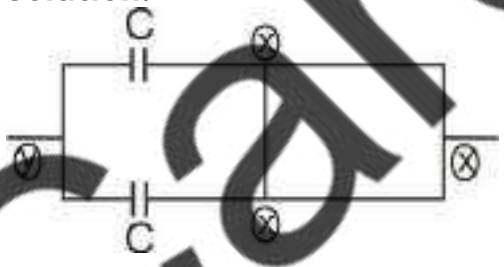
Option 2:
$2C$

Option 3:
$C/2$

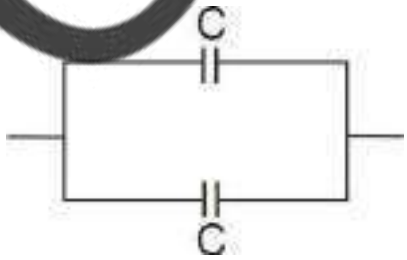
Option 4:
$3C/2$

Correct Answer:
$C/2$

Solution:



Capacitor is short circuited



$C_{eq} = 2C$

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Q. 15 The effective resistance of a parallel connection consists of four wires of equal length, equal area of cross-section, and the same material 0.25Ω. What will be the effective resistance if they are connected in series?

Option 1:
0.25Ω

Option 2:
0.5Ω

Option 3:
1Ω

Option 4:
4Ω

Correct Answer:
4Ω

Solution:
Four wires each of resistance for R parallel connection

$$R_{eq} = \frac{R}{4} = 0.25$$

$$\therefore R = 1$$

For series connection

$$R_{eq} = 4R = 4(1) = 4\Omega$$

Hence, the answer is the option (4).

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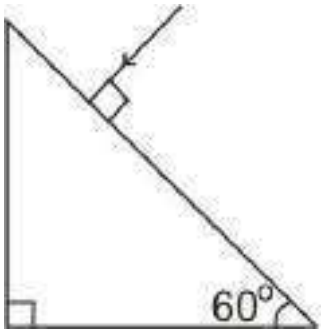
Q. 16 Find the value of the angle of emergency from the prism. Refractive index of the glass is $\sqrt{3}$.



Option 1:
60°

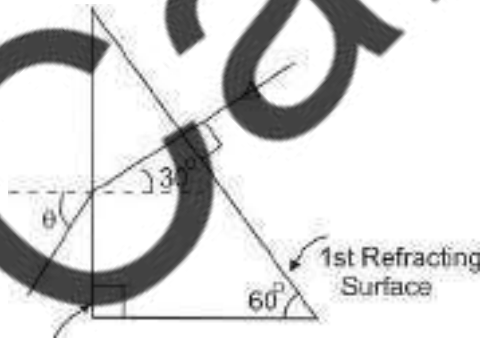
Option 2:
30°

Option 3:
45°

Option 4:
90°

Correct Answer:
60°

Solution:



2nd Refracting Surface

$$\mu_1 \sin 30^o = \mu_2 \sin \theta$$

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$$\sqrt{3} \times \frac{1}{2} = 1 \sin\theta$$

$$\theta = 60^o$$

$$\theta = Angle\, of\, ehergence = 60^o$$

Q. 17 A small block slides down on a smooth inclined plane, starting from rest at time t=0. Let Sn be the distance travelled by the block in the interval t=n-1 to t=n.

Then, the ratio $\frac{S_n}{S_{n+1}}$ is:

Option 1:
$\frac{2n-1}{2n}$

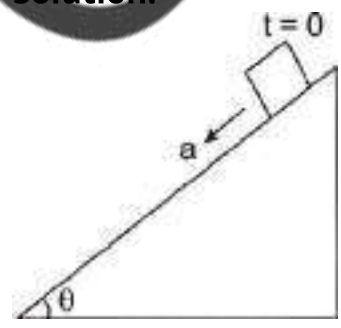
Option 2:
$\frac{2_n-1}{2n+1}$

Option 3:
$\frac{2_n+1}{2n-1}$

Option 4:
$\frac{2_n}{2n-1}$

Correct Answer:
$\frac{2_n-1}{2n+1}$

Solution:



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From $t = 0$ to $t = n - 1$

$$S_1 = 0 + \frac{1}{2}a\,(n-1)^2$$

From $t = 0\ to\ t = n$

$$S_2 = 0 + \frac{1}{2}a\,(n)^2$$

From $t = 0\ to\ t = n + 1$

$$S_3 = 0 + \frac{1}{2}a\,(n+1)^2$$

$$S_n = S_2 - S_1$$

$$S_n = \frac{1}{2}a\left[n^2 - \left(n^2 - 2n + 1\right)\right]$$

$$S_{n+1} = S_3 - S_2 = \frac{1}{2}a\left[n^2 + 2n + 1 - n^2\right]$$

$$\frac{S_n}{S_n + 1} = \frac{(2n-1)}{(2n+1)}$$

Q. 18 The escape velocity from the Earth's surface is V. The escape velocity from the surface of another planet having a radius, four times that of Earth and the same mass density is :

Option 1:
V

Option 2:
2V

Option 3:
3V

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Option 4:
4V

Correct Answer:
4V

Solution:

Escape velocity from the Earth's surface $= V = \sqrt{\frac{2GM}{R}}$

Escape velocity from a planet $= V' = \sqrt{\frac{2GM'}{R'}}$

$R' = 4R$

$M = 3 \times \frac{4}{3}\pi R^3$

$M' = 3 \times \frac{4}{3}\pi (R')^3 = 64M$

$\frac{V}{V'} = \sqrt{\frac{M}{M'} \times \frac{R'}{R}} = \sqrt{\frac{1}{64} \times \frac{4R}{R}}$

$\frac{V}{V'} = \frac{1}{4}$

$V' = 4V$

Hence, the answer is the option (4).

Q. 19 The half - life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be:

Option 1:
$\frac{1}{2}$

Option 2:
$\frac{1}{2\sqrt{2}}$

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Option 3:

$\frac{2}{3}$

Option 4:

$\frac{2}{3\sqrt{2}}$

Correct Answer:

$\frac{1}{2\sqrt{2}}$

Solution:

$T_{\frac{1}{2}} = 100 hours$

$\frac{N_t}{N_o} = 2^{-t/\frac{T}{2}} = 2^{-150/100}$

$\frac{N_t}{N_o} = 2^{-3/2} = \frac{1}{2\sqrt{2}}$

Q. 20 In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replace the first cell, then at what length of the wire, the balance point occurs?

Option 1:

60 cm

Option 2:

21.6 cm

Option 3:

64 cm

Option 4:

62 cm

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Correct Answer:
60 cm

Solution:

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{1.5}{2.5} = \frac{36}{l_2}$$

$$\frac{3}{5} = \frac{36}{l_2} \qquad l_2 = 60Cm$$

Q. 21 Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? ($g = 10m/s^2$)

Option 1:
10.2 kW

Option 2:
8.1 kW

Option 3:
12.3 kW

Option 4:
7.0 kW

Correct Answer:
8.1 kW

Solution:

$$\frac{dm}{dt} = 15\frac{kg}{s}$$

$$\frac{P_{out}}{P_{input}} = \eta = \frac{90}{100}$$

$$P_{output} = \eta P_{input}$$

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$$= 0.9 \times \left(\frac{dmgh}{dt}\right)$$

$$= 0.9 \times 15 \times 10 \times 60$$

$$P_{output} = 8.1kW$$

Hence, the answer is the option (2).

Q. 22 The electron concentration in an n-type semiconductor is the same as the hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

Option 1:
Current in n-type = current in p-type

Option 2:
Current in p-type > current in n-type

Option 3:
Current in n-type < current in p-type

Option 4:
No current will flow in p-type ,current will only flow in n-type

Correct Answer:
Current in n-type = current in p-type

Solution:
In both n-type and p-type semiconductors, electric current is due to a minority charge carrier when an electric field is applied across each of them. (as there is no diffusion)

Current in n-type =current in p-type

Hence, the answer is the option (1).

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Q. 23 Mathch Column-I and Column-II and Choose the correct match from the given choice

Column-I	Column-II
(A) Root Mean square speed of gas molecules	(P) $\frac{1}{3}nm$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3RT}{M}}$
(C) Average kinetic energy of a molecules	(R) $\frac{5}{2}RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2}KgT$

Option 1:
(A)-(R),(B)-(P),(C)-(S),(D)-(Q)

Option 2:
(A)-(Q),(B)-(R),(C)-(S),(D)-(P)

Option 3:
(A)-(Q),(B)-(P),(C)-(S),(D)-(R)

Option 4:
(A)-(R),(B)-(Q),(C)-(P),(D)-(S)

Correct Answer:
(A)-(Q),(B)-(P),(C)-(S),(D)-(R)

Solution:

Root Mean square speed of gas molecules=$V_{rms}=\sqrt{\frac{3RT}{M}}$

The pressure exerted by the ideal gas $=P=\frac{1}{3}nm\left(\bar{V}\right)^2$

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Average KE of a molecules$= \frac{3}{2}K_BT$

Total internal energy of 1 mole of a diatomic gas $= \frac{5}{2}RT$

Hence, the answer is the option (3).

Q. 24 A spring is stretched by 5 cm by a force of 10 N. The period of the oscillations when a mass of 2 kg is suspended by it is :

Option 1:
0.0628 s

Option 2:
6.28 s

Option 3:
3.14 s

Option 4:
0.628 s

Correct Answer:
0.628 s

Solution:

$x = 5cm$

$F = 10N$

$K = F/x$

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{200}}$

$T = \frac{\pi}{s} = 0.628s$

Hence, the answer is the option (4).

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Q. 25 The velocity of a small ball of mass M and density d,when dropped in a container filled with glycerine becomes constant after some time.if the density of glycerine is $\frac{d}{2}$,then the viscous force acting on the ball will be:

Option 1:
$\frac{Mg}{2}$

Option 2:
Mg

Option 3:
$\frac{3}{2}Mg$

Option 4:
$2Mg$

Correct Answer:
$\frac{Mg}{2}$

Solution:

$Mg = dvg$

$B = \frac{d}{2} \times vg = \frac{Mg}{2}$

$F_v \rightarrow viscous\ force$

$B + F_v = Mg$

$F_v = Mg - \frac{mg}{2}$

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$$F_v = \frac{mg}{2}$$

Q. 26 A body is executing simple harmonic motion with frequency 'n'. The frequency of its potential energy is:

Option 1:
n

Option 2:
$2n$

Option 3:
$3n$

Option 4:
$4n$

Correct Answer:
$2n$

Solution:

$x = A\sin\omega t$

$$PE = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2A^2\left[\frac{1-\cos 2\omega t}{2}\right]$$

$$pE = \frac{1}{4}m\omega^2A^2[1-\cos(20\omega t)]$$

As, $\omega = 2\pi n$

$$PE = \frac{1}{4}m\omega^2A^2[1-\cos(4\pi nt)]$$

The angular frequency of PE is 2ω

$\therefore$ Frequency=$2n$

Hence, the answer is the option (2).

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Q. 27 A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain the binding energy in the process is:

Option 1:
$0.9\ MeV$

Option 2:
$9.4MeV$

Option 3:
$804\ MeV$

Option 4:
$216\ MeV$

Correct Answer:
$216\ MeV$

Solution:
BE per nucleon of unfragmented nuclei $= 7.6MeV$

BE per nucleon of fragmented nuclei $= 8.5MeV$

An increase in BE per nucleon is $0.9MeV$

For 246 nucleons, total BE is $0.9 \times 240 = 216MeV$

Hence, the answer is the option (4).

Q. 28 A parallel plate capacitor has a uniform electric field $\vec{E}$ in the space between the plates. If the distance between the plates is 'd' and the area of each plate 'A', then the energy stored in the capacitor is :(ϵ_0 =permittivity of free space)

Option 1:
$\frac{1}{2}\epsilon_0 E^2$

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Option 2:
$\epsilon_0 EAd$

Option 3:
$\frac{1}{2}\epsilon_0 E^2 Ad$

Option 4:
$\frac{E^2 Ad}{\epsilon_0}$

Correct Answer:
$\frac{1}{2}\epsilon_0 E^2 Ad$

Solution:

$V = Ed, \quad C = \frac{A\varepsilon_0}{d}$

Energy stored in the capacitor $= \frac{1}{2}CV^2 = \frac{1}{2} \times \frac{A\varepsilon_0}{d} \times E^2 d^2 = \frac{1}{2}\varepsilon_0 E^2 Ad$

Hence, the answer is the option (3).

Q. 29 Polar molecules are the molecules:

Option 1:
having zero dipole moment .

Option 2:
acquire a dipole moment only in the presence of electric field due to displacement of charges.

Option 3:
acquire a dipole moment only when magnetic field is absent.

Option 4:
Having a permanent electric dipole moment .

Correct Answer:
Having a permanent electric dipole moment .

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Solution:
Polar molecules have permanent electric dipole moments.

Hence, the answer is the option (4).

Q. 30 A radioactive nucleus ${}^{A}_{Z}X$ undergoes spontaneous decay in the sequence ${}^{A}_{Z}X \rightarrow Z-1B \rightarrow Z-3 \rightarrow C \rightarrow Z-2D$,where Z is the atomic number of element X.The possible decay particles in the sequence are

Option 1:
α, β^-, β^+

Option 2:
α, β^+, β^-

Option 3:
β^+, α, β^-

Option 4:
β^-, α, β^+

Correct Answer:
β^+, α, β^-

Solution:
${}^{A}_{Z}X \rightarrow \quad B \rightarrow \quad C \rightarrow \quad D$

$Z-1 \quad Z-3 \quad Z-2$

With β^+ decay ,Z decrease by 1 with α-decay,Z decrease by 2(from z-1 to z-3)

with β^- decay , Z increase by 1(from z-3 to z-2)

decay particles in the sequence are β^+, α, β^-

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Q. 31 Consider the following statements (A) and (B) and identify the correct answer:

(A) A zener dipole is connected in reverse bias when used as a voltage regulator

(B) The potential barrier of the p-n junction lies between 0.1 V to 0.3 V

Option 1:
(A) and (B) are are correct

Option 2:
(A) and (B) both are incorrect

Option 3:
(A) is correct and (B) is incorrect .

Option 4:
(A) is incorrect but (B) is correct.

Correct Answer:
(A) is correct and (B) is incorrect .

Solution:
The Zener dipole is always connected in reverse based on the potential barrier of the p-n junction lies between 0.1 to 0.3 V nearly.

Hence, the answer is the option (3).

Q. 32 A capacitor of capacitance 'C' is connected across an AC source of voltage V given by $V = V_o \sin \omega t$. The displacement current between the plates of the capacitor would then be given by:

Option 1:
$I_d = V_0 \omega C \cos \omega t$

Option 2:
$I_d = \frac{V_0}{\omega C} \cos \omega t$

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Option 3:

$I_d = \frac{V_0}{\omega C} \sin \omega t$

Option 4:

$I_d = V_0 \omega C \sin \omega t$

Correct Answer:

$I_d = V_0 \omega C \sin \omega t$

Solution:

$V = V_o \sin \omega t$

Displacement current is in magnitude equal to conduction current.

$$I_d = I_c = \frac{V}{X_C} = \frac{V_0 \sin \omega_t}{(1/\omega C)}$$

$$I_d = V_0 \omega C (\sin \omega t)$$

Hence, the answer is the option (4).

Q. 33 If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of:

Option 1:

$[M^2]\ [L^{-1}]\ [T^0]$

Option 2:

$[M]\ [L^{-1}]\ [T^{-1}]$

Option 3:

$[M]\ [L^0]\ [T^0]$

Option 4:

$[M^2]\ [L^{-2}]\ [T^{-1}]$

Correct Answer:

$[M^2]\ [L^{-1}]\ [T^0]$

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Solution:

$$[E] = [M^1\ L^2\ T^{-2}]$$

$$[G] = \left[\frac{Fr^2}{m_1 m_2}\right] = [M^{-1}L^3T^{-2})$$

$$\left[\frac{E}{G}\right] = [M^2L^{-1}T^0]$$

Hence, the answer is the option (1).

Q. 34 A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. if a parallel beam of light falling on 'A' leaves'B' as a parallel beam ,then the distance 'd' in cm will be:

Option 1:
25

Option 2:
15

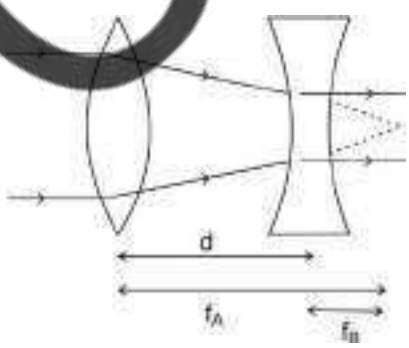
Option 3:
50

Option 4:
30

Correct Answer:
15

Solution:

$f_A = 20\ \text{cm} \quad f_B = 5\ \text{cm}$



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$f_A - f_B = d$

$d = 15 \text{ cm}$

Q. 35 If force [F], acceleration [A], and time [T] are chosen as the fundamental physical quantities, find the dimensions of energy.

Option 1:
$[F]\,[A]\,[T]$

Option 2:
$[F]\,[A]\,[T^2]$

Option 3:
$[F]\,[A]\,[T^{-1}]$

Option 4:
$[F]\,[A^{-1}]\,[T]$

Correct Answer:
$[F]\,[A]\,[T^2]$

Solution:

$[Energy] = [F]^x[A]^y[T]^z$

$[M^1L^2T^{-2}] = [M^1L^1T^{-2}]^x[L^1T^{-2}]^y[T^1]^z$

$[M^1L^2T^{-2}] = M^xL^{x+y}T^{-2y-2x+z}$

$x = 1$

$x + y = 2 \quad ; y = 1$

$-2x - 2y + z = -2$

$z = -2 + 4 = 2$

$[\text{Energy}] = [F]^1[A]^1[T]^2$

Hence, the answer is the option (2).

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Q. 36 A Step-Down Transformer connected to an AC mains supply of 220V is made to operate at 11 V,44W lamp. Ignoring power losses in the Transformer, what is current in the primary circuit?

Option 1:
0.2A

Option 2:
0.4A

Option 3:
2A

Option 4:
4A

Correct Answer:
0.2A

Solution:
Since power loss is ignored

$$\eta = 1 = \frac{(power)\,output}{(power)\,input}$$

$$P_{input} = P_{output}$$

$$V_p I_p = V_s I_s$$

$$I_p = \frac{44}{220} = \frac{1}{5} = 0.2A$$

Hence, the answer is the option (1).

Q. 37 A ball of mas 0.15kg is dropped from a height 10m. strikes to the ground and rebounded to the same height.The magnitude of impluse imparted to the ball is (g=10m/s^2) nearly:

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Option 1:
0 kg m/s

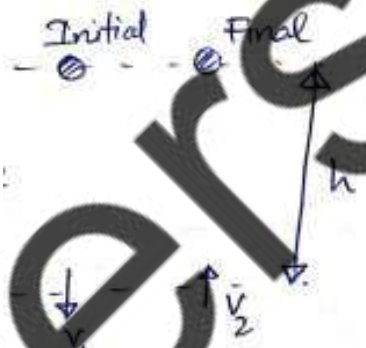
Option 2:
4.2 kg m/s

Option 3:
2.1 kg m/s

Option 4:
1.4 kg m/s

Correct Answer:
4.2 kg m/s

Solution:



$m = 0.15kg$

$h = 10m$

$\sin\ a$ ball rise to same height

$V_1 = V_2 = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 10}$

$V_1 = V_2 = 10\sqrt{2}$

Impulse=$\Delta P = P_f - P_i$

$= (+mv_2) - (-mv_1)$

$= (0.15)\,(v_2 + v_1)$

$= (0.15)\,20\sqrt{2} = 4.2kg\frac{m}{s}$

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Q. 38 A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle 'θ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection ,θ,is then given by :

Option 1:

$$\theta = \cos^{-1}\left(\frac{gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

Option 2:

$$\theta = \cos^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{\frac{1}{2}}$$

Option 3:

$$\theta = \sin^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{\frac{1}{2}}$$

Option 4:

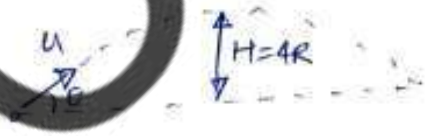
$$\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

Correct Answer:

$$\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

Solution:

$$T = \frac{2\pi R}{V}; V = \frac{2\pi R}{T}$$



$$H = 4R = \frac{u^2 \sin^2\theta}{2g}$$

$$u = v$$

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$$H = \frac{V^2 \sin^2\theta}{2g}$$

$$\sin^2 = \frac{2gh}{v^2} = \frac{2gH}{4\pi^2 R^2 \frac{}{T^2}}$$

$$\sin\theta = \left(\frac{2gHT^2}{4\pi^2 R^2}\right)^{\frac{1}{2}} = \left(\frac{2g\,(4g)\,T^2}{4\pi^2 R^2}\right)\frac{1}{2}$$

$$\sin\theta = \left(\frac{2gT^2}{\pi^2 R}\right)\frac{1}{2}$$

Q. 39 A particle of mass 'm' is projected with a velocity v=$Kv_e\,(K<1)$from the surface of the earth.$(v_e = escape - velocity)$. The maximum height above the surface reached by the particle is :

Option 1:

$$R\left(\frac{K}{1-K}\right)^2$$

Option 2:

$$R\left(\frac{K}{1+K}\right)^2$$

Option 3:

$$\frac{R^2K}{1+K}$$

Option 4:

$$\frac{RK^2}{1-K^2}$$

Correct Answer:

$$\frac{RK^2}{1-K^2}$$

Solution:

$$v = Kv_e\,(K<1)$$

$$TE_1 = TE_2$$

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$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right) = 0 + \left(\frac{-GMm}{R+h}\right)$$

$$\frac{1}{2}mxK^2 \times \frac{2GM}{R} - \frac{GMM}{R} = \frac{-GMm}{R+h}$$

$$\frac{GMm}{R}\left[K^2 - 1\right] = \frac{-GMm}{R+h}$$

$$\left[1 - K^2\right] = \frac{R}{R+h}$$

$$\frac{1}{(1-K^2)} = 1 + \frac{h}{R}$$

$$\frac{K^2R}{(1-K^2)} = h$$

Hence, the answer is the option (4).

Q. 40 A car starts from rest and accelerates at 5 m/s^2. At t=4s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at t=6s?

(Take $g = 10m/s^2$)

Option 1:
20 m/s ,5m/s^2

Option 2:
20 m/s,0

Option 3:
$20\sqrt{2}m/s, 0$

Option 4:
$20\sqrt{2}m/s, 10m/s^2$

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Correct Answer:

$20\sqrt{2}m/s, 10m/s^2$

Solution:

$u = 0, a = 5m/s^2$[along horizontal direction]

at$t = 4s, v_x = u_x + a_x t$

$$v_x = 20m/s$$

The ball when released at $t = 4s$

From $t = 4s$ to $t = 6s$

$v_y = 0; v_y = u_y + a_y t$

$$v_y = 0 + (10)(2) = 20m/s$$

$$a = \sqrt{a_x^2 + a_y^2}$$

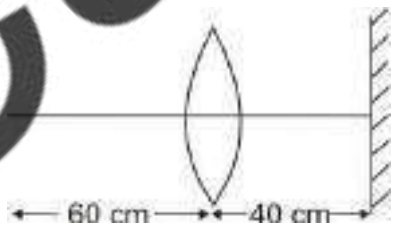
$$= \sqrt{0 + (10)^2} = 10m/s^2$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(20)^2 + (20)^2}$$

$$v = 20\sqrt{2}m/s$$

Hence, the answer is the option (4).

Q. 41 A point object is placed at a distance of 60 cm from a convex lens of focal lenght 30 cm . If a plane mirror were put perpendicular to the principle axis of the lens and at a distance of 40cm form it, the final image would be formed at the distance of :



Option 1:

20 cm form the lens,it would be real image.

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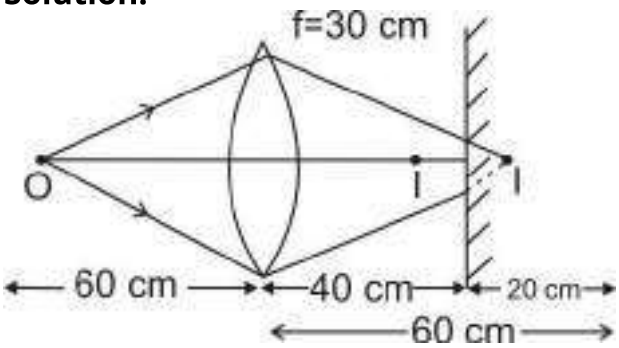
Option 2:
30 cm from the lens, it would be a real image .

Option 3:
30 cm from the plane mirror, it would be a virtual image .

Option 4:
20 cm from the plane mirror, it would be a virtual image .

Correct Answer:
20 cm form the lens,it would be real image.

Solution:



Object at 2f ,image at 2f

$I \rightarrow$ virtual object for plane mirror

$I' \rightarrow$ Real Image by plane mirror

Q. 42 Twenty seven drops of same size are charged at 220 V each.They combine to form a bigger drop.calculate the potential of the bigger drop.

Option 1:
660V

Option 2:
1320V

Option 3:
1520V

Option 4:
1980V

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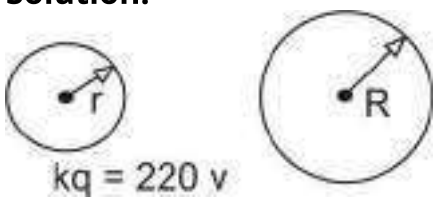
Correct Answer:
1980V

Solution:



$$V = \frac{kg}{r} = 220v$$

$$27 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$R = 3r$$

Q Total $27q$(by Change convervation)

$$V_{Bigger\ drop} = \frac{kQ_{total}}{R} = \frac{27kq}{3r}$$

$$= q\,(220)$$

$$V_{Bigger\ drop} = 1980v$$

Q. 43 In the product $\overrightarrow{F} = q\left(\overrightarrow{v} \times \overrightarrow{B}\right) = q\,\overrightarrow{v} \times \left(B\hat{i} + B\hat{j} + B_0\hat{k}\right)$.

For $q = 1 and\ \overrightarrow{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}\ \overrightarrow{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$. What will be the complete expression for $\overrightarrow{B}$?

Option 1:
$-8\hat{i} - 8\hat{j} - 6\hat{k}$

Option 2:
$-6\hat{i} - 6\hat{j} - 8\hat{k}$

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Option 3:

$8\hat{i}+8\hat{j}-6\hat{k}$

Option 4:

$6\hat{i}+6\hat{j}-8\hat{k}$

Correct Answer:

$-6\hat{i}-6\hat{j}-8\hat{k}$

Solution:

$$\bar{F}=q\left(\bar{v}\times\bar{B}\right)=q\bar{v}\times\left(B\hat{i}+B\hat{j}+B_0\hat{k}\right)$$

$$\left[4\hat{i}-20\hat{j}+12\hat{k}\right]=\left[2\hat{i}+4\hat{j}+6\hat{k}\right]\times\left[B\hat{i}+B\hat{j}+B_0\hat{k}\right]$$

$$=\hat{i}\left(4B_0-6B\right)-\hat{j}\left(2B_0-6B\right)+\hat{k}\left(-2B\right)$$

by equating the coefficients of equation ,we get

$-2B=12$

$B=-6$

$4B_0-6B=4$

$2B_0-6B=20$

$2B_0=-16, B_0=-8$

$B=-6$

$\bar{B}=B\hat{i}+B\hat{j}+B_0\hat{K}$

$\bar{B}=-6\hat{i}-6\hat{j}-8\hat{k}$

Hence, the answer is the option (2).

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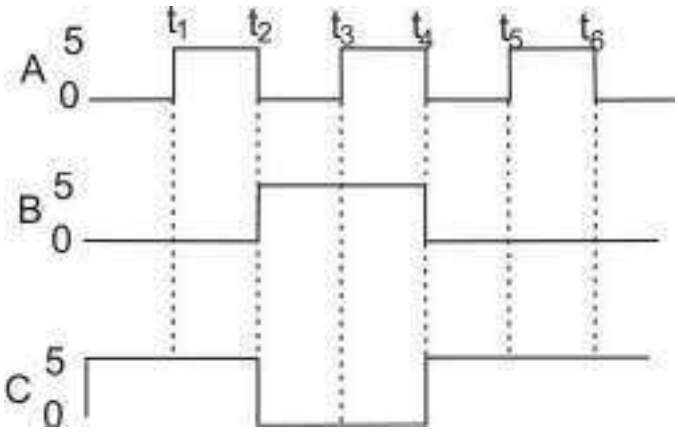
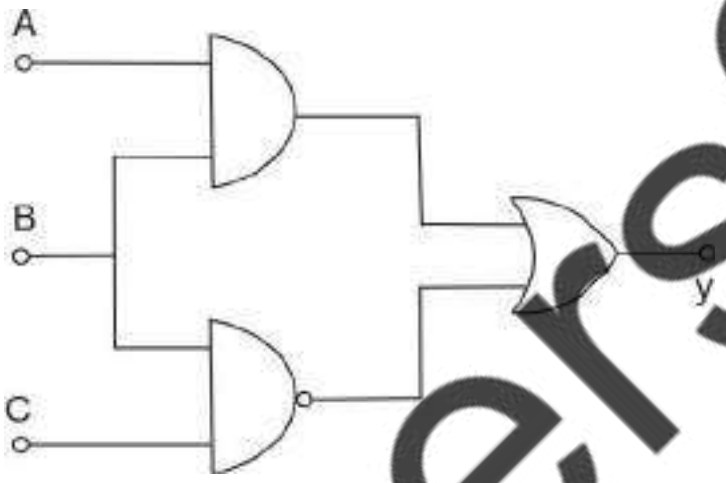
Q. 44 For the given circuit the input digital signals are applied at the terminals A,B and C ,What would be the output at the terminal y?





Option 1:

Option 2:

Option 3:

Option 4:

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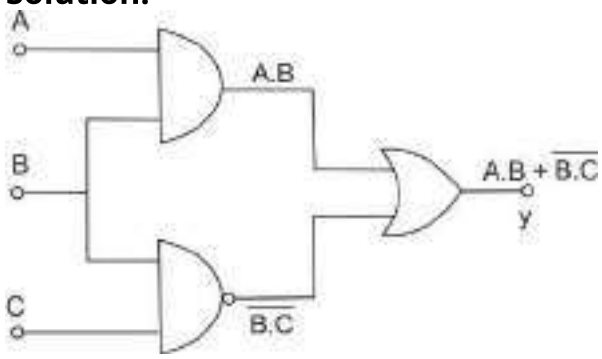
Correct Answer:

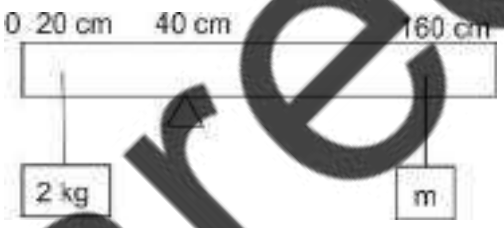
5 V

Solution:



A	B	C	A.B + $\overline{B.C}$ (Y)
0	0	1	1
1	0	1	1
0	1	0	1
1	1	0	1

Q. 45 A uniform rod of lenght 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark.A mass of 2 kg is suspended form the rod at 20 cm and another unkonwn mass 'm' is suspended from the rod at 160 cm mark as shown in the figure.Find the value of 'm' such that the rod is in equilibrium (g=10m/s^2)



Option 1:

$\frac{1}{2}kg$

Option 2:

$\frac{1}{3}kg$

Option 3:

$\frac{1}{6}kg$

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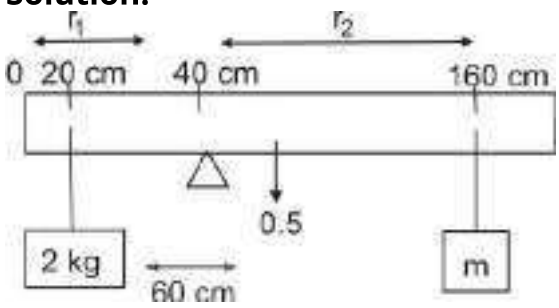
Option 4:

$\frac{1}{12}kg$

Correct Answer:

$\frac{1}{12}kg$

Solution:



for rotational equilibrium

$T_1 = T_2$

$(2 \times 10)\, r_1 = (m \times 10)\, r_2 + (0.5 \times 10)(0.6)$

$r_1 = 0.2m \quad r_2 = 1.2m$

$2 \times 0.2 - 0.3 = m\,(1.2)$

$\left(m = \frac{1}{12}kg\right)$

Q. 46 A uniform conducting wire of length 12a and resiatance 'R' is wound up as a current carrying coil in the shape of

(i) an equilateral triangle of side 'a'

(ii) a square of side 'a'

the magnetic dipole moments of the coil in each case respectively are

Option 1:

$\sqrt{3}Ia^2 and 3Ia^2$

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Option 2:

$3Ia^2 and Ia^2$

Option 3:

$3Ia^2 and 4Ia^2$

Option 4:

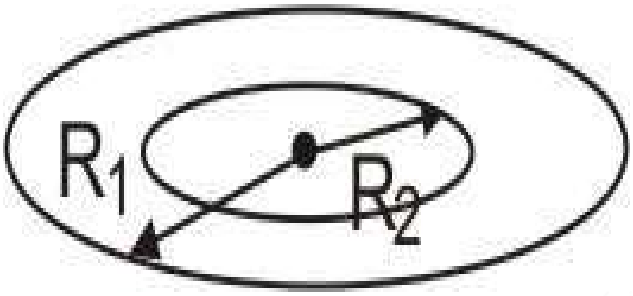
$4Ia^2 and 3Ia^2$

Correct Answer:

$\sqrt{3}Ia^2 and 3Ia^2$

Solution:

$Length = 12a$



$M_1 = n_1 I A_1$ $\qquad$ $M_2 = n_2 I A_2$

$4 \times I\left(\frac{1}{2} \times a \times \frac{\sqrt{3}}{2}a\right)$ $\qquad$ $= 3I\left(a^2\right)$

$M_1 = \sqrt{3}Ia^2$ $\qquad$ $M_2 = 3Ia^2$

Q. 47 Two conducting circular loops od raddi R_1 and R_2 are placed in the same plane with their centres coinciding .If $R_1 >> R_2$, the mutual inductance M between them will be directly proportional to :

Option 1:

$\frac{R_1}{R_2}$

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Option 2:

$\frac{R_2}{R_1}$

Option 3:

$\frac{R_1^2}{R_2}$

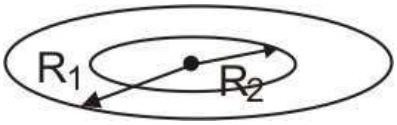
Option 4:

$\frac{R_2^2}{R_2}$

Correct Answer:

$\frac{R_2^2}{R_2}$

Solution:



$\phi = MI$

$BA = MI$

$\frac{\mu_0 I}{2R_1} \times TIR_2^2 = MI$

$M \propto \frac{R_2^2}{R_1}$

Q. 48 From a circular ring of mass 'M' and radius 'R' an are corresponding to a 90^0 sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the center of the ring and perpendicular to the plane of the ring is 'K'times MR^2, Then the value of 'K' is :

Option 1:

$\frac{3}{4}$

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Option 2:

$\frac{7}{8}$

Option 3:

$\frac{1}{4}$

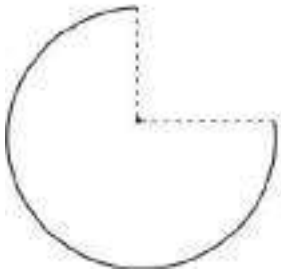
Option 4:

$\frac{1}{8}$

Correct Answer:

$\frac{3}{4}$

Solution:

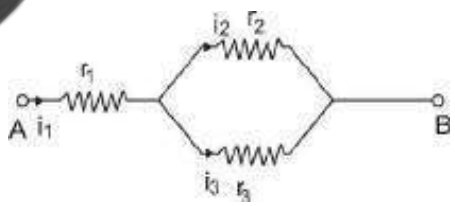


Remaining mass$= M' = \frac{3}{4}M$

$I = M'R^2 = \frac{3}{4}MR^2 = KMR^2$

$K = \frac{3}{4}$

Q. 49 Three resistance having resistances r_1, r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terme of resistances used in the circuit is :



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Option 1:

$\frac{r_1}{r_2+r_3}$

Option 2:

$\frac{r_2}{r_2+r_3}$

Option 3:

$\frac{r_1}{r_1+r_2}$

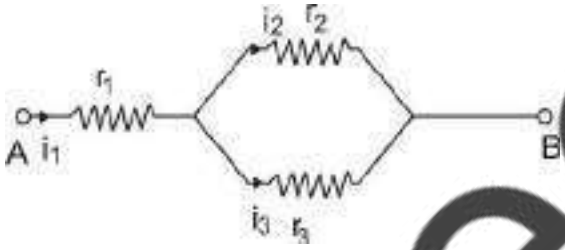
Option 4:

$\frac{r_2}{r_1+r_3}$

Correct Answer:

$\frac{r_2}{r_2+r_3}$

Solution:



$i_2 r_2 = i_3 r_3$ [resistance in parallel]

$i_2 + i_3 = i_1$[kirchoff's curret law]

$(i_1 - i_3)\, r_2 = i_3 r_3$

$i_1 r_2 = i_3 (r_3 + r_2)$

$\frac{i_3}{i_1} = \frac{r_2}{r_2+r_3}$

Q. 50 A series LCR circuit containing 5.0 H inductor $80\mu F$ capacitor and 40Ω resistor is connected to 230 V variable frequency AC source. The angular frequencies of the source at which power transferred to the circuit are half the power at the resonant angular frequency are likely to be :

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Option 1:
25 rad/s and 75 rad/s

Option 2:
50 rad/s and 25 rad/s

Option 3:
46 rad/s and 54 rad/s

Option 4:
42 rad/s and 58 rad/s

Correct Answer:
50 rad/s and 25 rad/s

Solution:

$L = 5H$

$C = 80\mu F$

$R = 40\Omega$

$P = \frac{1}{2}P_0$

$P_0 \rightarrow$ Power at resonance
$w_0 \rightarrow$ Resonance angular frequency
at resonance, $X_c = X_L$

$$w_0 C = \frac{1}{w_0 L}$$

$$w_0 = \frac{1}{\sqrt{LC}} = \frac{1}{20 \times 10^{-3}}$$

$$w_0 = 50\text{rad/s}$$

$P = V_{rms} I_{rms} \cos\phi_1$

$P_0 = V_{rms} I_{rms} \cos\phi_0$

$\cos\phi_0 = \frac{1}{2}$

$\cos\phi_1 = \frac{1}{2}$ $\cos\phi_o = \frac{R}{Z}$

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$$\frac{R}{Z_1} = \frac{1}{2} \qquad \phi_o = 0^o; \cos\phi_o = 1$$

$$80 = Z_1 = \sqrt{(X_L - X_C)^2 + R^2}$$

$$6400 = (X_L - X_C)^2 + 1600$$

$$4800 = (X_L - X_C)^2$$

$$4800 = \left(wL - \frac{1}{wC}\right)^2$$

$$40\sqrt{3} = \left(wL - \frac{1}{wC}\right)$$

$$w = 25 rad/s$$

$$w_o = 50\, rad/s \quad w = 25\ rad/s$$

Hence, the answer is the option (2).

Chemistry

Q. 1 Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them.

Option 1:
Noble gases are sparingly soluble in water.

Option 2:
Noble gases have very high melting and boiling point.

Option 3:
Noble gases have weak dispersion force.

Option 4:
Noble gases have large positive values of electron gain enthalpy.

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Correct Answer:
Noble gases have very high melting and boiling point.

Solution:
Noble gases are usually monoatomic and have very weak interaction forces. They usually have very low melting and boiling points.
The stronger the interaction between particles of gas more boiling and melting points.

Hence, the answer is the option (2).

Q. 2 The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is:

Option 1:
7

Option 2:
5

Option 3:
2

Option 4:
3

Correct Answer:
3

Solution:
Body-centred unit cells are possible for Orthorhombic, cubic and tetragonal types of Bravais lattice unit cells.

Hence, out of the 14 possible bravais lattice, body-centred unit cells exist for only 3.

Therefore, the correct option is (4).

Q. 3 Which are among the following is the correct option for right relationship between C_p and C_v for one mole of ideal gas?

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Option 1:
$C_p + C_v = R$

Option 2:
$C_p - C_v = R$

Option 3:
$C_p = RC_v$

Option 4:
$C_v = RC_p$

Correct Answer:
$C_p - C_v = R$

Solution:
The relation between $C_p \ and \ C_v$ for one mole of an ideal gas is given as

$$C_P - C_V = R.$$

Hence, the answer is the option (3).

Q. 4 Zr(Z=40) and Hf(Z=72) have similar atomic and ionic raddi because of:

Option 1:
Belonging to same group

Option 2:
Diagonal relationship

Option 3:
Lanthanoid contraction

Option 4:
Having Similar chemical properties

Correct Answer:
Lanthanoid contraction

Solution:

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Zr and Hf have an almost similar size. This is because of Lanthanoid contraction.

Hence, the answer is the option (3).

Q. 5 The Maximum Temperature that can be achieved in a blast furnace is :

Option 1:
Upto 1200 K

Option 2:
Upto 2200 K

Option 3:
Upto 1900 K

Option 4:
Upto 5000 K

Correct Answer:
Upto 2200 K

Solution:
The maximum temperature that can be achieved in a Blast furnace is 2200K.

The correct answer is option 2.

Q. 6 Statement I: Acid strength increases in the order given as
$HF << HCl << HBr << HI.$

Statement II: As the size of elements F, Cl, Br, I increases down the group, the bond strength of $HF, HCl, HBr, and\ HI$ decreases and so the acid strength increases

In the light of the above statements, choose the correct answer from the option given below

Option 1:
Both Statement I and Statement II are true

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Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but Statement II is false.

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true

Solution:
Statement I and Statement II are both correct.

As the size of the halogen increases, the acidic strength of the hydrogen halide increases.

Also, as the difference in the size of the halogen and hydrogen increases, the bond length increases and the bond dissociation energy decreases.

Lower is bond dissociation energy, more is acidic strength.
The correct answer is option 1.

Q. 7 The compound which shows metamerism is :

Option 1:
C_5H_{12}

Option 2:
C_3H_8O

Option 3:
C_3H_6O

Option 4:
$C_4H_{10}O$

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Correct Answer:

$C_4H_{10}O$

Solution:

Metamerism is shown by compounds that can have different alkylation around a central atom or a group.

$C_4H_{10}O$ can show metamerism which can be illustrated as

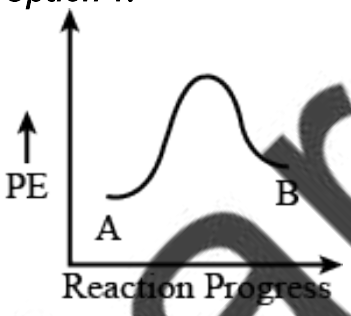
$C_2H_5 - O - C_2H_5 \ ; \ CH_3 - O - C_3H_7$

Here the alkylation is different in the compounds around the heteroatom O.

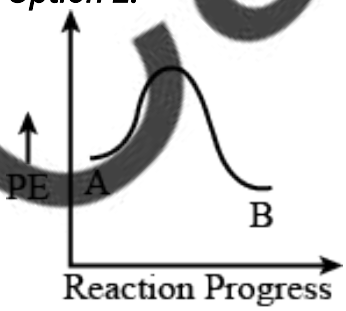
Hence, the answer is the option (4).

Q. 8 For a reaction $A \rightarrow B$, the enthalpy of reaction is $-4.2kJmol^{-1}$and enthalpy of activation is 9.6 kJ mol^{-1} The correct potential energy profile for the reaction is shown in the option.

Option 1:



Option 2:



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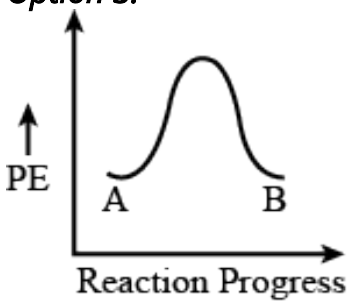
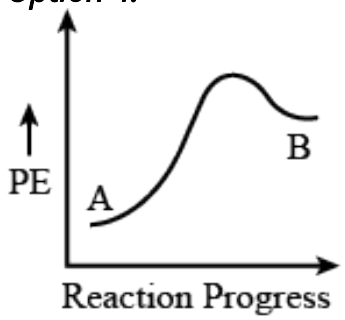
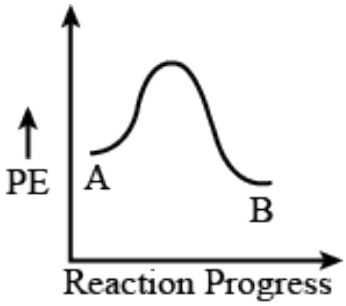
Option 3:



Option 4:

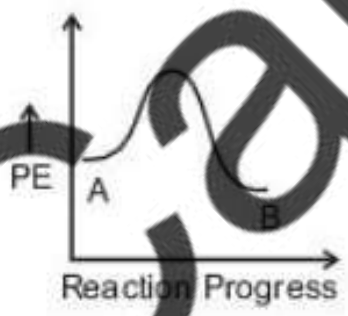


Correct Answer:



Solution:

For an exothermic reaction, the enthalpy of the reactant will be greater than the enthalpy of the product and as a result, the energy profile diagram will look like



The correct answer is option 2.

Q. 9 The right option for the statement "Tyndall effect is exhibited by" is :

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Option 1:
NaCl Solution

Option 2:
Glucose solution

Option 3:
Starch Solution

Option 4:
Urea Solution

Correct Answer:
Starch Solution

Solution:
Tyndall effect is shown by Colloidal solutions.

Out of the given solutions, only starch solution can have particles of colloidal size.

Urea, glucose, and NaCl form true solutions and hence, cannot show Tyndall Effect.

The correct answer is option 3.

Q. 10 BF_3 is a planar and electron-deficient compound. Hybridization and number of electrons around the central atom, respectively are:

Option 1:
$sp^3 \ and \ 4$

Option 2:
$sp^3 \ and \ 6$

Option 3:
$sp^2 \ and \ 6$

Option 4:
$sp^2 \ and \ 8$

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Correct Answer:
$sp^2 \ and \ 6$

Solution:
BF_3 is a trigonal planar molecule with 3 sigma bonds formed by Boron with the 3 Fluorine atoms.

$$H = \frac{3+3}{2} = 3$$

Hence, the hybridization is sp^2 and the number of electrons around boron is 6.

Hence, the answer is the option (3).

Q. 11 Match List -I with List -II

List I	List II
(a) Pcl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal Planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the corret Answer from the given below

Option 1:
(a)-(iv),(b)-(iii),(c)-(i) ,(d)-(ii)

Option 2:
(a)-(ii),(b)-(iii),(c)-(iv),(d)-(i)

Option 3:
(a)-(iii),(b)-(i),(c)-(iv),(d)-(ii)

Option 4:
(a)-(iv),(b)-(iii),(c)-(ii),(d)-(i)

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Correct Answer:
(a)-(iv),(b)-(iii),(c)-(i) ,(d)-(ii)

Solution:
To calculate hybridization

$$H = \frac{V + M - Q}{2}$$

Where V is the number of valence electrons in the central atom

M is the Number of surrounding atoms

Q is the charge present on the molecule (if any)
PCl_5 has trigonal bipyramidal shape and sp^3d hybridization.

$$H = \frac{5+5}{2} = 5 \quad sp^3d$$

SF_6 has sp^3d^2 hybridization and octahedral geometry.

$$H = \frac{6+6}{2} = 6 \quad sp^3d^2$$

BrF_5 has sp^3d^2 hybridization and square pyramidal geometry.

$$H = \frac{7+5}{2} = 6 \quad sp^3d^2$$

BF_3 has sp^2 hybridization and trigonal planar geometry.

$$H = \frac{3+3}{2} = 3 \quad sp^2$$

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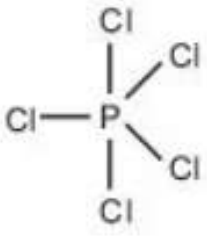
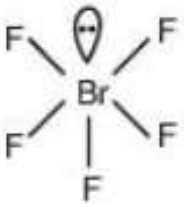
(a) 

Trigonal Bipyramidal

(b)

Octahedral

(c) 

Square Pyramidal

(d)

Trigonal Planar

Hence, the correct answer is option 1.

Q. 12 The structures of beryllium chloride in solid state and vapour phase, are

Option 1:
Chain and dimer, respectively

Option 2:
Linear in both

Option 3:
Dimer and Linear,respectively

Option 4:
Chain in both

Correct Answer:
Chain and dimer, respectively

Solution:
$BeCl_2$ exists as a polymeric solid and a dimeric gas. The structures in the solid state and the gaseous state is given below:

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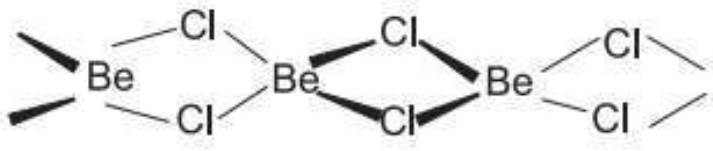
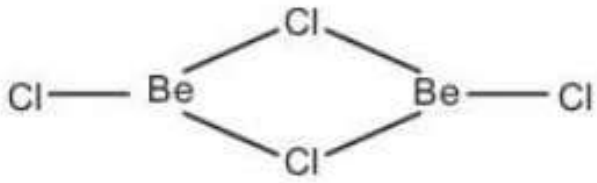


Structure of $BeCl_2$ Solid



Structure of $BeCl_2$ dimer in gaseous state

Hence, the correct answer is option 1.

Q. 13 The correct sequence of bond enthalpy of 'C-X' bond is :

Option 1:

$CH_3 - F < CH_3 - Cl < CH_3 - Br < CH_3 - I$

Option 2:

$CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$

Option 3:

$CH_3 - F < CH_3 - Cl > CH_3 - Br > CH_3 - I$

Option 4:

$CH_3 - Cl > CH_3 - F > CH_3 - Br > CH_3 - I$

Correct Answer:

$CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$

Solution:

Bond strength depends upon the extent of overlap. As the size difference increases, the bond energy or the bond strength decreases as the overlap becomes less effective.

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Thus, as the size of the halogen increases, the C-X bond energy decreases.

Therefore, the correct order is

$$CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$$

Hence, the answer is the option (2).

Q. 14 Which one of the following polymers is prepared by addition polymerisation?

Option 1:
Teflon

Option 2:
Nylon - 6,6

Option 3:
Novolac

Option 4:
Dacron

Correct Answer:
Teflon

Solution:
Teflon is Polytetrafluoroethene and it is prepared by the addition polymerisation of tetrafluoroethene.

Nylon, Novolac and Dacron are condensation polymers.

Hence, the correct answer is Option (1)

Q. 15 Dihedral angle of least stable conformer of ethane is :

Option 1:
120^0

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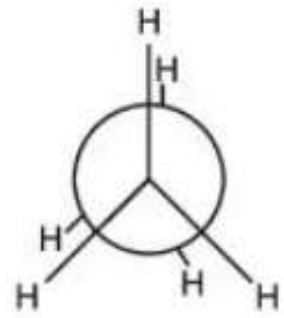
Option 2:
180^0

Option 3:
60^0

Option 4:
0^0

Correct Answer:
0^0

Solution:
The least stable conformer of ethane is the eclipsed form in which the dihedral angle is Zero.



Dihedral angle = 0°

Q. 16 An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is :[Atomic wt.C is 12, H is 1]

Option 1:
CH

Option 2:
CH_2

Option 3:
CH_3

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Option 4:
CH_4

Correct Answer:
CH_3

Solution:
The mass ratio of Carbon : Hydrogen = 78 : 22

$\therefore$ the molar ratio of Carbon : Hydrogen = 6.5 : 22

$\therefore$ the simplest ratio of Carbon : Hydrogen = 1 : 3.38

Thus, based on the given data, the most probable answer is $\mathrm{CH_3}$

Hence, the answer is the option (3).

Q. 17 The molar conductance of NaCl ,HCl,and $\mathrm{CH_2COONa}$ at infinite dilution are 126.45, 426.16 and $91.0\ \mathrm{S\ cm^2 mol^{-1}}$ respectively.The molar conductance of $\mathrm{CH_3COOH}$ at infinte dilution is _____.

Choose the right option for your answer:

Option 1:
$201.28\ \mathrm{S\ cm^2\ mol^{-1}}$

Option 2:
$390.71\ \mathrm{S\ cm^2\ mol^{-1}}$

Option 3:
$698.28\ \mathrm{S\ cm^2 mol^{-1}}$

Option 4:
$540.48\ \mathrm{S\ cm^2\ mol^{-1}}$

Correct Answer:
$390.71\ \mathrm{S\ cm^2\ mol^{-1}}$

Solution:
Given,

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$\Lambda_m^o, NaCl = 126.45$

$\Lambda_m^o, HCl = 426.16$

$\Lambda_m^o, CH_3COONa = 91$

$\Lambda_m^o, CH_3COOH = ?$

We know from Kohlrausch's Law of independent migration of ions,

$\Lambda_m^o, CH_3COOH = \Lambda_m^o, CH_3COONa + \Lambda_m^o, HCl - \Lambda_m^o, NaCl$

$\Lambda_m^o, CH_3COOH = 91 + 426.16 - 126.45 = 390.71$

Hence, the answer is the option (2).

Q. 18 Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is correct but **Statement II** is false.

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Option 4:
Statement I is incorrect but **Statement II** is true.

Correct Answer:
Both **Statement I** and **Statement II** are false.

Solution:
Aspirin and Paracetamol are Non Narcotic analgesics while Morphine and Heroin are Narcotic Analgesics.

Thus, Statement I and II are both incorrect.

Q. 19 The following solutions were prepared by dissolving 10 g of glucose $(C_6H_{12}O_6)$ in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2), and 10 g of sucrose $(C_{12}H_{22}O_{11})$ in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :

Option 1:
$P_2 > P_1 > P_3$

Option 2:
$P_1 > P_2 > P_3$

Option 3:
$P_2 > P_3 > P_1$

Option 4:
$P_3 > P_1 > P_2$

Correct Answer:
$P_2 > P_1 > P_3$

Solution:
We know that,

$$\pi = CRT$$

So under similar conditions of temperature and volume of solution, the species with the greater number of moles has a greater value of Osmotic Pressure.

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Order of moles : Urea > Glucose > Sucrose

Order of Osmotic Pressure:

$$P_2 > P_1 > P_3$$

Hence, the answer is the option (1).

Q. 20 Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :

Option 1:
Calcium chloride

Option 2:
Strontium chloride

Option 3:
Magnesium chloride

Option 4:
Beryllium chloride

Correct Answer:
Beryllium chloride

Solution:
For chlorides of group 2 elements, going down the group covalent character increases. This is because going down the group charge to size ratio decreases and the polarizing power of cation decreases and the lower is polarising power, the more ionic character.

Polarizing power of group 2 elements is $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

and, the covalent character is

$$BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$$

hence, the correct answer is option 4.

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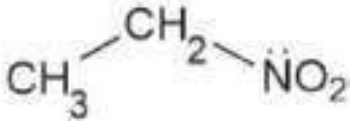
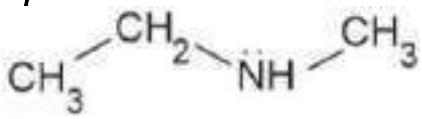
Q. 21 Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.

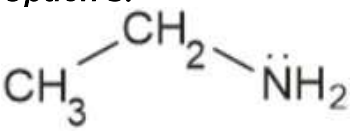
Option 1:



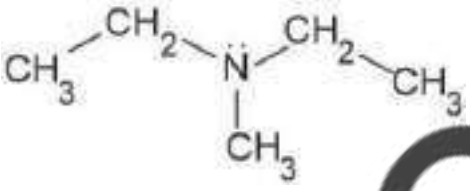
Option 2:



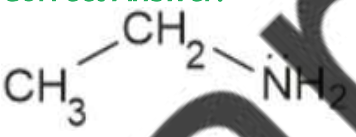
Option 3:



Option 4:



Correct Answer:



Solution:

Hinsberg's reagent test is given by both primary as well as secondary amines. However, it is the primary amines only that give a product which is soluble in Alkali.

This is because there is an acidic hydrogen ($N-H$) present in the addition product with Benzene Sulphonyl Chloride for a Primary amine.

Q. 22 Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are :

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Option 1:
$8, 4$

Option 2:
$6, 12$

Option 3:
$2, 1$

Option 4:
$12, 6$

Correct Answer:
$12, 6$

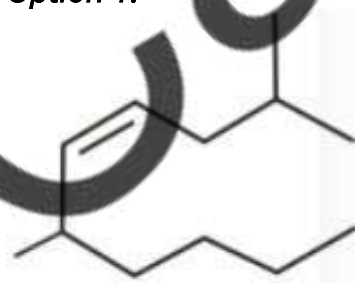
Solution:
In any closed-packed structure containing N number of particles, the number of Tetrahedral Voids is 2N and the number of Octahedral Voids is N.

Now since in a hexagonal closed packed structure, the number of particles is 6, the number of Tetrahedral and Octahedral Voids will be 12 and 6 respectively.

Therefore, the correct option is (4).

Q. 23 The correct structure of 2,6- Dimethyl-dec-4-ene is :

Option 1:



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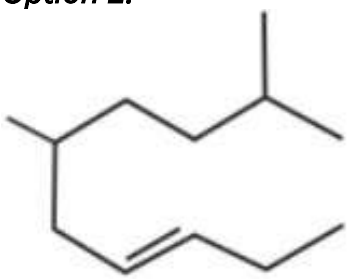
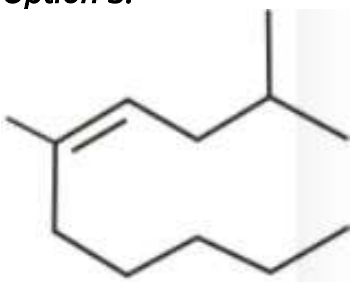
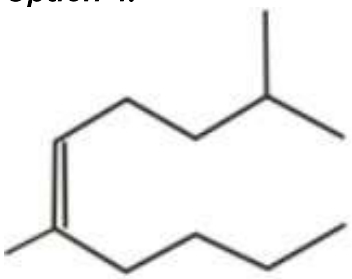
Option 2:



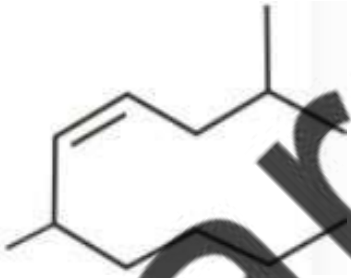
Option 3:



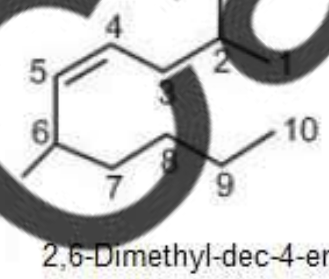
Option 4:



Correct Answer:



Solution:



2,6-Dimethyl-dec-4-ene

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Q. 24 The RBC deficiency is deficiency disease of :

Option 1:
Vitamin B_{12}

Option 2:
Vitamin B_6

Option 3:
Vitamin B_1

Option 4:
Vitamin B_2

Correct Answer:
Vitamin B_{12}

Solution:
Factual Question.

RBC deficiency is caused by deficiency of Vitamin B_{12}

Q. 25 Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature?

Option 1:
Electrolysis

Option 2:
Chromatography

Option 3:
Distillation

Option 4:
Zone refining

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Correct Answer:
Distillation

Solution:
Distillation is used as a process of refining metals having low boiling points like $Zn\ , Hg\ and\ Cd.$.

Hence, the correct answer is option 3.

Q. 26 What is the IUPAC name of the organic compound formed in the following chemical reaction ?

Acetone $\xrightarrow[(ii)H_2O,H^+]{(i)C_2H_5MgBr,\text{dry Ether}}$ product

Option 1:
2 methyl propan-2-ol

Option 2:
pentan - 2-ol

Option 3:
pantan-3-ol

Option 4:
2 - methyl butan- 2- ol

Correct Answer:
2 - methyl butan- 2- ol

Solution:

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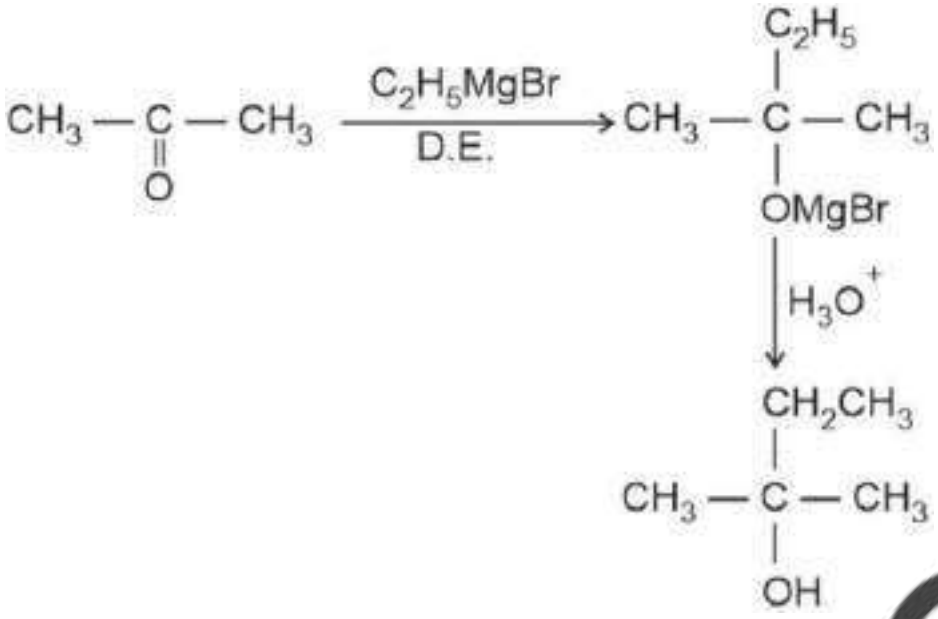
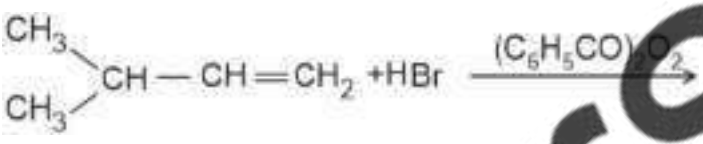
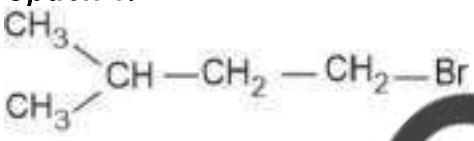


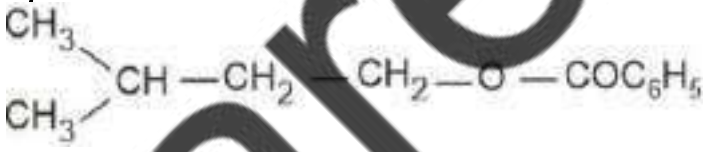
Q. 27 The major product of the following chemical reaction is :



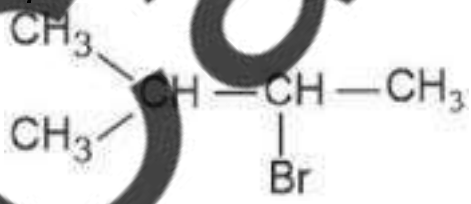
Option 1:



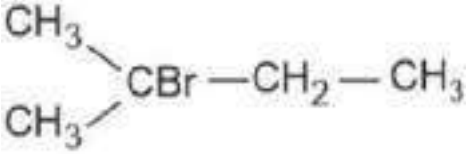
Option 2:



Option 3:



Option 4:



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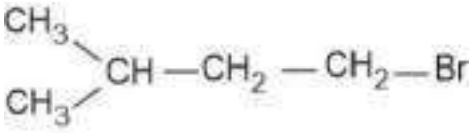
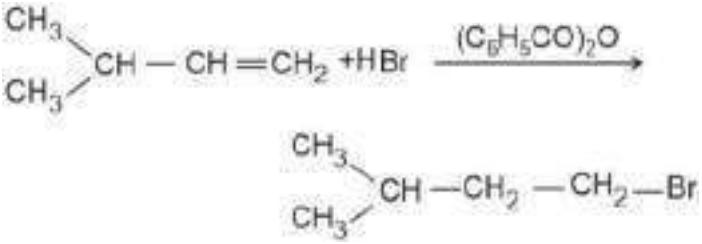
Correct Answer:



Solution:



according to antimarkovnikov Rules,

Q. 28 The **incorrect** statement among the following is :

Option 1:
Actionoid contraction is greater for element to element than Lanthanoid contraction.

Option 2:
Most of the trivalent Lanthanoid ions are colorless in the solid state.

Option 3:
Lanthanoids are good conductors of heat and electricity.

Option 4:
Actionoids are highly reactive metals, especially when finely divided.

Correct Answer:
Most of the trivalent Lanthanoid ions are colorless in the solid state.

Solution:
Most of the trivalent lanthanoids are Colored in both the Solid as well as the aqueous state.

Other statements are all correct.

Hence, the answer is the option (2).

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Q. 29 A particular station of All India Radio, New Dehli, broadcasts on a frequency of 1,368 kHz (kilohertz), the wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light, $\mathrm{e} = 3.0 \times 10^{8} \mathrm{ms}^{-1}$].

Option 1:
$219.3\ m$

Option 2:
$219.2\ m$

Option 3:
$2192\ m$

Option 4:
$21.92\ cm$

Correct Answer:
$219.3\ m$

Solution:
We know that,

$$\nu = \frac{c}{\lambda}$$

$$\therefore \lambda = \frac{\mathrm{c}}{\nu} = \frac{3 \times 10^{8}\ \mathrm{m/s}}{1368 \times 10^{3}\ \mathrm{s}} = 219.3\ \mathrm{m}$$

Hence, the answer is the option (1).

Q. 30 Which of the following reactions is the metal displacement reaction ? Choose the right option.

Option 1:
$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

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Option 2:

$$Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2CrCl_3$$

Option 3:

$$Fe + 2HCl \rightarrow FeCl_2 + H_2 \uparrow$$

Option 4:

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2 \uparrow$$

Correct Answer:

$$Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2CrCl_3$$

Solution:

Decomposition reaction leads to the breakdown of a compound into two or more components at least one of which must be in the elemental state.

Both reactions (1) and (4) are examples of decomposition reactions.

In a **displacement reaction**, an ion (or an atom) in a compound is replaced by an ion (or an atom) of another element. There are two types of displacement reactions:

Metal displacement - A metal in a compound can be displaced by another metal in the uncombined state

Non-metal displacement - The non-metal displacement redox reactions include hydrogen displacement and a rarely occurring reaction involving oxygen displacement.

Reactions (2) and (3), both are examples of displacement reactions,

But reaction (2) is an example of a metal displacement reaction because metal is displacing there.

Therefore, the correct option is (2).

Q. 31 Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures:

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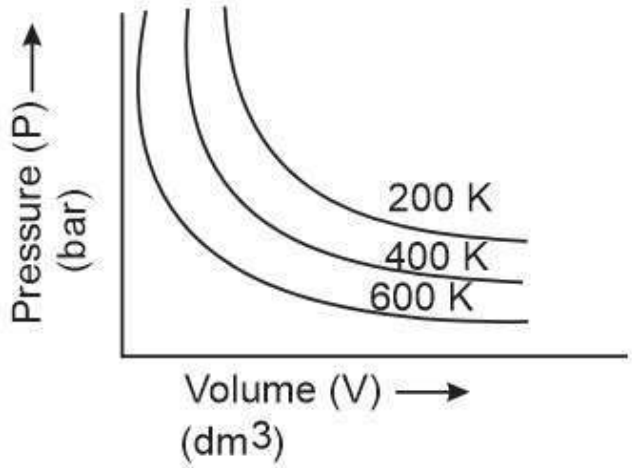
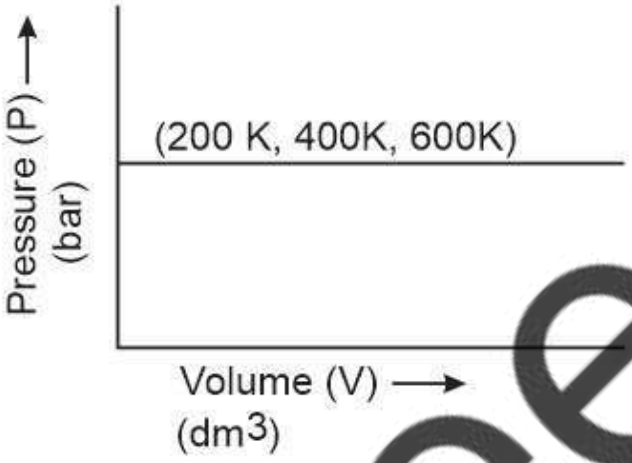
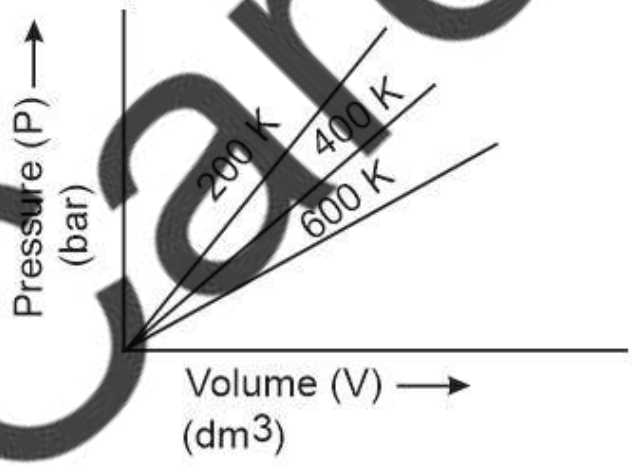
Option 1:



Option 2:



Option 3:



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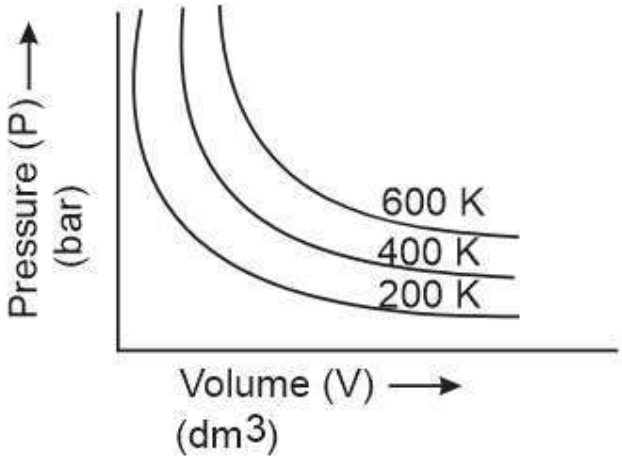
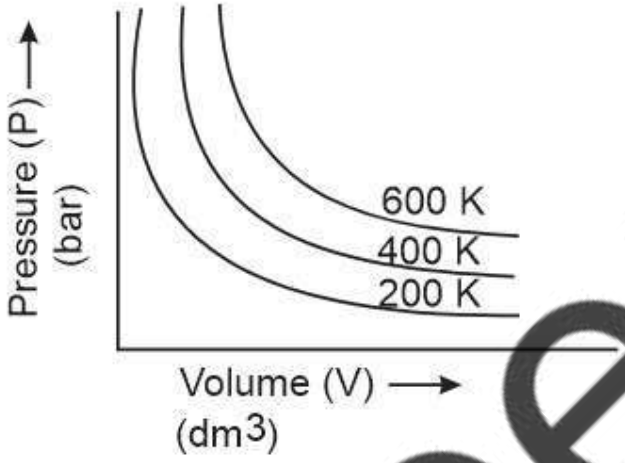
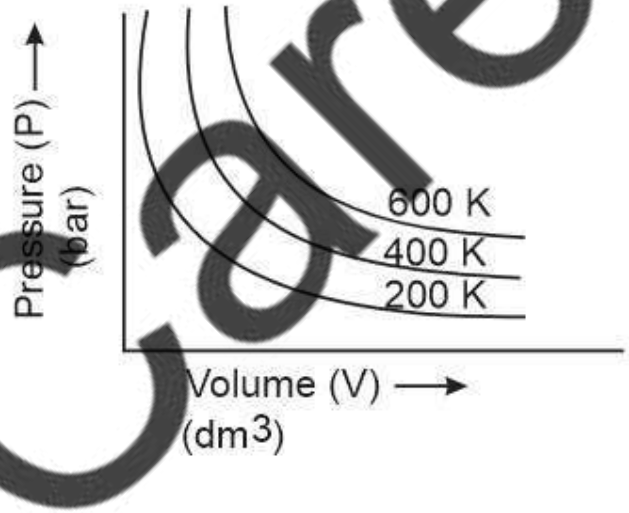
Option 4:



Correct Answer:



Solution:



Boyle's Law is -

at a constant temperature, the pressure of a fixed amount (i.e., number of moles n) of gas varies inversely with its volume.

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$$P \propto \frac{1}{V} \Rightarrow P = \frac{k}{V} \Rightarrow PV = k$$

Where k is proportionality constant and equal to nRT.

The graph between P vs. V should be a rectangular hyperbola.

If T increases the value of k increases then at the same value of V, P will increase due to k increases.

Therefore, the correct option is (4).

Q. 32 The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethylammonium acetate solution is :

Option 1:
8.50

Option 2:
5.50

Option 3:
7.75

Option 4:
6.25

Correct Answer:
7.75

Solution:
We know the formula for the pH of salt of the weak acid and weak base -

$$pH = 7 + \frac{1}{2}(pK_a - pK_b)$$

Given,

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$pK_a = 3.27$ and $pK_b = 4.77$

So,

$$pH = 7 + \frac{1}{2}(4.77 - 3.27)$$

$$pH = 7.75$$

Hence, the answer is the option (3).

Q. 33 The major product formed in dehydrohalogenation reaction of 2 - Bromo pentane is Pent - 2-ene. This product formation is based on ?

Option 1:
Saytzeff's Rule

Option 2:
Hund's Rule

Option 3:
Hofmann Rule

Option 4:
Huckel's Rule

Correct Answer:
Saytzeff's Rule

Solution:
Saytzeff's Rules:

" More substituted alkene is the major product"

Hence, the answer is the option (1).

Q. 34 Ethylene diaminetetraacetate (EDTA) ion is :

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Option 1:
Hexadentate ligand with four "O" and two "N" donor atoms

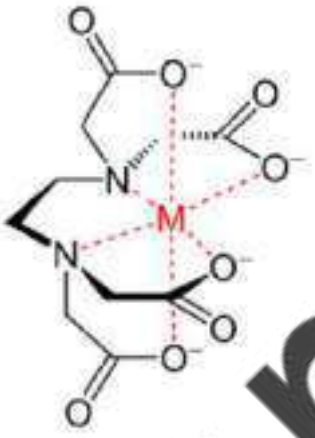
Option 2:
Unidentate ligand

Option 3:
Bidentate ligand with two "N" donor atoms

Option 4:
Tridentate ligand with three "N" donor atoms

Correct Answer:
Hexadentate ligand with four "O" and two "N" donor atoms

Solution:
The structure of EDTA is -



It is a Hexadentate ligand with four "O" and two "N" donor atoms.

Therefore, the correct option is (1).

Q. 35 Tritium, a radioactive isotope of hydrogen, emits which of the following particles?

Option 1:
Beta (β^-)

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Option 2:
Alpha (α)

Option 3:
Gamma (γ)

Option 4:
Neutron (n)

Correct Answer:
Beta (β^-)

Solution:
Tritium is a radioactive isotope of Hydrogen and it emits low-energy Beta particles (β^-).

Therefore, the correct option is (1).

Q. 36 Match List - I with List - II

List - I	List - II
(a) $[Fe(CN)_6]^{3-}$	(i) 5.92 BM
(b) $[Fe(H_2O)_6]^{3+}$	(ii) 0 BM
(c) $[Fe(CN)_6]^{4-}$	(iii) 4.90 BM
(d) $[Fe(H_2O)_6]^{2+}$	(iv) 1.73 BM

Choose the correct options from the given below:

Option 1:
(a) - (iv), (b) - (ii), (c)- (i), (d)- (iii)

Option 2:
(a) - (ii), (b) - (iv), (c)- (iii), (d)- (i)

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Option 3:
(a) - (i), (b) - (iii), (c)- (iv), (d)- (ii)

Option 4:
(a) - (iv), (b) - (i), (c)- (ii), (d)- (iii)

Correct Answer:
(a) - (iv), (b) - (i), (c)- (ii), (d)- (iii)

Solution:
CN^- is a strong field ligand and will cause pairing while H_2O is a weak field ligand and will not cause pairing

Thus, Fe^{3+} which has d^5 configuration and will have 1 unpaired electron in the complex with CN^- and 5 unpaired electrons in the complex with H_2O

And, Fe^{2+} which has d^6 configuration and will have 0 unpaired electron in the complex with CN^- and 4 unpaired electrons in the complex with H_2O

Complex	unpaired electrons (n)	Magnetic moment (BM)
$[Fe(CN)_6]^{3-}$	1	1.73
$[Fe(H_2O)_6]^{3+}$	5	5.92
$[Fe(CN)_6]^{4-}$	0	0
$[Fe(H_2O)_6]^{2+}$	4	4.9

Hence, the answer is the option (4).

Q. 37 In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?

Option 1:
$HF < HCl < HBr < HI$: Increasing acidic strength

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Option 2:
$H_2O < H_2S < H_2Se < H_2Te$: Increasing pK_a values

Option 3:
$NH_3 < PH_3 < AsH_3 < SbH_3$: Increasing acidic character

Option 4:
$CO_2 < SiO_2 < SnO_2 < PbO_2$: Increasing oxidizing power

Correct Answer:
$H_2O < H_2S < H_2Se < H_2Te$: Increasing pK_a values

Solution:
(1) is correct. On moving down the group, acidic strength increases

(2) is incorrect. The greater the acidic strength, the greater the Ka value and the lesser the pKa value.

(3) is correct. On moving down the group, acidic strength increases

(4) is correct. On moving down group 14, the stability of the +4 oxidation state decreases and the tendency to accept electrons increases. Thus, the oxidizing power increases

Hence, the answer is the option (2).

Q. 38 Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one liter at 0^oC is :

[Given $R = 0.082L \text{ atm } \text{ mol}^{-1}K^{-1}, T = 273 \ K$]

Option 1:
2.518

Option 2:
2.602

Option 3:
25.18

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Option 4:
26.02

Correct Answer:
25.18

Solution:
In the question, the pressure needed to find, Volume, and Temperature are given.

We can use PV =nRT formula.

V = 1 litre

T = 0^oC = 273 K

n = mole = mass/molar mass

mass is given of each. and molar mass we know that.

so,

$$n_{O_2} = \frac{4}{32} = \frac{1}{8}$$

$$n_{H_2} = \frac{2}{2} = 1$$

So, total moles will be -

$$n_t = \frac{1}{8} + 1 = \frac{9}{8}$$

The formula will be for mixture -

$$P_tV = n_tRT$$

$$P_t \times 1 = \frac{9}{8} \times 0.082 \times 273$$

$$P_t = 25.18 \text{ atm}$$

Therefore, the correct option is (3).

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Q. 39 Which of the following molecules is non-polar in nature ?

Option 1:
$POCl_3$

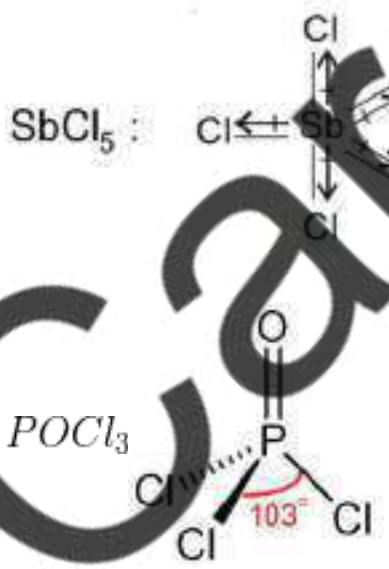
Option 2:
CH_2O

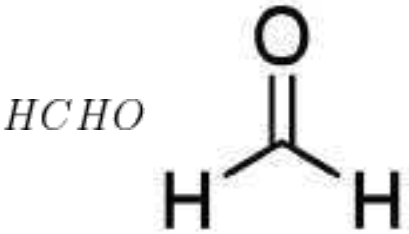
Option 3:
$SbCl_5$

Option 4:
NO_2

Correct Answer:
$SbCl_5$

Solution:
$SbCl_5$ molecule is non-polar in nature due to its symmetrical shape -





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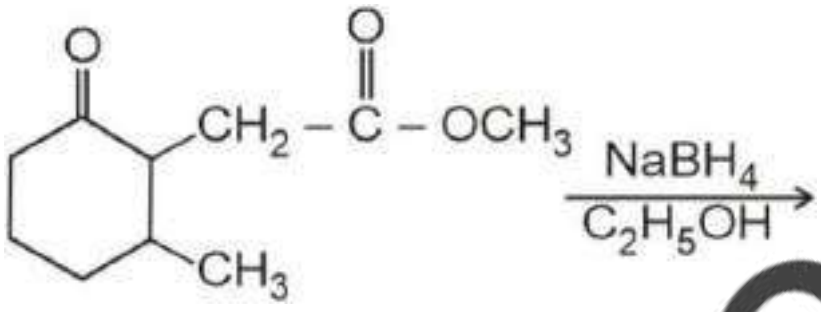
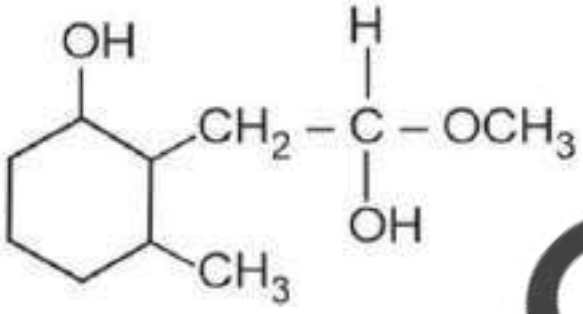
NO_2 has a bent shape and asymmetric hence is also polar in nature.,

Therefore, the correct option is (3).

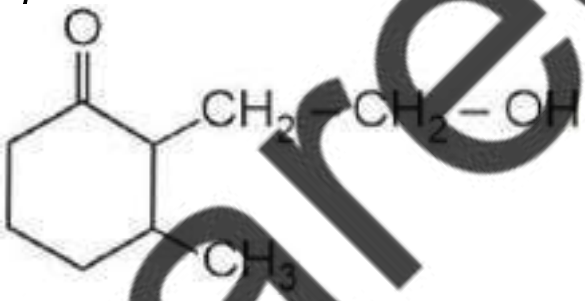
Q. 40 The product formed in the following chemical reaction is :



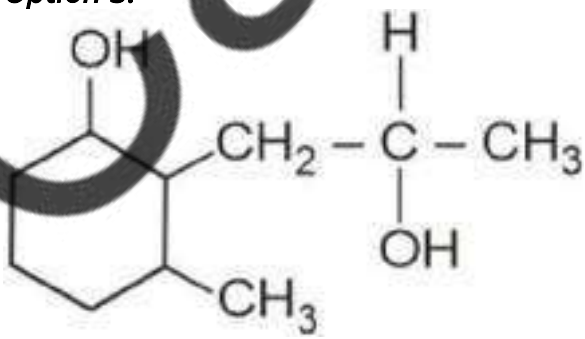
Option 1:



Option 2:



Option 3:



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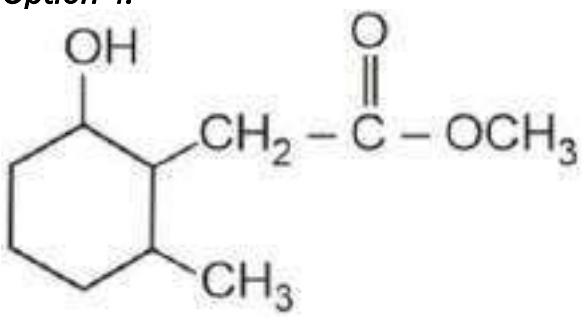
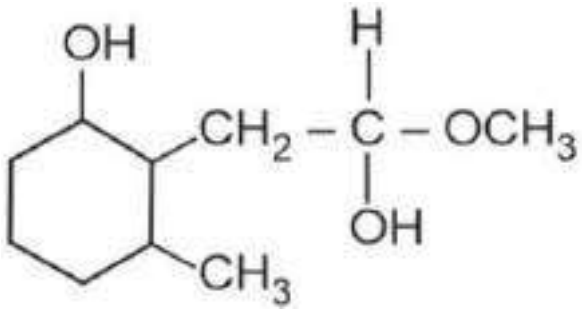
Option 4:

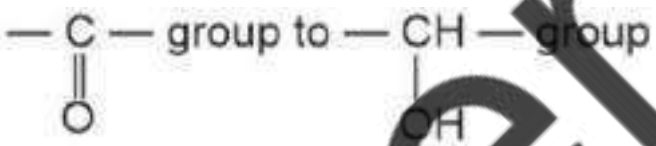


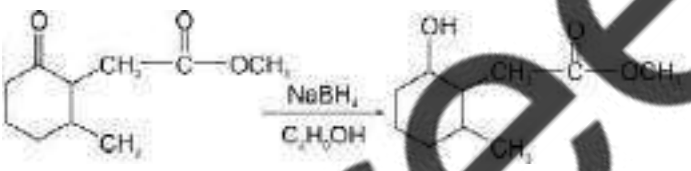
Correct Answer:



Solution:

$NaBH_4/C_2H_5OH$ is a weak reducing agent. It cannot reduce –COOR group

It reduces  group to group



Q. 41

The slope of the Arrhenius Plot $ln\ k\ v/s\ \frac{1}{T}$ the first-order reaction is $-5 \times 10^3 K$. The value of E_a of the reaction is. Choose the correct option for your answer.

$\left[\text{Given R} = 8.314\ \text{JK}^{-1}\text{mol}^{-1}\right]$

Option 1:

$41.5\ \text{kJ mol}^{-1}$

Option 2:

$83.0\ \text{kJ mol}^{-1}$

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Option 3:

$166\ \text{kJ mol}^{-1}$

Option 4:

$-83\ \text{kJ mol}^{-1}$

Correct Answer:

$41.5\ \text{kJ mol}^{-1}$

Solution:

We know the Arrhenius equation

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A + \ln e^{-E_a/RT}$$

$$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$$

The slope of $\left(\ln k \text{ v/s } \frac{1}{T}\right)$ of the above equation -

$$m = -\frac{E_a}{R}$$

Given, m = $-5 \times 10^3\ \text{K}$ and $R = 8.314 \text{JK}^{-1}\ \text{mol}^{-1}$

$$-5 \times 10^3 = -\frac{E_a}{8.314}$$

$$E_a = 5 \times 10^3 \times 8.314\ \text{J/mol}$$

$$E_a = 41.57 \times 10^3\ \text{J/mol}$$

$$E_a \simeq 41.5\ \text{kJ/mol}$$

Hence, the answer is the option (1).

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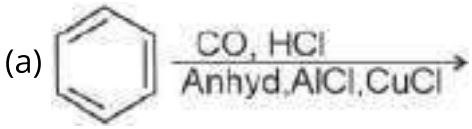
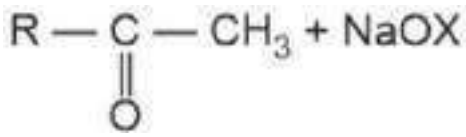
Q. 42 Match List - I with List - II

List - I	List - II
(a) 	(i) Hell - Volhard- Zelinsky reaction
(b) 	(ii) Gattermann- Koch reaction
(c) $R-CH_2-OH+R'COOH \xrightarrow{Conc.H_2SO_4}$	(iii) Haloform reaction
(d) $R-CH_2COOH \xrightarrow[(ii)H_2O]{(i)X_2/Red\ P)}$	(iv) Esterification

Choose the correct answer from the options given below.

Option 1:
(a)-(iv), (b)-(i), (c)- (ii), (d)-(iii)

Option 2:
(a) - (iii), (b)-(ii), (c)- (i), (d)-(iv)

Option 3:
(a) - (i), (b)-(iv), (c)- (iii), (d)-(ii)

Option 4:
(a) - (ii), (b)-(iii), (c)- (iv), (d)-(i)

Correct Answer:
(a) - (ii), (b)-(iii), (c)- (iv), (d)-(i)

Solution:

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List-I **List-II**

(a) $\xrightarrow[\text{Anhydrous } AlCl_3/CuCl]{CO,\ HCl}$ (ii) Gattermann-Koch reaction

(b) $R-\overset{\overset{O}{\|}}{C}-CH_3 + NaOX \longrightarrow$ (iii) Haloform reaction

(c) $R-CH_2OH + R'COOH \xrightarrow{\text{conc. } H_2SO_4}$ (iv) Esterification

(d) $R-CH_2COOH \xrightarrow[(ii)\ H_2O]{(i)\ X_2/\text{Red P}}$ (i) Hell-Volhard Zelinsky reaction

Correct answer is **option (4).**

Q. 43 For the irreversible expansion of an ideal gas under isothermal conditions, the correct option is :

Option 1:
$\Delta U = 0, \Delta S_{total} = 0$

Option 2:
$\Delta U \neq 0, \Delta S_{total} \neq 0$

Option 3:
$\Delta U = 0, \Delta S_{total} \neq 0$

Option 4:
$\Delta U \neq 0, \Delta S_{total} = 0$

Correct Answer:
$\Delta U = 0, \Delta S_{total} \neq 0$

Solution:
For irreversible expansion of an ideal gas under isothermal conditions -

We know, for a spontaneous process, $\Delta S_{\text{total}} > 0$ and since the irreversible process is always spontaneous, therefore $\Delta S_{\text{total}} > 0$.

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And $\Delta T = 0$ for the isothermal process.

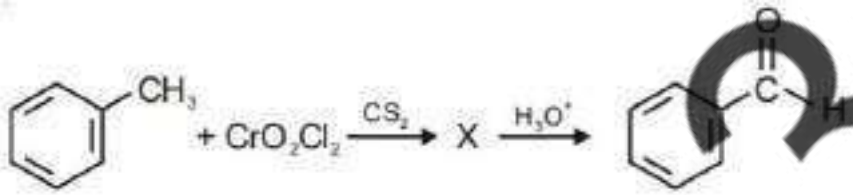
Now, $\Delta U = nC_V\Delta T$

So, $\Delta U = 0$

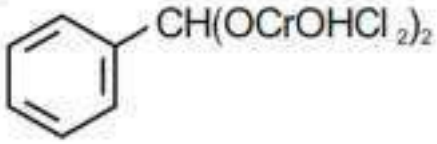
Correct option $\Delta U = 0, \Delta S_{total} \neq 0$.

Hence, the answer is the option (3).

Q. 44 The Intermediate compound 'X' in the following chemical reaction is :



Option 1:



Option 2:

Option 3:

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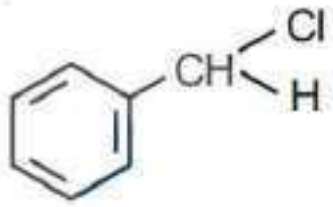
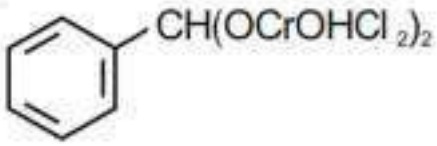
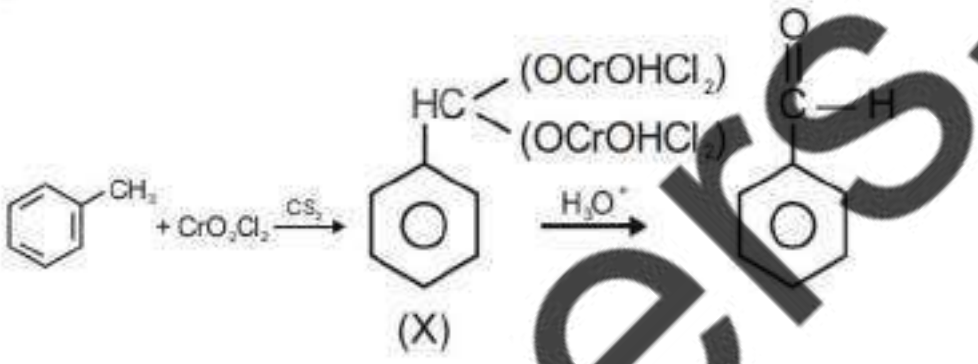
Option 4:



Correct Answer:



Solution:
The reaction will be -



Therefore, the correct option is (1).

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Q. 45 Match List - I with List - II.

List - I	List - II
(a) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(i) Acid rain
(b) $HOCl(g) \overset{h\nu}{\rightarrow} \dot{O}H + \dot{C}l$	(ii) Smog
(c) $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$	(iii) Ozone depletion
(d) $NO_2(g) \overset{h\nu}{\rightarrow} NO(g) + O(g)$	(iv) Troposphere pollution

Choose the correct answer from the optionsgiven below.

Option 1:
(a) - (i), (b)-(ii), (c)-(iii),(d)-(iv)

Option 2:
(a) - (ii), (b)-(iii), (c)-(iv),(d)-(i)

Option 3:
(a) - (iv), (b)-(iii), (c)-(i),(d)-(ii)

Option 4:
(a) - (iii), (b)-(ii), (c)-(iv),(d)-(i)

Correct Answer:
(a) - (iv), (b)-(iii), (c)-(i),(d)-(ii)

Solution:
We know about these reaction -

(a) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ - (iv) Troposphere pollution

(b) $HOCl(g) \overset{h\nu}{\rightarrow} \dot{O}H + \dot{C}l$ - (iii) Ozone depletion

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(c) $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$ - (i) Acid rain

(d) $NO_2(g) \overset{h\nu}{\rightarrow} NO(g) + O(g)$ - (ii) Smog

So, the correct match will be

(a) - (i), (b)-(ii), (c)-(iii),(d)-(iv)

Therefore, the correct option is (3)

Q. 46 The molar conductivity of 0.007 M acetic acid is $20\ \text{S cm}^2\text{mol}^{-1}$. What is the dissociation constant of acetic acid? Choose the correct option.

$$\left[\begin{array}{l} \Lambda^{\circ}_{H^+} = 350\ \text{S cm}^2\ \text{mol}^{-1} \\ \Lambda^{\circ}_{CH_3COO^-} = 50\ \text{S cm}^2\ \text{mol}^{-1} \end{array}\right]$$

Option 1:
$1.75 \times 10^{-4}\ \text{mol L}^{-1}$

Option 2:
$2.50 \times 10^{-4}\ \text{mol L}^{-1}$

Option 3:
$1.75 \times 10^{-5}\ \text{mol L}^{-1}$

Option 4:
$2.50 \times 10^{-5}\ \text{mol L}^{-1}$

Correct Answer:
$1.75 \times 10^{-5}\ \text{mol L}^{-1}$

Solution:
Given,

$$\Lambda_m = 20\ \text{S cm}^2\ \text{mol}^{-1}$$

$$\Lambda^{\circ}_m CH_3COOH = \Lambda^{0}_{CH_3COO^-} + \Lambda^{\circ}_{m^+}$$

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$$\Lambda_m^\circ CH_3COOH = 50 + 350 = 400\ S\ cm^2\ mol^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{20}{400} = \frac{1}{20}$$

$$K_a = \frac{C\alpha^2}{1-\alpha} \simeq C\alpha^2 = 7 \times 10^{-3} \times \left(\frac{1}{20}\right)^2$$

$$K_a = 7 \times 10^{-3} \times \frac{1}{4} \times 10^{-2}$$

$$K_a = 1.75 \times 10^{-5}\ mol\ L^{-1}$$

Hence, the answer is the option (3).

Q. 47 From the following pairs of ions which one is not an iso-electronic pair ?

Option 1:
O^{2-}, F^-

Option 2:
Na^+, Mg^{2+}

Option 3:
Mn^{2+}, Fe^{3+}

Option 4:
Fe^{2+}, Mn^{2+}

Correct Answer:
Fe^{2+}, Mn^{2+}

Solution:
The number of electrons present in the given ions-

$O^{2-} \Rightarrow 18$

$F^- \Rightarrow 18$

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$Na^+ \Rightarrow 10$

$Mg^{2+} \Rightarrow 10$

$Mn^{2+} \Rightarrow 23$

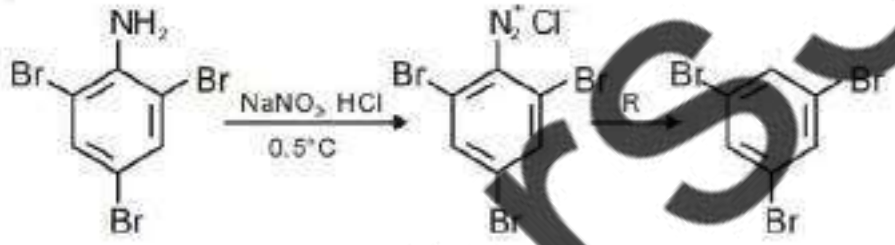
$Fe^{3+} \Rightarrow 23$

$Fe^{2+} \Rightarrow 24$

So, the pair of Fe^{2+}, Mn^{2+} ions is not iso-electronic.

Hence, the answer is the option (4).

Q. 48 The reagent 'R' in the given sequence of chemical reaction is:



Option 1:
H_2O

Option 2:
CH_3CH_2OH

Option 3:
HI

Option 4:
$CuCN/KCN$

Correct Answer:
CH_3CH_2OH

Solution:

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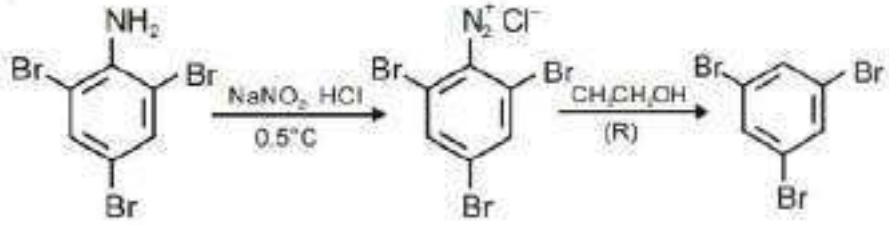

Here CH_3CH_2OH will be R.

Therefore, the correct option is (2).

Q. 49 The correct option for the value of vapour pressure of a solution at 45^oC with benzene to octane in molar ratio 3:2 is :

[At 45^oC vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]

Option 1:
160 mm of Hg

Option 2:
168 mm of Hg

Option 3:
336 mm of Hg

Option 4:
350 mm of Hg

Correct Answer:
336 mm of Hg

Solution:
We know this formula for vapour pressure of a solution -

$$\mathrm{P_{Total}} = \chi_{\mathrm{A}}\mathrm{P_A} + \chi_{\mathrm{B}}\mathrm{P_B}$$

At 45°C,

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vapour pressure of benzene (PA)= 280 mm Hg

vapour pressure of octane (PB) = 420 mm Hg

Given, benzene to octane in molar ratio 3:2

So,

$$\mathrm{P} = \frac{3}{3+2} \times 280 + \frac{2}{3+2} \times 420$$

$$\mathrm{P} = 168 + 168$$

$$\mathrm{P} = 336$$

Therefore, the correct option is (3).

Q. 50 $CH_3CH_2COO^-Na^+ \xrightarrow[Heat]{NaOH,+?} CH_3CH_3 + Na_2CO_3$

Consider the above reaction and identify the missing reagent/chemical.

Option 1:
B_2H_6

Option 2:
Red Phosphorus

Option 3:
CaO

Option 4:
DIBAL-H

Correct Answer:
CaO

Solution:
The reaction will be -

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$$CH_3CH_2COO^-Na^+ \xrightarrow[\text{Heat}]{NaOH+CaO} CH_3CH_3 + Na_2CO_3$$

by heating, the sodium salt of the carboxylic acid with soda lime (NaOH and CaO in the ratio of 3: 1), Alkane is produced.

Hence, the answer is the option (3).

Biology

Q. 1 Which of the following algae contains **mannitol** as reserve food material?

Option 1:
Ectocarpus

Option 2:
Graciliaria

Option 3:
Volvox

Option 4:
Ulothrix

Correct Answer:
Ectocarpus

Solution:
Ectocarpus, a type of brown algae, contains mannitol as a reserve food material. Mannitol is a sugar alcohol that serves as a storage carbohydrate in certain algae and plants.
Hence, the correct answer is option 1:

Q. 2 Amenasalism can be represented as:

Option 1:
Species A (-) : Species B (0)

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Option 2:
Species A (+) : Species B (+)

Option 3:
Species A (-) : Species B (-)

Option 4:
Species A (+) : Species B (0)

Correct Answer:
Species A (-) : Species B (0)

Solution:
Ammensalism: It is an interaction between two organisms of different species in which one species inhibits the growth of other species by secreting certain chemicals.

A(-) is inhibited

B(0) is unaffected.

Q. 3 Match List -I with List - II.

List -I	List - II
(a) Cristae	(i) Primary construction in chromosome.
(b) Thylakoids	(ii) Disc - shaped sacs in Golgi appartus
(c) Centromere	(iii) Infoldings in mitochondria
(d) Cisternae	(iv) Flattered membranes sacs in stromn in plastids

Choose the correct answer from the options given below.

Option 1:
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Option 2:
(a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

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Option 3:
(a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

Option 4:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Correct Answer:
(a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

Solution:
Cristae: Infoldings in mitochondria. **To increase the capacity of the mitochondrion to synthesize ATP**, the inner membrane of mitochondria is folded to form cristae. These folds allow a much greater amount of electron transport chain enzymes and ATP synthase to be packed into the mitochondrion.

Thylakoids: are **membrane-bound compartments inside chloroplasts and cyanobacteria**. They are the site of the light-dependent reactions of photosynthesis. Thylakoids consist of a thylakoid membrane surrounding a thylakoid lumen.

Centromere: Primary construction in **chromosome**.The two chromatids of chromosomes are attached to each other by a narrow area called centromere, also called **primary constriction**. This decides the ratio of arm length called **centromeric index.**

Cisternae: Disc - shaped sacs in **Golgi appartus**. These appear in the form of fla, curved, smooth membrane structures with swollen ends. They have a definite polarity. Their **convex pole** or **proximal pole** or **cis face** is associated with nuclear envelope. The **concave pole** or **distal pole** or **trans face**, is the oldest, which give rise to the **secondary vesicles.**

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Q. 4 The plant hormone used to destroy weeds in a field is :

Option 1:
IAA

Option 2:
NAA

Option 3:
2,4 -D

Option 4:
IBA

Correct Answer:
2,4 -D

Solution:
The 2,4-dichlorophenoxyacetic acid **(2,4-D)** is **a synthetic auxin that has been historically used as one of the most common herbicides to control broadleaf weeds**. At the same time, it also has good potential to enhance storage life and quality of fruits.

Q. 5 Diadelphous stamens are found in:

Option 1:
China Rose

Option 2:
Citrus

Option 3:
Pea

Option 4:
Chia rose and citrus

Correct Answer:
Pea

Solution:
Monadelphous stamens are seen in china rose, these stamens have a single bunch of filaments.

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The plants in which the stamen or anthers are present in three or more than three bundles are known as **polyadelphous stamen**. This type of arrangement is commonly observed in **citrus fruits like lemon.**

Diadelphous stamens are found in pea, having united filaments so that they are arranged in two groups.

Q. 6 A typical angiosperm embryo sac at maturity:

Option 1:
8 - nucleate and 7-celled

Option 2:
7 - nucleate and 8-celled

Option 3:
7- nucleate and 7-celled

Option 4:
8 - nucleate and 8-celled

Correct Answer:
8 - nucleate and 7-celled

Solution:
Nucleus of functional megaspore divides by three mitotic divisions to form 8 nuclei, out of which

(i) Three of them get organised as cells at micropylar end forming egg appartus. One is **egg cell** (n) and two are **synergids (n)** or co-operative cells.

(ii) Three cells at chalazal end form **antipodals** (n) or vegetative cells of female gametophyte.

(iii) Two nuclei (one from each pole) in the centre are called **polar nuclei** (n).

As the mebryo sac matures, these polar nuclei get fused to form a **secondary nucleus** (2n) or definite nucleus.

This constitutes an embryo sac as **7-celled and 8-nucleated** structure.

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Q. 7 Which of the following is correct sequence of steps in PCR (Polymearse Chain Reaction)?

Option 1:
Denaturation, Annealing , Extenstion

Option 2:
Denaturation, Extension, Annealing

Option 3:
Extension, Denaturation, Annealing

Option 4:
Annealing, Denaturation, Extension

Correct Answer:
Denaturation, Annealing , Extenstion

Solution:
PCR is based on three simple steps required for any DNA synthesis reaction:

(1) denaturation of the template into single strands;

(2) annealing of primers to each original strand for new strand synthesis; and

(3) extension of the new DNA strands from the primers.

Q. 8 The site of perception of light in plants during photoperiodism is:

Option 1:
Shoot apex

Option 2:
Stem

Option 3:
Axillary bud

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Option 4:
Leaf

Correct Answer:
Leaf

Solution:
The site of perception of photoperiod is **leaves**. The hormone **florigen**, which is responsible for flowering, induces flowering as it migrates from leaves to shoot apices on induction of required photoperiods.

Q. 9 Which of the following is incorrect statement?

Option 1:
Mature sieve tube elements posses a conspicous nucleus and usual cytoplasmic organelles.

Option 2:
Microbodies are present both in plants and animals cells.

Option 3:
The perinuclear space forms a barrier between the materials present inside the nucleus and the cytoplasm.

Option 4:
Nuclear pores act as passage for proteins and RNA molecules in both the directions between nucleus and cytoplasm.

Correct Answer:
Mature sieve tube elements posses a conspicous nucleus and usual cytoplasmic organelles.

Solution:
Mature sieve tube elements do not have nucleus but have cytoplasm known as a**nucleated** living cells.

Q. 10 The first stable product of CO_2 fixation in sorghum is:

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Option 1:
Pyruvic acid

Option 2:
Oxaloacteic acid

Option 3:
Succinic acid

Option 4:
Phosphoglyceric acid

Correct Answer:
Oxaloacteic acid

Solution:
The first stable product of CO_2 fixation in sorghum is Oxaloacetic acid (OAA), as sorghum is a C_4 plant.

Q. 11 When gene targetting involves gene amplification is attempted in an individual's tissue to treat disease, it is known as:

Option 1:
Biopiracy

Option 2:
Gene therapy

Option 3:
Molecular diagnosis

Option 4:
Safety testing

Correct Answer:
Gene therapy

Solution:

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In gene therapy technique the absent or faulty gene is replaced by a working gene, so that the body can make the correct enzyme or protein and consequently eliminate the root cause of the disease. The first clinical gene therapy was given in 1990 to a 4- year old girl with **adenosine deaminase (ADA) deficiency.**

Q. 12 Match List -I with List - II

List - I	List -II
(a) Protoplast fusion	(i) Totipotency
(b) Plant tissue culture	(ii) Pomato
(c) Meristem culture	(iii) Somaclones
(d) Micropropagation	(iv) Virus free plants

Choose the correct answer from the options given below:

Option 1:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Option 2:
(a) - (ii), (b) - (i) , (c) - (i), (d) - (iii)

Option 3:
(a) - (iii), (b) - (iv) , (c) - (i), (d) - (ii)

Option 4:
(a) - (iv), (b) - (iii) , (c) - (ii), (d) - (i)

Correct Answer:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Solution:
Somatic fusion, also called **protoplast fusion,** is a type of genetic modification in plants by which two distinct species of plants are fused together to form a new hybrid plant with the characteristics of both, a **somatic hybrid.**

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Plant tissue culture is defined as culturing plant seeds, organs, explants, tissues, cells, or protoplasts on a chemically defined **synthetic nutrient media** under sterile and controlled conditions of light, temperature, and humidity.

Meristem culture is defined as the tissue culture technique, which uses apical meristem with 1-3 leaf primordia to prepare clones of a plant by the vegetative propagation. This technique primarily involves the isolation of meristem by applying a V-Shape cut in the stem.

Micropropagation is **the clonal propagation of plants in closed vessels under aseptic conditions**. Inside the vessels, the plants are grown on culture media that contain nutrients and growth regulators, and are described as in vitro, which means 'in glass'. In contrast, soil-grown plants are described as in vivo.

Totipotency is defined in Wikipedia as the ability of a single cell to divide and produce all the differentiated cells in an organism, including extraembryonic tissues. Totipotent cells formed during sexual and asexual reproduction include spores and zygotes.

The **pomato** is a grafted plant that is produced by grafting together a tomato plant and a potato plant, both of which are members of the **Solanum genus** in the **Solanaceae family**.

Somaclones are the plants that are produced through tissue culture and are genetically identical to the parent plant from which they are produced and cultured.

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Q. 13 Which of the following stages of meiosis involves division of centromere?

Option 1:
Metaphase I

Option 2:
Metaphase II

Option 3:
Anaphase II

Option 4:
Telophase II

Correct Answer:
Anaphase II

Solution:
Anaphase II begins with simultaneous splitting of the centromere of each chroosome that allows them to move towards opposite poles of the cell.

Q. 14 Which of the following are not secondary metabolites in plants?

Option 1:
Morphine, codeine

Option 2:
Amino acids, glucose

Option 3:
Vinblastin, curcumin

Option 4:
Rubber, gums

Correct Answer:
Amino acids, glucose

Solution:
Amino acids and glucose are **primary metabolites** in plants.

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Primary metabolites are compounds that are directly involved in the growth and development of a plant whereas secondary metabolites are compounds produced in other metabolic pathways that, although important, are not essential to the functioning of the plant.

Q. 15 Which of the following plants is monoecious?

Option 1:
Carica papaya

Option 2:
Chara

Option 3:
Marchantia polymorpha

Option 4:
Cycas circinalis

Correct Answer:
Chara

Solution:
Monoecious or homothallic condition in Chara (green algae) is used to denote upper oogonium and lower antheridium on the same plant.

Q. 16 Mutations in plant cells can be induced by:

Option 1:
Kinetin

Option 2:
Infrared rays

Option 3:
Gamma rays

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Option 4:
Zeatin

Correct Answer:
Gamma rays

Solution:
Gamma rays are ionizing radiation and used in inducing mutations in seeds and other planting materials such as cuttings, pollen, or tissue-cultured calli.

Q. 17 During purification process for recombinant DNA technology, addition of chilled ethanol precipitates out:

Option 1:
RNA

Option 2:
DNA

Option 3:
Histones

Option 4:
Polysaccharides

Correct Answer:
DNA

Solution:
The purified DNA, after treatment with various enzymes, precipitates out after addition of chilled ethanol. This is viewed as a collection of fine threads in the suspension, and is easily collected. The process is known as **DNA spooling**.

Q. 18 Plants follow different pathways in response to environment or phases of life to form different kinds of structures.This ability is called:

Option 1:
Elasticity

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Option 2:
Flexibility

Option 3:
Plasticity

Option 4:
Maturity

Correct Answer:
Plasticity

Solution:
Plant plasticity refers to a plant's ability to adapt to and cope with changes in its environment. In contrast to animals, which are able to actively move away to avoid challenges such as a predators or a changing climate, plants have acquired the ability to biosynthesize an unprecedented array of structurally complex bioactive natural compounds with specialized roles in order to cope with environmental challenges.

For example, **genetically identical water flea (Daphnia) clones can differ in their morphology depending on whether reared in the absence or presence of a potential predator**.

Q. 19 Which of the following is not an application of PCR (Polymearse Chain Reaction)?

Option 1:
Molecular diagnosis

Option 2:
Gene amplification

Option 3:
Purification of isolated protein

Option 4:
Detection of gene mutation

Correct Answer:
Purification of isolated protein

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Solution:
Some of the **areas of application of PCR** are mentioned here:

(i) Diagnosis of pathogens.

(ii) Diagnosis of specific mutation.

(iii) DNA fingerprinting.

(iv) Detection of specific microorganisms.

(v) In prenatal diagnosis.

(vi) Diagnosis of plant pathogens.

(vii) In palaeontology.

Q. 20 The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is reffered as:

Option 1:
Climax

Option 2:
Climax community

Option 3:
Standing state

Option 4:
Standing crop

Correct Answer:
Standing state

Solution:
Standing state is **amount of inorganic nutrients found in an ecosystem**. It represents part of non-living matter. It circulates between living and non-living components of the ecosystem. It is being regularly depleted and replenished by the living matters. Standing crop is amount of biomass present in an ecosystem.

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Q. 21 The term used for transfer of pollen grains from anther of one plant to the stigma of a different plant which, brings genetically differnt types of pollen grains to stigma, is:

Option 1:
Xenogamy

Option 2:
Geitonogamy

Option 3:
Chasmogamy

Option 4:

Correct Answer:
Xenogamy

Solution:
Transfer of pollen grains from the anther to the stigma of same flower is known as **autogamy** or self pollination.

Geitonogamy is transfer of pollens from anther of one flower to stigma of another flower of the same plant

Xenogamy is the transfer of pollen grains from the anther to the stigma of a different plant.

Q. 22 The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram is called:

Option 1:
Bullet square

Option 2:
Punch square

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Option 3:
Punnett square

Option 4:
Net square

Correct Answer:
Punnett square

Solution:
The Punnett square is a square diagram that is used to predict the genotypes of a particular cross or breeding experiment. It is named after **Reginald C. Punnett**, who devised the approach in 1905. The diagram is used by biologists to determine the probability of an offspring having a particular genotype.

Q. 23 In the equation GPP - R = NPP

R represents:

Option 1:
Radiant energy

Option 2:
Retardation factor

Option 3:
Environment factor

Option 4:
Respiration losses

Correct Answer:
Respiration losses

Solution:
GPP is gross primary productivity that is the rate of total production of biomass by producers.

R is the respiratiory loss. A part of biomass produced by the plant utilized by them for their respiration.

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NPP is the balance of biomass or energy remaining after meeting the cost of respiration of producers.

Q. 24 DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as:

Option 1:
Yellow bands

Option 2:
Bright orange bands

Option 3:
Dark red bands

Option 4:
Bright blue bands

Correct Answer:
Bright orange bands

Solution:
DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as **Bright orange bands** as EtBr absorbs UV radiations at **210-285nm** wavelength and get excited at **605 nm** due to which it flourescence orange colour.

Q. 25 The factor that leads to founder effect in a population is:

Option 1:
Natural selection

Option 2:
Genetic recombination

Option 3:
Mutation

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Option 4:
Genetic drift

Correct Answer:
Genetic drift

Solution:
Founder effect and bottleneck effect are two kinds of Genetic drift.

Q. 26 When the centromere is situated in the middle of two equal arms of chromosomes, the chromosomes is reffered as:

Option 1:
Metacentric

Option 2:
Telocentric

Option 3:
Sub - metacentric

Option 4:
Acrocentric

Correct Answer:
Metacentric

Solution:
On the basis of the location of the centromere, chromosomes are classified into four types: **metacentric, submetacentric, acrocentric, and telocentric**.

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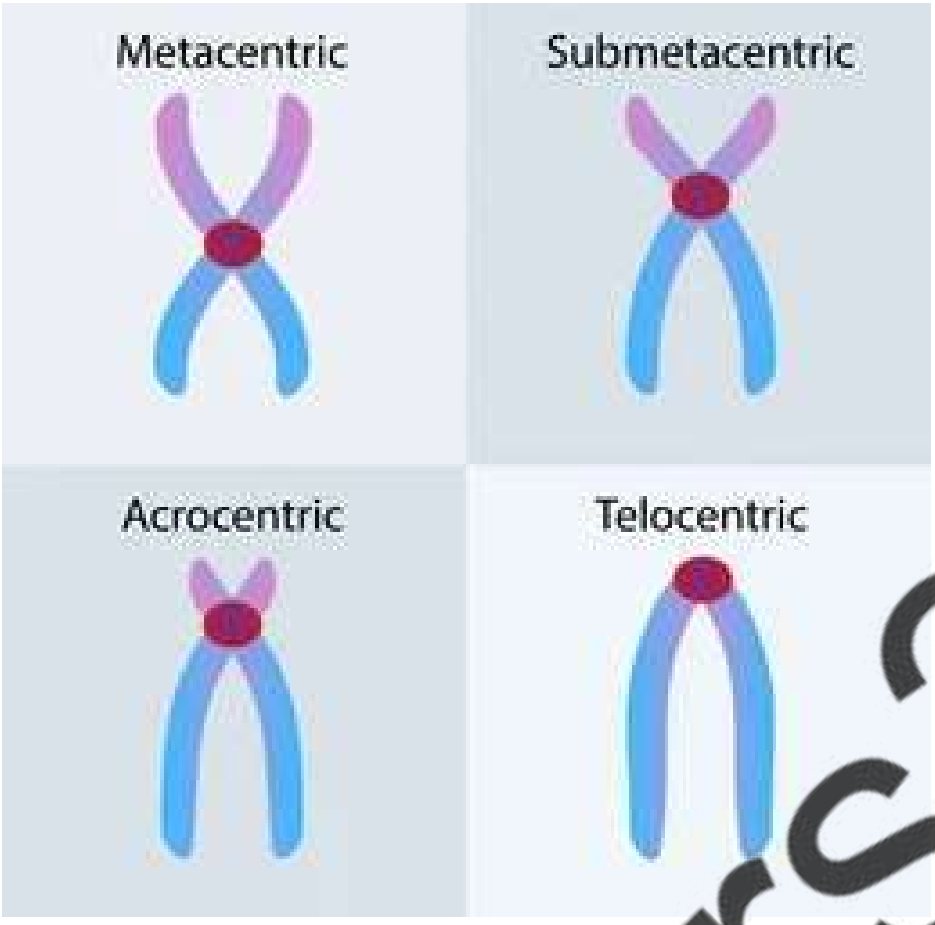

Q. 27 Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as:

Option 1:
Homosorus

Option 2:
Heterosorus

Option 3:
Homosporous

Option 4:
Heterosporous

Correct Answer:
Heterosporous

Solution:

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Heterospory is the production of spores of two different sizes and sexes by the sporophytes of land plants**.** The smaller of these, the microspore, is male and the larger megaspore is female.

Q. 28 Match List - I with List - II

List - I	List - II
(a) Cohesion	(i) More attraction in liquid plants
(b) Adhesion	(ii) Mutual attraction among water molecules
(c) Surface tension	(iii) Water loss in liquid phase
(d) Guttation	(iv) Attraction towards polar surfaces

Choose the correct answer from the options given below:

Option 1:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Option 2:
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Option 3:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Option 4:
(a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

Correct Answer:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Solution:
Cohesion, adhesion and surface tension are important proprties of water as they help the water from a tree's roots to be transported all the way to its leaves.

Cohesion holds hydrogen bonds together to create **surface tension** on water. Since water is attracted to other molecules, **adhesive forces** pull the water toward other molecules.

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Q. 29 Gemmae are present in:

Option 1:
Liverworts

Option 2:
Brown algae

Option 3:
Red algae

Option 4:
Blue - green algae

Correct Answer:
Liverworts

Solution:
Most **liverworts** can reproduce asexually by means of **gemmae**, which are disks of tissues produced by the gametophytic generation. The gemmae are held in special organs known as **gemma cups** and are dispersed by rainfall.

Q. 30 Which of the follwoing algae produces Carrageenan ?

Option 1:
Green algae

Option 2:
Brown algae

Option 3:
Red algae

Option 4:
Blue - green algae

Correct Answer:
Red algae

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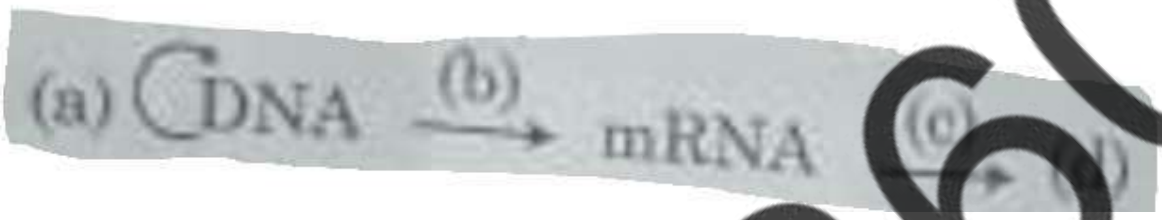
Solution:
Carrageenan is a **phycocolloids** produce in **red algae (Rhodophyceae)**. It is used as a clearing agent in liquors, leather finishing and as a emulsifier in chocolates, ice-creams, toothpastes, paints etc.

Q. 31 Complete the flow chart on central dogma.



Option 1:
(a) - Replication ; (b) - Transcription; (c) - Transduction; (d) - Protein

Option 2:
(a) -Translation; (b) - Replication ; (c) - Transcription ; (d) - Transduction

Option 3:
(a) -Replication; (b) - Transcription ; (c) - Translation ; (d) - Protein

Option 4:
(a) -Transduction ; (b) - Translation ; (c) - Replication ; (d) - Protein

Correct Answer:
(a) -Replication; (b) - Transcription ; (c) - Translation ; (d) - Protein

Solution:
The central dogma of molecular biology is **an explanation of the flow of genetic information within a biological system**. It is often stated as "DNA makes RNA, and RNA makes protein".

It was first stated by **Francis Crick** in 1957, then published in 1958.

Q. 32 Which of the following statements is not correct?

Option 1:
Pyramid of biomass in sea is generally inverted.

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Option 2:
Pyramid of biomass in sea is generally upright.

Option 3:
Pyramid of energy is always upright.

Option 4:
Pyramid of numbers in a grassland ecosystem is upright.

Correct Answer:
Pyramid of biomass in sea is generally upright.

Solution:
The pyramid of biomass in the sea is inverted because **the amount of biomass is least at the base of the pyramid and the amount of biomass is maximum at the apex of the pyramid**.

Eg: Biomass of zooplanktons is higher than that of phytoplanktons as life span of former is longer and the latter multiply much faster though having shorter life span. A number of generations of phytoplanktons may thus be consumed by single generation of zooplanktons.

Q. 33 Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival?

Option 1:
Resource partitioning

Option 2:
Competitive release

Option 3:
Mutualism

Option 4:
Predation

Correct Answer:
Resource partitioning

Solution:

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Resource partitioning or niche differentiation (**a process of natural selection that will force competitors to use resources differently**) is a way to avoid competition between species and allow for coexistence.

Species might evolve mechanism to live in the same niche by changing the **feeding time** or **foraging patterns.**

Q. 34 Match List -I with List -II

List - I	List - II
(a) Lenticels	(i) Phellogen
(b) Cork cambium	(ii) Suberin deposition
(c) Secondary cortex	(iii) Exchange of gases
(d) Cork	(iv) Phelloderm

Choose the correct answer from the options given below:

Option 1:
(a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Option 2:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Option 3:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Option 4:
(a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)

Correct Answer:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Solution:
Lenticels: In cork layer the lenticels are present which are meant for gaseous exchange. In cork, lenticels have loosely aggranged cells called **complementary cells** with intercelluar spaces.

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Cork cambium is **a** tissue found in many vascular plants as a part of the epidermis. It is one of the many layers of bark, between the cork and primary phloem. The cork cambium is a lateral meristem and is responsible for secondary growth that replaces the epidermis in roots and stems.

Secondary cortex is also known as **phelloderm**, produces from cork cambium on innerside of stem. Cells of phelloderm are thin walled, living and store food.

Cork also known as **phellem** produces from cork cambium on the outerside of the stem. The cells of phellem are dead, suberized and impervious to water.

Cork (phellem) + Cork cambium (phellogen) + Secondary cortex (Phelloderm)= Periderm.

Q. 35 Match the List - I and List -II

List - I	List - II
(a) Cells with active cell division capacity	(i) Vascular tissue
(b) Tissue having all cells similar in structure and function	(ii) Meristematic tissue
(c) Tissue having different types of cells	(iii) Sclereids
(d) Dead cells with highly thickned walls and narrow lumen	(iv) Simple tissue

Select the correct answer from the options given below:

Option 1:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Option 2:
(a) - (iv), (b) - (iii), (c) - (iii), (d) - (i)

Option 3:
(a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

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Option 4:
(a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

Correct Answer:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Solution:
Cells with active cell division capacity- **Meristematic tissue**

Tissue having all cells similar in structure and function- **Simple tissue**

Tissue having different types of cells- **Vascular tissue**

Dead cells with highly thickned walls and narrow lumen- **Sclereids**

Q. 36 Match Column - I with Column - II

Column - I	Column - II
(a) Nitrococcus	(i) Denitrification
(b) Rhizobium	(ii) Conversion of ammonia to nitrite
(c) Thiobacillus	(iii) Conversion of nitrite to nitrate
(d) Nitrobacter	(iv) Convesion of atmospheric nitrogen to ammonia

Chhose the correct answer from options given below:

Option 1:
(a) - (ii), (b) - (iv). (c) - (i), (d) - (iii)

Option 2:
(a) - (i), (b) - (ii). (c) - (iii), (d) - (iv)

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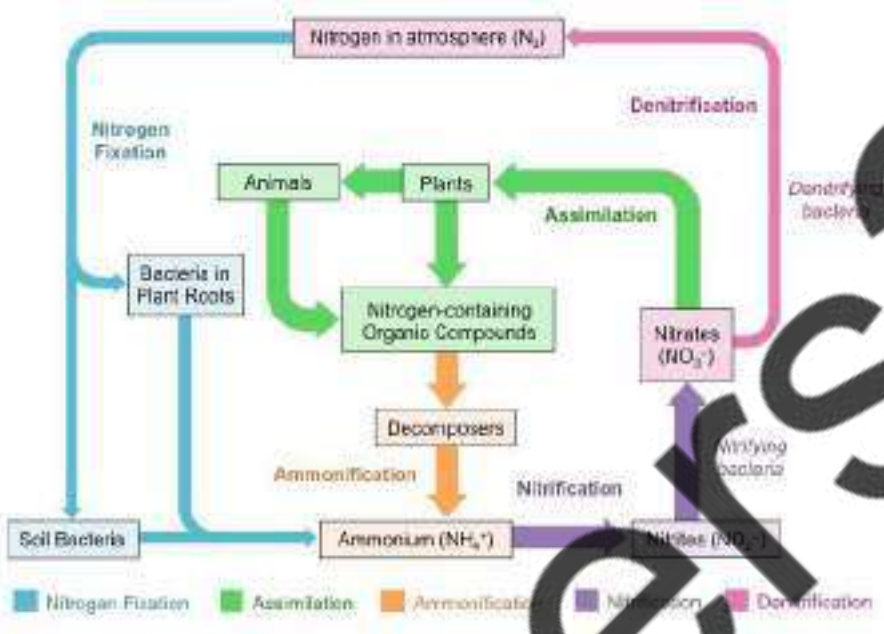
Option 3:
(a) - (iii), (b) - (i). (c) - (iv), (d) - (ii)

Option 4:
(a) - (iv), (b) - (iii). (c) - (ii), (d) - (i)

Correct Answer:
(a) - (ii), (b) - (iv). (c) - (i), (d) - (iii)

Solution:



Q. 37 Match List - I with List - II

List - I	List - II
(a) S phase	(i) Proteins and synthesized
(b) G_2 phase	(ii) Inactive phase
(c) Quiescent stage	(iii) Interval between mitosis and initaiation of DNA replication
(d) G_1 phase	(iv) DNA replication

Choose the correct answer from the option given below:

Option 1:
(a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

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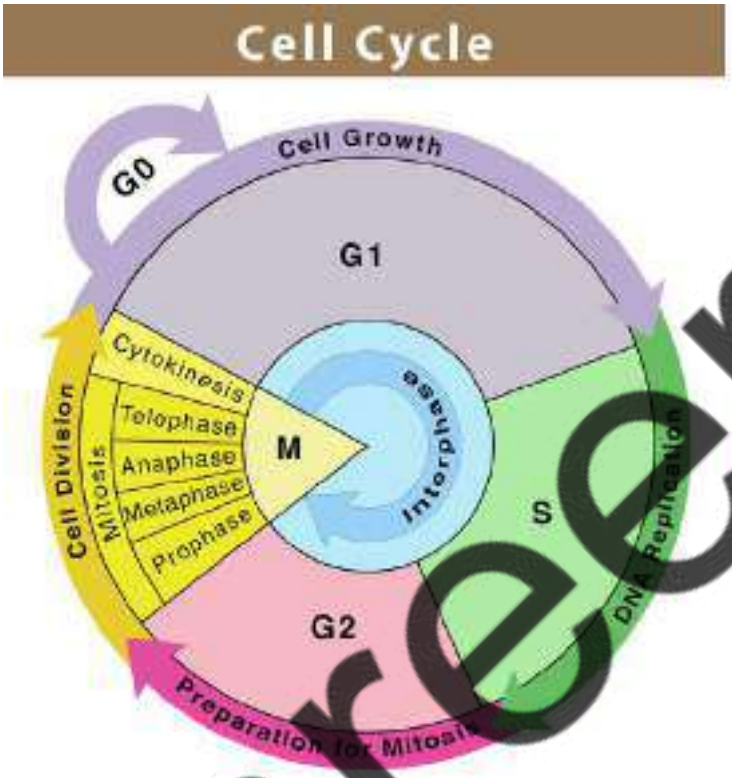
Option 2:
(a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

Option 3:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Option 4:
(a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

Correct Answer:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Solution:



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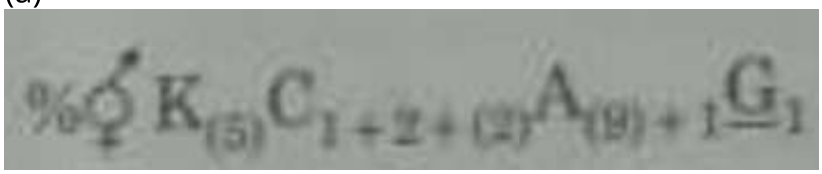
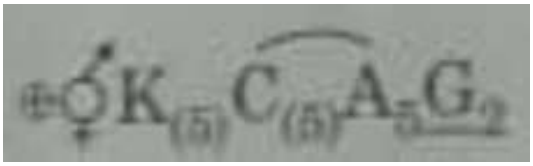
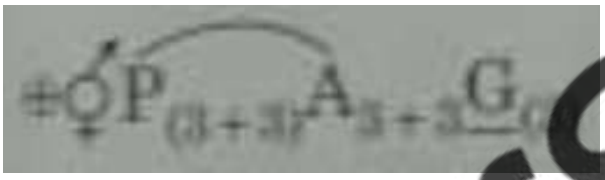
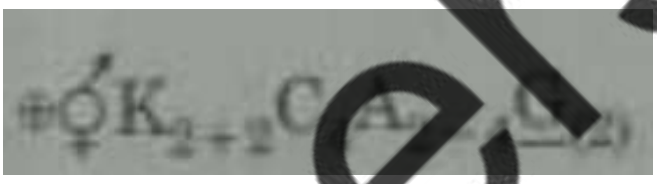
Q. 38 Match Column I with Column -II

Column - I	Column -II
(a) 	(i) Brassicaceae
(b) 	(ii) Liliaceae
(c) 	(iii) Fabaceae
(d) 	(iv) Solanaceae

Select the correct answer from the options given below:

Option 1:
(a) - (iii) , (b) - (iv), (c)- (ii), (d) - (i)

Option 2:
(a) - (i) , (b) - (ii), (c)- (iii), (d) - (iv)

Option 3:
(a) - (ii) , (b) - (iii), (c)- (iv), (d) - (i)

Option 4:
(a) - (iv) , (b) - (ii), (c)- (i), (d) - (iii)

Correct Answer:
(a) - (iii) , (b) - (iv), (c)- (ii), (d) - (i)

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Solution:
A Floral formula is **a notation for representing the structure of particular types of flowers**. Such notations use numbers, letters and various symbols to convey significant information in a compact form.

Q. 39 Which of the following statements is incorrect?

Option 1:
During aerobic respiration, role of oxygen is limited to the terminal stage.

Option 2:
In ETC (Electron Transport Chain), one molecule of NADH + H^+ gives rise to 2 ATP molecules, and $FADH_2$ gives rise to 3 ATP molecules.

Option 3:
ATP is synthesized through complex V.

Option 4:
Oxidation - reduction reactions produces proton gradient in respiration.

Correct Answer:
In ETC (Electron Transport Chain), one molecule of NADH + H^+ gives rise to 2 ATP molecules, and $FADH_2$ gives rise to 3 ATP molecules.

Solution:
Oxidation of one molecule of NADH gives rise to 3 molecules of ATP while that of one molecule of $FADH_2$ produces 2 molecules of ATP.

Q. 40 In the exponential growth equation

$N_t = N_0 e^{rt}$ represents:

Option 1:
The base number logarithms

Option 2:
The base of exponential logarithms

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Option 3:
The base of natural logarithms

Option 4:
The base of general logarithms

Correct Answer:
The base of natural logarithms

Solution:
The integral equation for exponential growth is $\mathbf{N_t=N0^{ert}}$. Where N_t=Population density after time t,No =Population density at time zero,r=intrinsic rate of natural increase,e=the base of natural logarithms.

Q. 41 Which of the following statements is incorrect?

Option 1:
Both ATP and NADPH + H^+ are synthesized during non-cyclic photophosphorylation

Option 2:
Stroma lamellae have PS I only and lack NADP reductase.

Option 3:
Grana lamellae have both PS I and PS II

Option 4:
Cyclic photophosphorylation involves both PS I and PS II

Correct Answer:
Cyclic photophosphorylation involves both PS I and PS II

Solution:
When only PS I is functional, the electron is circulated within the photosystem and the phosphorylation occurs, due to the cyclic flow of electrons. The membrane and lamella of the grana have both PS I and PS II, the stroma lamella membrane lack PS II as well as NADP reductase enzyme.

Q. 42 What is the role of RNA polymerase III in the process of transcription in eukaroytes?

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Option 1:
Transcribe rRNAs (28S, 18S and 5.8S)

Option 2:
Transcribes tRNA, 5S rRNAA and snRNA.

Option 3:
Transcribes precussor of mRNA

Option 4:
Transcribes only snRNAs

Correct Answer:
Transcribes tRNA, 5S rRNAA and snRNA.

Solution:
In eukaryote cells, RNA polymerase III (also called Pol III) **transcribes DNA to synthesize ribosomal 5S rRNA, tRNA and other small RNAs**. The genes transcribed by RNA Pol III fall in the category of "housekeeping" genes whose expression is required in all cell types and most environmental conditions.

Q. 43 Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hydridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because:

Option 1:
mutated gene partially appears on a photographic film

Option 2:
mutated gene completely and clearly appears on photographic films

Option 3:
mutated gene does not appear on a photographic film as the probe has no complumentary with it.

Option 4:
mutated gene does not appear on photographic film as the probe has complimentary with it.

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Correct Answer:
mutated gene does not appear on photographic film as the probe has complimentary with it.

Solution:
A single-stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography. The clone having the mutated gene will hence not appear on the photographic film, because the probe will not have complementarity with the mutated gene.

Q. 44 Which of the following statements is correct?

Option 1:
Fusion of two cells is called Karyogamy

Option 2:
Fusion of protoplasm between two motile on non -motile gametes is called plasogamy

Option 3:
Organism that depend on the living plants are called saprophytes.

Option 4:
Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.

Correct Answer:
Fusion of two cells is called Karyogamy

Solution:
Plasmogamy, **the fusion of two protoplasts (the contents of the two cells), brings together two compatible haploid nuclei**. At this point, two nuclear types are present in the same cell, but the nuclei have not yet fused.

Some of the organisms like **Nostoc and Anabaena** can fix atmospheric nitrogen in specialized cells called **Heterocyst**.

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Q. 45 Match List - I with List - II

List - I	List - II
(a) Protein	(i) C = C double bond
(b) Unsaturated fatty acid	(ii) Phosphodiester bonds
(c) Nucleic acid	(iii) Glycosidic bonds
(d) Polysaccharide	(iv) Peptide bond

Choose the correct answer from the options given below:

Option 1:
a - iv, b - i, c - ii , d- iii

Option 2:
a -i, b -iv, c -iii , d-ii

Option 3:
a -ii, b -i, c-iv, d - iii

Option 4:
a -iv, b - iii, c - i, d - ii

Correct Answer:
a - iv, b - i, c - ii , d- iii

Solution:
Protein- peptide bond: A **covalent bond** joining the α-amino group of one amino acid to the carboxyl group of another with the loss of a water molecule.

Unsaturated fatty acid- C=C double bond

Nucleic acid- **Phosphodiester bonds**: A phosphodiester bond occurs when exactly two of the hydroxyl groups in phosphoric acid react with hydroxyl groups on other molecules to form two ester bonds.

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Polysaccharide - **Glycosidic bonds**: glycosidic bond or glycosidic linkage is a type of covalent bond that joins a carbohydrate molecule to another group, which may or may not be another carbohydrate.

Q. 46 Select the correct pair

Option 1:
Large colorless empty cells in the epidemis of grass leaves - Subsidiary leaves

Option 2:
In dicot leaves, vascular bundles are surrounded by large thick walled cells - Conjuctive tissue

Option 3:
Cells of medullary rays yhat form part of cambial ring - Interfascicular cambium

Option 4:
Loose parenchyma cells rupturing epidermis and forming a lens-shaped opening in bark - Spongy parenchyma

Correct Answer:
Cells of medullary rays yhat form part of cambial ring - Interfascicular cambium

Solution:
Large colorless with thin flexible walls, empty cells in the epidemis of some grasses like Poa, Aropyron, Maize etc., epidermal cells are known as **Bulliform or motor cells.** These cells help in rolling and unrolling of leaves.

In dicot leaves, vascular bundles are surrounded by large thick walled cells known as **bundle sheath cells.**

Loose parenchyma cells rupturing epidermis and forming a lens-shaped opening in bark are called **lenticels**, which are meant for gaseous exchange.

Q. 47 DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as:

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Option 1:
Satellite DNA

Option 2:
Repetitive DNA

Option 3:
Single nucleotides

Option 4:
Polymorphic DNA

Correct Answer:
Repetitive DNA

Solution:
Repetitive DNA: Stretches of DNA that repeat themselves throughout a genome, either in tandem or interspersed along the genome. These stretches can comprise up to fifty percent or more of an organism's DNA.Hence, the correct option is (b).

Q. 48 Plasmid pBR322 has PstI restriction enzyme site within gene amp^R that confers ampicillin resistance.If this enzymes is used for inserting a gene β-galactoside production and the recombinant plasmid is inserted in an E.*coli* strain

Option 1:
it will not be able to confer ampicilin resistance to the host cell

Option 2:
the transformed cells will have the ability to resist ampicillin as well as produces β-galactoside

Option 3:
it will lead to lysis of host cell

Option 4:
it will be able to produce a novel protein with dual ability

Correct Answer:
it will not be able to confer ampicilin resistance to the host cell

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Solution:
Since the gene is inserted at Pst I of ampR region of pBR322, the ampR gene is inactivated which is called as **insertional inactivation.** Hence the genetically modified E. coli strain will not be able to confer ampicillin resistance.

Q. 49 Identify the correct statement:

Option 1:
In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA

Option 2:
RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

Option 3:
The coding strand in a transcription unit is copied to an mRNA

Option 4:
Split gene arrangement is characteristic of prokaryotes.

Correct Answer:
RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

Solution:
In the cell, Rho binds to untranslated naked RNAs and terminates the synthesis of mRNA at the end of a significant number of operons.

Q. 50 In some members of which of the following pairs of families, pollen grains retain their viability for months after release?

Option 1:
Poaceae: Rosaceae

Option 2:
Poaceae: Leguminosae

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Option 3:
Poaceae: Solanaceae

Option 4:
Rosaceae : Leguminosae

Correct Answer:
Rosaceae : Leguminosae

Solution:
In dicot families like **Solanaceae, Rosaceae and Leguminosae** the viability of pollen grains remain months together after their release.

Q. 51 Which of the following characteristics is incorrect with respect to cockroach ?

Option 1:
A ring of gastric caeca is present at the junction of midgut and hind gut

Option 2:
Hypopharynx lies within the cavity enclosed by the mouth parts.

Option 3:
In females, 7th , 9th sterna together form a genital pouch

Option 4:
10th abdominal segment in both sexes, bears a pair of anal cerci.

Correct Answer:
A ring of gastric caeca is present at the junction of midgut and hind gut

Solution:
Correct option is (a)

-Hepatic Caecae is between foregut and midgut

- Malphigian tubules is between midgut and hindgut.

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Q. 52 Match List - I with List - II.

List - I		List - II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the correct answer from the option given below.

Option 1:
(a) (b) (c) (d)
(iv) (iii) (i) (ii)

Option 2:
(a) (b) (c) (d)
(iii) (iv) (i) (ii)

Option 3:
(a) (b) (c) (d)
(iii) (iv) (ii) (i)

Option 4:
(a) (b) (c) (d)
(iv) (i) (ii) (iii)

Correct Answer:
(a) (b) (c) (d)
(iii) (iv) (ii) (i)

Solution:
Correct option is (c)

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Metamerism - Segmented body(Annelids)

Canal System - Porifera (Sponges)

Combi plates - Ctenophora

Cnidoblasts - Coelentetrata (Cnidoblasts)

Q. 53 A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :

Option 1:
Degenerate primer sequence

Option 2:
Okazaki sequences

Option 3:
Palindromic Nucleotide sequences

Option 4:
Poly (A) tail sequences

Correct Answer:
Palindromic Nucleotide sequences

Solution:
Palindrome sequence reads same from both the directions.It is specific recognition sequence identified by endonucleases to cut the DNA strands at the specific positions.

Q. 54 The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G_1 phase is 8, what would be the number of chromosomes after S phase ?

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Option 1:
8

Option 2:
16

Option 3:
4

Option 4:
32

Correct Answer:
8

Solution:
Correct option is (a)

It will be same as that of G1, hence answer is 8.

Q. 55 Persons with 'AB' blood group are called as "Universal recipients". This is due to :

Option 1:
Absence of antigens A and B on the surface of RBCs

Option 2:
Absence of antigens A and B in plasma

Option 3:
Presence of antibodies, anti- A and anti- B on RBCs

Option 4:
Absence of antibodies, anti-A and anti- B, in plasma

Correct Answer:
Absence of antibodies, anti-A and anti- B, in plasma

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Solution:
Correct option is (d)

ABO blood group is on the basis of Antigens present on the RBC surface.

It is either A blood group B, O or AB

AB blood group is due to the presence of two antigens A. Due to this no antibodies are present.

Q. 56 Which of the following statements wrongly represents the nature of smooth muscle?

Option 1:
These muscle have no striations

Option 2:
They are involuntry muscles

Option 3:
Communication among the cells is performed by intercalated discs

Option 4:
These muscles are present in the wall of blood vessels

Correct Answer:
Communication among the cells is performed by intercalated discs

Solution:
Correct option is (c)

Intercalated discs are feature of cardiac muscles only.

Q. 57 Which one of the following belongs to the family Muscidae?

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Option 1:
Fire fly

Option 2:
Grasshopper

Option 3:
Crockroach

Option 4:
House fly

Correct Answer:
House fly

Solution:
Correct option is (d)

House fly

Q. 58 Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?

Option 1:
DNA dependnet DNA polymerase

Option 2:
DNA dpendent RNA polymerase

Option 3:
DNA Ligase

Option 4:
DNase

Correct Answer:
DNA dpendent RNA polymerase

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Solution:

- In the process of transcription in prokaryotes, DNA-dependent RNA polymerase is the enzyme responsible for catalyzing initiation, elongation, and termination. It binds to the DNA template and synthesizes an RNA molecule complementary to the DNA strand, resulting in the production of RNA transcripts. Therefore, **option 2 is the correct answer.**
- Option 1 is incorrect because DNA-dependent DNA polymerase is involved in DNA replication.
- Option 3 is incorrect DNA ligase joins DNA fragments together
- Option 4 is incorrect DNase degrades DNA.

Q. 59 Which of the following RNAs is not required for the synthesis of protein ?

Option 1:
mRNA

Option 2:
tRNA

Option 3:
rRNA

Option 4:
aiRNA

Correct Answer:
aiRNA

Solution:
Correct option is (d)

mRNA for transcription tRNA and rRNA for translation.

Q. 60 For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection?

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Option 1:
Western Blotting Technique

Option 2:
Southern Blotting Technique

Option 3:
ELISA Technique

Option 4:
Hybridization Technique

Correct Answer:
ELISA Technique

Solution:
Correct option is (b)

Southern Blotting Technique

Q. 61 Erythropoietin hormone which stimulates R.B.C.formation is produced by :

Option 1:
Alpha cells of pancreas

Option 2:
The cells of rostral adenohypophysis

Option 3:
The cells of bone marrow

Option 4:
cells of the kidney

Correct Answer:
cells of the kidney

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Solution:
Correct option is (d)

Erythioprotein helps in the process of Erythroprocess.

Q. 62 The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are :

Option 1:
$pO_2 = 104 \; and \; pCO_2 = 40$

Option 2:
$pO_2 = 40 \; and \; pCO_2 = 45$

Option 3:
$pO_2 = 95 \; and \; pCO_2 = 40$

Option 4:
$pO_2 = 160 \; and \; pCO_2 = 80$

Correct Answer:
$pO_2 = 104 \; and \; pCO_2 = 40$

Solution:
Correct option is (a)

$pO_2 = 104 \; and \; pCO_2 = 40$

Q. 63 In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?

Option 1:
50%

Option 2:
75%

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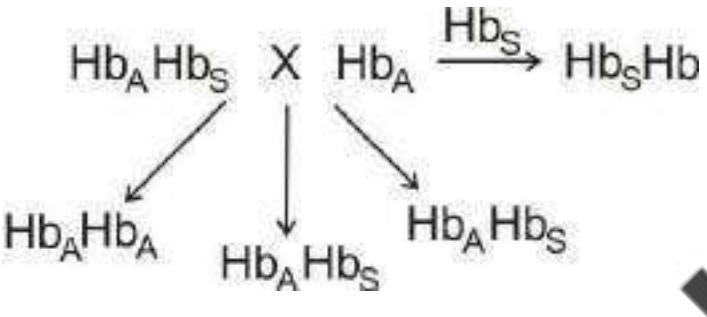
Option 3:
25%

Option 4:
100%

Correct Answer:
25%

Solution:
Correct option is (c)



Sickle cell is Autosomal receive. Only 25% of progeny is affected.

Q. 64 Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature?

Option 1:
Leptotene

Option 2:
Zygotene

Option 3:
Diakinesis

Option 4:
Pachytene

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Correct Answer:
Diakinesis

Solution:
Correct option is (c)

Terminalisation of Chiasmata is a feature of Diakeneses.

Q. 65 Which one of the following organisms bear hollow and pneumatic long bones ?

Option 1:
Neophron

Option 2:
Hemidactylus

Option 3:
Macropus

Option 4:
Ornithorhynchus

Correct Answer:
Neophron

Solution:
Correct option is (a)

This is a feature of Class Aves.(birds)

Neophron is Vulture

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Q. 66 Match the following :

List - I		List - II	
(a)	Physalia	(i)	Pearl oyester
(b)	Limulus	(ii)	Portuguese Man of War
(c)	Ancylostoma	(iii)	Living fossil
(d)	Pinctada	(iv)	Hookworm

Choose the correct answer from the options given below.

Option 1:
(a) (b) (c) (d)
(ii) (iii) (i) (iv)

Option 2:
(a) (b) (c) (d)
(iv) (i) (iii) (ii)

Option 3:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Option 4:
(a) (b) (c) (d)
(i) (iv) (iii) (ii)

Correct Answer:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Solution:
Correct option is (c)

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Physalia - Portuguese Man of War

Limulus - Living fossil

Ancylostoma - Hookworm

Pinctada - Pearl oyster

Q. 67 Which of the following is not an objective of Biofortification in crops ?

Option 1:
Improve protein content

Option 2:
Improve resistance to diseases

Option 3:
Improve vitamin content

Option 4:
and mineral content

Correct Answer:
Improve resistance to diseases

Solution:
Improving resistance to disease in crop is not an objective of Biofortification of crops.

Hence, the correct otpion is (b).

Q. 68 If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine, and Cytosine in it?

Option 1:
$T:20:G:30:C:20$

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Option 2:

$T : 20 : G : 20 : C : 30$

Option 3:

$T : 30 : G : 20 : C : 20$

Option 4:

$T : 20 : G : 25 : C : 25$

Correct Answer:

$T : 30 : G : 20 : C : 20$

Solution:

The correct option is (c)

$A = T$ hence $T = 30\%$

$A + T = 60\%$

$G + C = 100 - 60 = 40$

$\therefore G = C = 20\%$

Q. 69 Match **List-I** with **List - II**.

List - I		**List - II**	
(a)	Vaults	(i)	Entry of sperm through the Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of the fallopian tube

Choose the correct answer from the options given below.

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Option 1:

(a) (b) (c) (d)

(iv) (ii) (i) (iii)

Option 2:

(a) (b) (c) (d)

(i) (iii) (ii) (iv)

Option 3:

(a) (b) (c) (d)

(ii) (iv) (iii) (i)

Option 4:

(a) (b) (c) (d)

(iii) (i) (iv) (ii)

Correct Answer:

(a) (b) (c) (d)

(i) (iii) (ii) (iv)

Solution:

Correct option is (b)

Vaults - Entry of sperm through Cervix is blocked

IUDs - Phagocytosis of sperms within the Uterus

Vasectomy - Removal of Vas deferens

Tubectomy - Removal of fallopian tube

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Q. 70 Veneral diseases can spread through:

(a) Using sterile needles

(b) Transfusion of blood from infected person

(c) Infected mother of foetus

(d) Kissing

(e) Inheritance

Choose the correct answer from the options given below.

Option 1:
(a), (b) and (c) only

Option 2:
(b), (c) and (d) only

Option 3:
(b) and (c) only

Option 4:
(a) and (c) only

Correct Answer:
(b) and (c) only

Solution:
Veneral diseases are STD's. Hence, the Correct option is (c)

Q. 71 The centriole undergoes duplication during :

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Option 1:
S- phase

Option 2:
Prophase

Option 3:
Metaphase

Option 4:
G_2 Phase

Correct Answer:
S- phase

Solution:
Correct option is (a)

Centriole duplicates during S Phase

Q. 72 Receptors for sperm binding in mammals are present on :

Option 1:
Corona radiate

Option 2:
Vitelline membrane

Option 3:
Perovitelline space

Option 4:
Zona pellucida

Correct Answer:
Zona pellucida

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Solution:
Correct option is (d)

Zp (2) and ZP (3) are the receptors on Zona.

Q. 73 Dobson units are used to measure thickness of :

Option 1:
CFCs

Option 2:
Stratosphere

Option 3:
Ozone

Option 4:
Troposphere

Correct Answer:
Troposphere

Solution:
Correct option is (d)

Troposphere is a part of Ozone

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Q. 74 Match **List -I** with **List - II**.

List - I		**List - II**	
(a)	Aspergillus niger	(i)	Acetic Acid
(b)	Acetobacter aceti	(ii)	Lactic Acid
(c)	Clostridium butylicum	(iii)	Citric Acid
(d)	Lactobacillus	(iv)	Butyric Acid

Choose the correct answer from the options given below.

Option 1:
(a) (b) (c) (d)
(iii) (i) (iv) (ii)

Option 2:
(a) (b) (c) (d)
(i) (ii) (iii) (iv)

Option 3:
(a) (b) (c) (d)
(ii) (iii) (i) (iv)

Option 4:
(a) (b) (c) (d)
(iv) (ii) (i) (iii)

Correct Answer:
(a) (b) (c) (d)
(iii) (i) (iv) (ii)

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Solution:
Correct option is (a)

Aspergillus niger	(iii)	Citric Acid
Acetobacter aceti	(i)	Acetic Acid
Clostridium butylicum	(iv)	Butyric Acid
Lactobacillus	(ii)	Lactic Acid

Q. 75 Identify the **incorrect** pair.

Option 1:
Alkaloids - Codeine

Option 2:
Toxin - Abrin

Option 3:
Lectins - Concanavalin A

Option 4:
Drugs - Ricin

Correct Answer:
Drugs - Ricin

Solution:
Correct option is (d)

Ricin is s toxin

Q. 76 The organelles that are included in the endomembrane system are :

Option 1:
Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes

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Option 2:
Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

Option 3:
Golgi complex,Mitochondria,Ribosomes and Lysosomes

Option 4:
Golgi complex, Endoplasmic reticulum,Mitochondria and Lysosomes

Correct Answer:
Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

Solution:
Correct option is (b)

Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

Q. 77 Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :

Option 1:
Arthritis

Option 2:
Muscular dystrophy

Option 3:
Myasthenia gravis

Option 4:
Gout

Correct Answer:
Myasthenia gravis

Solution:
Correct option is (c)

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Myasthenia gravis

Q. 78 With regard to insulin choose correct options.

(a) C-peptide is not present in mature insulin.

(b) The insulin produced by rDNA technology has C-peptide.

(c) The pro-insulin has C-peptide

(d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

Option 1:
(b) and (d) only

Option 2:
(b) and (c) only

Option 3:
(a), (c) and (d) only

Option 4:
(a) and (d) only

Correct Answer:
(a), (c) and (d) only

Solution:
Correct option is (c)

(a) C-peptide is not present in mature insulin.

(c) The pro-insulin has C-peptide

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(d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Q. 79 During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?

Option 1:
Annealing

Option 2:
Extension

Option 3:
Denaturation

Option 4:
Ligation

Correct Answer:
Denaturation

Solution:
Correct option is (c)

Denaturation is the first step in PCR process

Q. 80 Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?

Option 1:
Thrombin

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Option 2:
Renin

Option 3:
Epinephrine

Option 4:
Thrombokinase

Correct Answer:
Thrombin

Solution:
Correct option is (a)

Thrombin

Q. 81 Succus entericus is referred to as :

Option 1:
Pancreatic juice

Option 2:
Intestinal juice

Option 3:
Gastric juice

Option 4:
Chyme

Correct Answer:
Intestinal juice

Solution:
Correct option is (b)

Intestinal juice

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Q. 82 Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

Option 1:
High pO_2, low pCO_2, less H^+, lower temperature

Option 2:
Low pO_2, high pCO_2, more H^+, higher temperature

Option 3:
High pO_2, high pCO_2, less H^+, higher temperature

Option 4:
Low pO_2, low pCO_2, more H^+, higher temperature

Correct Answer:
High pO_2, low pCO_2, less H^+, lower temperature

Solution:
Correct option is (a)

Oxyhaemoglobin is formed near alveoli. High pO_2, low pCO_2, low H^+, high pH^+. Low temperature are the conditions favourable for the same.

Q. 83 Which one of the following is an example of Hormone releasing IUD ?

Option 1:
CuT

Option 2:
LNG 20

Option 3:
Cu 7

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Option 4:
Multiload 375

Correct Answer:
LNG 20

Solution:
Correct option is (b)

LNG 20 is hormone containing IUD

Q. 84 Sphincter of oddi is present at :

Option 1:
Ileo-Caecal junction

Option 2:
Junction of hepato-pancreatic duct and duodenum

Option 3:
Gastro-oesophageal junction

Option 4:
Junction of jejunum and duodenum

Correct Answer:
Junction of hepato-pancreatic duct and duodenum

Solution:
Correct option is (b)

Junction of hepato-pancreatic duct and duodenum.

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Q. 85 Read the following statements:

(a) Metagenesis is observed in Helminths

(b) Echinoderms are triploblastic and coelomate animals

(c) Round worms have organ-system level of body organization.

(d) Comb plates present in ctenophores help in digestion.

(e) water vascular system is characterristic of Echinoderms.

Choose the **correct** answer from the options given below.

Option 1:
(c),(d) and (e) are correct

Option 2:
(a),(b) and (c) are correct

Option 3:
(a),(d) and (e) re correct

Option 4:
(b), (c) and (e) are correct

Correct Answer:
(b), (c) and (e) are correct

Solution:
Correct option is (d)

(b) Echinoderms are triploblastic and coelomate animals

(c) Round worms have organ-system level of body organization.

(e) water vascular system is characterristic of Echinoderms.

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Q. 86 Which of these is not an important component of initiation of parturition in humans?

Option 1:
Increases in estrogen and progesterone ratio

Option 2:
Synthesis of prostaglandins

Option 3:
Release of Oxytocin

Option 4:
Release of Prolactin

Correct Answer:
Increases in estrogen and progesterone ratio

Solution:
Correct option is (a)

Increases in estrogen and progesterone ratio is not an important factor for parturition.

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Q. 87 During muscular contraction which of following event occur?

(a) 'H' zone disappears

(b) 'A' band widens

(c) 'I' band reduces in width

(d) Myosine hydrolyzes ATP, releasing the ADP___and Pi

(e) Z-lines attached to actins are pulled in

Choose the correct answer from the option given below.

Option 1:
(a),(c),(d),(e) only

Option 2:
(a),(b),(c),(d) only

Option 3:
(b),(c),(d),(e) only

Option 4:
(b),(d),(e),(a) only

Correct Answer:
(a),(c),(d),(e) only

Solution:
Correct option is (a)

(a) 'H' zone disappears

(c) 'I' band reduces in width

(d) Myosine hydrolyzes ATP, releasing the ADPand Pi

(e) Z-lines attached to actins are pulled in

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Q. 88 Following are the statements with reference to 'lipids'.

(a) Lipids having only single bonds are called unsaturated fatty acids.

(b) Lecithin is a phospholipid.

(c) Trihydroxy propane is glycerol.

(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.

(e) Arachidonic acid has 16 carbon atoms.

Choose the correct answer from the options given below.

Option 1:
(a) and (b) only

Option 2:
(c) and (d) only

Option 3:
(b) and (c) only

Option 4:
(b) and (e) only

Correct Answer:
(b) and (c) only

Solution:
- **Statement (b) is correct. L**ecithin is indeed a phospholipid.
- **Statement (c) is also correct.** Trihydroxy propane is another name for glycerol, which is a common component of lipids.
- Statements (a), (d), and (e) are incorrect:
- Statement (a) is incorrect. Lipids having only single bonds are called saturated fatty acids, not unsaturated fatty acids.
- Statement (d) is incorrect. Palmitic acid has 16 carbon atoms, not 20.
- Statement (e) is incorrect. Arachidonic acid has 20 carbon atoms, not 16.
- Therefore, the correct answer is option 3.

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Q. 89 Which of the following is not a step in multiple Ovulation Embryo Transfer Technology (MOET) ?

Option 1:
Cow is administered hormone having LH like activity for super ovulation

Option 2:
Cow yeilds about 6-8 eggs at a time

Option 3:
Cow is fertilized by artificial insemination

Option 4:
Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage

Correct Answer:
Cow is administered hormone having LH like activity for super ovulation

Solution:
Correct option is (a)

In this method, cow is administered with FSH like hormone, **to induce follicular maturation** and then the super ovulation is brought about. In each cycle, 6−8 eggs mature simultaneously.

Q. 90 Identify the types of cell junctions that helps to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.

Option 1:
Gap junctions and Adhering junctions, respectively.

Option 2:
Tight junctions and Gap junctions, respectively.

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Option 3:
Adhering junctions and Tight junctions, respectively.

Option 4:
Adhering junctions and Gap junctions, respectively.

Correct Answer:
Tight junctions and Gap junctions, respectively.

Solution:
Correct option is (b)

Tight Junctions $\longrightarrow$ Brings about leakage of substances.

Gap Junctions $\longrightarrow$ Communication between cytoplasm.

Q. 91 The Adenosine deaminase deficiency results into:

Option 1:
Dysfunction of Immune system

Option 2:
Parkinson's disease

Option 3:
Digestive disorder

Option 4:
Addison's disease

Correct Answer:
Dysfunction of Immune system

Solution:
Correct option is (a)

Dysfunction of Immune system

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Q. 92 **Statement I:**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

'AAA' and 'AAG both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is correct but Statement II is false

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Statement I is incorrect but Statement II is true.

Solution:
AUG codes for only methionine but not tryptophan. (A given codon codes for only one amino acid but not more than one amino acids.). AAG and AAA are the two codons for the amino acid Lysine.

Hence,Correct option is (d) - Statement I is incorrect but Statement II is true.

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Q. 93 Match **List - I** with **List - II**.

List - I		**List - II**	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral Column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

Option 1:
(a) (b) (c) (d)
(i) (iii) (ii) (iv)

Option 2:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Option 3:
(a) (b) (c) (d)
(iv) (ii) (iii) (i)

Option 4:
(a) (b) (c) (d)
(iv) (iii) (ii) (i)

Correct Answer:
(a) (b) (c) (d)
(iv) (iii) (ii) (i)

Solution:
Correct option is (d)

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Scapula - Triangular flat bone

Cranium - Fibrous joints

Sternum - Flat bone

Vertebral column - Cartilaginous joints

Q. 94 Match **List - I** with **List - II**.

List - I		**List - II**	
(a)	Filariasis	(i)	Haemophilus Influenzae
(b)	Amoebiasis	(ii)	Trichophyton
(c)	Pneumonia	(iii)	Wuchereria bancrofti
(d)	Ringworm	(iv)	Entamoeba Histolytica

Choose the correct anwer from the options given below.

Option 1:
(a) (b) (c) (d)
(iv) (i) (iii) (ii)

Option 2:
(a) (b) (c) (d)
(iii) (iv) (i) (ii)

Option 3:
(a) (b) (c) (d)
(i) (ii) (iv) (iii)

Option 4:
(a) (b) (c) (d)
(ii) (iii) (i) (iv)

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Correct Answer:

(a) (b) (c) (d)

(iii) (iv) (i) (ii)

Solution:

Correct option is (b)

(a)	Filariasis	(i)	Wuchereria bancrofti
(b)	Amoebiasis	(ii)	Entamoeba Histolytica
(c)	Pneumonia	(iii)	Haemophilus Influenzae
(d)	Ringworm	(iv)	Trichophyton

Q. 95 Following are the statements about prostomium of earthworm.

(a) It serves as a covering for mouth

(b) It helps to open cracks in the soil into which it can crawl.

(c) It is one of the sensory structures.

(d) It is the first body segment.

Choose the correct answer from the options given below.

Option 1:

(a), (b) and (c) are correct

Option 2:

(a), (b) and (d) are correct

Option 3:

(a), (b), (c) and (d) are correct

Option 4:

(b) and (c) are correct

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Correct Answer:
(a), (b) and (c) are correct

Solution:
Correct option is (a)

(a) It serves as a covering for mouth

(b) It helps to open cracks in the soil into which it can crawl.

(c) It is one of the sensory structures.

Q. 96 Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?

Option 1:
Graafian follicle

Option 2:
Corpus luteum

Option 3:
Foetus

Option 4:
Uterus

Correct Answer:
Corpus luteum

Solution:
Correct option is (b)

Corpus luteum

- The **ovulation** (ovulatory phase) is followed by the **luteal phase** during which the remaining parts of the Graafian follicle transform as the corpus luteum.
- The corpus luteum secretes large amounts of progesterone which is essential for the maintenance of the endometrium.
- Such an endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy.

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- During pregnancy, all events of the menstrual cycle stop and there is no menstruation.
- In the absence of fertilisation, the corpus luteum degenerates.

Q. 97 Which one of the following statements about Histones is **wrong** ?

Option 1:
Histones are organized to form a unit of 8 molecules.

Option 2:
The pH of histones is slightly acidic

Option 3:
Histones are rich in amino acids- Lysine and Arginine

Option 4:
Histones carry positive charge in the side chain.

Correct Answer:
The pH of histones is slightly acidic

Solution:
Correct option is (b)

Histones are basic proteins

Q. 98 **Assertion (A) :**

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

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Option 1:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Option 2:
Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

Option 3:
(A) is true but **(R)** is false

Option 4:
(A) is false but **(R)** is true

Correct Answer:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Solution:
The correct option is (a)

Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Q. 99 Match **List-I** with **List - II**.

List - I		**List - II**	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaption	(ii)	Desert lizard
(c)	Behavioural adaption	(iii)	Marine fish at depth
(d)	Biochemical adaption	(iv)	Polar seal

Choose the **correct** answer from the options given below.

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Option 1:

(a) (b) (c) (d)

(iv) (ii) (iii) (i)

Option 2:

(a) (b) (c) (d)

(iv) (i) (iii) (ii)

Option 3:

(a) (b) (c) (d)

(iv) (i) (ii) (iii)

Option 4:

(a) (b) (c) (d)

(iv) (iii) (ii) (i)

Correct Answer:

(a) (b) (c) (d)

(iv) (i) (ii) (iii)

Solution:

Correct option is (c)

(a)	Allen's Rule	(i)	Polar seal
(b)	Physiological adaption	(ii)	Kangaroo rat
(c)	Behavioural adaption	(iii)	Desert Lizard
(d)	Biochemical adaption	(iv)	Marine fish at depth

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Q. 100 Match **List-I** with **List - II**.

List - I		**List - II**	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excrssive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

Option 1:

(a) (b) (c) (d)

(iv) (iii) (ii) (i)

Option 2:

(a) (b) (c) (d)

(iii) (ii) (i) (iv)

Option 3:

(a) (b) (c) (d)

(ii) (i) (iv) (iii)

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Option 4:

(a) (b) (c) (d)

(i) (iv) (iii) (ii)

Correct Answer:

(a) (b) (c) (d)

(iv) (iii) (ii) (i)

Solution:

Correct option is (a)

(a)	Adaptive radiation	(i)	Darwin Finches
(b)	Convergent evolution	(ii)	Wings of Butterfly and Bird
(c)	Divergent evolution	(iii)	Bones of forelimbs in Man and Whale
(d)	Evolution by anthropogenic action	(iv)	Selection of resistant varieties due to excrssive use of herbicides and pesticides

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Physics

Q. 1 The dimensions $\left[\mathrm{MLT}^{-2}\ \mathrm{A}^{-2}\right]$ belong to the:

Option 1:
self inductance

Option 2:
magnetic permeability

Option 3:
electric permittivity.

Option 4:
magnetic flux.

Correct Answer:
magnetic permeability

Solution:
We know that,

the force per unit length between two current-carrying conductors is

$$\frac{f}{l}=\frac{\mu_0 i_1 i_2}{2\pi a}$$

$$\therefore [\mu_0]=\left[\frac{F}{i_1 i_2}\right]\left[\frac{a}{l}\right]$$

$$[\mu_0]=\left[M^1L^1T^{-2}A^{-2}\right]$$

Hence, the answer is the option (2).

Q. 2 The area of a rectangular field $\left(\text{in } \mathrm{m}^2\right)$ of length $55.3\mathrm{m}$ and breadth $25\mathrm{m}$ after rounding off the value for correct significant digits is

Option 1:
1382

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Option 2:

1382.5

Option 3:

14×10^2

Option 4:

138×10^1

Correct Answer:

14×10^2

Solution:

$\mathrm{l} = 55.3\ \mathrm{m}$

$\mathrm{b} = 25\ \mathrm{m}$

$\text{Area } = \mathrm{l} \times \mathrm{b}$

$= 55.3 \times 25\ \mathrm{m}$

$= 14 \times 10^2$

$\therefore$ In product or division by arithmetic rules, the answer must have the least no of significant figures.

Hence, the answer is the option (3).

Q. 3 Plane angle and solid angle have:

Option 1:

Dimensions but no units

Option 2:

No units and no dimensions

Option 3:

Both units and dimensions

Option 4:

Units but no dimensions

Correct Answer:

Units but no dimensions

Solution:

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$$\text{plane angle} = \theta = \frac{\text{length of arc subtended}}{\text{radius of arc}}$$

$$\text{Solid angle}(\Omega) = \frac{\text{Area subtended}}{(\text{ radius })^2}$$

$\therefore$ Both plane angle and solid angle don't have dimensions, But the units of plane angle and solid angle are radian (rad) and steradian (sr) respectively.
Hence, the answer is the option (4).

Q. 4 A ball is projected with a velocity 10ms^{-1}, at an angle of 60° with the vertical direction . It's speed at the highest point of its trajectory will be:

Option 1:
$5\sqrt{3}\text{ms}^{-1}$

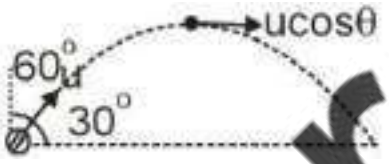
Option 2:
5ms^{-1}

Option 3:
10ms^{-1}

Option 4:
Zero

Correct Answer:
$5\sqrt{3}\text{ms}^{-1}$

Solution:

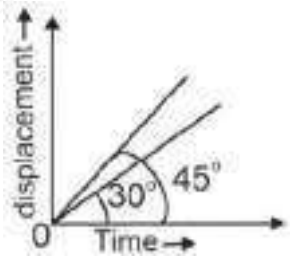


At the topmost position, the speed of the projectile is $\text{u}\cos\theta$

$$\text{u}\cos\theta = 10 \times \cos 30^\circ = 5\sqrt{3}\frac{\text{m}}{\text{s}}$$

Hence the correct option is 1

Q. 5 The displacement time graphs of two moving particles make angles of 30° and 45° with the x-axis as shown in the figure. the ratios of their respective velocity are



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Option 1:
1:1

Option 2:
1:2

Option 3:
$1 : \sqrt{3}$

Option 4:
$\sqrt{3} : 1$

Correct Answer:
$1 : \sqrt{3}$

Solution:
For the displacement-time graph,
Slope = velocity

$$\therefore V_1 = (\text{ slope })_1 = \tan 30^\circ$$
$$V_2 = (\text{ slope })_2 = \tan 45^\circ$$
$$\frac{V_1}{V_2} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}}$$

Q. 6 The ratio of the distances traveled by a freely falling body in the $1^{st}, 2^{nd}, 3^{rd}$ and 4^{th} second :

Option 1:
1: 4: 9: 16

Option 2:
1: 3: 5: 7

Option 3:
1: 1: 1: 1

Option 4:
1: 2: 3: 4

Correct Answer:
1: 3: 5: 7

Solution:
The initial speed of the body is zero i.e. u=0

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Distance travelled in the nth second is given by $S_{nth}=u+a/2(2n-1)$

$\Longrightarrow S_{nth}=(a/2)(2n-1)$ ($\because u=0$)

So, distance travelled in the first second, $S_1=(a/2)(2\times1-1)=a/2$

So, distance travelled in 2nd second, $S_2=(a/2)(2\times2-1)=3a/2$

So, distance travelled in 3rd second, $S_3=(a/2)(2\times3-1)=5a/2$

So, distance travelled in 4th second, $S_4=(a/2)(2\times4-1)=7a/2$

$\Longrightarrow S_1:S_2:S_3:S_4=1:3:5:7$

By Galileo's ratio, the distance travelled by a freely falling body is in the ratio 1: 3: 5: 7 (odd number)

$$\therefore s_1 : s_2 : s_3 : s_4 = 1 : 3 : 5 : 7$$

Hence, the answer is the option (2).

Q. 7 A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2:2:1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v, the speed of the third (lighter) fragment is :

Option 1:
$\sqrt{2}\mathrm{v}$

Option 2:
$2\sqrt{2}\mathrm{v}$

Option 3:
$3\sqrt{2}\mathrm{v}$

Option 4:
v

Correct Answer:
$2\sqrt{2}\mathrm{v}$

Solution:
The ratio of the masses of the fragments is

$$\mathrm{m_1 : m_2 : m_3 = 2 : 2 : 1}$$

$$\mathrm{m_1 + m_2 + m_3 = m}$$

$$\therefore \mathrm{m_1 = m_2 = 0.4\ m\ \&\ m_3 = 0.2\ m}$$

Since initially the shell is at rest and also external force is zero

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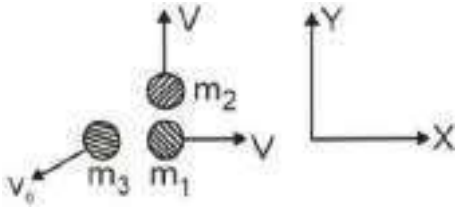
$\therefore \bar{P}_i = \bar{P}_f = 0$



$\bar{P}_f = m_1 v(\hat{i}) + m_2 v(\hat{j}) + m_3 \overline{v_0}$

$\bar{V}_0 = -(2v\hat{i} + 2v\hat{j})$

$V_0 = \left(\sqrt{(2)^2 + (2)^2}\right) = 2\sqrt{2}V$

The speed of the lighter fragment is $2\sqrt{2}V$

Hence the correct option is 2

Q. 8 An electric lift with a maximum load of $2000\ \text{kg}(\text{lift} + \text{passengers})$ is moving up with a constant speed of $1.5\ \text{ms}^{-1}$. The frictional force opposing the motion is $3000\ \text{N}$. The minimum power delivered by the motor to the lift in watts is: $(g = 10\ \text{ms}^{-2})$

Option 1:
20000

Option 2:
34500

Option 3:
23500

Option 4:
23000

Correct Answer:
34500

Solution:
For the minimum power delivered by the motor to the lift, lift is moving up with a constant speed i.e net acceleration is zero

$$\begin{aligned}\text{External force (by motor)} &= \text{Frictional force} + \text{weight} \\ &= 3000\ \text{N} + (2000) \times 10 \\ &= 23000\end{aligned}$$

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$$
\begin{aligned}
P_{\text{delivered}} &= F \cdot v \\
&= (23000)(1.5) \\
&= 34500 \text{ W}
\end{aligned}
$$

Hence, the answer is the option (2).

Q. 9 The energy that will be ideally radiated by a 100kW transmitter in an hour is:

Option 1:
$36 \times 10^4 \text{J}$

Option 2:
$36 \times 10^5 \text{J}$

Option 3:
$1 \times 10^5 \text{J}$

Option 4:
$36 \times 10^7 \text{J}$

Correct Answer:
$36 \times 10^7 \text{J}$

Solution:

$$
\begin{aligned}
\text{Energy} &= (\text{ power })(\text{ time }) \\
&= 100 \times 10^3 \left(\frac{\text{J}}{\text{s}}\right) \times (3600 \text{ s}) \\
&= 3600 \times 10^5 \text{ J} \\
&= 36 \times 10^7 \text{ J}
\end{aligned}
$$

Hence, the answer is the option (4).

Q. 10 Two objects of mass 10kg and 20kg respectively are connected to the two ends of a rigid rod of length 10m with negligible mass. The distance of the centre of mass system from the 10kg mass is :

Option 1:
$\frac{20}{3}\text{m}$

Option 2:
10m

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Option 3:

$5\mathrm{m}$

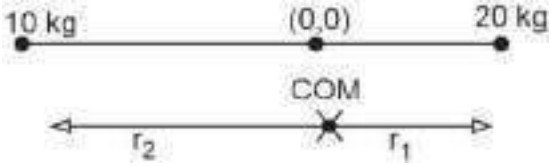
Option 4:

$\frac{10}{3}\mathrm{m}$

Correct Answer:

$\frac{20}{3}\mathrm{m}$

Solution:



Let the com be at the origin

$$r_{cm} = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$$

$$0 = \frac{20\,(r_1) + 10\,(-r_2)}{30}$$

$$r_1 = \frac{r_2}{2} \qquad \rightarrow 1$$

$$\because r_1 + r_2 = 10 \text{ m (given)}$$

$$\therefore r_1 = \frac{10}{3} \text{ m}$$

$$r_2 = \frac{20}{3} \text{ m}$$

the distance of the centre of mass of the system from the 10kg mass is $r_2 = \frac{20}{3}\mathrm{m}$

Hence the correct option is 1

Q. 11 The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the disc about its diameter is :

Option 1:

$\sqrt{2} : 1$

Option 2:

$4 : 1$

Option 3:

$1 : \sqrt{2}$

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Option 4:

$2:1$

Correct Answer:

$\sqrt{2}:1$

Solution:

Let the moment of inertia of a thin uniform disc about an axis normal to its plane and its diameter be I_1 & I_2 respectively

$$I_1 = \frac{MR^2}{2} = MK_1^2$$

$$I_2 = \frac{MR^2}{4} = MK_2^2$$

$$\frac{K_1}{K_2} = \sqrt{2}$$

Hence, the answer is the option (1).

Q. 12 The angular speed of a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/$_s^2$ is:

Option 1:

4π

Option 2:

12π

Option 3:

104π

Option 4:

2π

Correct Answer:

4π

Solution:

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$$\omega_{\text{initial}} = 1200\text{rpm} = 1200 \times \frac{2\pi}{60}$$
$$= 40\pi \frac{\text{rad}}{\text{s}}$$
$$\omega_{\text{final}} = 3120\text{rpm} = 3120 \times \frac{2\pi}{60}$$
$$= 104\pi$$
$$\text{Angular acceleration} = \frac{\omega_{\text{final}} - \omega_{\text{initial}}}{\Delta t}$$
$$(\alpha) = \frac{104\pi - 40\pi}{16}$$
$$= \frac{64\pi}{16} = 4\pi \frac{\text{rad}}{\text{s}^2}$$

Hence, the answer is the option (1).

Q. 13 Match list I with list II

	List I		List II
a	Gravitational constant(G)	i	$[\text{L}^2\text{T}^{-2}]$
b	Gravitational potential energy	ii	$[\text{M}^{-1}\text{L}^3\text{T}^{-2}]$
c	Gravitational potential	iii	$[\text{LT}^{-2}]$
d	Gravitational Intensity	iv	$[\text{ML}^2\text{T}^{-2}]$

Choose the correct answer from the options given below:

Option 1:
$(\text{a}) - (\text{ii}), (\text{b}) - (\text{iv}), (\text{c}) - (\text{i}), (\text{d}) - (\text{iii})$

Option 2:
$(\text{a}) - (\text{ii}), (\text{b}) - (\text{iv}), (\text{c}) - (\text{iii}), (\text{d}) - (\text{i})$

Option 3:
$(\text{a}) - (\text{iv}), (\text{b}) - (\text{ii}), (\text{c}) - (\text{i}), (\text{d}) - (\text{iii})$

Option 4:
$(\text{a}) - (\text{ii}), (\text{b}) - (\text{i}), (\text{c}) - (\text{iv}), (\text{d}) - (\text{iii})$

Correct Answer:
$(\text{a}) - (\text{ii}), (\text{b}) - (\text{iv}), (\text{c}) - (\text{i}), (\text{d}) - (\text{iii})$

Solution:
Gravitational Potential energy $= [\text{M}^1\text{L}^2\text{T}^{-2}]$

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Gravitational Potential $= \left[\mathrm{L}^2\mathrm{T}^{-2}\right]$

Gravitational Intensity $= \left[\dfrac{\mathrm{F}}{\mathrm{m}}\right] = \left[\mathrm{M}^0\mathrm{L}^1\mathrm{T}^{-2}\right]$

Gravitational Constant $= \left[\dfrac{\mathrm{Fr}^2}{\mathrm{m}_1\mathrm{m}_2}\right] = \left[\mathrm{M}^{-1}\mathrm{L}^3\mathrm{T}^{-2}\right]$

Hence, the answer is the option (1).

Q. 14 A body of mass 60g experiences a gravitational force $3.0\ \mathrm{N}$ when placed at a particular point. The magnitude of the gravitational field intensity at that point is:

Option 1:
$50\ \mathrm{N/kg}$

Option 2:
$20\ \mathrm{N/kg}$

Option 3:
$180\ \mathrm{N/kg}$

Option 4:
$0.05\ \mathrm{N/kg}$

Correct Answer:
$50\ \mathrm{N/kg}$

Solution:

$$\begin{aligned}\text{Gravitational field intensity} &= \frac{F}{m}\\ &= \frac{3\ \mathrm{N}}{60\times10^{-3}\ \mathrm{kg}}\\ &= 50\ \mathrm{N/kg}\end{aligned}$$

Hence, the answer is the option (1).

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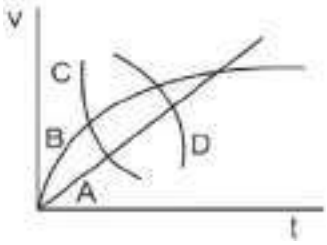
Q. 15 A spherical ball is dropped in a long column of a highly viscous liquid . The curve in the graph shown, which represents the speed of the ball as a function of time (t) is:



Option 1:
B

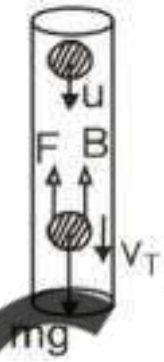
Option 2:
C

Option 3:
D

Option 4:
A

Correct Answer:
B

Solution:



Let the spherical ball is dropped with initial speed say, u
With the increase in speed, the friction drag force by stokes increase until it reaches the terminal speed

B $\rightarrow$ Buoyancy force

$$F + B = mg$$

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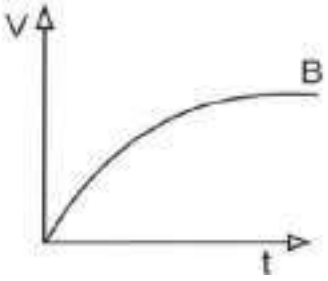

Q. 16 Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): The stretching of a spring is determined by the shear modulus of the material of the spring.

Reason (R): A coil spring of copper has more tensile strength than a steel spring of the same dimensions.

In light of the above statements choose the most appropriate answer from the options given below:

Option 1:
Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**

Option 2:
(A) is true but **(R)** is false

Option 3:
(A) is false but **(R)** is ture

Option 4:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Correct Answer:
(A) is true but **(R)** is false

Solution:
$\rightarrow$ During the stretching of spring, the shape of spring is changed. Therefore, we say that the shear modulus determines that.

$\rightarrow Y_{\text{steel}} > Y_{\text{copper}}$ (Tensile Strength αY)

$\therefore$ The assertion is true but the reason is not true

Hence, the answer is the option (2).

Q. 17 if a soap bubble expands, the pressure inside the bubble:

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Option 1:
increases

Option 2:

remains the same

Option 3:
is equal to the atmospheric pressure

Option 4:
decreases

Correct Answer:
decreases

Solution:
Inside the soap bubble, excess pressure is

$$\Delta P = \frac{4T}{R}$$

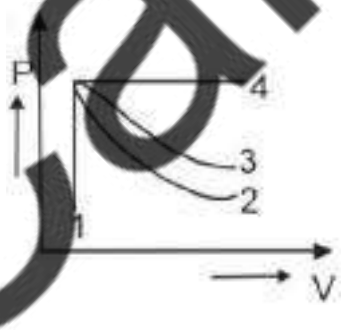
$$\Delta P = P_{\text{inside}} - P_{\text{outside}} = \frac{4T}{R}$$

$$\because P_{\text{outside}} = P_{\text{atmospheric}} = \text{constant}$$

With the increase in radius as the bubble expands, P_{inside} decreases.

Hence, the answer is the option (4).

Q. 18 An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1,2,3 and 4 is.



Option 1:
2

Option 2:
3

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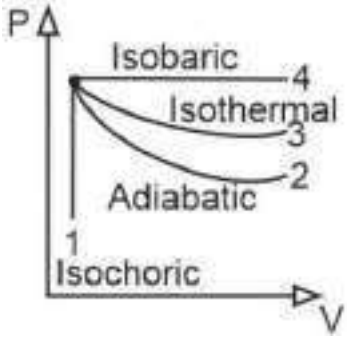
Option 3:
4

Option 4:
1

Correct Answer:
2

Solution:



curve 4 $\longrightarrow$ pressure is const constant, therefore it is an Isobaric process
curve 2 is more steeper than Curve 3
$\therefore$ curve 2 $\longrightarrow$ Adiabatic

curve 3 $\longrightarrow$ Isothermal
For curve 1, volume is constant
$\therefore$ curve 1 $\longrightarrow$ Isochoric process.
The correct option is (1)

Q. 19 The volume occupied by the molecules contained in $4.5kg$ water at STP, if the intermolecular forces vanish away is :

Option 1:
$5.6 \times 10^{3}\ \mathrm{m}^{3}$

Option 2:
$5.6 \times 10^{-3}\ \mathrm{m}^{3}$

Option 3:
$5.6\ \mathrm{m}^{3}$

Option 4:
$5.6 \times 10^{6}\ \mathrm{m}^{3}$

Correct Answer:
$5.6\ \mathrm{m}^{3}$

Solution:
The molar mass of water $= 18$

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Given mass, $M = 4500\text{gm}$

no of moles $= \frac{\text{m}}{\text{M}} = \frac{4500}{18}$

$\text{n} = 250\ \text{mol}$

At STP, 1 mol occupies 22.4dm^3 of volume

$1\text{mol} \rightarrow 22.4\text{dm}^3$

$250\ \text{mol} \rightarrow \text{V}$

$\text{V} = 250 \times 22.4\text{dm}^3$

$\text{V} = 5.6\ \text{m}^3 \qquad (\because (\text{dm})^3 = 10^{-3}\ \text{m}^3)$

Hence, the answer is the option (3).

Q. 20 Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is :

Option 1:
9

Option 2:
10

Option 3:
8

Option 4:
11

Correct Answer:
11

Solution:

$$\text{T}_1 = 2\pi\sqrt{\frac{\text{l}_1}{\text{g}}} = \frac{2\pi}{\omega_1} = \frac{2\pi \times \sqrt{1 \cdot 21}}{\sqrt{\text{g}}}$$

$$\text{T}_2 = 2\pi\sqrt{\frac{\text{l}_2}{\text{g}}} = \frac{2\pi}{\omega_2} = \frac{2\pi}{\sqrt{\text{g}}}(\sqrt{1})$$

Let θ_1 and θ_2 be the phase covered by the 1st and 2nd pendulum respectively. For the two pendulums to be in the same phase,

$\theta_2 - \theta_1 = 2n\pi$

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$(n = 1, 2, 3.....)$

For the minimum number of vibrations ie. $(n = 1)$

$\theta_2 - \theta_1 = \omega_2 t - \omega_1 t = 2\pi$

$t\left(\frac{2\pi}{T_2} - \frac{2\pi}{T_1}\right) = 2\pi$

$\frac{1}{t} = \frac{1}{T_2} - \frac{1}{T_1}$

$\because \frac{T_1}{T_2} = \frac{1.1}{1} = \frac{11}{10}$

$\therefore T_1 = \frac{11T_2}{10}$

$\frac{1}{t} = \frac{1}{T_2} - \frac{1}{\frac{11T_2}{10}}$

$\frac{1}{t} = \frac{1}{11T_2}$

$t = 11T_2$

Hence, the answer is the option (4).

Q. 21 If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is:

Option 1:
$\sqrt{2} : 1$

Option 2:
$1 : \sqrt{2}$

Option 3:
$1 : 2$

Option 4:
$1 : 1$

Correct Answer:
$1 : \sqrt{2}$

Solution:
The speed of a transverse wave along the string is

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$$V = \sqrt{\frac{T}{\mu}}$$
$$\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$$
$$T_2 = 2T_1 \ (\text{Given})$$
$$\frac{V_1}{V_2} = \frac{1}{\sqrt{2}}$$

Hence, the answer is the option (2).

Q. 22 Two hollow conducting spheres of radius $\mathrm{R_1}$ and $\mathrm{R_2}$ $(\mathrm{R_1} >> \mathrm{R_2})$ have equal charges. The potential would be:

Option 1:
more on smaller sphere

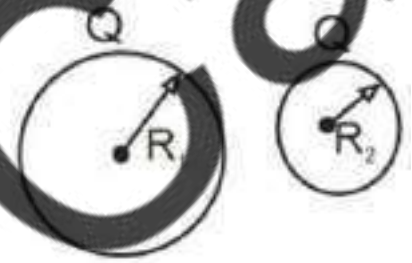
Option 2:
equal on both the spheres

Option 3:
dependent on the material property of the sphere

Option 4:
more on bigger sphere

Correct Answer:
more on smaller sphere

Solution:



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Let the potential of sphere of radius $\mathrm{R_1\ and\ R_2\ be\ V_1 \& V_2}$ respectively

$$V_1 = \frac{KQ}{R_1}$$

$$V_2 = \frac{KQ}{R_2}$$

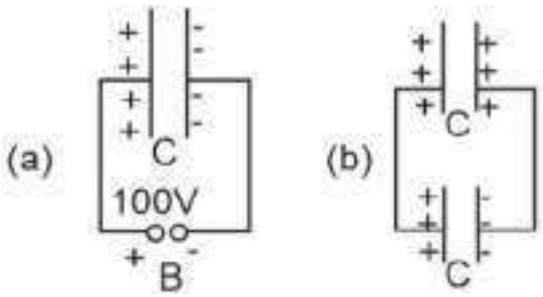
$$\because R_1 > R_2$$

$$\therefore V_1 < V_2$$

The potential would be more on smaller sphere.

The correct option is (1)

Q. 23 A capacitor of capacitance $\mathrm{C = 900pF}$ is charged fully by $100\mathrm{V}$ battery B as shown in fig (a). Then it is disconnected from the battery and connected to another uncharged capacitor of capacitance $\mathrm{C = 900pF}$ as shown in fig(b). The electrostatic energy stored by system (b) is :



Option 1:

$3.25 \times 10^{-6}\ \mathrm{J}$

Option 2:

$2.25 \times 10^{-6}\ \mathrm{J}$

Option 3:

$1.5 \times 10^{-6}\ \mathrm{J}$

Option 4:

$4.5 \times 10^{-6}\ \mathrm{J}$

Correct Answer:

$2.25 \times 10^{-6}\ \mathrm{J}$

Solution:

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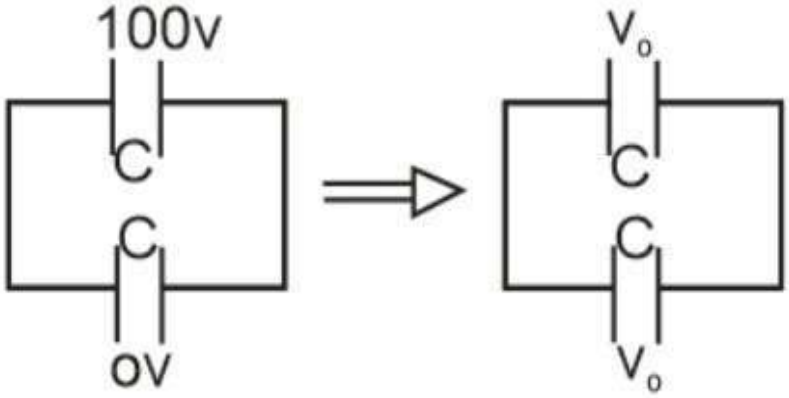


(charged capacitor just connected to uncharged capacitor)

(Steady state)

As both the capacitor are in a parallel connection they attained the same potential at the steady state

Common potential $= \mathrm{V_0} = \dfrac{\mathrm{C_1V_1 + C_2V_2}}{\mathrm{C_1 + C_2}}$

$$= \frac{100\mathrm{C} + 0}{2\mathrm{C}}$$

$$\mathrm{V_0} = 50\mathrm{volt}$$

$$\mathrm{U_{initial}} = \frac{1}{2}\mathrm{C(V)^2} = \frac{1}{2} \times 900 \times 10^{-12} \times 10^4$$

$$= 450 \times 10^{-8}\ \mathrm{J}$$

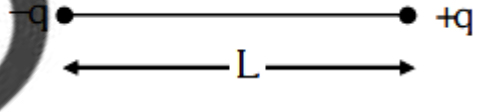
$$\mathrm{U_{final}} = \left(\frac{1}{2}\mathrm{CV_0^2} + \frac{1}{2}\mathrm{CV_0^2}\right)$$

$$= 2\left(\frac{1}{2} \times 900 \times 10^{-9} \times (50)^2\right)$$

$$= 225 \times 10^{-8}\ \mathrm{J}$$

Hence the correct option is 2

Q. 24 Two point charges -q and +q are placed at a distance of L, as shown in the figure.



The magnitude of electric field intensity at a distance $\mathrm{R}(\mathrm{R} >> \mathrm{L})$ varies as

Option 1:

$\dfrac{1}{R^3}$

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Option 2:

$\frac{1}{R^4}$

Option 3:

$\frac{1}{R^6}$

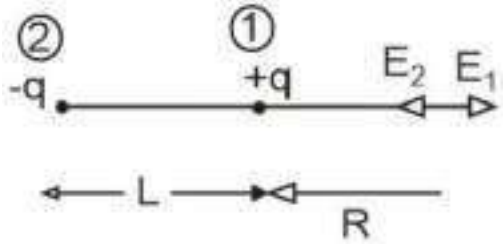
Option 4:

$\frac{1}{R^2}$

Correct Answer:

$\frac{1}{R^3}$

Solution:



$$E_1 = \frac{kq}{R^2}$$

$$E_2 = \frac{kq}{(R+L)^2}$$

$$E_{net} = E_1 - E_2$$

$$= \frac{kq}{R^2} - \frac{kq}{R^2\left(1+\frac{L}{R}\right)^2}$$

$$= \frac{kq}{R^2}\left[1-\left(1+\frac{L}{R}\right)^{-2}\right]$$

$$E_{net} = \frac{2kqL}{R^3}$$

$$= \frac{kq}{R^2}\left(1-\left(1-\frac{2L}{R}\right)\right) \quad (\text{ by binomial expantion })$$

Hence correct option is 1

Q. 25 The angle between the electric lines of force and the equipotential surface is :

Option 1:

45°

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Option 2:
90°

Option 3:
180°

Option 4:
0°

Correct Answer:
90°

Solution:
The angle between the equipotential surface and lines of force is always 90 degrees.

EXPLANATION

The angle between the equipotential surface and lines of force is always 90 degrees because when the electric potential becomes constant, the negative potential gradient becomes zero, and thus it makes it necessary for the Electric field to be always normal with the surface.

Hence, the answer is the option (2).

Q. 26 The angle between the electric lines of force and the equipotential surface is:

Option 1:
45°

Option 2:
90°

Option 3:
180°

Option 4:
0°

Correct Answer:
90°

Solution:
Electric field lines are always perpendicular to the equipotential surface.

The angle between an Electric field and an equipotential surface is always 90^0. This is because, when the potential becomes constant, the negative potential gradient also becomes zero, hence necessitating the need for the Electric field to be always normal with the surface.

Hence, the answer is the option (2).

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Q. 27 Two resistors of resistance, 100Ω and 200Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100Ω to that in 200Ω in a given time is:

Option 1:
2:1

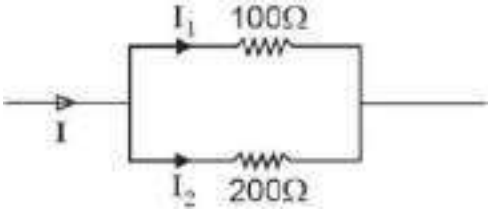
Option 2:
1:4

Option 3:
4:1

Option 4:
1:2

Correct Answer:
2:1

Solution:



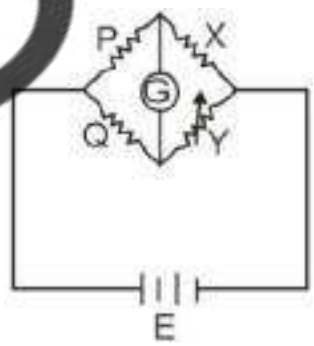
$$I_1 = \frac{I(200)}{(300)} = \frac{2I}{3}$$

$$I_2 = \frac{I(100)}{200+100} = \frac{I}{3}$$

$$\frac{H_1}{H_2} = \frac{I_1^2 R_1 t}{I_2^2 R_2 t} = \frac{2}{1}$$

The correct option is (1)

Q. 28 A Wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistances P and Q:



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Option 1:
should be approximately equal and are small

Option 2:
should be very large and unequal

Option 3:
do not play any significant role

Option 4:
should be approximately equal to 2X

Correct Answer:
should be approximately equal and are small

Solution:
In Wheatstone bridge for the balanced condition

$$\frac{P}{Q}=\frac{X}{Y}$$

$$P=\frac{XQ}{Y}$$

$$Q=\frac{YP}{X}$$

If $P \& Q$ are equal, then by balancing conditions for $X \& Y$ will be equal which is a possible case.

The correct option is 1

Q. 29 As the temperature increases, the electrical resistance

Option 1:
decreases for both conductors and semiconductors

Option 2:
increases for conductors but decreases for semiconductors

Option 3:
decreases for conductors but increases for semiconductors

Option 4:
increases for both conductors and semiconductors

Correct Answer:
increases for conductors but decreases for semiconductors

Solution:

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The temperature coefficient of resistance (α) is positive for conductors and negative for semiconductors as

$$\propto=\frac{\Delta R}{R\Delta T}$$

Therefore, with the increase in temperature, the electrical resistance increases for conductors but decreases for semiconductors

Hence, the answer is the option (2).

Q. 30 A copper wire of length $10\mathrm{m}$ and radius $\left(10^{-2}/\sqrt{\pi}\right)\,\mathrm{m}$ has an electrical resistance of 10Ω. The current density in the wire for an electric field strength of $10\,(\mathrm{V}/\mathrm{m})$ is :

Option 1:
$10^{6}\mathrm{A}/\mathrm{m}^{2}$

Option 2:
$10^{-5}\mathrm{A}/\mathrm{m}^{2}$

Option 3:
$10^{5}\mathrm{A}/\mathrm{m}^{2}$

Option 4:
$10^{3}\mathrm{A}/\mathrm{m}^{2}$

Correct Answer:
$10^{5}\mathrm{A}/\mathrm{m}^{2}$

Solution:
We know that,

$$\mathrm{J}=\frac{\mathrm{E}}{\rho}$$

$$\text{Also, } \mathrm{R}=\frac{\rho \mathrm{l}}{\mathrm{A}}$$

$$\therefore \mathrm{J}=\frac{\mathrm{E}}{\left(\frac{\mathrm{RA}}{\mathrm{l}}\right)}$$

$$\mathrm{J}=\frac{10}{\frac{10\times\pi\left(10^{-4}/\pi\right)}{10}}$$

$$\mathrm{J}=10^{5}\ \mathrm{A}/\mathrm{m}^{2}$$

Hence, the answer is the option (3).

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Q. 31 From Ampere's circuital law for a long straight wire of circular cross section carrying a steady current, the variation of magneic field in the inside and outside region of the wire is :

Option 1:
A linearly increasing function distance upto the boundary of the wire and then linearly decreasing for the outside region

Option 2:
A linearly increasing function distance r upto the boundary of the wire and then decreasing with 1/r for the outside region

Option 3:
A linearly increasing function distance upto the boundary of the wire and then linearly increasing one for the outside region

Option 4:
uniform and remains constant for both the regions.

Correct Answer:
A linearly increasing function distance r upto the boundary of the wire and then decreasing with 1/r for the outside region

Solution:

$$B_{inside} = \frac{\mu_0 ir}{2\pi R^2} \quad (r \leq R)$$

$$B_{outside} = \frac{\mu_0 i}{2\pi r} (r > R)$$

Hence the correct option is 2

Q. 32 Given below are two statements :
Statement I: Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (IdI) of a current-carrying conductor only.
Statement II: Biot-Savart's law is analogous to Coulomb's inverse square law of charge q, with the former being related to the field produced by a scalar source, Idl while the latter being produced by a vector source, q.
In light of the above statements choose the most appropriate answer from the options given below:

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Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct and Statement II is incorrect

Option 3:
Statement I is incorrect and Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Statement I is correct and Statement II is incorrect

Solution:
Statement (I) is correct

Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element (Idl) of a current-carrying conductor only

$$B = \frac{\mu_0}{4\pi}\frac{Idl\sin\theta}{r^2}$$

Statement (II) is incorrect
Biot-Savart's law is analogous to Coulomb's inverse square law of charge q with the former being related to the field produced by the vector source, Idl while the latter is produced by a ' scalar source ' q '.

Hence, the answer is the option (2).

Q. 33 A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is?

Option 1:
$12.56 \times 10^{-2}\mathrm{T}$

Option 2:
$12.56 \times 10^{-4}\mathrm{T}$

Option 3:
$6.28 \times 10^{-4}\mathrm{T}$

Option 4:
$6.28 \times 10^{-2}\mathrm{T}$

Correct Answer:
$12.56 \times 10^{-2}\mathrm{T}$

Solution:

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$$r = 1 \text{ mm}$$

$$n = \frac{N}{l} = 100 \text{ turns per mm}$$

$$i = 1 \text{ A}$$

$$B_{\text{centre}} = \mu_0 n i$$

$$= 4\pi \times 10^{-7} \times \frac{100}{10^{-3} \text{ m}} \times 1$$

$$= 4\pi \times 10^{-2} \text{ T}$$

$$B_{\text{centre}} = 12.56 \times 10^{-2} \text{ T}$$

Hence, the answer is the option (1).

Q. 34 A square loop of side 1m ad resistance 1Ω is placed in a magnetic field of 0.5T. If the plane of the loop is perpendicular to the direction of the magnetic field, the magnetic flux through the loop is :

Option 1:

0.5weber

Option 2:

1weber

Option 3:

zero weber

Option 4:

2weber

Correct Answer:

0.5weber

Solution:

$$\text{B} = 0.5\text{T}$$

$$\phi = \text{BA}\cos 0^\circ \qquad (\because \bar{\text{A}} \| \bar{\text{B}})$$

$$\phi = \text{BA} = (0.5)(1)^2$$

$$\phi = 0.5 \text{ weber}$$

Hence correct option is 1

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Q. 35 A big circular coil of 1000 turns and average radius $10m$ is rotating about its horizontal diameter at $2\ rad\ s^{-1}$. If the vertical component of the earth's magnetic field at the place is $2 \times 10^{-5}T$ and the electrical resistance of the coil is 12.56Ω, then the maximum induced current in the coil will be :

Option 1:
$1.5A$

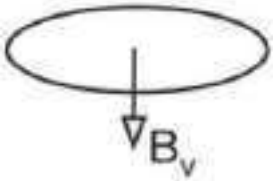
Option 2:
$1A$

Option 3:
$2A$

Option 4:
$0.25A$

Correct Answer:
$1A$

Solution:



$$e = e_0 \sin \omega t$$

$$\text{Peak emf induced} = e_0 = nAB\omega$$

$$i_{peak} = \text{Maximum induced current}$$

$$= \frac{e_0}{R} = \frac{nABW}{R}$$

$$i = \frac{1000 \times (\pi \times (10)^2 \times 2 \times 10^{-5} \times 2)}{12.56} = 1A$$

Hence correct option is 2.

Q. 36 A series LCR circuit with inductance 10H, capacitance $10\mu\text{F}$, and resistance 50Ω is connected to an AC source of voltage, $V = 200\sin(100t)$ volts, If the resonant frequency of the LCR circuit is ν_0 and the frequency of the AC source is ν, then :

Option 1:
$\nu_0 = \nu = \frac{50}{\pi}\text{Hz}$

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Option 2:

$\nu_0 = \frac{50}{\pi}\text{Hz}, \nu = 50\ \text{Hz}$

Option 3:

$\nu = 100\ \text{Hz}; \nu_0 = \frac{100}{\pi}\text{Hz}$

Option 4:

$\nu_0 = \nu = 50\ \text{Hz}$

Correct Answer:

$\nu_0 = \nu = \frac{50}{\pi}\text{Hz}$

Solution:

$\text{V} = 200\sin(100\text{t})$

$\nu = \frac{50}{\pi}$

$\nu_0 = \frac{1}{2\pi} \times \frac{1}{\sqrt{\text{LC}}} = \frac{1}{2\pi}\frac{1}{\sqrt{10 \times 10^{-5}}} = \frac{100}{2\pi} = \frac{50}{\pi}$

Hence, the answer is the option (1).

Q. 37 The peak voltage of the AC source is equal to:

Option 1:

the rms value of the ac source

Option 2:

$\sqrt{2}$ times the rms value of the ac source

Option 3:

$\frac{1}{\sqrt{2}}$ times the rms value of the ac source

Option 4:

the value of voltage supplied to the circuit

Correct Answer:

$\sqrt{2}$ times the rms value of the ac source

Solution:

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$$\frac{\text{Peak Voltage}}{\sqrt{2}} = \text{RMS Voltage}$$
$$\therefore \text{Peak voltage} = \sqrt{2}(\text{RMS voltage})$$

Hence, the answer is the option (2).

Q. 38 Match List-I with List-II

List-I	List-II
(Electromagnetic waves)	(Wavelength)
(a) AM radio waves	(i) 10^{-10}m
(b) Microwaves	(ii) 10^{2}m
(c) Infrared radiation	(iii) 10^{-2}m
(d) X-ray	(iv) 10^{-4}m

Choose the correct answer from the options given below:

Option 1:
(a)-(iii), (b)-(iii),(c)-(i),(d)-(iv)

Option 2:
(a)-(iii), (b)-(iv),(c)-(ii),(d)-(i)

Option 3:
(a)-(ii), (b)-(iii),(c)-(iv),(d)-(i)

Option 4:
(a)-(iv), (b)-(iii),(c)-(ii),(d)-(i)

Correct Answer:
(a)-(ii), (b)-(iii),(c)-(iv),(d)-(i)

Solution:

List-I	List-II
(Electromagnetic waves)	(Wavelength)
(a) AM radio waves	(i) 10^{2}m
(b) Microwaves	(ii) 10^{-2}m
(c) Infrared radiation	(iii) 10^{-4}m
(d) X-ray	(iv) 10^{-10}m

Hence, The answer is the option (3).

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Q. 39 When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r, the velocity of light v is given by : (c- velocity of light in vacuum)

Option 1:
$\mathrm{v}=\sqrt{\frac{\mu_r}{\varepsilon_r}}$

Option 2:
$\mathrm{v}=\sqrt{\frac{\varepsilon_r}{\mu_r}}$

Option 3:
$\mathrm{v}=\frac{\mathrm{c}}{\sqrt{\varepsilon_r \mu_r}}$

Option 4:
$\mathrm{v}=\mathrm{c}$

Correct Answer:
$\mathrm{v}=\frac{\mathrm{c}}{\sqrt{\varepsilon_r \mu_r}}$

Solution:
For light traveling in the medium of relative permeability and permittivity, μ_r and ϵ_r respectively the speed of light is

$$\mathrm{v}=\frac{\mathrm{c}}{\sqrt{\mu_r \varepsilon_r}}$$

Hence, the answer is the option (3).

Q. 40 A biconvex lens has radii of curvature, $20\ \mathrm{cm}$ each. If the refractive index of the material of the lens is 1.5, the power of the lens is

Option 1:
$+20\mathrm{D}$

Option 2:
$+5\mathrm{D}$

Option 3:
infinity

Option 4:
$+2\mathrm{D}$

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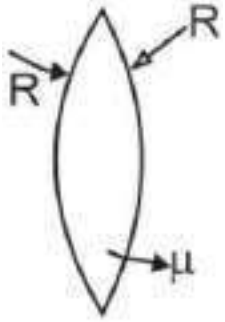
Correct Answer:
$+5\mathrm{D}$

Solution:



$R = 20\ \mathrm{cm}$
$\mu = 1.5$

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{(+R)} - \frac{1}{(-R)}\right)$$
$$\frac{1}{f} = \frac{2(\mu - 1)}{R} = \frac{1}{R} = \frac{1}{+R} = \frac{1}{20\ \mathrm{cm}} = +5D$$

The power of the lens is $5D$

The correct option is (2)

Q. 41 Two transparent media A and B are separated by a plane boundary. The speed of light in those media are $1.5 \times 10^8 m/s$ and $2.0 \times 10^8 m/s$ respectively. The critical angle for a ray of light for these two media is :

Option 1:
$\sin^{-1}(0.750)$

Option 2:
$\tan^{-1}(0.500)$

Option 3:
$\tan^{-1}(0.750)$

Option 4:
$\sin^{-1}(0.500)$

Correct Answer:
$\sin^{-1}(0.750)$

Solution:

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V_1 Medium 1 (Denser)

V_2 Medium 2 (Rarer)

$V_1 = 1.5 \times 10^8 \text{ m/s}$

$V_2 = 2 \times 10^8 \text{ m/s}$

$\mu \propto \frac{1}{V}$

$\mu_1 > \mu_2$

$\sin i_c = \frac{1}{\mu_D^R} = \frac{\mu_R}{\mu_D} = \frac{\mu_2}{\mu_1}$

$\sin_c = 0.75$

$i_c = \sin^{-1}(0.75)$

Hence correct option is 1

Q. 42 A light ray falls on a glass surface of the refractive index $\sqrt{3}$ at an angle $60°$. The angle between the refracted and reflected rays would be.

Option 1:
$60°$

Option 2:
$90°$

Option 3:
$120°$

Option 4:
$30°$

Correct Answer:
$90°$

Solution:

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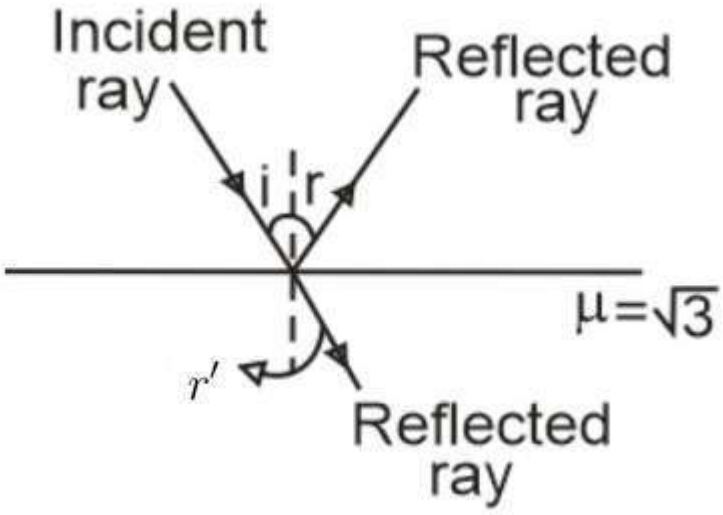

$\angle r = \angle i = 60^\circ (\text{Laws of reflection})$

$\mu_1 \sin i = \mu_2 \sin r'$

$1 \times \sin 60^\circ = \sqrt{3} \sin r'$

$\Rightarrow r' = 30^\circ$

The angle between the refracted and reflected rays would be

$(180 - (r + r')) = 90^\circ$

Q. 43 In Young's double slit experiment, a student observes 8 fringes in a certain segment of the screen when a monochromatic light of 600nm wavelength is used. If the wavelength of light is changed 400nm, then the number of fringes he would observe in the same region of the screen is :

Option 1:

8

Option 2:

9

Option 3:

12

Option 4:

6

Correct Answer:

12

Solution:

Let the length of the segment be y

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Since there are 8 fringes in that segment

$\therefore \quad 8\beta_1 = \mathrm{y}$ ....(1)

$\beta_1 = \dfrac{\lambda_1 \mathrm{D}}{\mathrm{d}} =$ fringe width for wavelength $(\lambda_1 = 600\mathrm{nm})$

Let's say there are n_2 fringes of wavelength $\lambda_2 = 400\mathrm{nm}$ in the segment y

$\therefore \mathrm{n}_2\beta_2 = \mathrm{y}$ .....(2)

From eqn (1)

$8\beta_1 = \mathrm{n}_2\beta_2$

$8 \times \lambda_1 = \mathrm{n}_2\lambda_2$

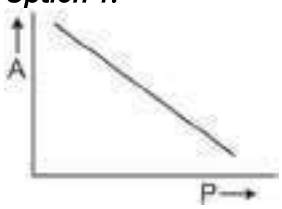
$8 \times 600 = \mathrm{n}_2(400)$

$\mathrm{n}_2 = 12$

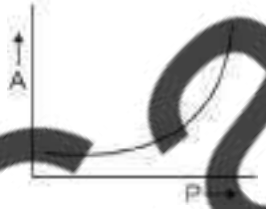
Hence, the answer is the option (3).

Q. 44 The graph which shows the variation of the De Broglie wavelength (A) of a particle and its associated momentum (p) is :

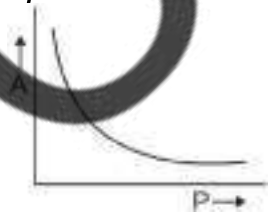
Option 1:



Option 2:



Option 3:



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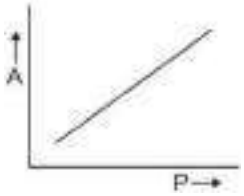
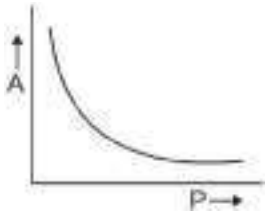
Option 4:



Correct Answer:



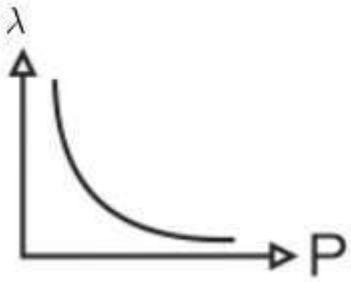
Solution:

The Broglie wavelength of a particle is $\lambda = \frac{\mathrm{h}}{\mathrm{p}}$

$\mathrm{p} \rightarrow$ momentum of particle

$\lambda \mathrm{p} = \mathrm{h} = \text{constant}$

$\therefore \lambda \text{ vs } \mathrm{p}$ the graph will be a rectangular hyperbola



Hence the correct option is 3

Q. 45 When two monochromatic lights of frequency, ν and $\frac{\nu}{2}$ are incident on a photoelectric metal, their stopping potential becomes $\frac{\mathrm{Vs}}{2}$ and Vs respectively. The threshold frequency for this metal is :

Option 1:

3ν

Option 2:

$\frac{2}{3}\nu$

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Option 3:
$\frac{3}{2}\nu$

Option 4:
2ν

Correct Answer:
$\frac{3}{2}\nu$

Solution:
By Einstein's photoelectric equation,

For the first source

$$\mathrm{h}\nu = \phi_0 + \frac{\mathrm{eV_s}}{2} \quad \text{.....(1)}$$

For the second source,

$$\frac{\mathrm{h}\nu}{2} = \phi_0 + \mathrm{eV_s} \quad \text{.....(2)}$$

$2\times$ Eqn (1) - Eqn(2)

$$\frac{3\mathrm{h}\nu}{2} = \phi_0 = \mathrm{h}\nu_0$$

$\nu_0 \rightarrow$ threshold frequency

$$\nu_0 \equiv \frac{3\nu}{2}$$

Hence, the answer is the option (3).

Q. 46 In the given nuclear reaction, the element X is : $^{22}_{11}Na \rightarrow X + e^{+} + v$

Option 1:
$^{23}_{10}\mathrm{Ne}$

Option 2:
$^{22}_{10}\mathrm{Ne}$

Option 3:
$^{22}_{12}\mathrm{Mg}$

Option 4:
$^{23}_{11}\mathrm{Na}$

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Correct Answer:

$^{22}_{10}\mathrm{Ne}$

Solution:

$^{22}_{11}Na \rightarrow {}^{A}_{z}X + {}^{0}_{1}e + v$

$22 = \mathrm{A} + 0$

$\therefore \mathrm{A} = 22$

$11 = \mathrm{Z} + 1$

$\therefore \mathrm{Z} = 10$

The atomic number and atomic mass number of X are 10 and 22 respectively

$\therefore \mathrm{X} = {}^{22}_{10}\mathrm{Ne}$

Hence the correct option is 2

Q. 47 Let T_1 and T_2 be the energy of an electron in the first and second excited states of the hydrogen atom, respectively. According to Bohr's model of an atom, the ratio $\mathrm{T}_1 : \mathrm{T}_2$ is :

Option 1:

$4 : 1$

Option 2:

$4 : 9$

Option 3:

$9 : 4$

Option 4:

$1 : 4$

Correct Answer:

$9 : 4$

Solution:

The energy of the electron in the nth level of the hydrogen atom,

$$\mathrm{E_n} = \frac{-13.6\mathrm{eV}}{\mathrm{n}^2}$$

For the first excited state, $\mathrm{n}_1 = 2$

For the second excited state, $\mathrm{n}_2 = 3$

$$\mathrm{T}_1 = -\frac{13.6\mathrm{eV}}{\mathrm{n}_1^2} = \frac{-13.6\mathrm{eV}}{4}$$

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$$T_2 = \frac{-13.6eV}{n_2^2} = \frac{-13.6eV}{9}$$

$$\frac{T_1}{T_2} = \frac{9}{4}$$

Hence, the answer is the option (3).

Q. 48 A nucleus of mass number 189 splits into two nuclei having mass numbers 125 and 64. The ratio of the radius of two daughter nuclei respectively is :

Option 1:
$4 : 5$

Option 2:
$5 : 4$

Option 3:
$25 : 16$

Option 4:
$1 : 1$

Correct Answer:
$5 : 4$

Solution:
$^{189}N \longrightarrow {}^{125}N_1 + {}^{64}N_2$

$A_1 = 125$

$A_2 = 64$

We know that the radius of nuclei and mass number relation is

$$R = R_0 A^{1/3}$$

$$\therefore \frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \frac{5}{4}$$

Hence, the answer is the option (2).

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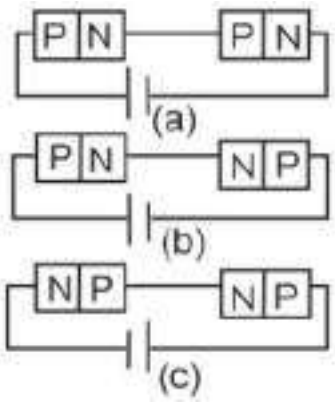
Q. 49



In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in :

Option 1:
Circuit (b) only

Option 2:
Circuit (c) only

Option 3:
Both circuits (a) and (c)

Option 4:
Circuit (a) only

Correct Answer:
Both circuits (a) and (c)

Solution:
(B) Both circuits (a) and (c) are forward-biased. The potential drop across both the diodes will be the same for circuits (a) & (c)
Circuit (b) is reverse-biased, so there is no current through the circuit.
The correct option is (3)

Q. 50

In half wave rectification, if the input frequency is $60\ \mathrm{Hz}$, then the output frequency would be:

Option 1:
$30\ \mathrm{Hz}$

Option 2:
$60Hz$

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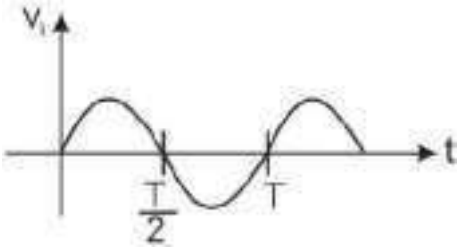
Option 3:
$120\ \text{Hz}$

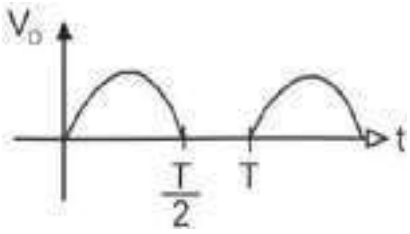
Option 4:
zero

Correct Answer:
$60Hz$

Solution:
Inpute voltage



After rectification, The output voltage is



It can be seen from the graph that frequency of output and input are same

$\text{f}_{\text{input}} = \text{f}_{\text{output}} = 60\text{Hz}$
The correct option is (2)

Q. 51

The truth table for the given logic circuit is :

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Option 1:

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

Option 2:

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

Option 3:

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

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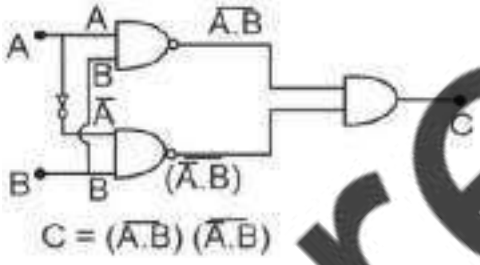
Option 4:

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

Correct Answer:

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

Solution:



A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

Hence correct option is 2

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Chemistry

Q. 1 What mass of 95% pure $CaCO_3$ will be required to neutralise

$50\ mL$ of $0.5M HCl$ solution according to the following reaction?

$$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(\ aq\ )} + CO_{2(\ g)} + 2H_2O_{(1)}$$

[Calculate upto second place of decimal point]

Option 1:
$1.32\ g$

Option 2:
$3.65\ g$

Option 3:
$9.50\ g$

Option 4:
$1.25\ g$

Correct Answer:
$1.32\ g$

Solution:
Give reaction

$$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2\ (aq)} + CO_2(\ g) + 2H_2O(l)$$

From above,

For 2 mole of HCl = 1 mole of required

$$1 \text{ mole of } HCl = \frac{1}{2} \text{ mole of } CaCO_3$$

$$(50ml \times 0.5M)HCl = \frac{50 \times 0.5}{2} m \text{ mole of } CaCO_3$$

$$= 12.5m \text{ mole of } CaCO_3$$

$$= 12.5 \times 10^{-3} \times 100$$

$$= 1.25\ g \text{ of } CaCO_3 \text{ required}$$

For some of $0.5M\ HCl$ solution $1.25g$ of $CaCO_3$ required

But $CaCO_3$ is 85% so,

$FaCO_3$ is 95% so,

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$$\% \text{ Purity } = \frac{\text{mass of pure } CaCO_3}{\text{Total Mass of impure } CaCO_3} \times 100$$

$$95 = \frac{1.25}{x} \text{ g}$$

$$x = \frac{1.25}{95} \text{ g} = 1.32 \text{ g}$$

Hence, the answer is the option (1).

Q. 2 Identify the incorrect statement from the following.

Option 1:
All the five $4d$ orbitals have shapes similar to the respective $3d$ orbitals.

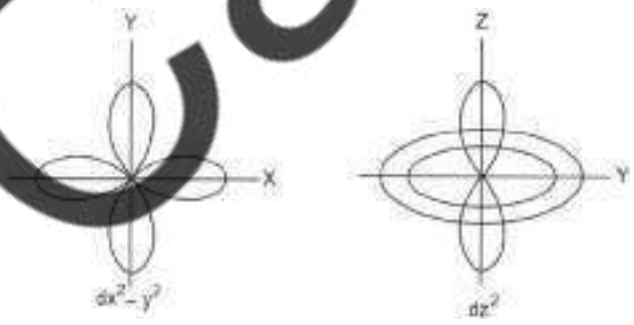
Option 2:
In an atom, all the five $3d$ orbitals are equal in energy in free state.

Option 3:
The shapes of d_{xy} d_{yz} and d_{zx} orbitals are similar to each other; and $d_{x^2} - y^2$ and $d_2 2$ are similar to each other.

Option 4:
All the five $5d$ orbitals are different in size when compared to the respective $4d$ orbitals.

Correct Answer:
The shapes of d_{xy} d_{yz} and d_{zx} orbitals are similar to each other; and $d_{x^2} - y^2$ and $d_2 2$ are similar to each other.

Solution:
The shapes of d-orbitals are given as



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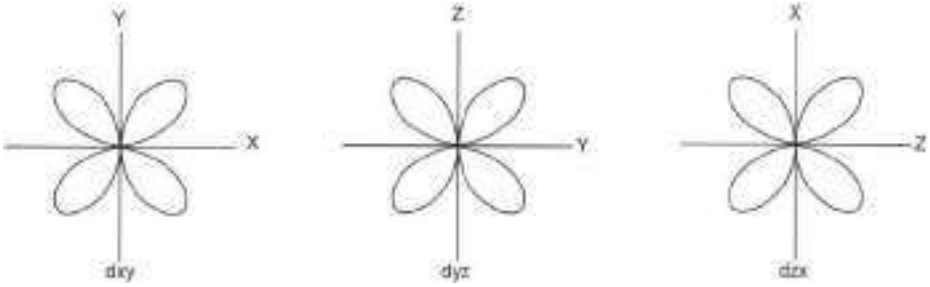

As evident from the orbital diagrams, the shapes of $\mathrm{dx^2 - y^2}$ and $\mathrm{dz^2}$ are not similar.

Thus, Statement in option (3) is not correct

Hence answer is option (3).

Q. 3 If radius of second Bohr orbit of the $\mathrm{He^+}$ ion is $105.8\ \mathrm{pm}$, what is the radius of third Bohr orbit of $\mathrm{Li^{2+}}$ion?

Option 1:
$15.87\mathrm{pm}$

Option 2:
$1.587\mathrm{pm}$

Option 3:
$158.7\mathrm{\AA}$

Option 4:
$158.7\mathrm{pm}$

Correct Answer:
$158.7\mathrm{pm}$

Solution:
We know,

$$\mathrm{r} \propto \frac{\mathrm{n^2}}{\mathrm{z}}$$

$$\therefore \frac{\mathrm{r\left(He^+, n=2\right)}}{\mathrm{r\left(Li^{2+}, n=3\right)}} = \frac{2^2 \times 3}{2 \times 3^2} = \frac{2}{3}$$

$$\therefore \mathrm{r\left(Li^{2+}, n=3\right)} = \frac{3}{2}\mathrm{r\left(He^+, n=2\right)}$$

$$= \frac{3}{2} \times 105.8$$

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$= 158.7\text{pm}$

Hence, the answer is the option (4).

Q. 4 The IUPAC name of an element with atomic number 119 is

Option 1:
unnilennium

Option 2:
unununnium

Option 3:
ununoctium

Option 4:
ununennium

Correct Answer:
ununennium

Solution:
IUPAC name of the element with atomic number 119 is ununennium

Hence, the answer is the option (4).

Q. 5 Amongst the following which one will have maximum ' lone pair - lone pair ' electron repulsions?

Option 1:
IF_5

Option 2:
SF_4

Option 3:
XeF_2

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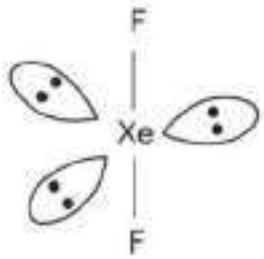
Option 4:
$\mathrm{ClF_3}$

Correct Answer:
$\mathrm{XeF_2}$

Solution:
The maximum number of lone pairs on the control atom is present on Xe in $\mathrm{XeF_2}$



Hence maximum lone pair - lone pair repulsion will be present in $\mathrm{XeF_2}$

Hence correct option is 3.

Q. 6 Which amongst the following is incorrect statement?

Option 1:
$\mathrm{C_2}$ molecule has four electrons in its two degenerate π molecular orbitals.

Option 2:
$\mathrm{H_2^+}$ ion has one electron.

Option 3:
$\mathrm{O_2^+}$ ion is diamagnetic.

Option 4:
The bond orders of $\mathrm{O_2^+}$, $\mathrm{O_2}$, $\mathrm{O_2^-}$ and $\mathrm{O_2^{2-}}$ are $2.5, 2, 1.5$ and 1 respectively.

Correct Answer:
$\mathrm{O_2^+}$ ion is diamagnetic.

Solution:
$\mathrm{O_2^+}$ ion will be paramagnetic due to the presence of unpaired electrons in $\pi - 2\mathrm{p}$ orbital of molecular orbital

Hence, the answer is the option (3).

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Q. 7 A $10.0\ \mathrm{L}$ flask contains $64\ \mathrm{g}$ of oxygen at $27^{\circ}\mathrm{C}$. (Assume $\mathrm{O_2}$ gas is behaving ideally). The pressure inside the flask in bar is
(Given $\mathrm{R} = 0.0831\ \mathrm{L\ bar\ K^{-1}\ mol^{-1}}$)

Option 1:
498.6

Option 2:
49.8

Option 3:
4.9

Option 4:
2.5

Correct Answer:
4.9

Solution:
From ideal gas equation, we have

$$\mathrm{PV} = \mathrm{nRT}$$

$$\Rightarrow \mathrm{P}(10) = \left(\frac{64}{32}\right)(0.0831)(300)$$

$$\Rightarrow \mathrm{P} = 4.986\ \mathrm{bar}.$$

Hence correct option is 3.

Q. 8 Which one is not correct mathematical equation for Dalton's Law of partial pressure? Here P = total pressure of the gaseous mixture.

Option 1:
$$\mathrm{P} = \mathrm{n_1}\frac{\mathrm{RT}}{\mathrm{V}} + \mathrm{n_2}\frac{\mathrm{RT}}{\mathrm{V}} + \mathrm{n_3}\frac{\mathrm{RT}}{\mathrm{V}}$$

Option 2:
$\mathrm{P_i} = \mathrm{x_i P}$, Where $\mathrm{P_i}$ =partial pressure of $\mathrm{i^{th}}$ gas
$\mathrm{X_i}$ =mole fraction of $\mathrm{i^{th}}$ gas in gaseous mixture

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Option 3:

$P_i = x_i P_i^0$, where x_i = mole fraction of i^{th} gas in gaseous mixture

P_i^0 = Pressure of i^{th} gas in pure state

Option 4:

$p = p_1 + p_2 + p_3$

Correct Answer:

$P_i = x_i P$, Where P_i = partial pressure of i^{th} gas

X_i = mole fraction of i^{th} gas in gaseous mixture

Solution:

Daltons law of partial pressure equations are :

$$P_1 = n_1\frac{RT}{V}, P_2 = n_2\frac{RT}{V}, P_3 = n_3\frac{RT}{V}$$

$$P_{total} = P_1 + P_2 + P_3 = n_1\frac{RT}{V} + n_2\frac{RT}{V} + n_2\frac{RT}{V}$$

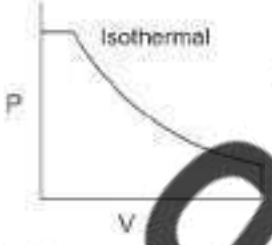
$$P_i = X_i P_{total}$$

but $P_i = X_i P_i^0$ is not the equation of Dalton's law of partial pressure

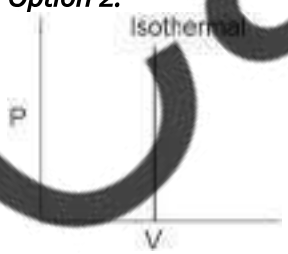
Hence correct option is 2

Q. 9 Which of the following p.V curve represents the maximum work done?

Option 1:



Option 2:



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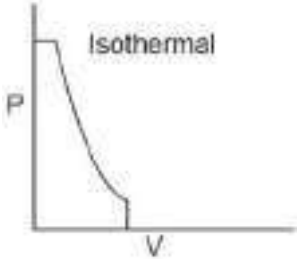
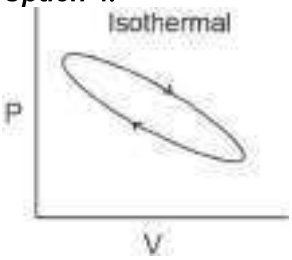
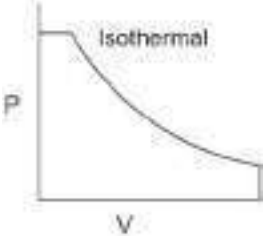
Option 3:



Option 4:



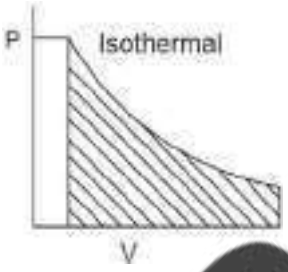
Correct Answer:



Solution:

In the P_V curve, the work is equal to the area under the P_V curve.

In graph (I) has the maximum area under the curve, so it will have maximum area under the curve



Hence the correct option is 1

Q. 10 $3O_2(g) \rightleftharpoons 2O_3(g)$

for the above reaction $298\mathrm{K}, \mathrm{K_c}$ is found to be 3.0×10^{-59}. If the concentration of $\mathrm{O_2}$ at equilibrium is $0.040\mathrm{M}$ then concentration of $\mathrm{O_3}$ in M is

Option 1:

1.9×10^{-63}

Option 2:

2.4×10^{31}

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Option 3:

1.2×10^{21}

Option 4:

4.38×10^{-32}

Correct Answer:

4.38×10^{-32}

Solution:

$3O_2(\ g) \rightleftharpoons 2O_3(g)$

$$K_c = \frac{[O_3]^2}{[O_2]^3}$$

$$\Rightarrow 3 \times 10^{-59} = \frac{[O_3]^2}{(0.04)^3}$$

$$\Rightarrow [O_3]^2 = 3 \times 10^{-59} \times (0.04)^3$$

$$\Rightarrow [O_3]^2 = 1.92 \times 10^{-63}$$

$$\Rightarrow [O_3] = 4.38 \times 10^{-32}$$

Hence, the answer is the option (4).

Q. 11 The pH of the solution containing $50\ \mathrm{mL}$ each of $0.10\mathrm{M}$ sodium acetate and $0.01\mathrm{M}$ acetic acid is [Given $\mathrm{pK}_a \text{of } \mathrm{CH_3COOH} = 4.57$]

Option 1:

3.57

Option 2:

4.57

Option 3:

2.57

Option 4:

5.57

Correct Answer:

5.57

Solution:

Given salt (sodium acetate), $\mathrm{M} = 0.10, \mathrm{V} = 50\mathrm{mL}$ Weak acid

$(\mathrm{CH_3COOH})\ , \mathrm{pKa} = 4.57$ and $\mathrm{M} = 0.01$ So, weak acid and salt form an

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acidic buffer

Then, $\mathrm{pH}=\mathrm{pK_a}+\log \frac{[\text{ salt}]}{[\text{ Acid}]}$

$$\mathrm{pH}=4.57+\log \frac{(0.1)}{(0.01)}$$

$$\mathrm{pH}=5.57$$

Trick: acetate ion is a base and if we add base in acid the pH solution will increase. So, pH of solution will be greater than 4.57

Hence, the answer is the option (4).

Q. 12 **Match List-I with List-II**

List-I(Hydrides)	List-II(Nature)
$(a)\mathrm{MgH_2}$	(i) Electron precise
$(b)\mathrm{GeH_4}$	(ii) Electron deficient
$(c)\mathrm{B_2H_6}$	(iii) Electron rich
$(d)\mathrm{HF}$	(iv) Ionic

Choose the correct answer from the options given below:

Option 1:
(a)-(iii),(b)-(i),(c)-(ii), (d)-(iv)

Option 2:
(a)-(i),(b)-(ii),(c)-(iv),(d)-(iii)

Option 3:
(a)-(ii),(b)-(iii),(c)-(iv),(d)-(i)

Option 4:
(a)-(iv),(b)-(i),(c)-(ii),(d)-(iii)

Correct Answer:
(a)-(iv),(b)-(i),(c)-(ii),(d)-(iii)

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Solution:
By the definition of Hydrides

$\mathrm{MgH_2} \rightarrow$ Ionic

$\mathrm{GeH_4} \rightarrow$ Electron precise

$\mathrm{B_2H_6} \rightarrow$ Electron deficient

$\mathrm{HF} \rightarrow$ Electron rich

(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

hence correct option is 4

Q. 13 Identify the incorrect statement from the following

Option 1:
The oxidation number of K in $\mathrm{KO_2}$ is $+4$.

Option 2:
Ionisation enthalpy of alkali metals decreases from top to bottom in the group.

Option 3:
Lithium is the strongest reducing agent among the alkali metals.

Option 4:
Alkali metals react with water to form their hydroxides.

Correct Answer:
The oxidation number of K in $\mathrm{KO_2}$ is $+4$.

Solution:
The oxidation number of K in $\mathrm{KO_2}$ is $+4$, is wrong because K oxidation number is $+1$,

$\mathrm{K^+ + O_2^- \rightarrow KO_2}$

superoxide

Hence correct option is 1.

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Q. 14 Match List-I with List-II

List-I	List-II
(a)Li	(i) Absorbent for carbon dioxide
(b)Na	(ii) electrochemical cells
(c)KOH	(iii) coolant in fast breeder reactors
(d)Cs	(iv) photoelectric cell

Choose the correct answer from the options given below:

Option 1:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Option 2:
(a) -(i),(b)-(iii),(c)-(iv),(d)-(ii)

Option 3:
(a)-(i),(b)-(iii),(c)-(i),(d)-(iv)

Option 4:
(a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Correct Answer:
(a)-(i),(b)-(iii),(c)-(i),(d)-(iv)

Solution:
1. Lithium is used to make electrochemical cells.

2. Liquid sodium metal is used as a coolant in fast breeder nuclear reactions.

3. KOH (Potassium hydroxide) is used in the manufacture of soft soap. It is also used as an excellent absorbent of carbon dioxide.

4. Caseium is used in deciding photoelectric cells.

So, (a) - (i), (b) - (iii), (c) - (i), (d) - (iv)

Hence the correct option is 3.

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Q. 15 Which of the following statement is not correct about diborane?

Option 1:
The four terminal B-H bonds are two centre two electron bonds.

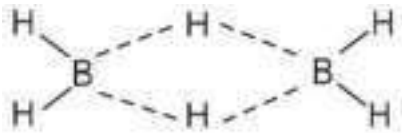
Option 2:
The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.

Option 3:
Both the Boron atoms are sp^2 hybridised.

Option 4:
There are two 3 -centre-2-electron bonds.

Correct Answer:
Both the Boron atoms are sp^2 hybridised.

Solution:
The structure of Diaborone is given below :



The bridge B-H-B bonde are 3c-2e bonds while the terminal B-H bonds are 2c-2e bonds

Boron atoms are sp^2 hybridized.

Hence correct option is 3

Q. 16 Choose the correct statement:

Option 1:
Diamond is covalent and graphite is ionic.

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Option 2:
Diamond is sp^3 hybridised and graphite is sp^2 hybridized.

Option 3:
Both diamond aat graphite are used as dry lubricants.

Option 4:
Diamond and graphite have two dimensional network.

Correct Answer:
Diamond is sp^3 hybridised and graphite is sp^2 hybridized.

Solution:
In diamond, each carbon atom undergoes sp^3 hybridization and linked to form other carbon atoms by using hybridised orbitals in tetrahedral fashion. In graphite, each carbon atom in hexagonal ring undergoes sp^2 hybridization and makes three sigma bonds with the neighbouring carbon atoms.

Fourth electron forms π bond.

Diamond and graphite both of them are covalent.

Only graphite is used as dry lubricant.

Diamond form 3D network and graphite forms 2D network

hence correct option is 2.

Q. 17 Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : ICl is more reactive than I_2.
Reason (R): $\mathrm{I-Cl}$ bond is weaker than $\mathrm{I-I}$ bond.
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both (A) and (R) are correct but (R) is not the correct explanation of (A).

Option 2:
(A) is correct but (R) is not correct.

Option 3:
(A) is not correct but (R) is correct.

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Option 4:
Both (A) and (R) are correct and (R) is the cornect explanation of (A).

Correct Answer:
Both (A) and (R) are correct and (R) is the cornect explanation of (A).

Solution:
Interhalogen compounds are more reactive than halogens (except fluorine). This is because $X-X'$ bond in interhalogens is weaker than $X-X$ bond in halogens except $F-F$ bond .

So, ICl is more reactive than I_2 and because of $I-Cl$ bond is weaker than $I-I$ bond.

Hence the correct option is 4.

Q. 18 Given below are two statements.

Statement I:
The boiling points of the following hydrides of group 16 elements increase in the orders-
$H_2O < H_2\ S < H_2Se_e < H_2Te$
Statement II:
The boiling points of these hydrides increases with increase in molar mass.

In the light of the above statements choose the most appropriate answer from the options given below

Option 1:
Both Statement I and Statement II are incorrect.

Option 2:
Statement I is correct but Statement II is incorrect.

Option 3:
Statement I is incorrect but Statement II is correct.

Option 4:
Both Statement I and Statement II are correct.

Correct Answer:
Statement I is incorrect but Statement II is correct.

Solution:
The boiling points of the following hydrides :

$$H_2S < H_2Se < H_2Te < H_2O$$

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The boiling points of their hydrides increase with the increase in molar mass from H_2S to

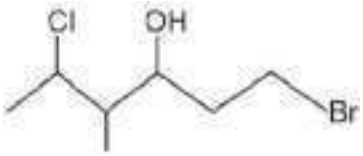
H_2Te, but due to the presence of H-bonding in H_2O and that forms intermolecular

H-bonding which increases the boiling point of H_2O

So, statement I is incorrect and statement II is correct

Hence, the answer is the option (3).

Q. 19 The correct IUPAC name of the following compound is :



Option 1:

$6-\text{bromo}-2-\text{chloro}-4-\text{methylhexan}-4-\text{ol}$

Option 2:

$1-\text{bromo}-4-\text{methyl}-5-\text{chlorohexan}-3-\text{ol}$

Option 3:

$6-\text{bromo}-4-\text{methyl}-2-\text{chlorohexan}-4-\text{ol}$

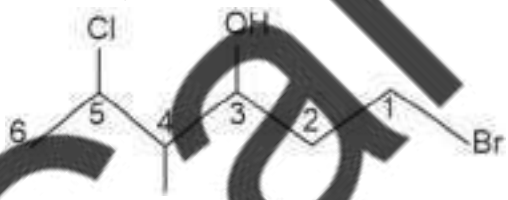
Option 4:

$1-\text{bromo}-5-\text{chloro}-4-\text{methylhexan}-3-\text{ol}$

Correct Answer:

$1-\text{bromo}-5-\text{chloro}-4-\text{methylhexan}-3-\text{ol}$

Solution:



Convert IUPAC name is

1 - Bromo - 5 - Chloro - 4 - methylhexan - 3 - ol

hence correct option is 4.

Q. 20 The Kjeldahl's method for the estimation of nitrogen can be used to estimate the amount of nitrogen in which one of the following compounds?

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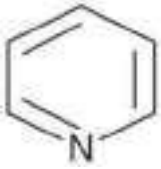
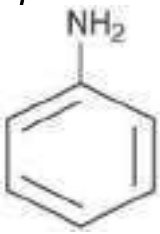
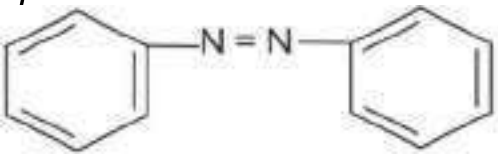
Option 1:



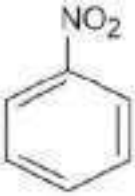
Option 2:



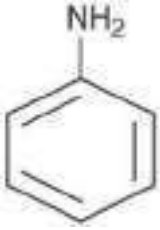
Option 3:



Option 4:



Correct Answer:



Solution:

As we have learnt,

Nitro compounds, Azo compounds and compounds containing nitrogen in aromatic rings do not respond to Kjeldahl's test.

Thus, among the given compounds, the one which response to Kjeldehl's test is aniline

Hence the correct option is 2

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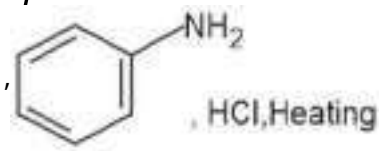
Q. 21 Which of the following is suitable to synthesize chlorobenzene ?

Option 1:

$\text{Phenol}, NaNO_2, HCl, CuCl$

Option 2:

Option 3:



Option 4:

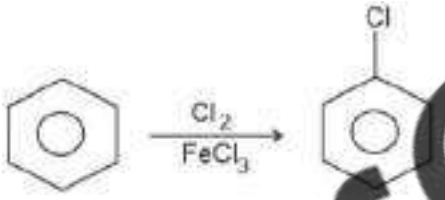
$\text{Benzene}, Cl_2, \text{anhydrous } FeCl_3$

Correct Answer:

$\text{Benzene}, Cl_2, \text{anhydrous } FeCl_3$

Solution:

Chlorobenzene can be obtained from benzene upon reaction with $Cl_2/FeCl_3$



hence correct option is 4

Q. 22 Which compound amongst the following is not an aromatic compound ?

Option 1:

Option 2:

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Option 3:

Option 4:

Correct Answer:

Solution:

(1) $6\pi e^{(-)}$, planer, cyclic $\Rightarrow$ Aromatic

(2) $2\pi e^{(-)}$, planer, cyclic $\Rightarrow$ Aromatic

(3) $6\pi e^{(-)}$, not Planar, not cyclic $\Rightarrow$ Non-Aromatic

(4) $6\pi e^{(-)}$, planar, cyclic $\Rightarrow$ Aromatic

hence correct option is 3.

Q. 23 The pollution due to oxides of sulphur gets enhanced due to the presence of :
(a) particulate matter
(b) ozone
(c) hydrocarbons
(d) hydrogen peroxide
Choose the most appropriate answer from the options given below:

Option 1:
(a), (b), (d) only

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Option 2:
(b), (c), (d) only

Option 3:
(a), (c), (d) only

Option 4:
(a), (d) only

Correct Answer:
(a), (b), (d) only

Solution:
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The oxidation of SO_2 to SO_3 is catalysed in the presence of particulate matter

$$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$$

The reaction can be promoted by Ozone and hydrogen peroxide.

$$SO_2(g) + O_3(g) \rightarrow SO_3(g) + O_2(g)$$

$$SO_2(g) + H_2O_2(l) \rightarrow H_2SO_4(aq).$$

Hence correct option is 1

Q. 24 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) :
In a particular point defect, an ionic solid is electrically neutral, even if few of its cations are missing from its unit cells.

Reason (R) :
In an ionic solid, Frenkel defect arises due to dislocation of cation from its lattice site to interstitial site, maintaining overall electrical neutrality.

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 2:
(A) is correct but (R) is not correct

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Option 3:
(A) is not correct but (R) is correct

Option 4:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

Correct Answer:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Solution:
Assertion (A) is correct as ionic solids are always electrically neutral. The given example is of metal deficiecy defect.

Reason (R) is also correct as frenkel defect is a dislocation defect

however **(R)** does not explain **(A)**

Hence correct option is 1

Q. 25 Copper crystallises in fcc unit cell with cell edge length of $3.608 \times 10^{-8} \mathrm{cm}$. The density of copper is $8.92\ \mathrm{g\ cm^{-3}}$. Calculate the atomic mass of copper.

Option 1:
$31.55\ \mathrm{u}$

Option 2:
$60\ \mathrm{u}$

Option 3:
$65\ \mathrm{u}$

Option 4:
$63.1\ \mathrm{u}$

Correct Answer:
$63.1\ \mathrm{u}$

Solution:
As we have learnt ,

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$$d = \frac{Z \times M_0}{N_A \times a^3}$$

Putting values in the above expression we have

$$8.92 = \frac{4 \times M_0}{6.022 \times 10^{23} \times (3.608 \times 10^{-8})^3}$$

$$\Rightarrow \quad M_0 = \frac{8.92 \times 6.022 \times 10^{23} \times (3.608 \times 10^{-8})^3}{4}$$

$$\Rightarrow \quad M_0 = 63.1$$

Hence correct option is 4

Q. 26 In one molal solution that contains 0.5 mole of a solute, there is

Option 1:
500g of solvent

Option 2:
100mL of solvent

Option 3:
$1000\ \text{g}$ of solvent

Option 4:
500mL of solvent

Correct Answer:
500g of solvent

Solution:
1 molal solution means

1 mole of solvent is present in 1kg of solute

Thus 500g of solvent will contain 0.5 mole of solute

Hence, the answer is the option (1).

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Q. 27 Find the emf of the cell in which the following reaction takes place at $298\ \mathrm{K}$

$$\mathrm{Ni(s) + 2Ag^{+}(0.001M) \rightarrow Ni^{2} + (0.001M) + 2Ag(s)}$$

(Given that $\mathrm{E^{0}_{cell} = 1.05V}, \dfrac{2.303\ \mathrm{RT}}{\mathrm{F}} = 0.0285\ \text{at}\ 298\mathrm{K}$)

Option 1:

$1.385\ \mathrm{V}$

Option 2:

$0.9615\ \mathrm{V}$

Option 3:

$1.05\ \mathrm{V}$

Option 4:

$1.0385\ \mathrm{V}$

Correct Answer:

$0.9615\ \mathrm{V}$

Solution:

The cell reaction is

$$\mathrm{N_i(s) + 2Ag^{+}(aq) \rightarrow Ni^{2+}(aq) + 2Ag}$$

$$\mathrm{n = 2\ and\ Q = \frac{[Ni^{2+}]}{[Ag^{+}]^{2}}}$$

$\therefore$ From the nearest equation, we have

$$\mathrm{E = E^{0}_{cell} - \frac{0.059}{2}\log\frac{[Ni^{2+}]}{[Ag^{+}]^{2}}}$$

$$= 1.05 - 0.0295\log\frac{10^{-3}}{(10^{-3})^{2}}$$

$$= 1.05 - 0.0295\log 10^{3}$$

$$= 1.05 - 0.0295 \times 3$$

$$= 0.9615\ \mathrm{V}$$

Hence, the answer is the option (2).

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Q. 28 At $298\ \mathrm{K}$, the standard electrode potentials of $\mathrm{Cu^{2+}/Cu, Zn^{2+}/Zn, Fe^{2+}/Fe}$ and $\mathrm{Ag^{+}/Ag}$ are $0.34\ \mathrm{V}, -0.76\ \mathrm{V}, -0.44\ \mathrm{V}$ and $0.80\ \mathrm{V}$, respectively.
On the basis of standard electrode potential, predict which of the following reaction can not occur?

Option 1:
$\mathrm{CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu(s)}$

Option 2:
$\mathrm{FeSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Fe(s)}$

Option 3:
$\mathrm{2CuSO_4(aq) + 2Ag(s) \rightarrow 2Cu(s) + Ag_2SO_4(aq)}$

Option 4:
$\mathrm{CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)}$

Correct Answer:
$\mathrm{2CuSO_4(aq) + 2Ag(s) \rightarrow 2Cu(s) + Ag_2SO_4(aq)}$

Solution:
Given Standard electrical potentials so order will be

$$\mathrm{E^0_{Zn^{2+}/Zn} < E^0_{Fe^{2+}/Fe} < E^0_{Cu^{2+}/Cu} < E^0_{Ag^{2+}/Ag}}$$

$$-0.76 \qquad -0.44 \qquad 0.34 \qquad 0.80$$

From the above data,

$\mathrm{Ag^{2+}}$ can oxidize others and then get reduced to

$\mathrm{Zn^{2+}}$ cannot oxidize others

$\mathrm{Fe^{2+}}$ can oxidize only Zn and Fe metals but not Cu and Ag

$\mathrm{Cu^{2+}}$ can oxidize only Zn and Fe metals but it cannot oxidize Ag metal due to high value

In reaction $\mathrm{2CuSO_4(aq) + 2Ag(s) \rightarrow 2Cu_4(s) + Ag_2SO_4(aq)}$

$\mathrm{Cu^{2+}}$ is oxidizing Ag which is not possible, so this reaction can not occur.

Hence, the answer is the option (3).

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Q. 29 Given below are half cell reactions:

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$E_{Mn^{2+}}/MnO_4 = -1.510\ V$$

$$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O,$$

$$E_{O2}/H_2O = 1.223\ V$$

Will the permanganate ion, MnO_4^- liberate O_2 from Water in the presence of an acid?

Option 1:
No, because $E^0{}_{cell} = -0.287\ V$

Option 2:
Yes, because $E^o{}_{cell} = +2.733\ V$

Option 3:
No, because $E^o{}_{cell} = -2.733\ V$

Option 4:
Yes, because $E^0{}_{cell} = +0.287\ V$

Correct Answer:
Yes, because $E^0{}_{cell} = +0.287\ V$

Solution:
The E^0 values corresponding to the reactions are given below:

$$MnO^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O, E^0 = 1.510$$

$$H_2O \longrightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-, E^0 = -1.223$$

$$\therefore E^\circ Cell = (1.570 - 1.223)V$$

$$= 0.287\ V$$

Thus, MnO_4^- will be able to liberate O_2 from water in the presence of H^+

Hence, the answer is the option (4).

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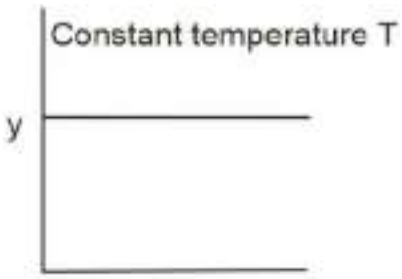
Q. 30 The given graph is a representation of kinetics of a reaction



The y and x axes for zero and first order reactions, respectively are

Option 1:
zero order (y= concentration and x= time), first order (y= rate constant and x= concentration)

Option 2:
zero order (y=rate and x=concentration), first order ($y = t_{1/2}$ and x= concenration)

Option 3:
zero order (y= rate and x=concentration), first order (y= rate and $x = t_{1/2}$)

Option 4:
zero order (y= concentration and x=time), first order ($y = t_{1/2,}$ and x= concentration)

Correct Answer:
zero order (y=rate and x=concentration), first order ($y = t_{1/2}$ and x= concenration)

Solution:
As we have learnt,

For a zero order reaction,

$\text{rate} \propto [A]^0$

$\Rightarrow$ rate = K = constant.

$\Rightarrow$ rate is independent of concentration

For a first order reaction

$$t_{1/2} = \frac{\ln 2}{K}.$$

$\Rightarrow$ Half life is independent of concentration

Hence correct option is 2

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Q. 31 For a first order reaction $\mathrm{A} \rightarrow$ Products, initial concentration of A is $0.1\ \mathrm{M}$, which become $0.001\ \mathrm{M}$ after 5 minutes. Rate constant for the reaction in min^{-1} is

Option 1:
0.9212

Option 2:
0.4606

Option 3:
0.2303

Option 4:
1.3818

Correct Answer:
0.9212

Solution:
For a first-order reaction, the integrated rate law expression is

$$\ln\left(\frac{\mathrm{A}_0}{\mathrm{A}_t}\right) = \mathrm{Kt}.$$

$$\Rightarrow \quad \ln\left(\frac{0.1}{0.001}\right) = \mathrm{K}(5)$$

$$\Rightarrow \mathrm{K} = \frac{\ln 100}{5} = \frac{4.606}{5} = 0.9212\ \mathrm{min}^{-1}$$

Hence, the answer is the option (1).

Q. 32 The incorrect statement regarding enzymes is:

Option 1:
Like chemical catalysts, enzymes reduce the activation energy of bio processes.

Option 2:
Enzymes are polysaccharides.

Option 3:
Enzymes are very specific for a particular reaction and substrate.

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Option 4:
Enzymes are biocatalysts.

Correct Answer:
Enzymes are polysaccharides.

Solution:
Enzymes are not polysaccharides

Statement 2 is incorrect and others are correct

Hence the correct option is 2

Q. 33 Given below are two statements:

Statement I:
In the coagulation of a negative solution, the flocculating power of the three given ions is in the order-

$$Al^{3}+ > Ba^{2}+ > Na^{+}$$

Statement II:
In the coagulation of a positive sol, the flocculating power of the three given salts is in the order-

$$NaCl > Na_2SO_4 > Na_3PO_4$$

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are incorrect.

Option 2:
Statement I is correct but Statement II is incorrect.

Option 3:
Statement I is incorrect but Statement II is correct.

Option 4:
Both Statement I and Statement II are correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
Hardly Schulze rule's say the greater the valence of the flocculating ion added, the greater is it's power to cause precipitation.

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In the coagulation of a negative solution, the flocculating power is in the order :

$$Al^{3+} > Ba^{2+} > Na^{+}$$

In the coagulation of a positive solution, the fluctuating power is in the order :

$$[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^{-}$$

So, a statement I is correct but statement II is incorrect.

Hence the correct option is 2.

Q. 34 Match List -I with List -II

List-I (Ores)	List-II (Composition)
(a) Haematite	(i) Fe_3O_4
(b) Magnetite	(ii) $ZnCO_3$
(c) Calamine	(iii) Fe_2O_3
(d) Kaolinite	(iv) $[Al_2(OH)_4Si_2O_5]$

Choose the correct answer from the options given below:

Option 1:
(a)-(iii),(b)-(i),(c)-(ii),(d)-(iv)

Option 2:
(a)-(iii),(b)-(i),(c)-(iv),(d)-(ii)

Option 3:
(a)-(i),(b)-(iii),(c)-(ii),(d)-(iv)

Option 4:
(a)-(i),(b)-(ii),(c)-(iii),(d)-(iv)

Correct Answer:
(a)-(iii),(b)-(i),(c)-(ii),(d)-(iv)

Solution:
The correct match of Ores and their composition is given below:

(a) Hanematite : (iii) Fe_2O_3

(b) Magnetite : (i) Fe_3O_4

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(c) Calamine : (ii) $ZnCO_3$

(d) Kaolinite : (iv) $[Al_2(OH)_4Si_2O_5]$

Hence the correct option is 1.

Q. 35 In the neutral or faintly alkaline medium, $KMnO_4$ oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is from

Option 1:
$+6$ to $+4$

Option 2:
$+7$ to $+3$

Option 3:
$+6$ to $+5$

Option 4:
$+7$ to $+4$

Correct Answer:
$+7$ to $+4$

Solution:
In a neutral or faintly alkaline medium $KMnO_4$ is converted to MnO_2

$$MnO_4^- \rightarrow MnO_2$$

$(+7) \qquad (+4)$

Hence, the answer is the option (4).

Q. 36 Gadolinium has a low value of third ionisation enthalpy because of

Option 1:
high exchange enthalpy

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Option 2:
high electronegativity

Option 3:
high basic character

Option 4:
small size

Correct Answer:
high exchange enthalpy

Solution:
Gd^{2+} has a configuration of $4f^75d^1$. Thus, Gd^{3+} has a half-filled $4f^7$ configuration which has a high value of exchange enthalpy

Hence the third ionization energy of Gd is low

Hence, the answer is the option (1).

Q. 37 The $IUPAC$ name of the complex - $[AgH_2O)_2]\,[Ag(CN)_2]^2$ is :

Option 1:
diaquasilver(II) dicyanidoargentate(II)

Option 2:
dicyanidosilver(I) diaquaargentate(I)

Option 3:
diaquasilver(I) dicyanidoargentate(I)

Option 4:
dicyanidosilver(II) diaquaargentate(II)

Correct Answer:
diaquasilver(I) dicyanidoargentate(I)

Solution:
$$[Ag\,(H_2O)_2]\,[Ag(CN)_2] \rightarrow [Ag\,(H_2O)_2]^+ + [Ag\,(CN)_2]^-$$

$$\downarrow \qquad\qquad \downarrow$$

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Cation complex Anion complex

Cation Complex $\Rightarrow$ Metal name: Silver

Anion Complex $\Rightarrow$ Metal name (Suffix-ate or Latin names)

$\Rightarrow$ Argentate (Latin name of Ag)

$H_2O \rightarrow$ aqua, $2\,(H_2O) \rightarrow$ diaqua

$CN^- \rightarrow$ cyanide, $2\,(CN^-) \rightarrow$ dicyanido

Oxidation state of $Ag = +1$

So IUPAC Name : Diaquasilver (I) dicyanidoargentate (I)

Hence, the answer is the option (3).

Q. 38 The order of energy absorbed which is responsible for the color of complexes

$(A)\ [Ni\,(H_2O)_2\,(en)_2]^{2+}$

$(B)[Ni(H_2O)_4(en)]^{2+}$ and

$(C)[Ni(en)_3]^{2+}$ is

Option 1:
(C)>(B)>(A)

Option 2:
(C)>(A)>(B)

Option 3:
(B)>(A)>(C)

Option 4:
(A)>(B)>(C)

Correct Answer:
(C)>(A)>(B)

Solution:
As we have learnt,

Energy absorbed by complexes depends upon the sitting energy.

Metal ion remaining save, splitting is greater for strong field chelating ligands.

$\therefore$ The order of splitting in given complexes is

$[Ni(en)_3]^{2+} > [Ni(H_2O)_2en_2]^{2+} > [Ni(H_2)_4en]^{2+}$

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(C) (A) (B)

Hence, the answer is the option (2).

Q. 39 The incorrect statement regarding chirality is :

Option 1:
The product obtained by S_N2 reaction of haloalkane having chirality at the reactive site shows inversion of configuration.

Option 2:
Enantiomers are superimposable mirror images on each other.

Option 3:
A racemic mixture shows zero optical rotation.

Option 4:
S_N1 reaction yields $1:1$ mixture of both enantiomers.

Correct Answer:
Enantiomers are superimposable mirror images on each other.

Solution:
Enantiomers are non-superimposable mirror images of each other

Statement (2) is incorrect

Hence, the answer is the option (2).

Q. 40 Given below are two statements:
Statement I:
The acidic strength of monosubstituted nitrophenol is higher than phenol because of electron withdrawing nitro group.
Statement II:
o-nitrophenol, m-nitrophenol and p-nitrophenol will have same acidic strength as they have one nitro group attached to the phenolic ring

In the light of the above statements, choose the most appropriate answer from the options given below:

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Option 1:
Both Statement I and Statement II are incorrect.

Option 2:
Statement I is correct but Statement II is incorrect.

Option 3:
Statement I is incorrect but Statement II is correct.

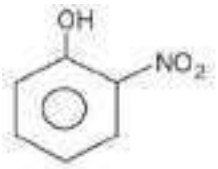
Option 4:
Both Statement I and Statement II are correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

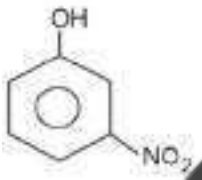
Solution:
Electron withdrawing group o Phenol increases it's acidic strength

Statement I is correct

now, the pka values of nitro substituted Phenols are given below:



: 7.23



: 8.35

: 7.14

thus acidic strength of nitrophenols is

Para > Ortho > Meta

Hence statement I is correct while statement II is incorrect

hence correct option is 2

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Q. 41 Given below are two statements:

Statement I :

In Lucas test, primary, secondary and tertiary alcohols are distinguished on the basis of their reactivity with conc. $HCl + ZnCl_2$, known as Lucas Reagent.

Statement II :

Primary alcohols are most reactive and immediately produce turbidity at room temperature on reaction with Lucas Reagent.

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:

Both Statement I and Statement II are incorrect.

Option 2:

Statement I is correct but Statement II is incorrect.

Option 3:

Statement I is incorrect but Statement II is correct.

Option 4:

Both Statement I and Statement II are correct

Correct Answer:

Statement I is correct but Statement II is incorrect.

Solution:

As we have learnt

In the Lucas test for alcohols, tertiary alcohols produce instant turbidity, secondary alcohols produce turbidity after $(5 - 10)$ min. While primary alcohol does not produce turbidity.

Thus, Statement I is correct while Statement II is incorrect

Hence, the answer is the option (2).

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Q. 42 Given below are two statements :

Statement I:

The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular masses because of weak molecular association in aldehydes and ketones due to dipole- dipole interactions,

Statement II:

The boiling points of aldehydes and ketones are lower than the alcohols of similar molecular masses due to the absence of H-bonding.

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:

Both Statement I and Statement II are incorrect

Option 2:

Statement I is correct but Statement II is incorrect.

Option 3:

Statement I is incorrect but Statement II is correct.

Option 4:

Both Statement I and Statement II are correct.

Correct Answer:

Both Statement I and Statement II are correct.

Solution:

The boiling points of aldehydes and ketones are higher than hydrocarbons and ethers of comparable molecular masses. It is due to weak molecular association in Aldehydes and Ketones arising out of the dipole-dipole interactions. Also, their boiling points are lower than those of alcohols of similar molecular masses due to the absence of intermolecular hydrogen bonding.

Hence, the answer is the option (4).

Q. 43 Compound X on reaction with $\mathrm{O_3}$ followed by $\mathrm{Zn/\ H_2O}$ gives formaldehyde and 2-methyl propanal as products. The compound X is :

Option 1:

$2-\mathrm{Methylbut}-1-\mathrm{ene}$

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Option 2:

$2-\text{Methylbut}-2-\text{ene}$

Option 3:

$\text{Pent}-2-\text{ene}$

Option 4:

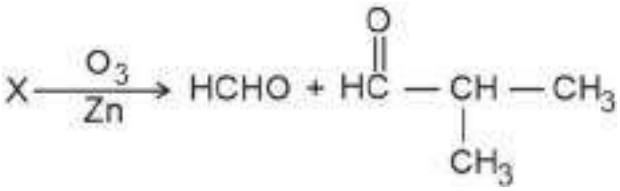
$3-\text{Methylbut}-1-\text{ene}$

Correct Answer:

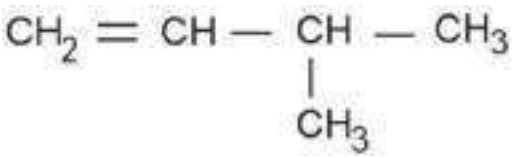
$3-\text{Methylbut}-1-\text{ene}$

Solution:

Given,



$\therefore$ The alkene X is

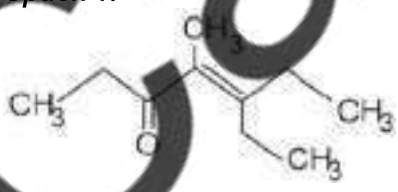


It's IUPAC name is 3-Methylbut-1-ene

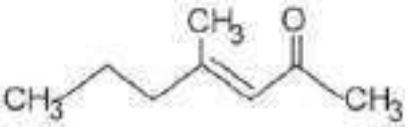
hence correct option is 4

Q. 44 Which one of the following is not formed when acetone reacts with 2-pentanone in the presence of dilute NaOH followed by heating?

Option 1:



Option 2:



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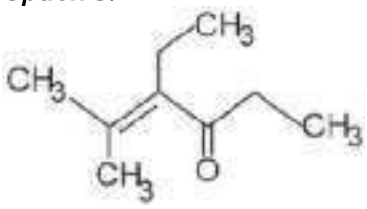
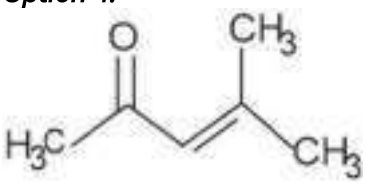
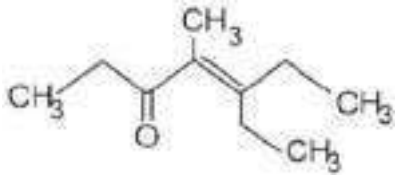
Option 3:



Option 4:

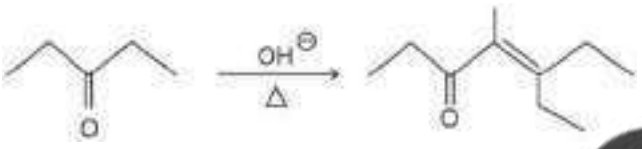


Correct Answer:



Solution:

The compound in option 1 is a self-aldol product of 3-Pentanome



Hence correct option is 1.

Q. 45 Match List-I with List-II

List-I (products formed)	List-II(Reaction of carbonyl compound with)
(a) Cyanohydrin	(i) NH_2OH
(b) Acetal	(ii) RNH_2
(c) Schiffs base	(iii) alcohol
(d) Oxime	(iv) HCN

Choose the correct answer from the options given below:

Option 1:

(a) -(ii),(b)-(iii),(c)-(iv),(d)-(i)

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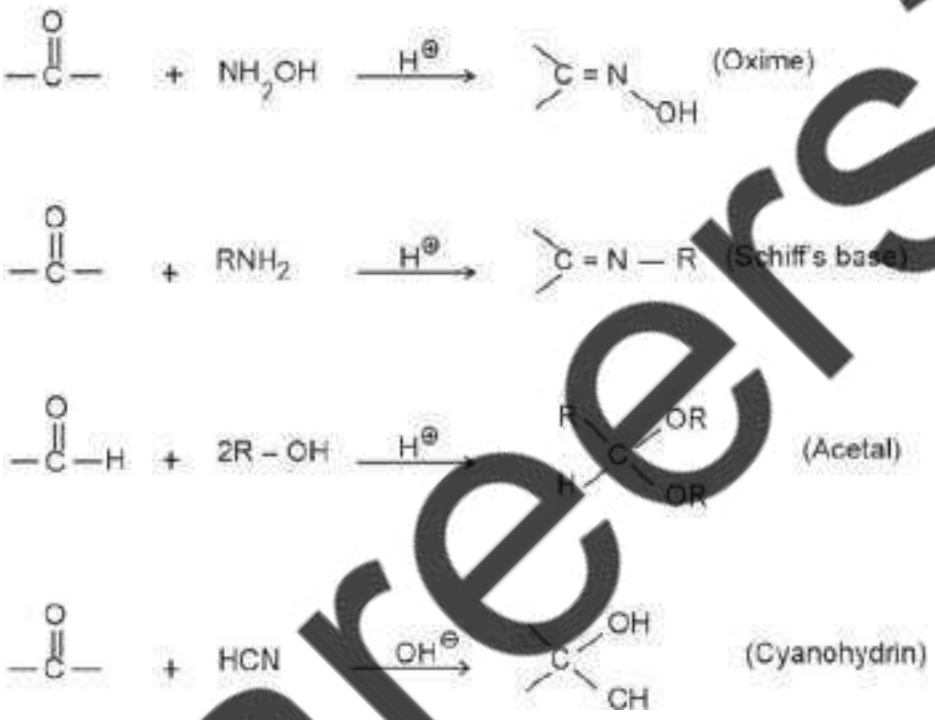
Option 2:
(a)-(i),(b)-(iii),(c)-(ii),(d)-(iv)

Option 3:
(a)-(iv),(b)-(iii),(c)-(ii),(d)-(i)

Option 4:
(a)-(iii),(b)-(iv),(c)-(ii),(d)-(i)

Correct Answer:
(a)-(iv),(b)-(iii),(c)-(ii),(d)-(i)

Solution:
Reaction of carbonyl and formed products.



So, (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Hence the correct option is 3.

Q. 46 $\mathrm{RMgX} + \mathrm{CO_2} \xrightarrow[ether]{dry} \mathrm{Y} \xrightarrow{H_2O} \mathrm{RCOOH}$

What is Y in the above reaction?

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Option 1:
$R_3CO^-Mg^+X$

Option 2:
$RCOO^-X^+$

Option 3:
$(RCOO)_2Mg$

Option 4:
$RCOO^-Mg^+X$

Correct Answer:
$RCOO^-Mg^+X$

Solution:
The given reaction occurs as

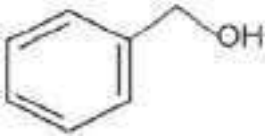
$$RMgX + CO_2 \xrightarrow{\text{ether}} RCOO^{(-)}Mg^{(+)} \xrightarrow[H_2O]{H^+} RCOOH$$

Hence, the answer is the option (4).

Q. 47 The product formed from the following reaction sequence is

Option 1:

Option 2:

Option 3:


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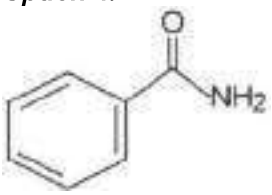
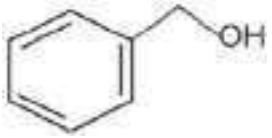
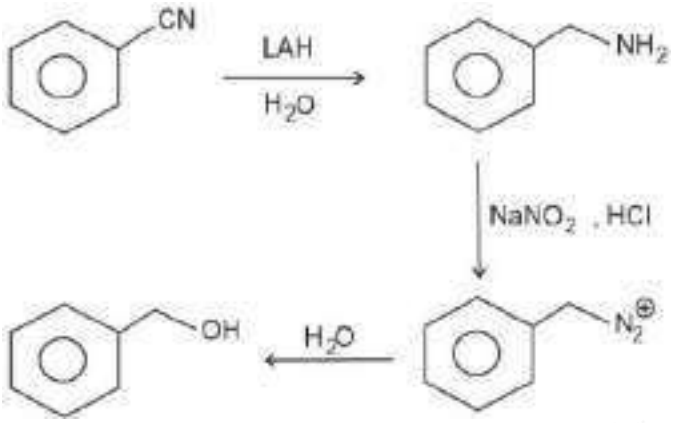
Option 4:



Correct Answer:



Solution:
The reaction sequence occurs as



hence correct option is 3.

Q. 48 Given below are two statements :

Statement I:
Primary aliphatic amines reacts with HNO_2 to give unstable diazonium salts.

Statement II:
Primary aromatic amines reacts with HNO_2 to form diazonium salts which are stable even above $300\ K$.
In the light of the above statement, choose the most appropriate answer from the options given below:

Option 1:
Both statement I and Statement II are incorrect.

Option 2:
Statement I is correct but Statement II is incorrect.

Option 3:
Statement I is incorrect but Statement II is correct.

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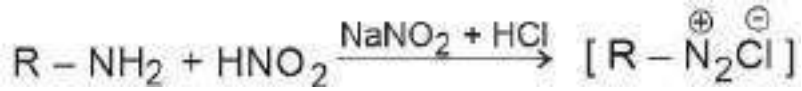
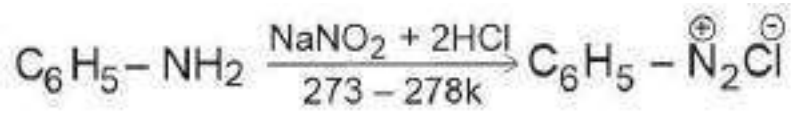
Option 4:
Both Statement I and Statement II are correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
Primary aliohatic amines react with nitrous acid to form aliphatic diazonium saltswhich being unstable.



Primary aminesreact with nitrous acid at low temperature $273-278\mathrm{K}$ to form diazonium salts



But above $300\mathrm{K}$ diazonium salt is very unstable

So, statement I is correct but Statement II is incorrect

hence correct option is 2

Q. 49 Which statement regarding polymers is not correct?

Option 1:
Fibres possess high tensile strength.

Option 2:
Thermoplastic polymers are capable of repeatedly softening and hardening on heating and cooling respectively.

Option 3:
Thermosetting polymers are reusable.

Option 4:
Elastomers have polymer chains held together by weak intermolecular forces.

Correct Answer:
Thermosetting polymers are reusable.

Solution:
1. Fibers possess a high tensile strength

2. Thermoplastic polymers are capable of repeatedly softening and hardening on heating

and cooling respectively.

3. Thermosetting polymers cannot be reused

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4. Elastomers have polymer chains held together by weak intermolecular forces.

Hence the correct option is 3.

Q. 50 Match List-I with List-II.

List-I(Drug Class)	List-II(Drug molecule)
(a) Antacids	(i) Salvarsan
(b) Antihistamines	(ii) Morphine
(c) Analgesics	(iii) Cimetidine
(d) Antimicrobials	(iv) Seldane

Choose the correct answer from the options given below:

Option 1:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Option 2:
(a) - (i), (b) - (iv), (c) - (ii), (d) - (iii)

Option 3:
(a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

Option 4:
(a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

Correct Answer:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Solution:
The correct matching of drug class with drug molecule is given below:

(a) Antacids : Cimetidine (iii)

(b) Antihistamines : Seldane (iv)

(c) Analgesics : Morphine (iii)

(d) Antimicrobials : Salvoarson (i)

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Hence correct option is 1

Biology

Q. 1 Exoskeleton of arthropods is composed of :

Option 1:
Cellulose

Option 2:
Chitin

Option 3:
Glucosamine

Option 4:
Cutin

Correct Answer:
Chitin

Solution:
The exoskeleton of arthropods is made up of Chilin.

Hence, the correct answer is (2)

Q. 2 Which of the following is incorrectly matched?

Option 1:
Ulothrix - Mannitol

Option 2:
Porphyra - Floridian Starch

Option 3:
Volvox-Starch

Option 4:
Ectocarpus - Fucoxanthin

Correct Answer:
Ulothrix - Mannitol

Solution:

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Ulothrix is a member of chlorophycean (green algae) with reserved food material, starch.

Mannitol is the stored food material of Phaeophyceae (brown algae).

Hence, the correct option is (1).

Q. 3 Match the plant with the kind of life cycle it exhibits :

	List I		List II
a	Spirogyra	i	Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
b	Fern	ii	Dominant haploid free-living gametophyte
c	Funaria	iii	Dominant diploid sporophyte alternating with reduced gametophyte called prothallus
d	Cycas	iv	Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte

Choose the correct answer from the options given below:

Option 1:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Option 2:
(a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

Option 3:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Option 4:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Correct Answer:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Solution:
→ Spirogyra is an alga. It shows the haplontic life cycle.

→ Fern is a pteridophyte. The dominant phase of the life cycle is diploid sporophyte. Its gametophyte is called prothallus

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→ funaria is a bryophyte. Its gametophyte is a leafy stage

→ Cycas is a gymnosperm. The main plant body is sporophyte (2n). They have highly reduced gametophyte stage.

Hence, the correct option is (1).

Q. 4 In the taxonomic categories which hierarchial arrangement in ascending order is correct in case of animals ?

Option 1:
Kingdom, Class, Phylum, Family, Order, Genus, Species

Option 2:
Kingdom, Order, Class, Phylum, Family, Genus, Species

Option 3:
Kingdom, Order, Phylum, Class, Family, Genus, Species

Option 4:
Kingdom, Phylum,Class, Order, Family, Genus, Species

Correct Answer:
Kingdom, Phylum,Class, Order, Family, Genus, Species

Solution:
Correction order is

Kingdom
↓
Phylum
↓
Class
↓
Order
↓
Family
↓
Genus
↓
Species

Q. 5 In which of the following animals, digestive tract has additional chambers like crop and gizzard ?

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Option 1:
Bufo, Balaenoptera, Bangarus

Option 2:
Carla, Columbia, Crocodilus

Option 3:
Paco,Psittacula, Corous

Option 4:
Corpus, Columba, Chameleon

Correct Answer:
Paco,Psittacula, Corous

Solution:
Crop and Gizzard are part of class Aves and Phylum Arthropoda. Pavo, Pssittacula and Corvus all are birds (Aver) and hence the covert answer

Option 1- Bufo is Amphibian

2 - Crocodilus is Reptile

4 - chameleon is Reptile

Q. 6 -Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) :

All vertebrates are chordates but all chordates but all chordates are not vertibrates.

Reason (R) :

Notochord is replaced by vertebral column in the adult vertebrates

In light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 2:
(A) is correct but (R) is not correct

Option 3:
(A) is not correct but (R) is correct

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Option 4:
Both (A) and (R) are correct but (R) is the correct explanation of (A)

Correct Answer:
Both (A) and (R) are correct but (R) is the correct explanation of (A)

Solution:
Assertion is True. Chordates can be vertebrates & Invertebrates. Thus, all vertebrates are definitely chordates.

Reason: Chordates are vertebrates when Notochord is replaced by a vertebral Column. Hence Reason is the correct explanation of Assertion.

Hence, Option (4) is correct.

Q. 7 Which of the following is a correct statement ?

Option 1:
Bacteria are exclusively heterotrophic organisms

Option 2:
Slime moulds are saprophytic organisms classified under Kingdown Monera

Option 3:
Mycoplasma have DNA, Ribosome and cell wall

Option 4:
Cyanobacteria are a group of autotorphic organisms classified under Kingdown Monera

Correct Answer:
Cyanobacteria are a group of autotorphic organisms classified under Kingdown Monera

Solution:
1) Bacteria can be photosynthetic as well

2) Slime moulids belong to the kingdom Protista

3) Mycoplasma Lack cell wall

Hence, Option (4) is correct.

Q. 8 Hydrocolloid carrageen is obtained from :

Option 1:
Phaeophyceae and Rhodophyceae

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Option 2:
Rhodophyceae only

Option 3:
Phaeophyceae only

Option 4:
Chlorophyceae and Phaeophyceae

Correct Answer:
Rhodophyceae only

Solution:
Carrageenan is a polysaccharide obtained from red seaweed (Rhodophyta) It is a gelatin agent, makes the food via cous & use to make the capsule of medicines.

Hence, the correct option is (2).

Q. 9 Give below are two statement :

Statement I :

Mycoplasma ca pass through less than 1 micron filter size

Statement II :

Mycoplasma are bacteria with cell wall

In light of the above statement s, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statemet II is incorrect

Option 3:
Statement I is incorrect but Statemet II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Statement I is correct but Statemet II is incorrect

Solution:
Myoplasm are the smallest cells and hence can pass through filters less than 1 μ m. These are 'Jokers of Microbiology' as these can change their shape because they lack cell walls. Hence statement 1 is correct and 2 is incorrect.

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Q. 10 Identify the correct set of statements :
(a) The leaflets are modified into pointed harc thorns in Citrus and Bougainvillea
(b) Axillary buds form slender and spirally coilec tendrils in cucumber and pumpkin
(c) Stem is flattened and fleshy in Opuntia anc modified to perform the function of leaves
(d) Rhizophora shows vertically upward growing roots that help to get oxygen for respiration
(e) Subaerially growing stems in grasses and strawberry help in vegetative propagation
Choose the correct answer from the options given below:

Option 1:
(a) and (d) Only

Option 2:
(b), (c), (d) and (e) Only

Option 3:
(a), (b), (d) and (e) Only

Option 4:
(b) and (c) Only

Correct Answer:
(b), (c), (d) and (e) Only

Solution:
Adventitious buds and no leaflets are modified into pointed hard thorns in Citrus and bougainvillea. The thorns not only check the rate of transpiration but also protect the plants from herbivorous animals.

Hence, the correct option is (2)

Q. 11 Match **List I** with **List II**

	List I		**List II**
(a)	Bronochioles	(i)	Dense Regular Connective Tissue
(b)	Goblet cell	(ii)	Loose Connective Tissue
(c)	Tendons	(iii)	Glandular Tissue
(d)	Adiopose Tissue	(iv)	Ciliated Epithelium

Choose the correct answer from the options given below:

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Option 1:
(a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

Option 2:
(a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

Option 3:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Option 4:
(a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

Correct Answer:
(a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

Solution:
a) Bronchioles – Ciliated epithelium

b) Goblet cell – Glandular

c) Tendons – Dense regular CT

d) Adipose – Loose connective Tissues

Hence, Option (4) is correct

Q. 12 Which of the following is not a connective tissue ?

Option 1:
Adipose tissue

Option 2:
Cartilage

Option 3:
Neuroglia

Option 4:
Blood

Correct Answer:
Blood

Solution:
Adipose Tissue $\longrightarrow$ Loose Connective Tissue

Cartilage $\longrightarrow$ specialised Connective Tissue (skeletal)

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Blood ⟶ specialised Connective Tissue

Neuroglia ⟶ supporting cells of Nervous Tissue.

Q. 13 Which of the following is present between the adjacent bones of the vertebral column ?

Option 1:
Cartilage

Option 2:
Areolar tissue

Option 3:
Smooth muscle

Option 4:
Intercalated discs

Correct Answer:
Cartilage

Solution:
White Fibrous cartilage is present between two vertebrae.

This provides limited movement.

Q. 14 Which one of the following plants shows vexillary aestivation and diadelphous stamens?

Option 1:
Pisum sativum

Option 2:
Allium cepa

Option 3:
Solanum nigrum

Option 4:
Colchicum autumnale

Correct Answer:
Pisum sativum

Solution:

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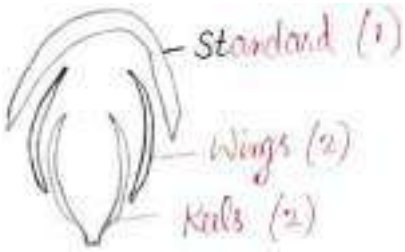
Vexillary aestivation: - It is a unique arrangement where the other smaller petals are covered by one large petal. The larger petal is called the vexillum or standard while the other two curved petals laterally positioned are called wings. The two inmost, boat-shaped petals are called reels. E.g.: - In pea.



Stamens are the male reproductive structure of flowers. They are either free or united, when they are free, they are known as polyandrous. When they are united, they are referred to according to the number of bunches they form: -

So, the pea (Pisum sativum) is the plant that shows both vexillary aestivation 8 amadelphous stamens.

Hence the correct option is (a).

Q. 15 The flowers are Zygomorphic in:
(a) Mustard
(b) Gulmohar
(c) Cassia
(d) Datura
(e) Chilly
Choose the correct answer from the options given below.

Option 1:
(b), (c) Only

Option 2:
(d), (e) Only

Option 3:
(c), (d), (e) Only

Option 4:
(a), (b), (c) Only

Correct Answer:
(b), (c) Only

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Solution:
Examples of zygomorphic flowers: Gulmohar and Cassia

Examples of actinomorphic flowers: Mustard, Datura and Chilly

Zygomorphic -Flowers with only a single line of symmetry like humans. humans.

Actinomorphic - flowers have multiple lines of symmetry (like a starfish) are radial Symmetrical, also called actinomorphic.

Hence, the correct option is (1)

Q. 16 The anatomy of springwood shows some peculiar features. Identify the correct set of statements about springwood.
(a) It is also called as the earlywood
(b) In spring season cambium produces xylem elements with narrow vessels
(c) It is lighter in colour
(d) The springwood along with autumnwood shows alternate concentric rings forming annual rings
(e) It has lower density
Choose the correct answer from the options given below:

Option 1:
(a), (c), (d) and (e) Only

Option 2:
(a), (b) and (d) Only

Option 3:
(c), (d) and (e) Only

Option 4:
(a), (b), (d) and (e) Only

Correct Answer:
(a), (c), (d) and (e) Only

Solution:
In the spring season, Cambium produces xylem elements with wider lumen instead of narrow ones, as the rate of transpiration in plants is very high at - that time. So to fulfill the requirement of water, the xylem with a wider lumen transports more water.

Hence, the correct option is (1).

Q. 17 Tegmina in cockroach, arises from :

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Option 1:
Mesothorax

Option 2:
Metathorax

Option 3:
Prothorax and Mesothorax

Option 4:
Prothorax

Correct Answer:
Metathorax

Solution:
Cockroaches show 2 pairs of wings 1st pair is the forewings. These are also known as Tegmina. These arise from Mesothorax. These are for the protection of the hind wings.2nd pair is Hind wing. These are for flying.

Q. 18 "Girdling Experiment" was performed by Plant Physiologists to identify the plant tissue through which :

Option 1:
food is transported

Option 2:
for both water and food transportation

Option 3:
osmosis is observed

Option 4:
Water is transported

Correct Answer:
food is transported

Solution:
Girdling/Ringing experiment was performed by 'Malpighi' in 1672, to identify the tissues through which food is transported. It involves the removal of all the tissue outside to vascular cambium (bark, cortex and phloem) in the woody stem except for the xylem. In the absence of downward movement of food, the portion of the bark above the ring on the stem becomes swollen after a few weeks. This simple experiment shows that Phloem is the tissue responsible for the translocation of food and that transport takes place in one direction, i.e., towards the roots.

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Hence, the correct option is (a).

Q. 19 Read the following statements about the vascular bundles:
(a) In roots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii.
(b) Conjoint closed vascular bundles do not possess cambium
(c) In open vascular bundles, cambium is present in between xylem and phloem
(d) The vascular bundles of dicotyledonous stem possess endarch protoxylem
(e) In monocotyledonous root, usually there are more than six xylem bundles present
Choose the most appropriate answer from the options given below:

Option 1:
(a),(c),(d) and (e) Only

Option 2:
(a),(b),(c) and (d) Only

Option 3:
(b), (c), (d) and (e) Only

Option 4:
(a), (b) and (d) Only

Correct Answer:
(a),(c),(d) and (e) Only

Solution:

- Statement (a) In roots, the xylem and phloem in a vascular bundle are alternately organized along various radii. This statement is correct and reflects how the xylem and phloem are arranged in root vascular bundles.
- Statement (b) Conjoint closed vascular bundles lack cambium. This statement is not included in the correct option.
- Statement (c) Cambium is found between the xylem and phloem in open vascular bundles. This statement is correct because cambium is present in open vascular bundles.
- Statement (d) Dicotyledonous stem vascular bundles contain endarch protoxylem. This statement is true and refers to the arrangement of the protoxylem towards the center of the stem.
- Statement (e) In the monocotyledonous root, usually there are more than six xylem bundles present. This statement is true and describes the typical number of xylem bundles in monocotyledonous roots.
- Therefore, the correct option is Option 1, which includes statements (a), (c), (d), and (e) only.

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Q. 20 In old trees the greater part of secondary xylem is dark brown and resistant to insect attack due to :
(a) Secretion of secondary metabolities and their deposition in the lumen of vessels.
(b) Deposition of organic compounds like tannins and resins in the central layers of stem.
(c) Deposition of suberin and aromatic substances in the outer layer of stem.
(d) Deposition of tannins, gum, resin and aromatic substances in the peripheral layers of stem.
(e) Presence of parenchyma cells, functionally active xylem elements and essential oils.
Choose the correct answer from the options given below:

Option 1:
(c) and (d) Only

Option 2:
(d) and (e) Only

Option 3:
(b) and (d) Only

Option 4:
(a) and (b) Only

Correct Answer:
(a) and (b) Only

Solution:
In old trees, the greater part of the secondary xylem is dark brown due to the deposition of organic compounds like tannins, Resins, Oils, Gums and aromatic substances in the central or innermost layers of the stem. These substances make it hard, durable and resistant to the attacks of micro-organisms and insects.Hence, the correct option is (4).

Q. 21 Which one of the following never occurs during mitotic cell division?

Option 1:
Movement of centrioles towards opposite poles

Option 2:
Pairing of homologous chromosomes

Option 3:
Coiling and condensation of the chromatids

Option 4:
Spindle fibres attach to kinetochores of chromosomes

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Correct Answer:
Pairing of homologous chromosomes

Solution:
Pairing of homologous chromosomes accurate prophase, I of meiosis.

Movement of centrioles towards opposite holes, coiling and condensation of the chromatics and attachment of spindle fibers to the kinetochores of chromosomes Occur in both meiosis and mitosis.

Hence, the correct option is (2).

Q. 22 Regarding Meiosis, which of the statement is incorrect ?

Option 1:
DNA replication occurs in S phase of Meiosis-II

Option 2:
Pairing of homologous chromosomes and recombination occurs in Meiosis - I

Option 3:
Four haploid cells are formed at the end of Meiosis - II

Option 4:
There are two stages in Meiosis, Meiosis - I and II

Correct Answer:
DNA replication occurs in S phase of Meiosis-II

Solution:
DNA replication occurs during S-phase. This is the part of Interphase which is before Meiosis I

Hence, Option (1) is correct.

Q. 23 The appearance of recombination nodules on homologous chromosomes during meiosis characterizes :

Option 1:
Bivalent

Option 2:
Sites at which crossing over occurs

Option 3:
Terminalization

Option 4:
Synaptonemal complex

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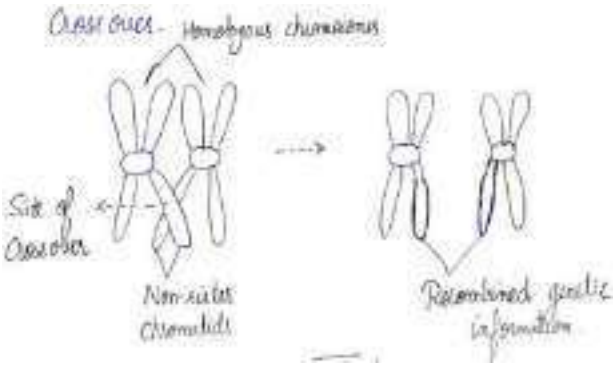
Correct Answer:
Sites at which crossing over occurs

Solution:
Prophase I of meiosis I is a unique phase that involves the recombination of genetic material between homologous chromosomes. This occurs in the pachytene stage where the non-sister chromatics of the homologous chromosomes cross over and the exchange of genetic material takes place. Recombination modules are the sites at which the non-sister chromatids Cross over.



Hence, the correct option is (2)

Q. 24 Match List - I with List - II.

	List I		**List II**
a	Metacentric chromosome	i	Centromere situated close to the end forming one extremely short and one very long arms
b	Acrocentric chromosome	ii	Centromere at the terminal end
c	Sub-metacentric	iii	Centromere in the middle forming two equal arms of chromosomes
d	Telocentric chromosome	iv	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the correct answer from the options given below:

Option 1:
(a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Option 2:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

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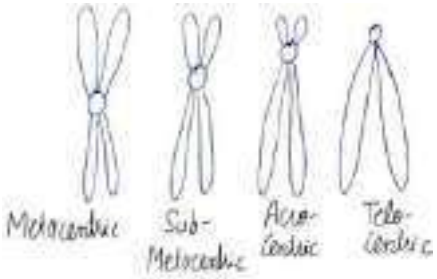
Option 3:
(a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

Option 4:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Correct Answer:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Solution:



Metacentric ⟶ Centromere lies in the middle forming two equal arms of chromosomes.

Sub-Metacentric ⟶ Centromere slightly away from the middle forming one shorter arm and one longer arm.

Acrocentric ⟶ Centromere situated close to the end forming one extremely short and one very long arm.

Telocentric ⟶ Centromere at the terminal end.

Hence, the correct option is (4)

Q. 25 Read the following statements on lipids and find out correct set of statements :
(a) Lecithin found in the plasma membrane is a glycolipid
(b) Saturated fatty acids possess one or more c=c bonds
(c) Gingely oil has lower melting point, hence remains as oil in winter
(d) Lipids are generally insoluble in water but soluble in some organic solvents
(e) When fatty acid is esterified with glycerol, monoglycerides are formed
Choose the correct answer from the options given below:

Option 1:
(a), (d) and (e) only

Option 2:
(c), (d) and (e) only

Option 3:
(a), (b) and (d) only

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Option 4:
(a), (b) and (c) only

Correct Answer:
(c), (d) and (e) only

Solution:
Lecithin is a type of phospholipid found in the plasma membrane.

Saturated fatty acids have a single bond (c-c), whereas unsaturated fatty acids have double (c=c) and triple bonds (C = C)

Hence statements a and b are incorrect.

Hence, the correct option is (2)

Q. 26 Select the incorrect statement with reference to mitosis :

Option 1:
Spindle fibers attach to centromere of chromosomes

Option 2:
Chromosomes decondense at telophase

Option 3:
Splitting of centromere occurs at anaphase

Option 4:
All the chromosomes lie at the equator at metaphase

Correct Answer:
Spindle fibers attach to centromere of chromosomes

Solution:
In Mitosis, spindle fibers attach to the kinetochore of the chromatids.

Hence option 1 is incorrect

Q. 27 A dehydration reaction links two glucose molecules to produce maltose. If the formula for glucose is $C_6H_{12}O_6$ then what is the formula for maltose ?

Option 1:
$C_{12}H_{24}O_{12}$

Option 2:
$C_{12}H_{22}O_{11}$

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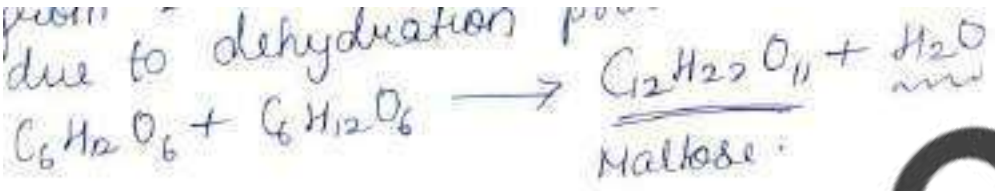
Option 3:
$C_{12}H_{24}O_{11}$

Option 4:
$C_{12}H_{20}O_{10}$

Correct Answer:
$C_{12}H_{22}O_{11}$

Solution:
As mentioned, Maltose is formed from 2 Glucose Molecules. This is due to the dehydration process. Thus



Hence, Option (2) is correct.

Q. 28 Which of the following statements with respect to Endoplasmic Reticulum is incorrect ?

Option 1:
SER is devoid of ribosomes

Option 2:
In prokaryotes only RER are present

Option 3:
SER are the sites for lipid synthesis

Option 4:
RER has ribosomes attached to ER

Correct Answer:
In prokaryotes only RER are present

Solution:
ER is absent in prokaryotes. Hence RER (Rough ER) and SER (smooth ER) are both absent. Option 2 is correct

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Q. 29 Match List - I with List - II.

List I		List II	
a	Manganese	i	Activates the enzyme catalase
b	Magnesium	ii	Required for pollen germination
c	Boron	iii	Activates enzymes of respiration
d	Iron	iv	Functions in splitting of water during photosynthesis

Choose the correct answer from the options given below:

Option 1:
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Option 2:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Option 3:
(a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)

Option 4:
(a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

Correct Answer:
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Solution:
Manganese - Functions in splitting of water during photosynthesis.

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Magnesium - Activates enzymes of respiration

Boron - Required for pollen germination

Iron - Activates the enzyme catalase.

Hence the correct option is (1).

Q. 30 Production of Cucumber has increased manifold in recent years. Application of which of the following phytohormones has resulted in this increased yield as the hormone is known to produce female flowers in the plants :

Option 1:
Gibberellin

Option 2:
Ethylene

Option 3:
Cytokinin

Option 4:
ABA

Correct Answer:
Ethylene

Solution:
In cucumber, ethylene is responsible for promoting female flower production and thereby increasing the yield. Ethylene is a gaseous phytohormone.

Hence, the correct option is (2)

Q. 31 What is the role of large bundle shealth cells found around the vascular bundles in C4 plants ?

Option 1:
To increase the number of chloroplast for the operation of Calvin cycle

Option 2:
To enable the plant to tolerate high temperature

Option 3:
To protect the vascular tissue from high light intensity

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Option 4:
To provide the site for photorespiratory pathway

Correct Answer:
To increase the number of chloroplast for the operation of Calvin cycle

Solution:
C4 plants are evolved C3 plants. They have large bundle sheath cells around the mesophyll cell to avoid a wasteful process i.e. photorespiration. Rubisco enzyme is present in the bundle sheath cells to perform Calvin Cycle and release sugar.

Hence, the correct option is (1).

Q. 32 Which one of the following is not true regarding the release of energy during ATP synthesis through chemiosmosis? It involves :

Option 1:
Breakdown of proton gradient

Option 2:
Movement of protons across the membrane to the stroma

Option 3:
Reduction of NADP to $NADPH_2$ on the stroma side of the membrane

Option 4:
Breakdown of proton gradient

Correct Answer:
Breakdown of proton gradient

Solution:
- Chemiosmosis is a process involved in ATP synthesis.

- During this process, protons in the Strom decrease in number, while in the lumen there is an accumulation of protons. This creates a proton gradient across the thylakoid membrane, and no formation of an electron gradient occurs.

The breakdown of this proton gradient leads to the synthesis of ATP.

Hence the correct option is (a)

Q. 33 Which one of the following produces nitrogen fixing nodules on the roots of Alnus?

Option 1:
Frankia

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Option 2:
Rhodospirillum

Option 3:
Beijernickia

Option 4:
Rhizobium

Correct Answer:
Frankia

Solution:
Frankia - produces nitrogen-fixing nodules on the roots of the non-leguminous plants (e.g., Alnus)

Beijerinckia & Rhodospirillum - Free-living nitrogen fixing aerobic microbes

Rhizobium - Free-living in soil, but as symbionts can fix atmospheric nitrogen in the roots of the leguminous plants.

Hence the correct option is (a)

Q. 34 What amount of energy is released from glucose during lactic acid fermentation ?

Option 1:
More than 18%

Option 2:
About 10%

Option 3:
Less than 7%

Option 4:
Approximately 15%

Correct Answer:
Less than 7%

Solution:
During Aerobic respiration, 38 ATP are releases per glucose molecule.

During lactic acid fermentation, less than 7 % of the total energy in glucose

During alcoholic fermentation, 2 ATP is releases per glucose molecule.

Hence, the correct answer is Option (3)

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Q. 35 The gaseous plant growth regulator is used in plants to :

Option 1:
promote root growth and roothair formation to increase the absorption surface

Option 2:
help overcome apical dominance

Option 3:
kill dicotyledonous weeds in the fields

Option 4:
speed up the malting process

Correct Answer:
promote root growth and roothair formation to increase the absorption surface

Solution:
→The gaseous plant growth regulator i.e. ethylene promotes root growth and root hair formation to increase the absorption surface.

→The presence of cytokinin in an area causes preferential movement of nutrients towards it. When applied to lateral buds, they helps in their growth despite the presence of apical bud .they thus act antagonistically to auxin which promotes apical dominance. Therefore cytokines can overcome apical dominance, caused by auxins.

→The compound 2,4-dichloro phenoxy acetic acid or 2, 4-D is a herbicide used-lo kill any dicot plant weed. The substance is a synthetic auxin, which is a phytohormone.

→Gibberellins are used to speed up the malting process in brewing in dust. They increase the yield of malt from barley grains.

Hence, the correct option is (1).

Q. 36 What is the net gain of ATP when each molecule of glucose is converted to two molecules of pyruvic acid ?

Option 1:
Six

Option 2:
Two

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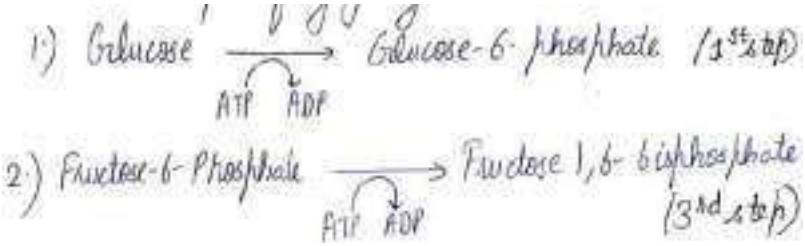
Option 3:
Eight

Option 4:
Four

Correct Answer:
Two

Solution:
One molecule of glucose undergoes glycolysis to form 4 molecules of ATP, two are used in the activation phase or initial steps of glycolysis i.e. when



So, the Net gain is 2 ATP.

Hence, the correct option is (2)

Q. 37 Given below are two statements:
Statement I : The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells.
Statement II : Mesophyll cells of C_4 plants lack RuBisCo enzyme. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:

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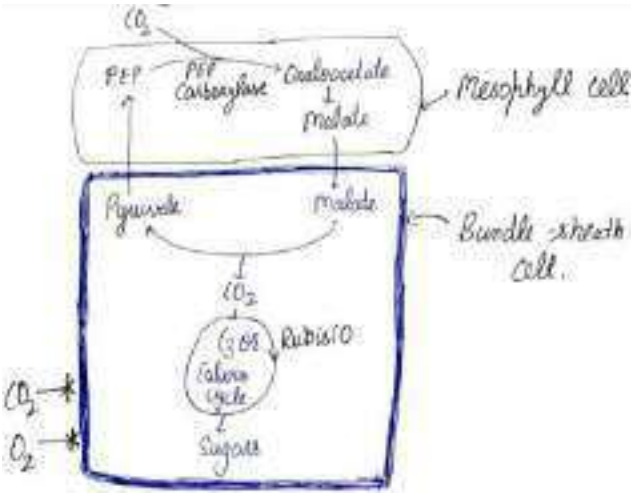
C4 plants are evolved C3 plants to avoid the process of photorespiration. C4 plants have mesophyll cells and bundle sheath cells. In mesophyll cells, primary CO_2 acceptor is PEP (Phosphoenolpyruvate), whereas the Rubisco enzyme is present in the bundle sheath cells. Due to the impervious walls of bundle sheath cells, CO_2 cannot reach Rubisco and photorespiration does not occur.



Hence, the correct option is (4)

Q. 38 Which one of the following plants does not show plasticity ?

Option 1:
Coriander

Option 2:
Buttercup

Option 3:
Maize

Option 4:
Cotton

Correct Answer:
Maize

Solution:
Plants follow different pathways in response to the environment or phases of beef to form different kinds of structures. This ability is called plasticity. e.g., heterophylly in cotton, coriander and larkspur in such plants, leaves of the juvenile plants are different in shape from those in mature plants. Maize does not show plasticity.

Hence, the correct option is (3).

Q. 39 Addition of more solutes in a given solution will :

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Option 1:
lower its water potential

Option 2:
make its water potential zero

Option 3:
not affect the water potential at all

Option 4:
raise its water potential

Correct Answer:
lower its water potential

Solution:
If some solute is dissolved in pure water, the solution has lower free water and the concentration of water decrease, reducing its water potential. The magnitude of this lowering due to the dissolution of a solute is called solute potential.

Hence, the correct option is (1).

Q. 40 Which of the following is not observed during apoplastic pathway?

Option 1:
The movement does not involve crossing of cell membrane

Option 2:
The movement is aided by cytoplasmic streaming

Option 3:
Apoplast is continuous and does not provide any barrier to water movement.

Option 4:
Movement of water occurs through intercellular spaces and wall of the cells.

Correct Answer:
The movement is aided by cytoplasmic streaming

Solution:
Pathway of water movement in the root

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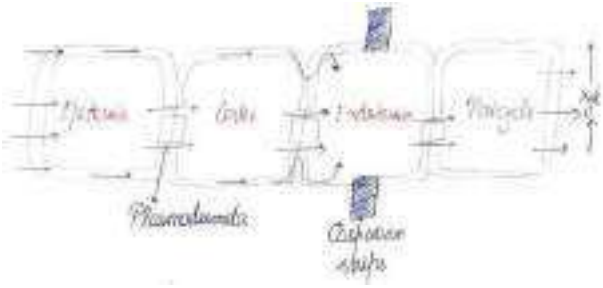

Apoplast = Away from cell

Symplast = Via cytoplasm

The apoplast is the system of adjacent cell walls that is continuous throughout the plant, except at the Casparian strips of the endodermis in the roots.

Therefore, it is not aided by cytoplasmic streaming.

Hence the correct option is (b).

Q. 41 Which of the following functions is not performed by secretions from salivary glands ?

Option 1:
Digestion of complex carbohydrates

Option 2:
Lubrication of oral cavity

Option 3:
Digestion of disaccharides

Option 4:
Control bacterial population in mouth

Correct Answer:
Digestion of disaccharides

Solution:
Salivary amylase brings about the digestion of starch (complex Carbohydrates into disaccharides (Maltose). It cannot digest Maltose further.
Hence option 3 is not a function.

Q. 42 Which of the following is correct match for disease and it's symptoms ?

Option 1:
Tetany - high Ca^{2+} level causing rapid spasms

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Option 2:
Myasthenia gravis - Genetic disorder resulting in weakening and paralysis of skeletal muscle

Option 3:
Muscular dystrophy - An auto immune disorder causing progressive degeneration of sketel muscle.

Option 4:
Arthritis - Inflammed joints

Correct Answer:
Arthritis - Inflammed joints

Solution:
(1) $\rightarrow$ Tetany is due to low Ca^{+2} levels.

(2) Myasthenia Gravis is an Autoimmune disorder where Ab destroys the body cells.

(3) Muscular dystrophy - It is a Genetic disorder.

(4) Arthritis $\rightarrow$ Inflamed joints.

Q. 43 Which of the following is not the function of conducting part of respiratory system?

Option 1:
Inhaled air is humudified

Option 2:
Temperature of inhaled air is brought to body temperature

Option 3:
Provides surface for diffusion of O_2 and CO_2

Option 4:
It clears inhaled air from foreign particles

Correct Answer:
Provides surface for diffusion of O_2 and CO_2

Solution:
Respiratory system is divided into conducting part and Respiratory Part. Conducting Part is for transporting the gases. Respiratory is for exchange. Alveoli are for exchange that brings about diffusion.

Hence, Option (3) is correct.

Q. 44 Nitrogenous waste is excreted in the form of pellet or paste by :

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Option 1:
Salamamdra

Option 2:
Hippocampus

Option 3:
Pavo

Option 4:
Ornithorhyuchus

Correct Answer:
Pavo

Solution:
Nitrogenous waste excreted in the form of pallets is uric acid (since it is insoluble in water). This is seen in class Aues. Hence option 3 pave is the answer

Hence, Option (3) is correct.

Q. 45 Which of the following statements is correct ?

Option 1:
The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria.

Option 2:
Blood moves freely from atrium to the ventricle during joint diastole.

Option 3:
Increased ventricular pressure causes closing of the semilunar valves.

Option 4:
The atrio-ventricular node (AVN) generates an action potential to stimulate aerial contraction

Correct Answer:
Blood moves freely from atrium to the ventricle during joint diastole.

Solution:
1) Tricuspid & Bicuspid open due to blood & arial systole

2) True

3) Increased ventricular pressure causes the opening of semilunar valves

4) Action potentials are generated by the SA node as it is the pacemaker

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Hence, Option (2) is correct.

Q. 46 Which of the following are not the effects of Parathyroid hormone ?

(a) Stimulates the process of bone resorption

(b) Decrease Ca^{2+} level in blood

(c) Reabsorption of Ca^{2+} by renal tubules

(d) Decrease the absorption of Ca^{2+} from digested food

(e) Increase metabolism of carbohydrates

Choose the most appropriate answer from the options given below :

Option 1:
(b), (d) and (e) only

Option 2:
(a) and (e) only

Option 3:
(b) and (c) only

Option 4:
(a) and (c) only

Correct Answer:
(b), (d) and (e) only

Solution:
Parathyroid hormone (PTH) or Collop's hormone increases Ca+2 levels in the blood plasma.

This is achieved by

1. Increasing reabsorption by the bones
2. Increasing reabsorption by Kidney
3. Increasing absorption by intestines

Hence Option (1) is correct.

Q. 47 Select the incorrect statement regarding synapses:

Option 1:
Electrical current can flow directly from one neuron into the other across the electrical synapse.

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Option 2:
Chemical synapses use neuotrasmitters

Option 3:
Impulse transmission across a chemical synapse is always faster than an electrical synapse.

Option 4:
The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.

Correct Answer:
Impulse transmission across a chemical synapse is always faster than an electrical synapse.

Solution:
The electrical synapse is faster than the chemical synapse.

Hence, Option (3) is correct.

Q. 48 Under normal physiological conditions in human being every 100ml of oxygenated blood can deliver _____ ml of O_2 to the tissues.

Option 1:
5ml

Option 2:
4ml

Option 3:
10ml

Option 4:
2ml

Correct Answer:
5ml

Solution:
Every 100ml of oxygenated blood can deliver 5ml of O_2 to the tissues. Option 1 is correct

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Q. 49 Match List I with List II

	List I (Biological Molecules)		List II (Biological functions)
(a)	Glycogen	(i)	Hormone
(b)	Globulin	(ii)	Biocatalyst
(c)	Steroids	(iii)	Antibody
(d)	Thrombin	(iv)	Storage product

Choose the correct answer from the options given below:

Option 1:
$(a)-(iv),(b)-(ii),(c)-(i),(d)-(iii).$

Option 2:
$(a)-(ii),(b)-(iv),(c)-(iii),(d)-(i)$

Option 3:
$(a)-(iv),(b)-(iii),(c)-(i),(d)-(ii)$

Option 4:
$(a)-(iii),(b)-(ii),(c)-(iv),(d)-(i)$

Correct Answer:
$(a)-(iv),(b)-(iii),(c)-(i),(d)-(ii)$

Solution:
a) Glycogen – Storage product

b) Globulin – AB

c) Steroid – Hormone

d) Thrombin – Biocatalyst

Hence, Option (3) is correct.

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Q. 50 Given below are two statements

Statement I: Fatty acids and glycerols cannot be observed in the blood.

Statement II: Specialized lymphatic capillaries called lacteals carry chylomicrons into lymphatic vessels and ultimately into the blood.

In light of the above statements, choose the **most appropriate** answer from the options given below:

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:
Statement 1 is correct because Fatty acids and Glycerol are insoluble in water. Hence these cannot be absorbed directly into the blood. Hence these cannot be absorbed directly into the blood.

Statement 2 is correct because Fatty acid and Glycerol are first converted into micelles. These move from the intestinal lumen into mucosa These are then transformed into very small protein-coated fat globules called chylomicrons. These enter lacteals (lymphatic capillaries) because chylomicrons are bigger and cannot enter blood capillaries. Inside the lacteals, they break down and finally enter blood:

Hence, option 4 is the correct answer.

Q. 51 Given below are two statement s:

Statement I: The coagulum is formed of a network of threads called thrombins.

Statement II: The spleen is the graveyard of erythrocytes.

In light of the statements, choose the most appropriate answer from the options given below :

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Option 1:
Both the Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both the Statement I and Statement II are correct

Correct Answer:
Statement I is incorrect but Statement II is correct

Solution:
Statement 1 is incorrect because Coagulum is a blood clot. It is formed of a network of threads called fibrins.

Statement 2 is correct because RBCs have a life cycle of 120 days after which they die in the spleen Thus spleen is the graveyard is correct

Therefore, the correct answer is option 3.

Q. 52 Match List I with List II with respect to methods of contraception and their respective actions

	List I		**List II**
(a)	Diaphragms	(i)	Inhibit ovulation and Impkantation
(b)	Contraceptive pills	(ii)	Increase phagocytosis of sperm with Uterus
(c)	Intra uterine Devices	(iii)	Absence of Mentrual cycle and ovulation following parturition
(d)	Lactational Amenorrhea	(iv)	They cover the cervix blocking the entry of sperms

Choose the correct answer from the options given below :

Option 1:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iv)

Option 2:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Option 3:
(a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

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Option 4:
(a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Correct Answer:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iv)

Solution:
Diaphragms – Cover the cervix blocking the entry of sperms

Contraceptive Pills – Inhibit Ovulation and Implantation

Iu D – Increase phagocytosis of sperm with Uterus

Lactational Amenorrhea – Absence of Menstrual Cycle & ovulation following patriation

Hence, Option (1) is correct.

Q. 53 Identify the incorrect statement related to Pollination:

Option 1:
Pollination by wind is more common amongst abiotic pollination

Option 2:
Flowers produce foul odours to attract flies and beetles to get pollinated

Option 3:
Moths and butterflies are the most dominant pollinating agents among insects

Option 4:
Pollination by water is quite rare in flowering plants

Correct Answer:
Moths and butterflies are the most dominant pollinating agents among insects

Solution:
Among the animals, insects, particularly are the dominant biotic pollinating agents.

Hence, the correct answer is (c)

Q. 54 Given below are two statements:
Statement I: Cleistogamous flowers are invariably autogamous
Statement II: Cleistogamy is disadvantageous as there is no chance for cross-pollination In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are incorrect

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Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

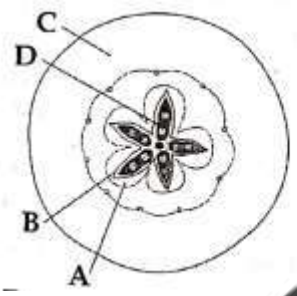
Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:
Qeistogamous flowers do not open at all. In such flowers, the anthers and stigma lie close to each other. When anthers dehisce in the flower buds, pollen grains come in contact with the stigma to affect pollination. Thus, Cleistogamous flowers are in variably autogenous as there is no chance of cross-pollen landing on the stigma.

Hence, the correct answer is (4)

Q. 55 Which part of the fruit, labelled in the given figure makes it a false fruit?



Option 1:
B → Endocarp

Option 2:
C → Thalamus

Option 3:
D → Seed

Option 4:
A → Mesocarp

Correct Answer:
C → Thalamus

Solution:
True fruits ⟶ Fruits develop from the ovary.

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False fruits $\rightarrow$ Fruits develop from other floral parts and the thalamus along with the development of the ovary wall.

Eg $\rightarrow$ Apple, pears, strawberries, pineapple etc.

Hence, the correct option is (2)

Q. 56 Given below are two statements :

Statement I :

The release of sperms into the seminiferous tubules is called spermiation.

Statement II :

Spermiogenesis is the process of formation of sperms from spermatogonia .

In light of the above statements choose the most appropriate answer from the options given below :

Option 1:
Both Statements I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statements I and Statement II are correct

Correct Answer:
Both Statements I and Statement II are incorrect

Solution:
Spermiogenesis $\rightarrow$ Formation of sperms from spermatids. This is a morphologic change

Spermiation $\rightarrow$ The sperm once formed is embedded in the Sertoli cells where these gain maturation and are then released into the lumen.

Hence both statements are False.

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Q. 57 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) :

Osteoporosis is characterized by decreased bone mass and increased chances of fractures.

Reason (R) :

Common cause of osteoporosis is increased levels of estrogen.

In light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 2:
(A) is correct but (R) is not correct

Option 3:
(A) is not correct but (R) is correct

Option 4:
Both (A) and (R) are correct but (R) is the correct explanation of (A)

Correct Answer:
(A) is correct but (R) is not correct

Solution:
Estragon is important for bone growth and maturation. It is also important for overall bone turnover. Hence, in the post-menopausal stage, bones become brittle or chances of osteoporosis increase. (As estrogen levels decrease after menopause).

Osteoporosis is decreased bone mass So the Assertion is correct but the Reason is wrong.

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Q. 58 Which of the following Statemets are true for spermatogeesis it do it hold true for Oogeesis ?

(a) It results in the formation of haploid gametes

(b) Differentiation of gamete occurs after the completion of meiosis

(c) Meiosis occurs continuously in a mitotically dividing stem cell population

(d) It is controlled by the Luteinising hormone (LH) and Follic Stimulating Hormone (FSH) screted by the anterior pituitary

(e) It is initiated at puberty

Choose the most appropriate answer from the options given below :

Option 1:
(b) and (c) only

Option 2:
(b) , (d) and (e) only

Option 3:
(b), (c) and (e) only

Option 4:
(c) and (e) only

Correct Answer:
(b), (c) and (e) only

Solution:
(a) $\rightarrow$ True for both spermatogenesis and Oogenesis

(b) $\rightarrow$ This is true for spermatogenesis I Primary spermatocyte forms 4 sperms. In Oogenesis, when gametes are formed, they remain arrested at Meiosis - II (Metaphases II)

(c) $\rightarrow$ Meiosis is continuous in Spermatogenesis. In Oogenesis, it is arrested.

(d) This is true for both:

(e) Spermatogenesis initiates at Puberty Oogenesis is before birth

Hence b, c, and, e are true for spermatogenesis and not for Oogenesis

Q. 59 At which stage of life the oogenesis process is initiated ?

Option 1:
Embryonic development stage

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Option 2:
Birth

Option 3:
Adult

Option 4:
Puberty

Correct Answer:
Embryonic development stage

Solution:
Oogenesis is the formation of ova. It begins in intra-utrine life when cells from the yolk sac enter the ovary and develop into primary oogonia. Thus, the answer is the Embryonic development Stage.

Hence, Option (1) is correct.

Q. 60 Lippe's loop is a type of contraceptive used as :

Option 1:
Vault barrier

Option 2:
Non-Medicated IUD

Option 3:
Copper realising IUD

Option 4:
Cervical barrier

Correct Answer:
Non-Medicated IUD

Solution:
Lippes Loop is a non-medicated IUD. It helps in the phagocytosis of sperms in the uterus.

Hence, Option (2) is correct.

Q. 61 Identify the asexual reproductive structure associated with Penicillium :

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Option 1:
Conidia

Option 2:
Gemmules

Option 3:
Buds

Option 4:
Zoospores

Correct Answer:
Conidia

Solution:
In Penicillium, asexual spores are Conidia. These are seen exogenously and these produce mycelium.

Hence, Option (1) is correct.

Q. 62 XO type of sex determination can be found in:

Option 1:
Birds

Option 2:
Grasshoppers

Option 3:
Monkeys

Option 4:
Drosophila

Correct Answer:
Grasshoppers

Solution:
Grasshopper - XO, XX Males females

X X X

Birds - Males females

ZZ ZW

Drosophila - Male females

XY XX

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Monkey - Male females

XY XX

Hence the correct option is (2).

Q. 63 The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome?

Option 1:
d,b,a,c

Option 2:
a,b,c,d

Option 3:
a,c,b,d

Option 4:
a,d,b,c

Correct Answer:
a,c,b,d

Solution:
Distance between genes is measured by frequency of recombination.

According to the question,

-Recombination frequency between a & d is highest. This means they are farthest.

-So, option 2 & 3 can be our answer

a_____b

-Recombination frequency between

a and b = 20%

a and c = 5%

so, a and c are close.

Hence the sequence is

a c b d

Hence, Option (3) is correct.

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Q. 64 Read the following statements and choose the set of correct statements :
(a) Euchromatin is loosely packed chromatin
(b) Heterochromatin is transcriptionally active
(c) Histone octomer is wrapped by negatively charged DNA in nucleosome
(d) Histones are rich in lysine and arginine
(e) A typical nucleosome contains 400bp400bp of DNA helix
Choose the correct answer from the options given below:

Option 1:
(a), (c), (d) Only

Option 2:
(b), (e) Only

Option 3:
(a), (c), (e) Only

Option 4:
(b), (d), (e) Only

Correct Answer:
(a), (c), (d) Only

Solution:

Euchromatin	Heterochromatin
1. lightly stained	1. Darkly stained area
2. Participates in the translation process	2. Do not participate in the translation process.
3. Loosely packed form of DNA	3.A tightly packed form of DNA
4. Low DNA density	4. High DNA density
5. Found in both prokaryotes and eukaryotes	5. Found in eukaryotes only

àA typical nucleosome contains 200 bp of DNA helix.

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Q. 65 The process of translation of mRNA to proteins begins as soon as :

Option 1:
The larger subunit of ribosome encounters mRNA

Option 2:
Both the subunits join together to bind with mRNA

Option 3:
The tRNA is activated and the larger subunit of ribosome encounters mRNA

Option 4:
The small subunit of ribosome encounters mRNA

Correct Answer:
The small subunit of ribosome encounters mRNA

Solution:
When the smaller submit of ribosome encounter m-RNA, the process of translation of m-RNA to protein begins. This process is followed by the binding of a larger subunit. The activated t-RNA molecule carrying the amine acid methionine binds to the start codon of the mRNA sequence.

Hence, the correct option is (4).

Q. 66 Given below are two statements:
Statement I : Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance
Statement II : Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

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Correct Answer:
Both Statement I and Statement II are correct

Solution:
George J. Mendel conducted hybridization experiments on garden pea and selected 14 true-breeding pea plant verities (seven contrasting traits). Contrasting traits

Studied were: - 1.) Smooth or winkled seeds

2.) Yellow or green seeds

3.) Inflated or constricted pods

4.) green or yellow pods.

5.) tall or dwarf plants

6) Violet or white flowers

7.) Axial or terminal flower positions.

Hence, the correct option is (4).

Q. 67 Which of the following occurs due to the presence of autosome linked dominant trait?

Option 1:
Myotonic dystrophy

Option 2:
Haemophilia

Option 3:
Thalessemia

Option 4:
Sickle cell anaemia

Correct Answer:
Myotonic dystrophy

Solution:
Hemophilia is a disorder in which blood doesn't clot normally and it is a X-linked recessive disorder.

Thalassemia is a blood disorder involving a lower than normal amount of an-oxygen carrying protein .it is an autosomal recessive disorder.

Sickle cell anemia is a group of disorders that causes red blood cells to become misshapen and break down it is an autosomal recessive disorder.

Myotonic dystrophy is an autosomal dominant disorder i.e.it occurs due to the presence of autosomal linked dominant trail.

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Hence, the correct option is (1)

Q. 68 If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called as:

Option 1:
Gene mapping

Option 2:
Expressed sequence tags

Option 3:
Bioinformatics

Option 4:
Sequence annotation

Correct Answer:
Sequence annotation

Solution:
Sequencing the whole set of genomes that contained all the coding and non-coding sequences and later assigning different regions in the sequence with functions is called sequence annotation.

The second approach is EST (Expressed Sequence Tags) in which the focus is on identifying all the genes that are expressed as RNA.

Hence, the correct option is (4).

Q. 69 Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason(R).
Assertion (A) : Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.
Reason (R): Closely located genes assort independently.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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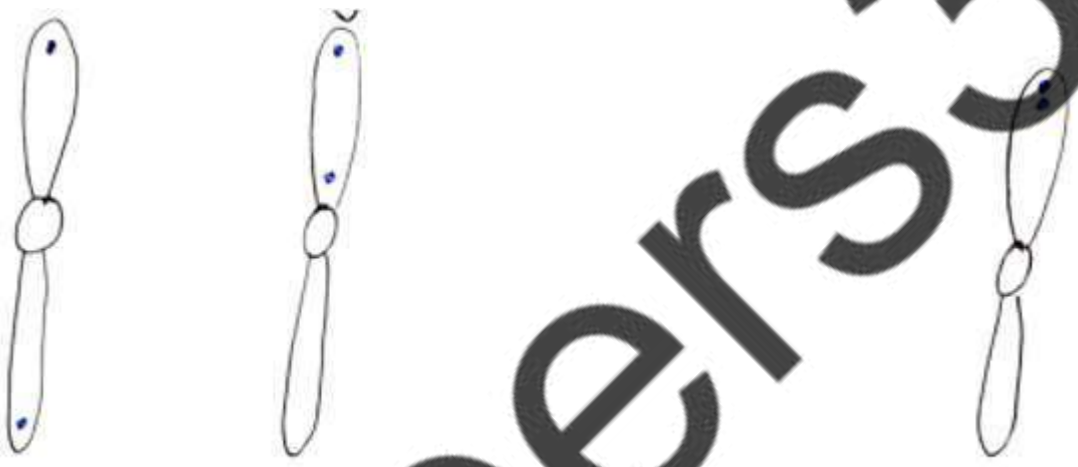
Option 2:
(A) is correct but (R) is not correct

Option 3:
(A) is not correct but (R) is correct

Option 4:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

Correct Answer:
(A) is correct but (R) is not correct

Solution:
If genes are closely located on the chromosome, they won't assort independently and do not show recombination. Mendel's law of independent assortment holds good only for the genes that are located on the same chromosome, not for the closely related genes. Hence statement 1 is right, and 2nd is incorrect.



Case I	Case II	Case III
very far, therefore, show Independent assortment and recombination Occur with high percentage. Linkage Recombination	same arm, therefore shows independent assortment but the chance of Chances of recombination Will be less. Linkage Recombination	no independent assortment occur and no chance of recombination Linkage Recombination

Hence, the correct option is (b)

Q. 70 In an Ecoli Strain i gene gets mutated and it's product can not bind the inducer molecule. If growth medium is provided with lactose, what be the outcome ?

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Option 1:
z,y,a a genes will be transcribed

Option 2:
z,y,a genes will not be translated

Option 3:
RNA polymerase will bind thepromoter region

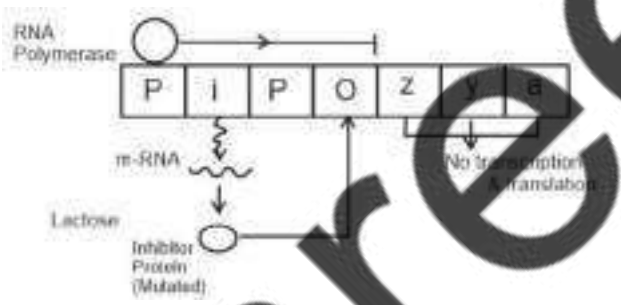
Option 4:
Only z gene will get transcribed

Correct Answer:
RNA polymerase will bind thepromoter region

Solution:
Normally, in On condition i.e. when lactose is present in the medium, the inhibitor protein binds to the lactose and not to the operator gene. This is because the inhibitor protein has more affinity with lactose as compared to the operator gene. This will allow the z, and y & genes to undergo transcription & then translation.

In this case, 'i' gene' has been mutated due to which, even in the presence of lactose, inhibitor protein binds to the operator gene which will block the movement of RNA polymerase and z, y and genes will not undergo transcription and translation.

Hence the correct option is C.



Q. 71 Natural selection where more individuals acquire specific character value other than the mean character value, leads to :

Option 1:
Directional Change

Option 2:
Disruptive change

Option 3:
Random change

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Option 4:
Stabilising change

Correct Answer:
Directional Change

Solution:
- If in Natural selection, individuals do not acquire mean value, it can be Disruptive/Directional.

- Hence question is about a specific value other than the mean, hence it has to be Directional

- In Disruptive, peripheral values are selected (at both ends) and this is nonspecific.

Hence, Option (1) is correct.

Q. 72 If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness ?

Option 1:
50%

Option 2:
75%

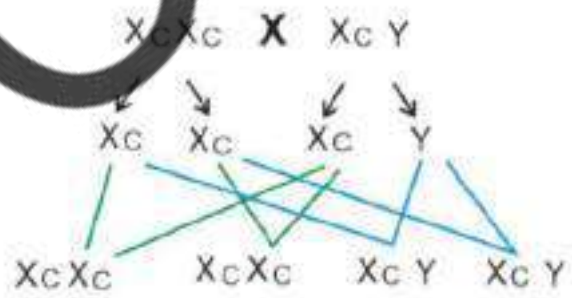
Option 3:
100%

Option 4:
25%

Correct Answer:
100%

Solution:
As given, the Female is colorblind i.e., XcXc & Male's mother was colorblind this means the male is XcY

So,



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So, all the boys & girls will be colorblind. Hence 100% is the answer

Hence, Option (3) is correct.

Q. 73 Which of the following statement is not true ?

Option 1:
Sweet Potato and Potato is an example of analogy

Option 2:
Homology indicates common ancestry

Option 3:
Flippers of penguins and dolphins are a pair of homlogous organs

Option 4:
Analogous structures are a result of convergent evolution

Correct Answer:
Flippers of penguins and dolphins are a pair of homlogous organs

Solution:
-Flippers of Penguins and Dolphins are pairs of analogous organs

-Analogous organs have similar functions but dissimilar structures

Hence, Option (3) is correct.

Q. 74 If the length of a DNA molecule is 1.1 meters, what will be the approximate number of base pairs ?

Option 1:
$6.3 \times 10^{-9}\text{bp}$

Option 2:
$3.6 \times 10^{-9}\text{bp}$

Option 3:
$6.6 \times 10^{9}\text{bp}$

Option 4:
$3.3 \times 10^{-9}\text{bp}$

Correct Answer:
$3.3 \times 10^{-9}\text{bp}$

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Solution:

$$\text{Length of DNA} = 1.1 \text{ m}$$

$$\text{Distance bet}^{\text{n}} \text{ 2 base pairs} = 0.34 \text{ nm}$$

$$= 0.34 \times 10^{-9}\text{m}$$

$$\therefore \text{N Q of Base pairs} = \frac{1.1}{0.34} \times 10^{-9}$$

$$= 3.23 \times 10^{-9}\text{bp}$$

Hence, the correct answer is option 4.

Q. 75 Given below are two statements:

Statement I :

Autoimmune disorder is a condition where body defense mechanism recognizes it's own cells as foreign bodies

Statement II :

Rheumatoid arthritis is a condition where body does not attack self cells.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Statement I is correct but Statement II is incorrect

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Solution:
Statement 1: It is the definition of Autoimmune diseases

Statement 2: RA is an autoimmune and hence it is wrong

Hence, Option (2) is correct.

Q. 76 Breading crops with higher levels of vitamins and minerals or higher protiens and healthier fats is called :

Option 1:
Bio-remediation

Option 2:
Bio-fortification

Option 3:
Bio-accumulation

Option 4:
Bio-magnification

Correct Answer:
Bio-fortification

Solution:
Biofortification is process of breeding staple crops to have high levels of important nutrients like Zn in golden rice etc.

Hence, Option (2) is correct.

Q. 77 Select the incorrect statement with respect to acquired immunity

Option 1:
Anamnestic response is elicited on subsequent encounters with the same pathogen

Option 2:
Anamnestic response is due to memory of first encounter

Option 3:
Acquired immunity is non-specific type of defense present at the time of birth

Option 4:
Primary response is produced when our body encounters a pathagen for the first time.

Correct Answer:
Acquired immunity is non-specific type of defense present at the time of birth

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Solution:
Acquired immunity is Adaptive immunity. This is specific immunity

Hence, Option (3) is correct.

Q. 78 identify the microorganism which is responsible for the production of an immunousuppressive molecule cyclosporin A :

Option 1:
Clostridium butylicum

Option 2:
Aspergillus niger

Option 3:
Streptococcus cerevisiae

Option 4:
Trichoderma polysporum

Correct Answer:
Trichoderma polysporum

Solution:
Cyclosporin A is used to treat RA. It reduces the activity of T-cells & thus suppresses the immune system. It is isolated from the Trichoderma polyparium.

Hence, Option (4) is correct.

Q. 79 Which one of the following statement is not true regarding gel electrophoresis technique?

Option 1:
The separated DNA fragments are stained by using ethidium bromide.

Option 2:
The presence of chromogenic substrate gives blue coloured DNA bands on the gel.

Option 3:
Bright orange coloured bands of DNA can be observed in the gel when exposed to UV light.

Option 4:
The process of extraction of separated DNA strands from gel is called elution.

Correct Answer:
The presence of chromogenic substrate gives blue coloured DNA bands on the gel.

Solution:

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In agarose gel electrophoresis, bands are visualized as orange bands with EtBr (Ethiclium Bromide). Etbr on intercalation with nitrogenous bases gives orange colored bands on gel and it shows fluorescence upon UV exposure. Protein bands stain blue on SDS-PAGE.

Hence, the correct option is (2).

Q. 80 Which of the following is not a descriable feature of a cloning vector ?

Option 1:
Presence of a maker gene

Option 2:
Presence of single restriction enzyme site

Option 3:
Presence of two or move recognition sites

Option 4:
Presence of origin of replication

Correct Answer:
Presence of a maker gene

Solution:
Only 1 recognition site is important

Hence, Option (1) is correct.

Q. 81 DNA polymorphism forms the basis of :

Option 1:
DNA finger printing

Option 2:
Both genetic mapping and DNA finger printing

Option 3:
Translation

Option 4:
Genetic mapping

Correct Answer:
Both genetic mapping and DNA finger printing

Solution:
DNA polymorphism is a difference in DNA sequence among individuals, groups, or populations.

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Genetic mapping refers to the process of determining the location of genes on chromosomes.

DNA fingerprinting is a laboratory technique used to determine the probable identity of a person based on the nucleotide sequences of certain regions of human DNA that are unique to individuals.

Therefore, DNA polymorphism forms the basis of genetic mapping and DNA fingerprinting.

Hence, the correct option is (2)

Q. 82 In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme?

Option 1:
5' GAATTC3'; 3' CT T A A G 5'

Option 2:
5'CTCAGT3'; 3' GAGTCA 5'

Option 3:
5' G T A T T C3' ; 3' CATA A G5'

Option 4:
5' GATACT3'; 3' CTATGA 5'

Correct Answer:
5' GAATTC3'; 3' CT T A A G 5'

Solution:
Palindromic sequences are the sequences which are read back and forth in the same way.

Therefore, only option 1 has a palindromic sequence, the rest are not palindromic sequences.

Hence the correct option is 1.

Q. 83 Transposons can be used during which one of the following ?

Option 1:
Gene silencing

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Option 2:
Autoradiography

Option 3:
Gene sequencing

Option 4:
Polymerase Chain Reaction

Correct Answer:
Gene silencing

Solution:
Option 1 is correct because the source of complementary RNA for RNAi (RNA interference) or gene silencing could be transposons (mobile genetic elements).

Option 2 is incorrect as autoradiography usually follows hybridization

Option 3 is incorrect as transposons are not required in gene sequencing.

Option 4 is also incorrect as PCR (polymerase Chain Reaction) is a process to make copies of DNA and does not need transposons.

Hence, the correct option is (1).

Q. 84 Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Polymerase chain reaction is used in DNA amplification
Reason (R): The ampicillin resistant gene is used as a selectable marker to check transformation In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 2:
(A) is correct but (R) is not correct

Option 3:
(A) is not correct but (R) is correct

Option 4:
Both (A) and (R) are correct and (R) is the correct explanation of (A)

Correct Answer:
Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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Solution:
$\rightarrow$PCR stands for Polymerase Chain Reaction. In this reaction, multiple copies of the gene (or DNA) of interest is synthesized in irtro using two sets of primers and the enzyme DNA polymerase.

In this reaction, each cycle has 3 steps:-

i) Denaturation

ii) Primer annealing

iii) Extension of primers.

$\rightarrow$ Selectable markers help in identifying and eliminating non-transformants and selectively permitting the Growth of the transformants.

Transformation is a procedure through which a piece of DNA is in introduced in a host bacterium.

The genes encoding resistance to antibiotics are Ampallang, chloramphenicol, tetracycline, Kanamycin, etc.

Hence, the correct option is an option (a).

Q. 85 Statements related to human Insulin are given below :

Which statement(s) is /are correct about genetically engineered Insulin?

(a) Pro-hormone insulin contains extra stretch of C-peptide

(b) A-peptide and B-peptide chains of insulin were produced separately in E.coli, extracted and combines by creating disulphide bond between them

(c) Insulin used for treating Diabetes was extracted from Cattles and Pigs

(d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone

(e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below:

Option 1:
(b) only

Option 2:
(c) and (d) only

Option 3:
(c), (d) and (e) only

Option 4:
(a) ,(b) and (d) only

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Correct Answer:
(b) only

Solution:
Only b is correct - A-peptide and B-peptide chains of insulin were produced separately in E.coli, extracted and combines by creating disulphide bond between them

Hence, Option (1) is correct.

Q. 86 Given below are two statements:

Statement I :

Restriction endonucleases recognize specific sequence to cut DNA known as palindromic nucleotide sequence.

Statement II :

Restriction endonucleases cut the DNA strand a little away from the center of the palindromic site.

In light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statament II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statament II are correct

Correct Answer:
Both Statement I and Statament II are correct

Solution:
Restriction Endonucleases are 'Molecular Sussors'. These recognize special sequences known as Palindromic sequences. (Eg. MAM is a Palindrome word). Thus statement 1 is correct. These enzymes cut the strand a little away from center. **Hence statement 2 is correct.**

Q. 87 In gene therapy of Adeosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because:

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Option 1:
Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages.

Option 2:
Lymphocytes from patient's blood are grown in culture outside the body

Option 3:
Genetically engineered lymphocytes are not immortal cells.

Option 4:
Retroviral vector is introduced into these lymphocytes.

Correct Answer:
Lymphocytes from patient's blood are grown in culture outside the body

Solution:
In ADA deficiency, lymphocytes are given but these cells are immortal and hence periodic infusion is needed. Option 2 is correct.

Q. 88 The entire fleet of buses in Delhi were converted to CNG from diesel. In reference to this, which one of the following statements is false?

Option 1:
The same diesel engine is used in CNG buses making the cost of conversion low

Option 2:
It is cheaper than diesel

Option 3:
It can not be adulterated like diesel

Option 4:
CNG burns more efficiently than diesel

Correct Answer:
The same diesel engine is used in CNG buses making the cost of conversion low

Solution:
CNG is cheaper than petrol and it burns more efficiently unlike petrol or diesel.it also cannot be adulterated like diesel and petrol. The same diesel engine cannot be used in CNG buses for making the cost conversion low.

Hence, the correct option is (1).

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Q. 89 If $'8'$ Drosophila in a laboratory population of $'80'$ died during a week, the death rate in the population is _____individuals per Drosophila per week

Option 1:
10

Option 2:
1.0

Option 3:
Zero

Option 4:
0.1

Correct Answer:
0.1

Solution:

$$\begin{aligned}\text{Death rate} &= \frac{\text{No of Death}}{\text{Total Population}} \\ &= \frac{8}{80} \\ &= \frac{1}{10} \\ &= 0.1\end{aligned}$$

Hence correct option is 4

Q. 90 Which one of the followings statements cannot be connected to predation ?

Option 1:
it might lead to extinetion of a species

Option 2:
Both the interacting species are negatively. impacted

Option 3:
It is necessitated by nature to maintaine the ecological balance

Option 4:
It helps in maintaining species diversity in a community

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Correct Answer:
Both the interacting species are negatively. impacted

Solution:
Predation is a biological interaction where one organism, the predator, kills and eats other organisms, its prey. It is denoted as: -

	Species A (Predator)	Species B (Prey)
Predation	+	-

When both species are negatively impacted then it is known as competition.

	Species A	Species B
Competition	-	-

Hence the correct option is (b).

Q. 91 The device which can remove particulate matter present in the exhaust from a thermal power plant is :

Option 1:
Incinerator

Option 2:
Electrostatic Precipitator

Option 3:
Catalytic Convertor

Option 4:
STP

Correct Answer:
Electrostatic Precipitator

Solution:
Incinerator: - An Incinerator is a furnace for burning waste.

Electrostatic precipitator: - It is the most effective device to remove fine particulate matter present in the exhaust from a thermal power plant. It works on the principle of electrical charging of the dust particles and collecting it on a differently changed platform.

Catalytic converter: - Device which converts automobile exhaust into less harmful gases.

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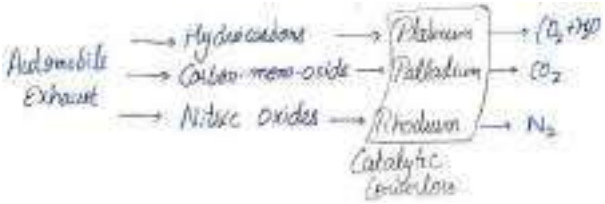

STP (Sewage Treatment Plant): -

It is the process of removing contaminants from wastewater primarily from household sewage. It includes physical, chemical & biological processes to remove these Contaminants & produce environmentally safe treated wastewater.

Q. 92 Given below are two statements:
Statement I : Decomposition is a process in which the detritus is degraded into simpler substances by microbes.
Statement II : Decomposition is faster if the detritus is rich in lignin and chitin In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Statement I is correct but Statement II is incorrect

Solution:
Decomposition is the process of conversion of complex organic matter into simple inorganic matter. Under different environmental conditions, the rate of decomposition may vary. If detritus is rich in lignin & chitin, the rate of decomposition is low and for nitrogen-rich detritus rate of decomposition is fast.

Hence, the correct option is (2)

Q. 93 Habitat loss and fragmentation, over exploitation, alien species invasion and co-extinction are causes for:

Option 1:
Competition

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Option 2:
Biodiversity loss

Option 3:
Natality

Option 4:
Population explosion

Correct Answer:
Biodiversity loss

Solution:
The evil quartet for biodiversity loss is:-

1) Habitat loss and fragmentation

2.) Over exploitation

3) Alien species invasion

4) Co-extinction.

Hence, the correct option is (2)

Q. 94 Which of the following is not a method of ex situ conservation?

Option 1:
National Parks

Option 2:
Micropropagation

Option 3:
Cryopreservation

Option 4:
In vitro fertilization

Correct Answer:
National Parks

Solution:

In-situ conservation	Ex-situ conservation

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It means 'Onsite' conservation	It means 'offsite' conservation.
It is the conservation of species in their natural habitats i.e. their maintenance ad recovery	This is the conservation of endangered species in man-made habitat that imitates their natural habitat
Egà National parks wildlife sanctuaries, biosphere reserves etc.	Eg.zoo, aquarium, seed bank botanical garden micropropagation, cryopreservation etc.

Micropropagation is the rapid vegetative propagation of plants under in vitro conditions.

Cryopreservation is a process that preserves Organelles, cells, tissues, or any other biological Construct by cooling the samples to a very low temperature.

Hence, the correct option is (1)

Q. 95 Which one of the following will accelerate phosphorus cycle?

Option 1:
Volcanic activity

Option 2:
Weathering of rocks

Option 3:
Rain fall and storms

Option 4:
Burning of fossil fuels

Correct Answer:
Weathering of rocks

Solution:
The phosphorus cycle is a sedimentary cycle. The Reservoir pool of phosphorus in the ecosystem is the earth's crust or lithosphere. Weathering of rocks accelerates the phosphorus cycle.

Hence, the correct option is (2).

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Q. 96 While explaining interspecific interaction of population, (+)sign is assigned for beneficial interaction, (−) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (−) for another species involved in the interaction?

Option 1:
Amensalism

Option 2:
Commensalism

Option 3:
Competition

Option 4:
Predation

Correct Answer:
Predation

Solution:
(+) Sign indicates the population is benefitted

(-) Sign indicates the population is harmed

(0) sign indicates no effect on the population

→In predation, one species is benefitted and the other is harmed. It is (+-) type of population interaction.

→In anmensalism, one species is neutral where the other species is harmed. It is (0 -) type of population interaction.

→ In Commensalism, One species is benefited whereas the other species is neutral. It is (+0) type of population interaction.

→ In competition, both the species are harmed. It is (- -) type of population interaction.

Hence the correct option is (4).

Q. 97 Detritivores breakdown detritus into smaller particles. The process is called :

Option 1:
Fragmentation

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Option 2:
Humification

Option 3:
Decomposition

Option 4:
Catabolism

Correct Answer:
Fragmentation

Solution:
Humification ⟶ Accumulation of Humus

Decomposition ⟶ Breakdown of bigger substances into simpler parts especially by the action of living things

Catabolism ⟶ Degradation of detritus into simple inorganic substances by bacterial and fungal enzymes.

Fragmentation ⟶ Breakdown of detritus by detritivores

Hence option 1 is correct.

Q. 98 In-situ conservation refers to :

Option 1:
Conserve only high risk species

Option 2:
Conserve only endangered species

Option 3:
Conserve only extinct species

Option 4:
Protect and conserve the whole ecosystem

Correct Answer:
Protect and conserve the whole ecosystem

Solution:
In Situ conservation is to conserve Organisms at their natural habitat. It helps to protect the organism as well as the ecosystem where they live.

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Q. 99 Given below are two statements:

Statement I :

In a scrubber the exhaust from the thermal plant is passed through the electric wires to charge the dust particles

Statement II :

Particles matter $(\mathrm{PM}\ 2.5)$ can not be removed by scrubber but can be removed by an electrostatic precipitator.

In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are incorrect

Option 2:
Statement I is correct but Statement II is incorrect

Option 3:
Statement I is incorrect but Statement II is correct

Option 4:
Both Statement I and Statement II are correct

Correct Answer:
Both Statement I and Statement II are incorrect

Solution:
Statement 1: Scrubber removes gases like Sulphur dioxide

Statement 2: Precipitators cannot remove particular matter (2.5)

Hence bother is incorrect

Hence, Option (1) is correct.

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Physics

Q. 1 The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are

Option 1:
Instrumental errors

Option 2:
Personal errors

Option 3:
Least count errors

Option 4:
Random errors

Correct Answer:
Random errors

Solution:
The cause is not known so, It is a Random error.

Hence, the answer is the option (4).

Q. 2 A metal wire has mass $(0.4 \pm 0.002)g$, radius $(0.3 \pm 0.001)\text{mm}$ and length $(5 \pm 0.02)\text{cm}$. The maximum possible percentage error in the measurement of density will nearly be:

Option 1:
1.2%

Option 2:
1.3%

Option 3:
1.6%

Option 4:
1.4%

Correct Answer:
1.6%

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Solution:

$$\rho = \frac{\text{Volume}}{\text{Vos}} = \frac{\text{m}}{\pi r^2 \ell}$$

$$\frac{\Delta \text{m}}{\text{m}} \times 100 = \frac{0.002}{0.4} \times 100 = 0.5\%$$

$$\frac{\Delta \text{r}}{\text{r}} \times 100 = \frac{0.001}{0.3} \times 100 = \frac{1}{3}\%$$

$$\frac{\Delta \ell}{\ell} \times 100 = \frac{0.02}{5} \times 100 = 0.4\%$$

$$\frac{\Delta \rho}{\rho} \times 100 = \left(\frac{\Delta \text{m}}{\text{m}} \times 100\right) + 2\frac{\Delta \text{r}}{\text{r}} \times 100 + \frac{\Delta \ell}{\ell} \times 100$$

$$= 1.56\%$$

$$\simeq 1.6\%$$

Hence, the answer is the option (3).

Q. 3 A bullet is fired from a gun at the speed of $280\ \text{m s}^{-1}$ in the direction 30° above the horizontal. The maximum height attained by the bullet is $\left(g = 9.8\ \text{m s}^{-2}, \sin 30^\circ = 0.5\right)$:

Option 1:
$2800\,\text{m}$

Option 2:
$2000\,\text{m}$

Option 3:
$1000\,\text{m}$

Option 4:
$3000\,\text{m}$

Correct Answer:
$1000\,\text{m}$

Solution:

$$\text{H} = \frac{\text{u}^2 \sin^2\theta}{2\ \text{g}}$$

$$\text{u} = 280\ \text{m/s}, \sin 30^\circ = 0.5$$

$$\text{g} = 9.8\ \text{m/s}^2$$

$$\text{H} = 1000\ \text{m}$$

Hence, the answer is the option (3).

Q. 4 The angular acceleration of a body, moving along the circumference of a circle, is :

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Option 1:
along the radius, away from centre

Option 2:
along the radius towards the centre

Option 3:
along the tangent to its position

Option 4:
along the axis of rotation

Correct Answer:
along the axis of rotation

Solution:
Angular acceleration is an axial vector.

Hence, the answer is the option (4).

Q. 5 A vehicle travels half the distance with speed v and the remaining with speed 2v. Its average speed is :

Option 1:
$\frac{\mathrm{v}}{3}$

Option 2:
$\frac{2\mathrm{v}}{3}$

Option 3:
$\frac{4\mathrm{v}}{3}$

Option 4:
$\frac{3\mathrm{v}}{4}$

Correct Answer:
$\frac{4\mathrm{v}}{3}$

Solution:

$$\mathrm{v_{avg}} = \frac{2v_1v_2}{v_1+v_2} \quad [when\ distance\ is\ same]$$

$$v_{\mathrm{avg}} = \frac{2v(2v)}{3v} = \frac{4v}{3}$$

Hence, the answer is the option (3).

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Q. 6 A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity $4\,\mathrm{ms}^{-1}$. The ball strikes the water surface after 4s. The height of bridge above water surface is $\left(\text{ Take } g=10\ \mathrm{ms}^{-2}\right)$:

Option 1:
$56\,\mathrm{m}$

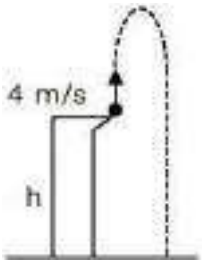
Option 2:
$60\,\mathrm{m}$

Option 3:
$64\,\mathrm{m}$

Option 4:
$68\,\mathrm{m}$

Correct Answer:
$64\,\mathrm{m}$

Solution:



$$-\mathrm{h}=\mathrm{ut}-\frac{1}{2}\mathrm{gt}^2 \quad \Rightarrow \quad -\mathrm{h}=4\times 4-\frac{1}{2}\times 10\times 16 \quad \Rightarrow \quad h=64\ \mathrm{m}$$

Q. 7 A bullet from a gun is fired on a rectangular wooden block with velocity u. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$ .Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is

Option 1:
$27\ \mathrm{cm}$

Option 2:
$24\ \mathrm{cm}$

Option 3:
$28\ \mathrm{cm}$

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Option 4:

$30\ \text{cm}$

Correct Answer:

$27\ \text{cm}$

Solution:

$$v^2 = u^2 - 2as$$

$$\left(\frac{u}{3}\right)^2 = u^2 - 2a(24)....(1)$$

$$Now\ 0 = \left(\frac{u}{3}\right)^2 - 2a(x)....(2)$$

$$x = 3\ \text{cm}$$

$$\text{total length} = 24 + x$$

$$= 27\ \text{cm}$$

Hence, the answer is the option (1).

Q. 8 A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is :

Option 1:

along eastward

Option 2:

along northward

Option 3:

along north-east

Option 4:

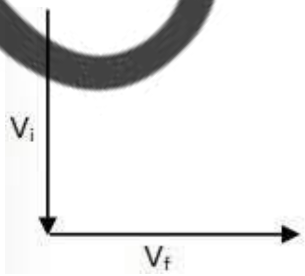
along south-west

Correct Answer:

along north-east

Solution:

$$\text{Force} = \frac{\Delta P}{\Delta t}$$



direction of force = direction of change in velocity.

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$$\Delta\vec{V} = \vec{V}_f - \vec{V}_i$$
$$= v\hat{i} + v\hat{j}$$
$$North - East$$

Q. 9 Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is $0.15 (g = 10\ \text{ms}^{-2})$.

Option 1:
$1.2\ \text{ms}^{-2}$

Option 2:
$150\ \text{ms}^{-2}$

Option 3:
$1.5\ \text{ms}^{-2}$

Option 4:
$50\ \text{ms}^{-2}$

Correct Answer:
$1.5\ \text{ms}^{-2}$

Solution:
$a_{\text{max}} = \mu g$
$= 0.15 \times 10$
$a_{\text{max}} = 1.5\ \text{m/s}^2$

Hence, the answer is the option (3).

Q. 10 The potential energy of a long spring when stretched by 2 cm is U. If the spring stretched by 8 cm, potential energy stored in it will be :

Option 1:
$2U$

Option 2:
$4U$

Option 3:
$8U$

Option 4:
16U

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Correct Answer:

$16\mathrm{U}$

Solution:

$$U = \frac{1}{2}kx^2$$

$$U = \frac{1}{2}k(2)^2$$

$$U' = \frac{1}{2}k(2 \times 4)^2$$

$$U' = \frac{1}{2}k(2)^2 \times 16$$

$$U' = 16U$$

Hence, the answer is the option (4).

Q. 11 The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is :

Option 1:

$3 : 5$

Option 2:

$5 : 3$

Option 3:

$2 : 5$

Option 4:

$5 : 2$

Correct Answer:

$3 : 5$

Solution:

$$\frac{k_1}{k_2} = \frac{\sqrt{2/5}R}{\sqrt{2/3}R}$$

$$\frac{k_1}{k_2} = \sqrt{3} : \sqrt{5}$$

Hence, the answer is the option (1).

Q. 12 A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents:

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Option 1:
T

Option 2:
T^2

Option 3:
T^3

Option 4:
$\sqrt{T}$

Correct Answer:
T^2

Solution:

$$\mathrm{T}^2 = \frac{4\pi^2}{\mathrm{GM}}\left(\mathrm{R}^3\right)$$

$$\mathrm{T}^2 = \frac{4\pi^2 \times \mathrm{R}^3}{\mathrm{Gd}\left(\frac{4}{3}\pi\mathrm{R}^3\right)}$$

$$\mathrm{T}^2 = \frac{3\pi}{\mathrm{Gd}}$$

Hence, the answer is the option (2).

Q. 13 Two bodies of mass m and 9m are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant) :

Option 1:
$-\frac{8\mathrm{Gm}}{\mathrm{R}}$

Option 2:
$-\frac{12\mathrm{Gm}}{\mathrm{R}}$

Option 3:
$-\frac{16\mathrm{Gm}}{\mathrm{R}}$

Option 4:
$-\frac{20\mathrm{Gm}}{\mathrm{R}}$

Correct Answer:
$-\frac{16\mathrm{Gm}}{\mathrm{R}}$

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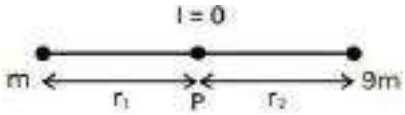
Solution:
Let point P is the point where the gravitational field equals zero



$$r_1 = \left(\frac{\sqrt{m}}{\sqrt{9m}+\sqrt{m}}\right) R = \frac{R}{4}$$

$$r_2 = \frac{3R}{4}$$

$$V = -\frac{Gm}{\left(\frac{R}{4}\right)} - \frac{G(9m)}{\left(\frac{3R}{4}\right)}$$

$$V = -\frac{16\mathrm{Gm}}{R}$$

Q. 14 Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is :

Option 1:
2W/A

Option 2:
W/A

Option 3:
W/2A

Option 4:
Zero

Correct Answer:
W/A

Solution:

$$\text{Longitudinal stress} = \frac{\mathrm{F}_{\perp}}{\mathrm{A}} = \frac{\mathrm{Mg}}{\mathrm{A}} \Rightarrow \frac{\mathrm{W}}{\mathrm{A}}$$

Hence, the answer is the option (2).

Q. 15 The venturi-meter works on :

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Option 1:
Huygen's principle

Option 2:
Bernoulli's principle

Option 3:
The principle of parallel axes

Option 4:
The principle of perpendicular axes

Correct Answer:
Bernoulli's principle

Solution:
Venturi meter-

- It is a device is used for measuring the rate of flow of liquid through pipes.
- This device is based on the application **of Bernoulli's theorem.**

Hence, the answer is the option (2).

Q. 16 The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly : $(\text{surface tension of soap solution } = 0.03 \text{ N m}^{-1})$

Option 1:
30.16×10^{-4} J

Option 2:
5.06×10^{-4} J

Option 3:
3.01×10^{-4} J

Option 4:
50.1×10^{-4} J

Correct Answer:
3.01×10^{-4} J

Solution:

$$Energy Required = 8\pi R^2 \text{ T}$$

$$= 8 \times \frac{22}{7} \times 4 \times 10^{-4} \times 3 \times 10^{-2} = 3.01 \times 10^{-4} \text{ J}$$

Hence, the answer is the option (3).

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Q. 17 A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is :

Option 1:
$27^{\circ}\mathrm{C}$

Option 2:
$15^{\circ}\mathrm{C}$

Option 3:
$100^{\circ}\mathrm{C}$

Option 4:
$200^{\circ}\mathrm{C}$

Correct Answer:
$27^{\circ}\mathrm{C}$

Solution:

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\frac{1}{2} = 1 - \frac{T_2}{600}$$

$$T_2 = 300\ \mathrm{K}$$

$So, \mathrm{T}_2 = 27^{\circ}\mathrm{C}$

Q. 18 The temperature of a gas is -50° C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

Option 1:
$669\,^{\circ}C$

Option 2:
$3295\,^{\circ}C$

Option 3:
$3097K$

Option 4:
$223\,K$

Correct Answer:
$3295\,^{\circ}C$

Solution:

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$v_{rms} \propto \sqrt{T}$

let initial speed is v As speed is increased by 3 times so final speed becomes 4v

$$\frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}}$$
$$V_2 = 4\ V_1$$
$$T_2 = 3568\ K$$

or $3295^{\circ}C$

Hence, the answer is the option (2).

Q. 19 The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is :

Option 1:
$1 : 2$

Option 2:
$2 : 1$

Option 3:
$1 : 3$

Option 4:
$3 : 1$

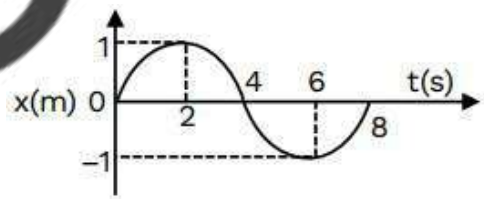
Correct Answer:
$2 : 1$

Solution:

$$\frac{n_{oop}}{n_{cop}} = \frac{\frac{v}{2\ell}}{\frac{v}{4\ell}} = \frac{2}{1}$$

Hence, the answer is the option (2).

Q. 20 The x-t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at t = 2 s is



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Option 1:

$\frac{\pi^2}{8}\ \mathrm{ms}^{-2}$

Option 2:

$-\frac{\pi^2}{8}\ \mathrm{ms}^{-2}$

Option 3:

$\frac{\pi^2}{16}\ \mathrm{ms}^{-2}$

Option 4:

$-\frac{\pi^2}{16}\ \mathrm{ms}^{-2}$

Correct Answer:

$-\frac{\pi^2}{16}\ \mathrm{ms}^{-2}$

Solution:

$a = -\omega^2 x$

$a = -\left(\frac{2\pi}{T}\right)^2 \times 1$

$a = -\frac{\pi^2}{16} m/s^2$

Q. 21 The equivalent capacitance of the system shown in the following circuit is :

Option 1:

$2\,\mu\mathrm{F}$

Option 2:

$3\,\mu\mathrm{F}$

Option 3:

$6\,\mu\mathrm{F}$

Option 4:

$9\,\mu\mathrm{F}$

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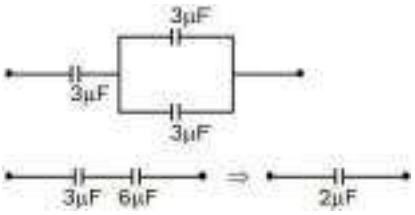
Correct Answer:

$2\,\mu\mathrm{F}$

Solution:



Q. 22 If $\oint_s \vec{E} \cdot \vec{ds} = 0$ over a surface, then:

Option 1:

the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

Option 2:

the magnitude of electric field on the surface is constant.

Option 3:

all the charges must necessarily be inside the surface.

Option 4:

the electric field inside the surface is necessarily uniform

Correct Answer:

the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

Solution:

$$\oint \vec{\mathrm{E}} \cdot \mathrm{d}\vec{\mathrm{S}} = \phi_{\mathrm{in}} - \phi_{\mathrm{out}} = 0 \Rightarrow \phi_{\mathrm{in}} = \phi_{\mathrm{out}}$$

Hence, the answer is the option (1).

Q. 23 An electric dipole is placed at an angle of 30° with an electric field of intensity $2 \times 10^5 \mathrm{NC}^{-1}$. It experiences a torque equal to 4 Nm. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.

Option 1:

$8\,\mathrm{mC}$

Option 2:

$6\,\mathrm{mC}$

Option 3:

$4\,\mathrm{mC}$

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Option 4:

$2\,\mathrm{mC}$

Correct Answer:

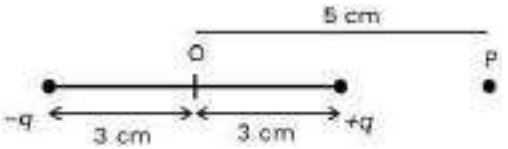
$2\,\mathrm{mC}$

Solution:

$$\because \mathrm{q} = \frac{\tau}{\mathrm{dE}\sin\theta} = \frac{4}{0.02 \times 2 \times 10^5 \times \sin 30^\circ} = 2 \times 10^{-3}\mathrm{C} = 2\mathrm{mc}$$

Hence, the answer is the option (4).

Q. 24 An electric dipole is placed as shown in the figure.



The electric potential $(\text{in } 10^2 \text{ V})$ at point P due to the dipole is

$(\epsilon_0 = \text{ permittivity of free space and } \frac{1}{4\pi\epsilon_0} = K)$

Option 1:

$\left(\frac{3}{8}\right)\mathrm{qK}$

Option 2:

$\left(\frac{5}{8}\right)\mathrm{qK}$

Option 3:

$\left(\frac{8}{5}\right)qK$

Option 4:

$\left(\frac{8}{3}\right)qK$

Correct Answer:

$\left(\frac{3}{8}\right)\mathrm{qK}$

Solution:

$$V = \frac{k(-q)}{8 \times 10^{-2}} + \frac{k(q)}{2 \times 10^{-2}} = \left(\frac{3}{8}\right)qk \times 10^2$$

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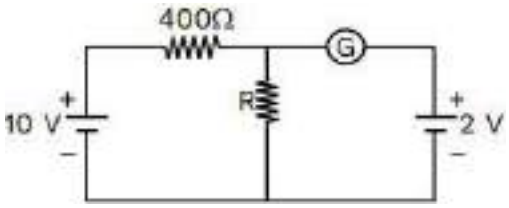
Q. 25 If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by :



Option 1:
$200\,\Omega$

Option 2:
$50\,\Omega$

Option 3:
$100\,\Omega$

Option 4:
$400\,\Omega$

Correct Answer:
$100\,\Omega$

Solution:
potential difference across r is given by

$$iR = V$$

$$\frac{10}{400+R} \times R = 2 \Rightarrow 5R = 400 + R$$

$$\Rightarrow 4R = 400$$

$$\Rightarrow R = 100\Omega$$

Q. 26 Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%)\Omega$. The colour of third band must be:

Option 1:
Red

Option 2:
Green

Option 3:
Orange

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Option 4:
Yellow

Correct Answer:
Orange

Solution:

$$\mathrm{R} = 22000 \pm 5\%$$

$$\mathrm{R} = 22 \times 10^{3} \pm 5\%$$

$$(\mathrm{I}, \mathrm{II} \times 10^{\mathrm{III}} \pm \mathrm{IV})$$

$$\mathrm{III} = \mathrm{Orange}$$

Q. 27 The resistance of platinum wire at $0^{\circ}\mathrm{C}$ is 2Ω and 6.8Ω at $80^{\circ}\mathrm{C}$. The temperature coefficient of resistance of the wire is:

Option 1:
$3 \times 10^{-4}{}^{\circ}\mathrm{C}^{-1}$

Option 2:
$3 \times 10^{-3}{}^{\circ}\mathrm{C}^{-1}$

Option 3:
$3 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$

Option 4:
$3 \times 10^{-1}{}^{\circ}\mathrm{C}^{-1}$

Correct Answer:
$3 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$

Solution:

$$R_T = R_{\circ}(1 + \alpha T)$$

$$6.8 = 2[1 + \alpha(80)]$$

$$\alpha = \frac{3.4 - 1}{80} = \frac{2.4}{80} = 3 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$$

Hence, the answer is the option (3).

Q. 28 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is :

Option 1:
10

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Option 2:

100

Option 3:

1

Option 4:

1000

Correct Answer:

100

Solution:

$$I_1 = \frac{E}{10R}$$

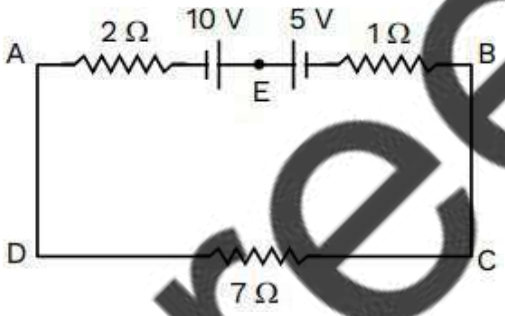
$$I_2 = \frac{E}{R/10} = \frac{10E}{R}$$

$$\Rightarrow I_2 = nI_1$$

$$\Rightarrow \frac{10E}{R} = n\left(\frac{E}{10R}\right) \Rightarrow n = 100$$

Hence, the answer is the option (2).

Q. 29 The magnitude and direction of the current in the following circuit are:



Option 1:

$0.2\ \text{A}\ from\ B\ to\ A\ through\ E$

Option 2:

$0.5\ \text{A}\ from\ A\ to\ B\ through\ E$

Option 3:

$\frac{5}{9}\ A\ from\ A\ to\ B\ through\ E$

Option 4:

$1.5\ \text{A}\ from\ B\ to\ A\ through\ E$

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Correct Answer:

$0.5\ \text{A}\ from\ A\ to\ B\ through\ E$

Solution:

$$I = \frac{\Delta V}{R_{\text{net}}}$$

$$\Rightarrow I = \frac{10-5}{2+7+1} \Rightarrow 0.5\ \text{A}$$

$from\ A\ to\ B\ through\ E$

Q. 30 The net magnetic flux through any closed surface is :

Option 1:
Zero

Option 2:
Positive

Option 3:
Infinity

Option 4:
Negative

Correct Answer:
Zero

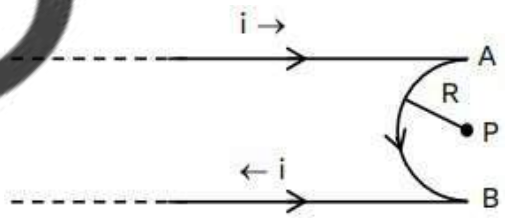
Solution:

$\because \oint \vec{B} \cdot d\vec{s} = 0$

Because magnetic field lines make closed loops. So, for any closed surface. Net magnetic flux is always zero.

Hence, the answer is the option (1).

Q. 31 A very long conducting wire is bent in semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by:



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Option 1:

$\frac{\mu_0 i}{4R}$ *pointed into the page*

Option 2:

$\frac{\mu_0 i}{4R}$ *pointed away from the page*

Option 3:

$\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left[1-\frac{2}{\pi}\right]$ *pointed away from page*

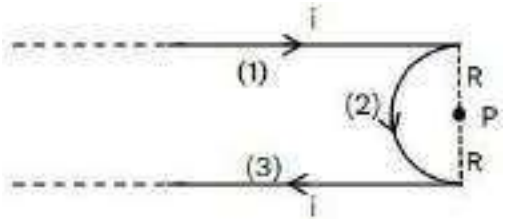
Option 4:

$\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left[1-\frac{2}{\pi}\right]$ *pointed into the page*

Correct Answer:

$\frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left[1-\frac{2}{\pi}\right]$ *pointed away from page*

Solution:



$$\mathrm{B}_1 = \mathrm{B}_2 - \left(\begin{array}{c} \mathrm{B}_1 + \mathrm{B}_3 \\ \otimes \end{array} \right)$$

$$B_p = \frac{\mu_0 i}{4R} - \left(\frac{\mu_0 i}{4\pi R} + \frac{\mu_0 i}{4\pi R}\right)$$

$$\mathrm{B_p} = \frac{\mu_0 \mathrm{i}}{4\mathrm{R}}\left(1-\frac{2}{\pi}\right)\odot$$

Q. 32 A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\vec{\mathrm{B}} = (2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} - 4\hat{\mathrm{k}})\mathrm{T}$. The magnitude of the magnetic force acting on the wire is:

Option 1:

$3\ \mathrm{IL}$

Option 2:

$\sqrt{5}\ \mathrm{IL}$

Option 3:

$5\ \mathrm{IL}$

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Option 4:

$\sqrt{3}\ IL$

Correct Answer:

$5\ \mathrm{IL}$

Solution:

$\vec{F} = I(\vec{L} \times \vec{B})$

$= I[(L\hat{i}) \times (2\hat{i} + 3\hat{j} - 4\hat{k})]$

$\vec{F} = IL(3\hat{k} + 4\hat{j})$

$|\vec{F}| = 5IL$

Hence, the answer is the option (3).

Q. 33 The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is :

Option 1:

$4\,\mu\mathrm{J}$

Option 2:

$4\ \mathrm{mJ}$

Option 3:

$8\ \mathrm{mJ}$

Option 4:

$8\,\mu\mathrm{J}$

Correct Answer:

$8\,\mu\mathrm{J}$

Solution:

$U = \frac{1}{2}Li^2$

$= \frac{1}{2} \times 4 \times 10^{-6} \times 2^2$

$= 8 \times 10^{-6}\ \mathrm{J}$

$= 8\mu\mathrm{J}$

Hence, the answer is the option (4).

Q. 34 A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding ?

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Option 1:
$0.27\,\mathrm{A}$

Option 2:
$2.7\,\mathrm{A}$

Option 3:
$3.7\,\mathrm{A}$

Option 4:
$0.37\,\mathrm{A}$

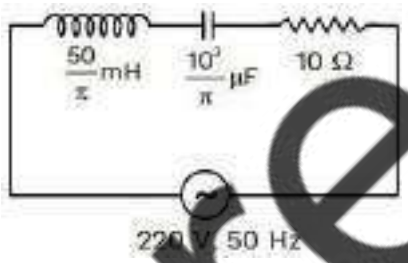
Correct Answer:
$0.27\,\mathrm{A}$

Solution:
For an ideal transformer,

$$P_{\text{in}} = P_{\text{out}}$$
$$\Rightarrow V_{\text{in}}\, I_{\text{in}} = V_{\text{out}}\, I_{\text{out}}$$
$$\Rightarrow 220 \times I_{\text{in}} = 60 \quad \Rightarrow I_{\text{in}} = 0.27\ \text{A}$$

Hence, the answer is the option (1).

Q. 35 The net impedance of circuit (as shown in figure) will be :



Option 1:
$10\sqrt{2}\Omega$

Option 2:
15Ω

Option 3:
$5\sqrt{5}\Omega$

Option 4:
25Ω

Correct Answer:
$5\sqrt{5}\Omega$

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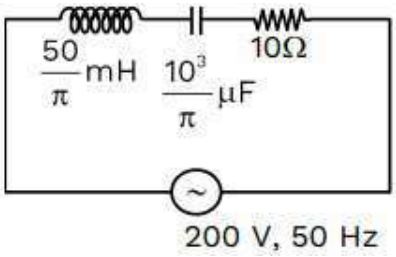
Solution:

$X_L = 2\pi fL$

$= 2\pi \times 50 \times \frac{50}{\pi} \times 10^{-3}$

$X_L = 5\Omega$



$X_C = \frac{1}{2\pi fC} = \frac{10^6}{2\pi \times 50 \times \frac{10^3}{\pi}}$

$X_C = 10\Omega$

$Z = \sqrt{(10)^2 + (10-5)^2}$

$Z = \sqrt{125} = 5\sqrt{5}\Omega$

Q. 36 In a series LCR circuit, the inductance L is 10 mH, capacitance C is 1μF and resistance R is $100\ \Omega$. The frequency at which resonance occurs is?

Option 1:

$15.9\ rad/s$

Option 2:

$15.9\ \text{kHz}$

Option 3:

$1.59\ \text{rad/s}$

Option 4:

$1.59\ \text{kHz}$

Correct Answer:

$1.59\ \text{kHz}$

Solution:

$f_R = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10 \times 10^{-3} \times 10^{-6}}}$

$= \frac{10^4}{2\pi} \approx 1.59 \times 10^3\ \text{Hz} = 1.59\text{kHz}$

Hence, the answer is the option (4).

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Q. 37 In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude $48\ Vm^{-1}$. Then the amplitude of oscillating magnetic field is :
(Speed of light in free space $= 3 \times 10^8\ \text{m s}^{-1}$)

Option 1:
$1.6 \times 10^{-9}\ \text{T}$

Option 2:
$1.6 \times 10^{-8}\ \text{T}$

Option 3:
$1.6 \times 10^{-7}\ \text{T}$

Option 4:
$1.6 \times 10^{-6}\ \text{T}$

Correct Answer:
$1.6 \times 10^{-7}\ \text{T}$

Solution:
$\because C = \dfrac{E_0}{B_0}$

$\Rightarrow \quad B_0 = \dfrac{E_0}{C} = \dfrac{48}{3 \times 10^8} = 16 \times 10^{-8} = 1.6 \times 10^{-7}\ \text{T}$

Hence, the answer is the option (3).

Q. 38 An ac source is connected to a capacitor C. Due to decrease in its operating frequency:

Option 1:
capacitive reactance decreases.

Option 2:
displacement current increases.

Option 3:
displacement current decreases.

Option 4:
capacitive reactance remains constant

Correct Answer:
displacement current decreases.

Solution:

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$$Because\ I_c = I_d = \frac{v}{X_c}\left(\because x_c = \frac{1}{2\pi fc}\right)$$
$$Here\ if\ f \downarrow x_c \uparrow \Rightarrow I_c\ or\ I_d \downarrow (So\ displacement\ Decreases)$$

Hence, the answer is the option (3).

Q. 39 For Young's double slit experiment, two statements are given below:

Statement I : If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II : If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements; choose the correct answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is true but **Statement II** is false.

Option 4:
Statement I is false but **Statement II** is true.

Correct Answer:
Statement I is true but **Statement II** is false.

Solution:

$$\because \theta\ (\text{angular fringe width}) = \frac{\lambda}{d}$$

(i) In statement I : $\text{if } D \uparrow \text{ then } \theta \rightarrow \text{ same}$

(ii) In statement II : $\text{if } \lambda \uparrow \text{ then } \theta \uparrow$

Hence, the answer is the option (3).

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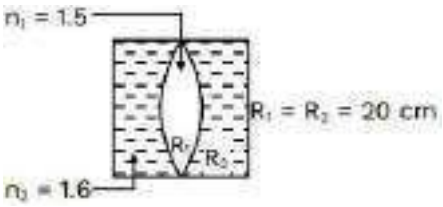
Q. 40 In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that all layers are thin)?



Option 1:
$40\ \text{cm}$

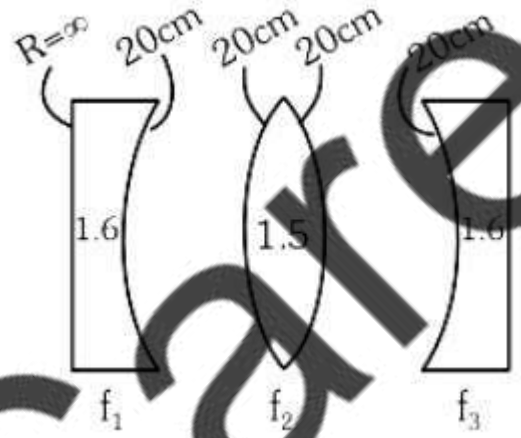
Option 2:
$-40\ \text{cm}$

Option 3:
$-100\ \text{cm}$

Option 4:
$-50\ \text{cm}$

Correct Answer:
$-100\ \text{cm}$

Solution:



Use $\frac{1}{f} = [\mu - 1]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

$$\frac{1}{f_1} = [1.6 - 1]\left[\frac{1}{\infty} - \frac{1}{20}\right] = \frac{-3}{100}$$

$$\frac{1}{f_2} = [1.5 - 1]\left[\frac{1}{20} + \frac{1}{20}\right] = \frac{1}{20}$$

$$\frac{1}{f_3} = \frac{-3}{100}$$

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$$\frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

$$\frac{1}{f_{eq}} = -\frac{3}{100} + \frac{1}{20} - \frac{3}{100} = \frac{-1}{100}$$

$$f = -100 \text{ cm} \Rightarrow P = -1D$$

Q. 41 Two thin lenses are of the same focal lengths (f), but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be:

Option 1:
$Zero$

Option 2:
$f/4$

Option 3:
$f/2$

Option 4:
$Infinite$

Correct Answer:
$Infinite$

Solution:

$$\frac{1}{f_{\text{net}}} = \frac{1}{f} - \frac{1}{f} \quad \Rightarrow \quad \frac{1}{f_{\text{net}}} = 0 \quad \Rightarrow \quad f_{\text{net}} = \infty$$

Hence, the answer is the option (4).

Q. 42 Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?

Option 1:
$\sin^{-1}\left(\frac{t_2}{t_1}\right)$

Option 2:
$\sin^{-1}\left(\frac{10t_2}{t_1}\right)$

Option 3:
$\sin^{-1}\left(\frac{t_1}{10t_2}\right)$

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Option 4:

$\sin^{-1}\left(\frac{10t_1}{t_2}\right)$

Correct Answer:

$\sin^{-1}\left(\frac{10t_1}{t_2}\right)$

Solution:

$$\sin\theta_c = \frac{v_2}{v_1} = \left(\frac{10x/t_2}{x/t_1}\right) = \frac{10t_1}{t_2}$$

$$\theta_c = \sin^{-1}\left(\frac{10t_1}{t_2}\right)$$

Hence, the answer is the option (4).

Q. 43 The work functions of Caesium (Cs), Potassium (K) and Sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV, which of these photosensitive surfaces may emit photoelectrons?

Option 1:

Cs only

Option 2:

Both Na and K

Option 3:

K only

Option 4:

Na only

Correct Answer:

Cs only

Solution:

For photoelectric effect

$$\left(E_{\text{photon incident}} \geq \phi \text{ (work function)} \right)$$

So the answer is Cs only.

Hence, the answer is the option (1).

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Q. 44 The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:

Option 1:
$\sqrt{V}$

Option 2:
$\frac{1}{V}$

Option 3:
$\frac{1}{\sqrt{V}}$

Option 4:
V^2

Correct Answer:
$\frac{1}{V}$

Solution:
If the whole energy of e^- is converted into x-rays then the corresponding wavelength will be minimal.

$$\because \mathrm{eV} = \frac{\mathrm{hc}}{\lambda_{\min}}$$

$$\lambda_{\min} = \frac{\mathrm{hc}}{\mathrm{eV}} \Rightarrow \lambda_{\min} \propto \frac{1}{\mathrm{V}}$$

Hence, the answer is the option (2).

Q. 45 The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $\left(\frac{1}{16}\right)^{\mathrm{th}}$ of its initial value ?

Option 1:
20 minutes

Option 2:
40 minutes

Option 3:
60 minutes

Option 4:
80 minutes

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Correct Answer:
80 minutes

Solution:

$$\left(\frac{1}{2}\right)^n = \frac{1}{16} \Rightarrow n = 4$$

$$\text{So, } \Delta T = 4 \times T_{\text{half}}$$

$$= 4 \times 20 \text{ min}$$

$$= 80 \text{ min}$$

Q. 46 The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom?

Option 1:
$0.53\mathring{A}$

Option 2:
$1.06\mathring{A}$

Option 3:
$1.59\mathring{A}$

Option 4:
$4.77\mathring{A}$

Correct Answer:
$4.77\mathring{A}$

Solution:

$$r_n \propto n^2 \qquad (n = \text{ orbit No. })$$

$$\because r_1 = 5.3 \times 10^{-11} = 0.53\mathring{A}$$

$$then r_3 = (3)^2(0.53) = 4.77\mathring{A}$$

Hence, the answer is the option (4).

Q. 47 In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Bracket series is :

Option 1:
2λ

Option 2:
4λ

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Option 3:
9λ

Option 4:
16λ

Correct Answer:
4λ

Solution:
For Balmer series

$$\frac{1}{\lambda_{\min}} = \mathrm{R}\left(\frac{1}{(2)^2} - \frac{1}{\infty}\right) = \frac{\mathrm{R}}{4}$$

$$\Rightarrow \lambda_{\min} = \lambda = \frac{4}{\mathrm{R}} \quad ...(i)$$

For Bracket series

$$\frac{1}{\lambda'_{\min}} = \mathrm{R}\left(\frac{1}{(4)^2} - \frac{1}{\infty}\right) = \frac{\mathrm{R}}{16}$$

$$\Rightarrow \lambda'_{\min} = \frac{16}{\mathrm{R}} \qquad ...(ii)$$

$$\Rightarrow \lambda'_{\min} = 4\lambda$$

Hence, the answer is the option (2).

Q. 48 Given below are two statements:

Statement I : Photovoltaic devices can convert optical radiation into electricity.

Statement II : Zener diode is designed to operate under reverse bias in breakdown region.
In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both **Statement I** and **Statement II** are correct ·

Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

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Correct Answer:
Both **Statement I** and **Statement II** are correct ·

Solution:
Both Statements I and Statement II are correct.

(I) Photo voltaic devices convert EMW into electricity

(II) Zener diode works as a voltage stabilizer in the reverse biased breakdown region.

Hence, the answer is the option (1).

Q. 49 A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output ?

Option 1:
A centre-tapped transformer

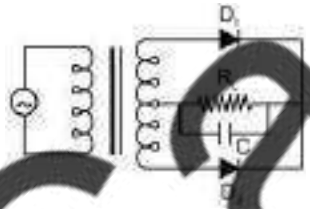
Option 2:
p-n junction diodes

Option 3:
Capacitor

Option 4:
Load resistance

Correct Answer:
Capacitor

Solution:



Capacitor works here as a filter means remove ripples from rectified DC wave .

The role of the capacitor is to filters out the AC ripple and provides pure DC output.

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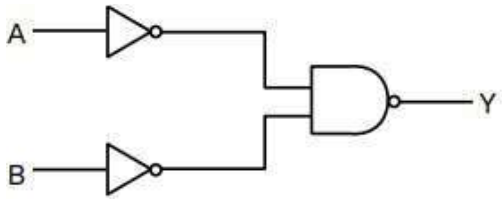
Q. 50 For the following logic circuit, the truth table is:



Option 1:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Option 2:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Option 3:

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

Option 4:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Correct Answer:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Solution:

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relative table is

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Chemistry

Q. 1 The right option for the mass of CO_2 produced by 20 g of 20% pure limestone is (Atomic mass of Ca =40)

$$CaCO_3 \overset{1200K}{\rightarrow} CaO + CO_2$$

Option 1:
1.12g

Option 2:
1.76g

Option 3:
2.64g

Option 4:
1,32g

Correct Answer:
1.76g

Solution:

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$$\% \text{ purity of a reactant } \frac{(\text{ amount of pure reactant })}{\text{total amount of sample}} \times 100$$

$$20 = \frac{\text{amount of pure reactant } \times 100}{20 \text{ g}}$$

$$\therefore \text{ pure substance } = 4 \text{ g} CaCO_3$$

$$CaCO_3 \longrightarrow CaCO + CO_2$$

$$100 \text{ g}$$

$$\therefore 100 \text{ of } CaCO_3 \longrightarrow 44 \text{ g of } CO_2 \text{ is produced}$$

$$\therefore 4 \text{ g of } CaCO_3 \longrightarrow \frac{44}{100} \times 4 \text{ g of } CO_2 \text{ will be produced}$$

$$= 1.76 \text{ g}$$

Hence, the answer is the option (2).

Q. 2 Select the correct statement from the following:

A. Atoms of all elements are composed of two fundamental particles.

B. The mass of the electron is $9.10939 \times 10^{-31} kg$.

C. All the isotopes of given elements show the same chemical properties.

D. Protons and electrons are collectively known as nucleons

E. Dalton's atomic theory regarded the atom as an ultimate particle of matter.

Choose the correct answer from the option given below:

Option 1:
A, B and C only

Option 2:
C, D and E only

Option 3:
A and E only

Option 4:
B, C and E only

Correct Answer:
B, C and E only

Solution:

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Statements (A) and (D) are incorrect.
Because → Atoms of all elements are composed of their fundamental particles (e, p, n)
(D) 'Protons' & 'Neutrons' are collectively known as nucleons.

Hence, the answer is the option (4).

Q. 3 Which of the following reaction will NOT given primary amine as the product?

Option 1:

$$CH_3CONH_2 \xrightarrow{Br_2/KOH} \text{product}$$

Option 2:

$$CH_3CN \xrightarrow[(ii)H_3O+]{(i)LiAlH_3} \text{Product}$$

Option 3:

$$CH_3NC \xrightarrow[(ii)H_3O+]{(i)LiAlH_3} \text{Product}$$

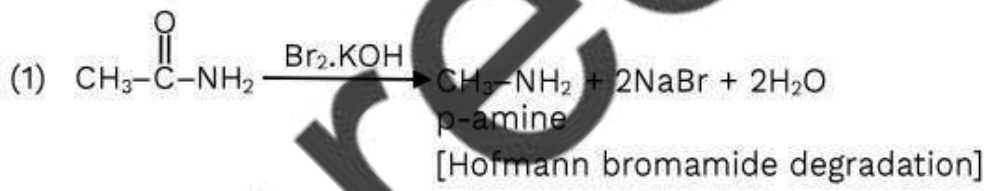
Option 4:

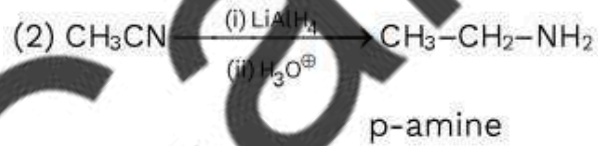
$$CH_3CONH_2 \xrightarrow[(ii)H_3O+]{(i)LiAlH_3} \text{Product}$$

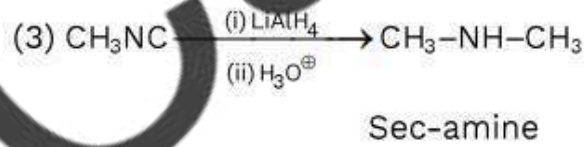
Correct Answer:

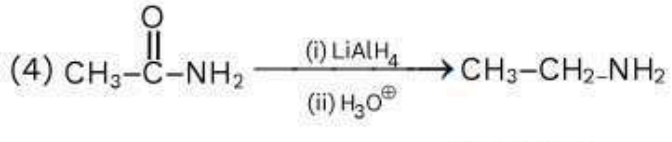
$$CH_3NC \xrightarrow[(ii)H_3O+]{(i)LiAlH_3} \text{Product}$$

Solution:









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Q. 4 The relation between
$n_m, (n_m = \text{the number of permissible values of magnetic quantum number})$
for a given value of azimuthal quantum number (l), is

Option 1:
$l = \frac{n_m - 1}{2}$

Option 2:
$l = 2n_m + 1$

Option 3:
$n_m = 2l^2 + 1$

Option 4:
$n_m = l + 2$

Correct Answer:
$l = \frac{n_m - 1}{2}$

Solution:
m (magnetic quantum No.)
Can have values -l to +l (including zero)
Total values of m in a given

Subshell = (2l + 1)
↓
n_m = 2l + 1

$l = \frac{n_m - 1}{2}$

Q. 5 The element expected to form the largest ion to achieve the nearest noble gas configuration is:

Option 1:
O

Option 2:
F

Option 3:
N

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Option 4:
Na

Correct Answer:
N

Solution:
$\mathrm{N}^{-3} > \mathrm{O}^{2-} > \mathrm{F}^{-} > \mathrm{Na}^{+}$
Hence, the answer is the option (3).

Q. 6 The correct order of energies of molecular orbitals of N_2 molecule is:

Option 1:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Option 2:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_x < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_x$

Option 3:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$

Option 4:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_x) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$

Correct Answer:
$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Solution:
$\mathrm{N}_2 = 14$ electron
$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2P_x^2 = \pi 2p_y^2 < \sigma 2p_z^2 < \pi \star 2P_x^0 = \pi \star 2p_y^0 < \sigma^* 2p_z^0$

Hence, the answer is the option (1).

Q. 7 Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is $NH_3, \mathrm{AlCl}_3, \mathrm{BeCl}_2, \mathrm{CCl}_4, \mathrm{PCl}_5$:

Option 1:
3

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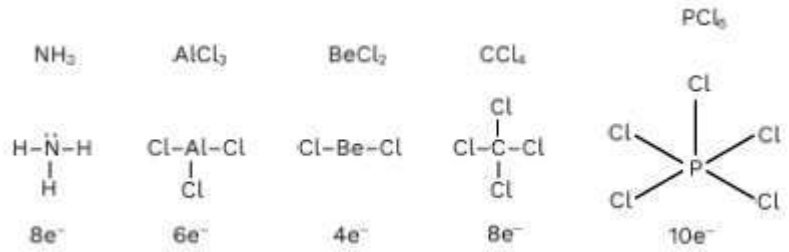
Option 2:
2

Option 3:
4

Option 4:
1

Correct Answer:
3

Solution:



So, $AlCl_3, BeCl_2$ and PCl_5 do not follow octet rule.

Q. 8 Intermolecular forces are forces of attraction and repulsion between interacting particles that will include:
A. Dipole - dipole forces
B. dipole - induced dipole forces.
C. hydrogen bonding
D. covalent bonding
E. dispersion forces.

Choose the most appropriate answer from the options given below:

Option 1:
B, C, D, E are correct

Option 2:
A, B, C, D are correct

Option 3:
A, B, C, E are correct

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Option 4:
A, C, D, E are correct

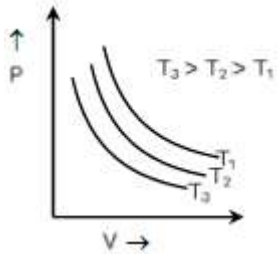
Correct Answer:
A, B, C, E are correct

Solution:
Covalent bonding is not Intermolecular forces, it is a bond.

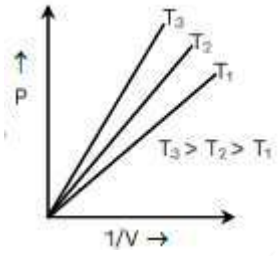
A, B, C, and E are correct

Q. 9 Which amongst the following options is correct graphical representation of Boyle's Law?

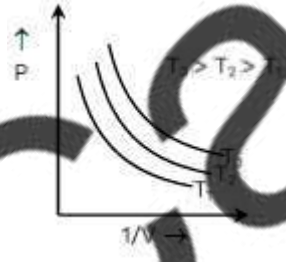
Option 1:



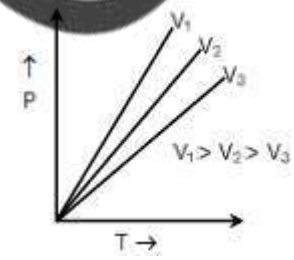
Option 2:



Option 3:



Option 4:



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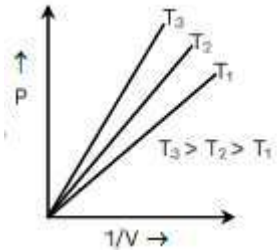
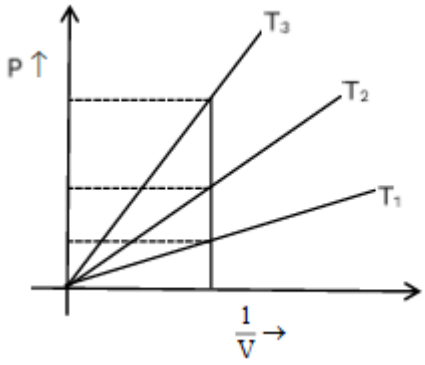
Correct Answer:



Solution:
Acoording to Boyle's Law,

$$P \propto \frac{1}{V}$$

At a constant T, P vs $\frac{1}{V}$ graph should be straight line.



At high P, temperature will be higher at given volume, so $T_3 > T_2 > T_1$

Q. 10 Which of the following options is the correct relation between change in enthalpy and change in internal energy?

Option 1:
$\Delta \mathrm{H} = \Delta U - \Delta n_g RT$

Option 2:
$\Delta \mathrm{H} = \Delta \mathrm{U} + \Delta \mathrm{n_g} RT$

Option 3:
$\Delta \mathrm{H} - \Delta \mathrm{U} = -\Delta n RT$

Option 4:
$\Delta \mathrm{H} + \Delta \mathrm{U} = \Delta \mathrm{nR}$

Correct Answer:
$\Delta \mathrm{H} = \Delta \mathrm{U} + \Delta \mathrm{n_g} RT$

Solution:
Relation between $\Delta \mathrm{H}$ & $\Delta \mathrm{U}$,

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$$\Delta H = \Delta U + \Delta n_g RT$$

Hence, the answer is the option (2).

Q. 11 Amongst the given options which of the following molecules/ion acts as a Lewis acid?

Option 1:
NH_3

Option 2:
H_2O

Option 3:
BF_3

Option 4:
OH^-

Correct Answer:
BF_3

Solution:
Boron trihalides (BX_3) are electron deficient in nature, hence acting as Lewis acid.

Hence, the answer is the option (3).

Q. 12 The equilibrium concentrations of the species in the reaction $A + B \rightleftharpoons C + D$ are $2, 3, 10$ and $6 \text{ mol } L^{-1}$, respectively at $300 \text{ K}. \Delta G^\circ$ for the reaction is $(R = 2 \text{cal/mol K})$

Option 1:
1372.60 cal

Option 2:
-137.26 cal

Option 3:
-1381.80 cal

Option 4:
-13.73 cal

Correct Answer:
-1381.80 cal

Solution:

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$$A + B \rightleftharpoons C + D$$

$$2M \quad 3M \quad 10M \quad 6M$$

$$K_C = \frac{10 \times 6}{2 \times 3} = 10$$

$$\Delta G^\circ = -2.303RT \log K_C$$

$$= -2.303 \times 2 \times 300 \log 10$$

$$= -1381.80 \text{cal}$$

Hence, the answer is the option (3).

Q. 13 On balancing the given redox reaction,

$$aCr_2O_7^{2-} + bSO_3^{2-}(aq) + cH^+(aq) \rightarrow 2aCr^{3+}(aq) + bSO_4^{2-}(aq) + \frac{c}{2}H_2O(l)$$

Option 1:

$1, 3, 8$

Option 2:

$3, 8, 1$

Option 3:

$1, 8, 3$

Option 4:

$8, 1, 3$

Correct Answer:

$1, 3, 8$

Solution:

$Cr_2O_7^{2-} \longrightarrow 2Cr^{3+}$ (Reduction half cell)

$14H^+ + Cr_2O_7^{2-} \longrightarrow 2Cr^{3+} + 7H_2O$ (balance O & H)

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ ... (1) (balance O & H)

$H_2O + SO_3^{2-} \longrightarrow SO_4^{2-} + 2H^+$ (balance O & H)

$H_2O + SO_3^{2-} \longrightarrow SO_4^{2-} + 2H^+ + 2e$ ... (2) (charge balance)

Multiply equation 2 by 3 and then adding to equation 1,

$$Cr_2O_7^{2-} + 14H^+ + 3SO_3^{2-} + 8H^+ \longrightarrow 2Cr^{3+} + 4H_2O + 6SO_4^{2-}$$

$\downarrow$ $\downarrow$ $\downarrow$

a b c

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Q. 14 Which of the following statements are NOT correct?

A. Hydrogen is used to reduce heavy metal oxides to metals.

B. Heavy water is used to study reaction mechanism.

C. Hydrogen is used to make saturated fats from oils.

D. The H-H bond dissociation enthalpy is lowest as compared to a single bond between two atoms of any element.

E. Hydrogen reduces oxides of metals that are more active than iron.

Choose the most appropriate answer from the options given below.

Option 1:
B, C, D, E only

Option 2:
B, D only

Option 3:
D, E only

Option 4:
A, B, C only

Correct Answer:
D, E only

Solution:
The H-H bond dissociation enthalpy is NOT the lowest as compared to a single bond between two atoms of any element.

Hydrogen Cannot reduce oxides of metals that are more active than iron.

D and E are incorrect.

Q. 15 Which one of the following statements is correct?

Option 1:
The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g

Option 2:
All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor.

Option 3:
The bone in human body is an inert and unchanging substance.

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Option 4:
Mg plays roles in neuromuscular function and interneuronal transmission.

Correct Answer:
The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g

Solution:
The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g

Correct option 1

Q. 16 Match List-I with List-II

List - I	List - II
A. Coke	I. Carbon atoms are sp^3 hybridised.
B. Diamond	II. Used as a dry lubricant
C. Fullerene	III. Use as a reducing agent
D. Graphite	IV. Cage like molecules

Choose the correct answer from the options given below:

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
Group - 14 (allotropes of carbon)
A. Coke : III. Use as a reducing agent.
B. Diamond : I. Carbon atoms are sp^3 hybridised.
C. Fullerene : IV. Cage like molecules
D. Graphite : II. Used as a dry lubricant

So, the correct match will be
A-III, B-I, C-IV, D-II

Correct option 3

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Q. 17 Taking stability as the factor, which one of the following represents the correct relationship?

Option 1:
$TlCl_3 > TlCl$

Option 2:
$Inl_3 > Inl$

Option 3:
$AlCl > AlCl_3$

Option 4:
$TlI > TlI_3$

Correct Answer:
$TlI > TlI_3$

Solution:
Due to the inert pair effect, Tl^{+} is more stable than Tl^{3+}.

Hence, the answer is the option (4).

Q. 18 Given below are two statements: one labelled as Assertion A and the other labelled as Reason R:
Assertion A: Helium is used to dilute oxygen in diving apparatus.
Reason R: Helium has high solubility in O_2.

In the light of the above statements choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true and R is NOT the correct explanation of A.

Option 3:
A is true but R is false

Option 4:
A is false but R is true

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Correct Answer:
Both A and R are true and R is NOT the correct explanation of A.

Solution:
Assertion & Reason both are correct because, helium is used to dilute oxygen for scuba diving but reason is not correct explanation, because Helium has high solubility in O_2 as gases are high miscible with each other.

Q. 19 Given below are two statements : one is labelled as Assertion A and the other is labelled a Reason R:

Assertion A : Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which paramagnetic.
Reasons R : The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is true but R is false.

Correct Answer:
A is true but R is false.

Solution:
Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic. The deep blue solution is due to the formation of ammoniated electrons.

A is true, But R is false.

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Q. 20

List -I	(Oxoaciids of Sulphur)	List - II	(Bonds of Sulphur)
A.	Peroxodisulphuric acid	I.	Two S – OH, Four S = O, One S – O – S
B.	Sulphuric acid	II.	$\text{Two S} - \text{OH, One S} = \text{O}$
C.	Pyrosulphuric acid	III.	Two S – OH, Four S = O, One S – O – O – S
D.	Sulphurous acid	IV.	$\text{Two S} - \text{OH, Two S} = \text{O}$

Choose the correct answer from the options given below:

Option 1:
$A - I, B - III, C - II, D - IV$

Option 2:
$A - III, B - IV, C - I, D - II$

Option 3:
$A - I, B - III, C - IV, D - II$

Option 4:
$A - III, B - IV, C - II, D - I$

Correct Answer:
$A - III, B - IV, C - I, D - II$

Solution:
A. Peroxodisulphuric acid :

$\text{III. Two S} - \text{OH, Four S} = \text{O, One S} - \text{O} - \text{O} - \text{S}$
B. Sulphuric acid :

$\text{IV. Two S} - \text{OH, Two S} = \text{O}$
C. Pyrosulphuric acid :

$\text{II. Two S} - \text{OH, One S} = \text{O}$
D. Sulphurous acid : $\text{I. Two S} - \text{OH, Four S} = \text{O, One S} - \text{O} - \text{S}$

$(2)\ \text{A-III, B-IV, C-I, D-II}$ is correct.

Q. 21 In Lassaigne's extract of an organic compound, both nitrogen and sulfur are present, which gives blood red color with Fe^{3+} due to the formation of -

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Option 1:
$Fe_4[Fe(CN)_6]_3.xH_2O$

Option 2:
$NaSCN$

Option 3:
$[Fe(CN)_6NOS]^{4-}$

Option 4:
$[Fe(SCN)]^{2+}$

Correct Answer:
$[Fe(SCN)]^{2+}$

Solution:
In case, nitrogen and sulfur both are present in an organic compound, sodium thiocyanate is formed. It gives blood a red color.

$$\mathrm{Na + C + N + S \rightarrow NaSCN}$$

$$\mathrm{Fe^{3+} + SCN^{-} \rightarrow \underset{\text{Blood Red}}{[Fe(SCN)]^{2+}}}$$

Hence, the answer is the option (4).

Q. 22 The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are:

Option 1:
11,12,0

Option 2:
12,3,0

Option 3:
11,3,1

Option 4:
12,2,1

Correct Answer:
11,3,1

Solution:

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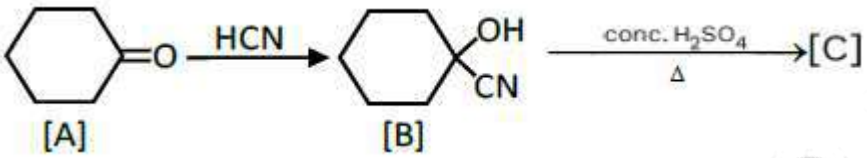
Pyridine

σ bond ⇒ 11

π bond ⇒ 3

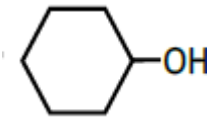
Lone pair ⇒ 1

Q. 23 Complete the following reaction:

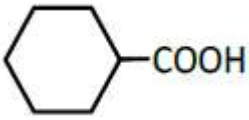


[C] is _____

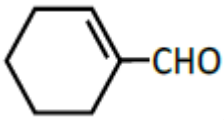
Option 1:



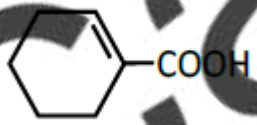
Option 2:



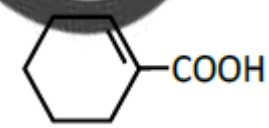
Option 3:



Option 4:



Correct Answer:



Solution:

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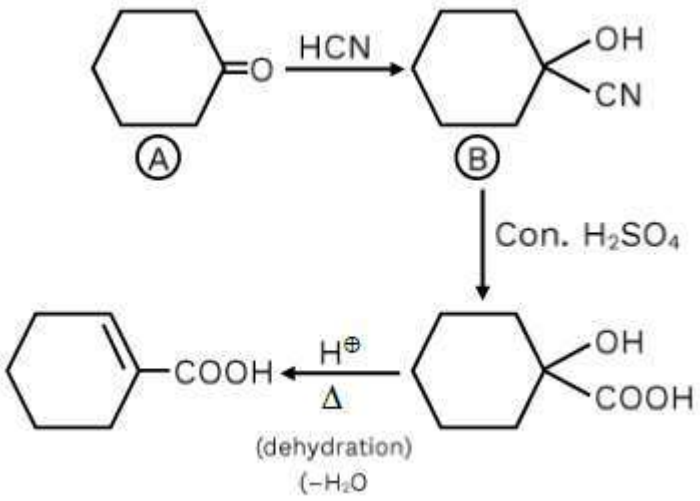


Q. 24 Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in the presence of calcium oxide is:

Option 1:
16

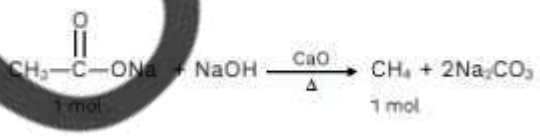
Option 2:
32

Option 3:
30

Option 4:
18

Correct Answer:
32

Solution:



$$\text{Weight of 1 mole } CH_4 \Rightarrow 16gm$$
$$\text{So weight of 2 mole } CH_4 \Rightarrow 32gm$$

Q. 25 Which amongst the following will be most readily dehydrated under acidic conditions ?

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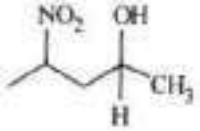
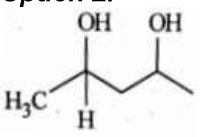
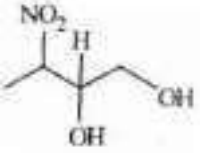
Option 1:



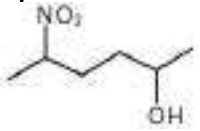
Option 2:



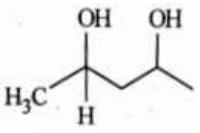
Option 3:



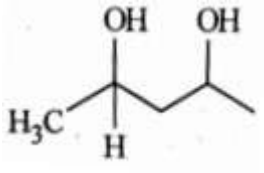
Option 4:



Correct Answer:



Solution:



This compound will be most readily dehydrated under acidic conditions due to the presence of conjugation in the product.

Correct option 2

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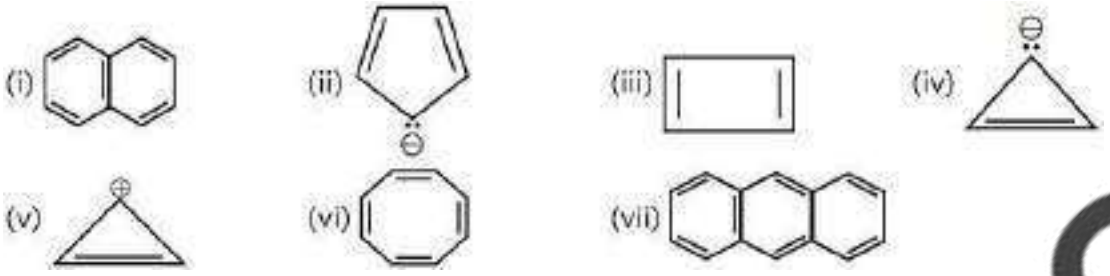
Q. 26 Consider the following compounds/ species:



The number of compounds/species which obey Huckel's rule is _____.

Option 1:

4

Option 2:

6

Option 3:

2

Option 4:

5

Correct Answer:

4

Solution:

According to Huckel's rule compound is aromatic if it is cycle, planar and have $4n + 2\pi$ electrons.
According to it number of compounds species which obey huekel's rule is.

(i)

4n + 2 = 10
4n = 8
n = 2

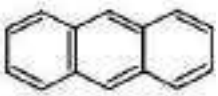
(ii)

4n + 2 = 6
4n = 4
n = 1

(v)

4n + 2 = 2
4n = 0
n = 0

(ii)



4n + 2 = 14
4n = 12
n = 3

n = 0, 1, 2, 3 .....

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Q. 27 Given below are two statements:
Statement I: The nutrient deficient water bodies lead to eutrophication.
Statement II: Eutrophication leads to decrease in the level of oxygen in the water bodies.
Choose the light of the above statements choose the correct answer from the options given below:

Option 1:
Both Statement-I and Statement-II are true

Option 2:
Both Statement-I and Statement-II are false

Option 3:
Statement-I is correct but Statement-II is false

Option 4:
Statement-I is incorrect but Statement-II is true

Correct Answer:
Statement-I is incorrect but Statement-II is true

Solution:
Nutrient enriched water bodies support a dense plant population which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as Eutrophication

Q. 28 A compound is formed by two elements A and B. The elements B forms cubic close packed structure and atoms of A occupy $\frac{1}{3}$ of tetrahedral voids. If the formula of the compound is A_xB_y, then the value of x+y is in option

Option 1:
5

Option 2:
4

Option 3:
3

Option 4:
2

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Correct Answer:
5

Solution:
$\mathrm{B} \rightarrow \mathrm{CCP} \rightarrow 4$
$\mathrm{A} \rightarrow \frac{1}{3}$ of $\mathrm{THV} \rightarrow \frac{1}{3} \times 8$
$\mathrm{A_{B/3}B_4} \rightarrow$ formula of compound
or $\mathrm{A_8B_{12}}$
or $\mathrm{A_2\ B_3}$ (simple ratio)
So, $x = 2$ & $y = 3$
$x + y = 2 + 3 = 5$

Q. 29 What fraction of one edge centred octahedral void lies in one unit cell of fcc ?

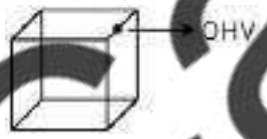
Option 1:
$\frac{1}{2}$

Option 2:
$\frac{1}{3}$

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{1}{12}$

Correct Answer:
$\frac{1}{4}$

Solution:


In Fcc unit cell, OHV at Edge centre is equally shared in 4 unit cell so in one unit cell, only $\frac{1}{4}th$ part is present.

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Q. 30 Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R :
Assertion A: In equation $\Delta G = -nFE_{\text{cell}}$, value of ΔG depends on n.
Reasons R : E_{cell} is an intensive property and ΔG is an extensive property.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true and R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
$\Delta G_r = -F E_{cell}$
$\Delta G_r = \text{depends upon 'n'}$

$So, \Delta_r G$ is an extensive property

Both A and R are true and R is the correct explanation of A.

Hence, the answer is the option (1).

Q. 31 The conductivity of the centimolar solution $\text{KCl at } 250^{\circ}\text{C}$ is $0.0210\text{ohm}^{-1}\text{ cm}^{-1}$ and resistance of the cell containing the solution at 25°C is 60 ohms. The value of cell cons is

Option 1:
1.34 cm^{-1}

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Option 2:

$3.28\ \mathrm{cm}^{-1}$

Option 3:

$1.26\ \mathrm{cm}^{-1}$

Option 4:

$3.34\ \mathrm{cm}^{-1}$

Correct Answer:

$1.26\ \mathrm{cm}^{-1}$

Solution:

$$\mathrm{K} = \frac{1}{R}\mathrm{G}^*$$

$$0.0210 = \frac{1}{60} \times \mathrm{G}^*$$

$$\mathrm{G}^* = 1.26\ \mathrm{cm}^{-1}$$

Hence, the answer is the option (3).

Q. 32 Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R :

Assertion A: A reaction can have zero activation energy.

Reasons R: The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to the threshold value, is called activation energy.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:

Both A and R are true and R is the correct explanation of A.

Option 2:

Both A and R are true and R is NOT the correct explanation of A.

Option 3:

A is true but R is false.

Option 4:

A is false but R is true.

Correct Answer:

Both A and R are true and R is NOT the correct explanation of A.

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Solution:
The free radical reactions can have zero activation energy. In free radical combination reaction in termination, steps have $E_a = 0$

Hence, the answer is the option (2).

Q. 33 For a certain reaction, the rate $= k[A]^2[B]$, when the initial concentration of A is tripled keeping the concentration of B constant, the initial rate would

Option 1:
decrease by a factor of nine.

Option 2:
increase by a factor of six.

Option 3:
increase by a factor of nine

Option 4:
increase by a factor of three

Correct Answer:
increase by a factor of nine

Solution:
$r = K[A]^2[B]$
$r \propto [A]^2, \text{ if } [B] \text{ remains constant}$
$r_f \propto [3A]^2$
$r_f \propto 9 \times [A]^2 \propto 9 \times r_i$

So, the rate will increase 9 times of initial rate.

Hence, the answer is the option (3).

Q. 34 Which one is an example of heterogenous catalysis?

Option 1:
Oxidation of sulphur dioxide into sulphur trioxide in the presence of oxides of nitrogen.

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Option 2:
Hydroiysis of sugar catalysed by H^+ions.

Option 3:
Decomposition of ozone in presence of nitrogen monoxide.

Option 4:
Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.

Correct Answer:
Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.

Solution:
$$N_{2(\ g)} + 3H_{2(\ g)} \longrightarrow 2NH_{3(\ g)}$$

Since reactants and catalysts are in different phases (gas and solid respectively) so it is called as heterogeneous catalysis.

Q. 35 Pumice stone is an example of

Option 1:
sol

Option 2:
gel

Option 3:
solid sol

Option 4:
foam

Correct Answer:
solid sol

Solution:
Pumice stone is an example of solid sol with 'gas' as 'Dispersed phase' and solid as 'Dispersion medium.

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Q. 36 The reaction that does NOT take place in a blast furnace between $900\ \mathrm{K}$ to $1500\ \mathrm{K}$ temperature range during extraction of iron is:

Option 1:
$Fe_2O_3 + CO \rightarrow 2FeO + CO_2$

Option 2:
$FeO + CO \rightarrow Fe + CO_2$

Option 3:
$C + CO_2 \rightarrow 2CO$

Option 4:
$CaO + SiO_2 \rightarrow CaSiO_3$

Correct Answer:
$Fe_2O_3 + CO \rightarrow 2FeO + CO_2$

Solution:

At 500-800K

$CaCO_3 \rightarrow CaO + CO_2$
(Limestone)

$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$
(Iron ore)

$Fe_3O_4 + CO \rightarrow 3FeO + CO_2$

At 900-1500K

$FeO + CO \rightarrow Fe + CO_2$

$C + CO_2 \rightarrow 2CO$

$CaO + SiO_2 \rightarrow CaSiO_3$
(Slag)

Burning of Coke

$C + O_2 \rightarrow CO_2$

$FeO + C \rightarrow Fe + CO$

Correct option 1

Q. 37 The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solution due to -

Option 1:
first ionisation enthalpy

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Option 2:
enthalpy of atomization

Option 3:
hydration energy.

Option 4:
second ionisation enthalpy.

Correct Answer:
hydration energy.

Solution:
The stability of Cu^{2+} is more than Cu^{+} salts in an aqueous solution due to hydration energy.

Hence, the answer is the option (3).

Q. 38 Which of the following statements is incorrect?

A. All the transition metals except scandium form MO oxides which are ionic.
B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7.
C. Basic character increases from V_2O_3 to $V_2O_4 to V_2O_5$.
D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.
E. CrO is basic but Cr_2O_3 is amphoteric
Choose the correct answer from the options given below :

Option 1:
A and E only

Option 2:
B and D only

Option 3:
C and D only

Option 4:
B and C only

Correct Answer:
C and D only

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Solution:
Acidic character increases with an increase in oxidation state because the tendency to gain electrons increases with the increase in oxidation state.

$\text{Basic character decreases from } V_2O_3 \text{ to } V_2O_4 \text{ to } V_2O_5.$

$V_2O_4 \text{ dissolves in acids to give } VO^{2+} \text{ not } VO_4^{3-} \text{ salts.}$

C and D are incorrect.

Hence, the answer is the option (3).

Q. 39 Homoleptic complex from the following complexes is :

Option 1:
Potassium trioxalatoaluminate (III)

Option 2:
Diamminechloridonitrito-N-platinum (II)

Option 3:
Pentaamminecarbnonatocobalt (III) chloride

Option 4:
Triamminetriaquachromium (III) chloride

Correct Answer:
Potassium trioxalatoaluminate (III)

Solution:
Coordination compounds

$(1) K_3[Al(C_2O_4)_3]$

$(2) [Pt(NH_3)_2(Cl)(NO_2)]$

$(3) [Co(NH_3)_5(CO_3)]Cl$

$(4) [Cr(NH_3)_3(H_2O)_3]Cl_3$

In the complex of 2,3,4 two or more types of ligands are present hence they are heteroleptic complexes.

Hence, the answer is the option (1).

Q. 40 Which complex compound is most stable?

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Option 1:
$[Co\,(NH_3))_4\,(H_2O)\,Br]\,(NO_3)_2$

Option 2:
$[Co\,(NH_3)_3\,(NO_3)_3]$

Option 3:
$[CoCl_2(en)_2]\,NO_3$

Option 4:
$[Co\,(NH_3)_6]_2\,(SO_4)_3$

Correct Answer:
$[CoCl_2(en)_2]\,NO_3$

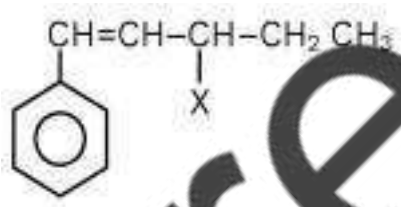
Solution:
We know,

$$\text{Stability} \propto \text{Chelation}$$

So, $[CoCl_2(en)_2]\,NO_3$ has an "en" ligand which provides chelation to the complex which makes it stable.

Hence, the answer is the option (3).

Q. 41 The given compound



Is an example of _____

Option 1:
Benzylic halide

Option 2:
Aryl Halide

Option 3:
Allylic halide

Option 4:
Vinylic halide

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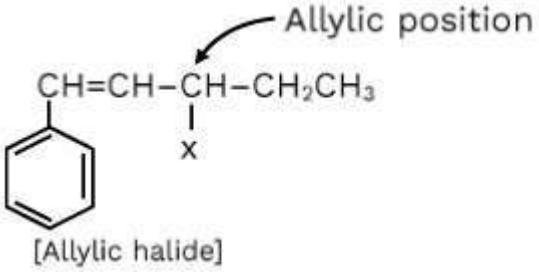
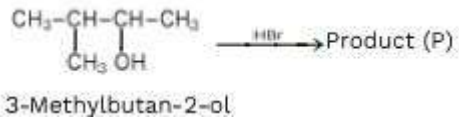
Correct Answer:
Allylic halide

Solution:

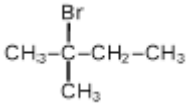


It is an example of Allylic halide.

Q. 42 Consider the following reaction and identify the product (P).



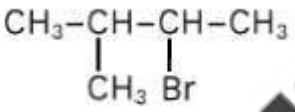
Option 1:



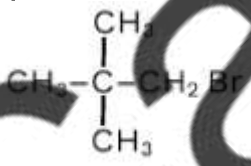
Option 2:

$CH_3CH = CH - CH_3$

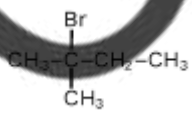
Option 3:



Option 4:



Correct Answer:



Solution:

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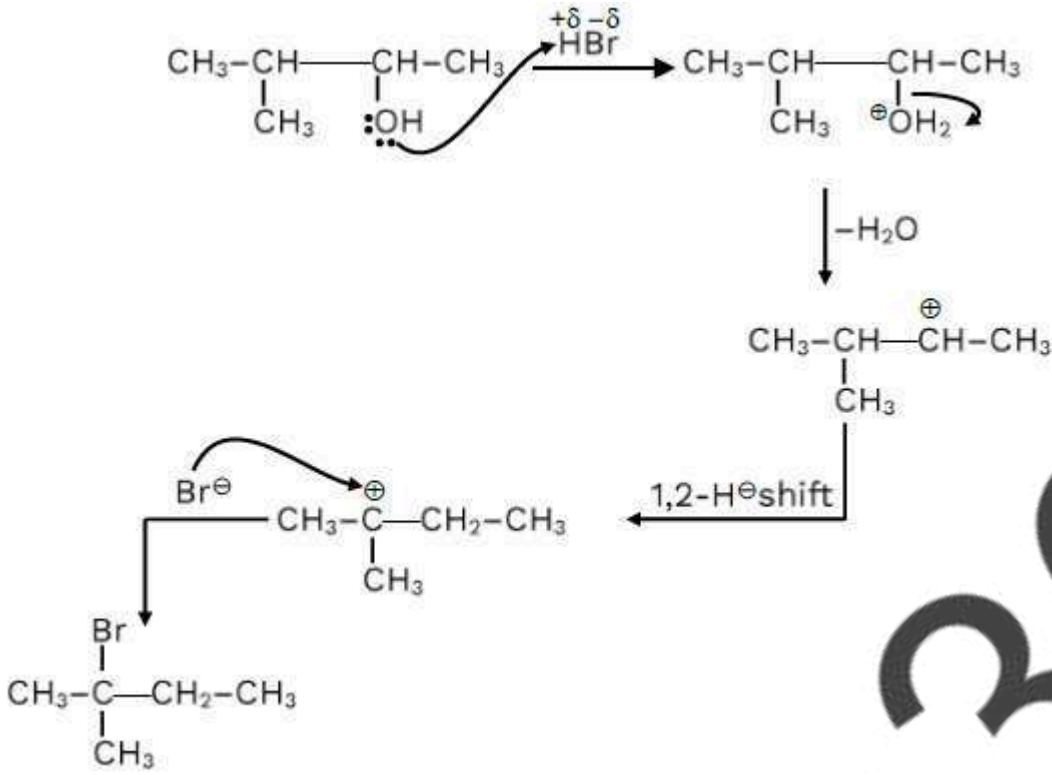
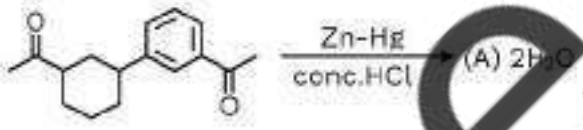
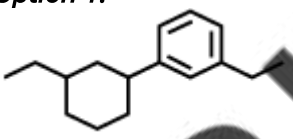


Correct option 1.

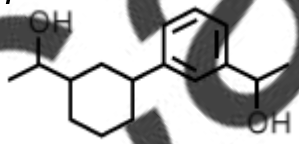
Q. 43 Identify product (A) in the following reaction.



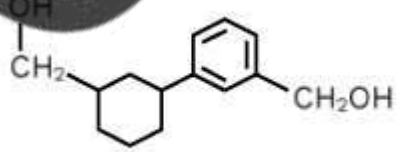
Option 1:



Option 2:



Option 3:



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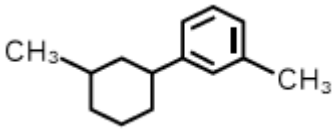
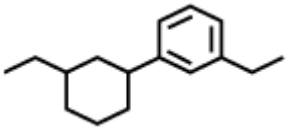
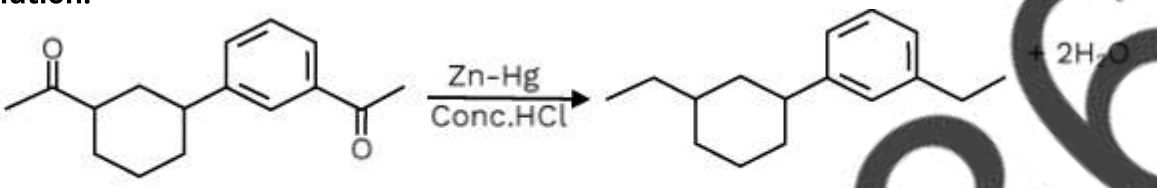
Option 4:



Correct Answer:

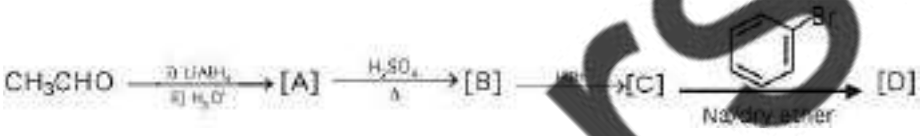


Solution:



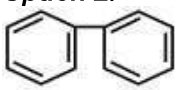
(Clemmensen reduction)

Q. 44 Identify the final product [D] obtained in the following sequence of reactions.



Option 1:

Option 2:



Option 3:

C_4H_{10}

Option 4:

$HC \equiv C \ominus Na^+$

Correct Answer:

Solution:

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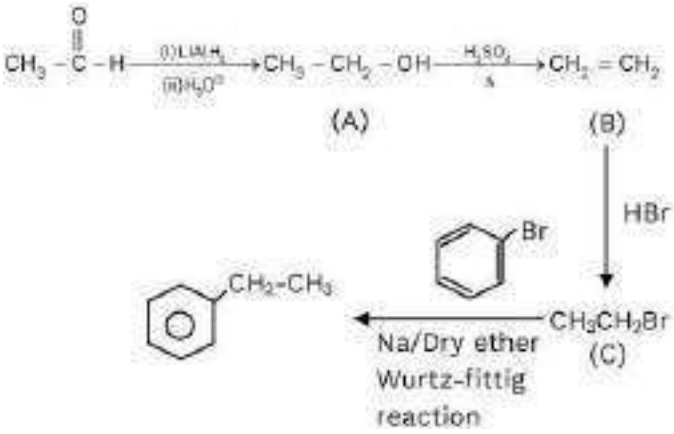
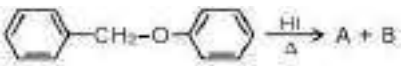


Q. 45 Consider the following reaction:



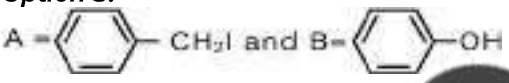
Identify products A and B

Option 1:

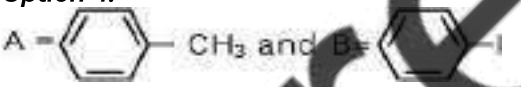
Option 2:

a

Option 3:

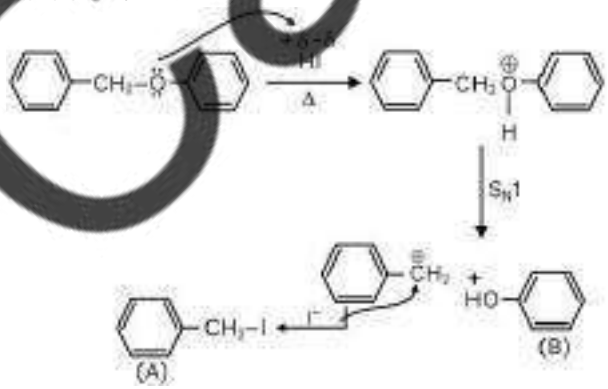


Option 4:



Correct Answer:

Solution:



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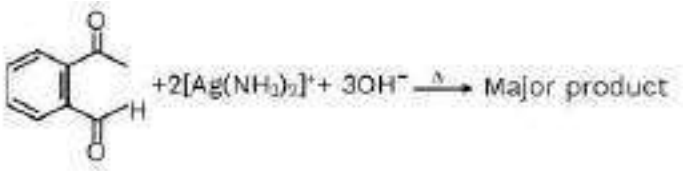
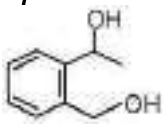
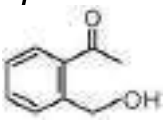
Q. 46 Identify the major product obtained in the following reaction:



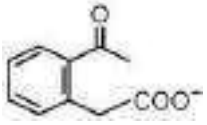
Option 1:



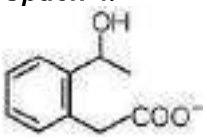
Option 2:



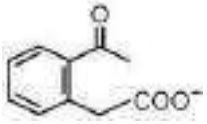
Option 3:



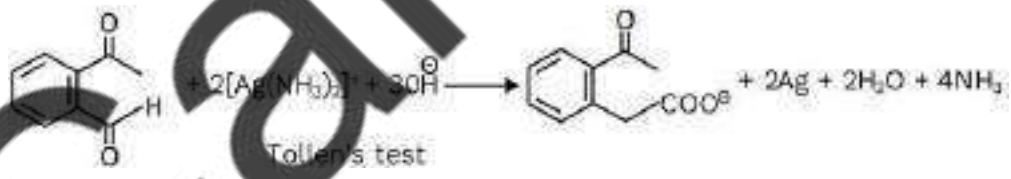
Option 4:



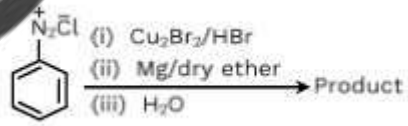
Correct Answer:



Solution:



Q. 47 Identify the product in the following reaction:



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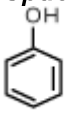
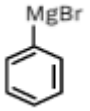
Option 1:



Option 2:

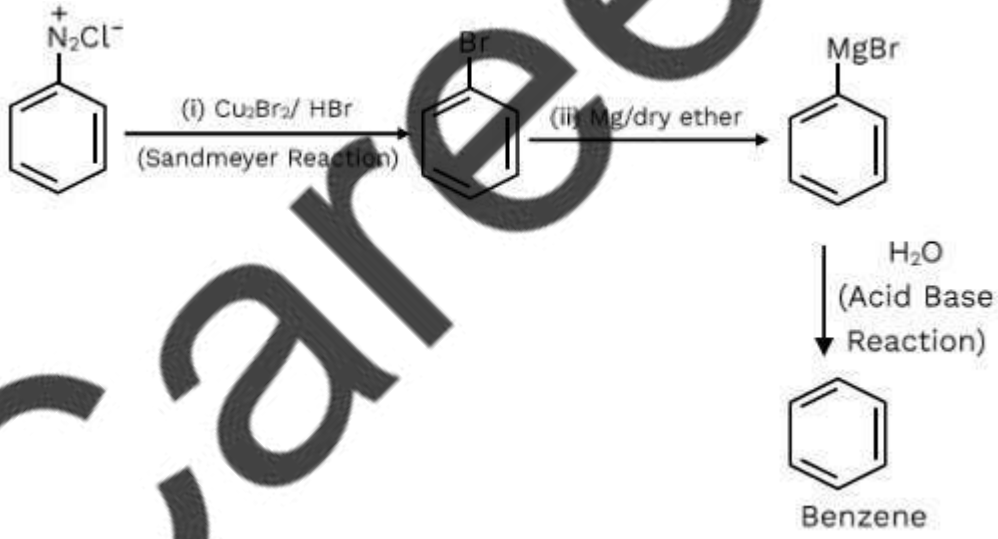
Option 3:



Option 4:

Correct Answer:

Solution:



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Q. 48 Given below are two statements:
Statement I: a unit formed by the attachment of a base of 1' position of sugar is known as nucleoside.

Statement II: When nucleoside is linked to phosphorous acid at 5' -position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are true.

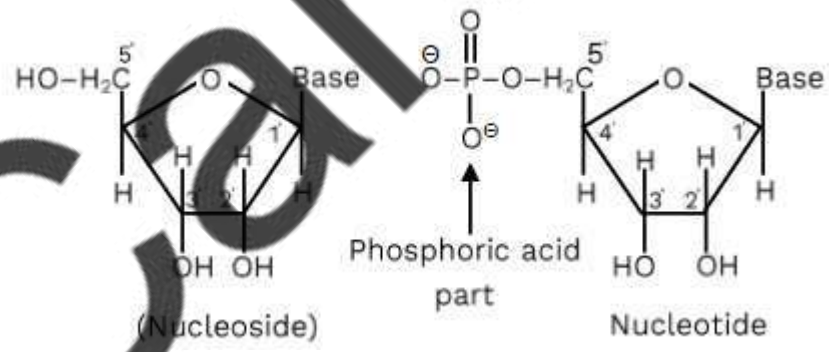
Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:



Q. 49 Which amongst the following molecules polymerization produces neoprene?

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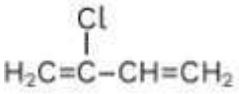
Option 1:

$H_2C = CH - CH = CH_2$

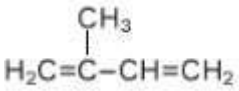
Option 2:



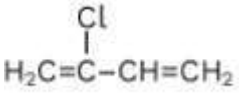
Option 3:

$H_2C = CH - C \equiv CH$

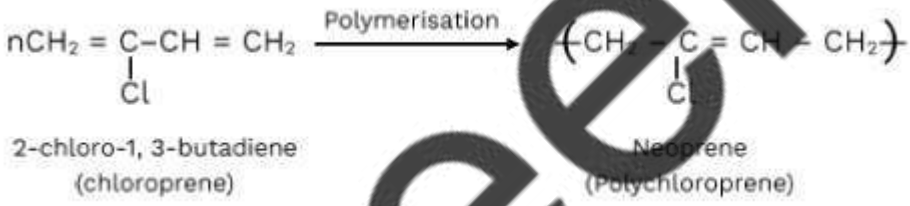
Option 4:



Correct Answer:



Solution:



Q. 50 Some tranquilizers are listed below which one from the following belongs barbiturates?

Option 1:

Chlordiazepoxide

Option 2:

Meprobamate

Option 3:

Valium

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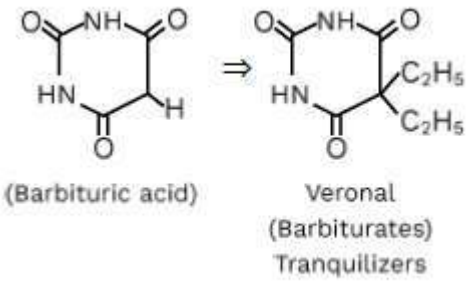
Option 4:
Veronal

Correct Answer:
Veronal

Solution:



Biology

Q. 1 In recent years, DNA sequences (nucleotide sequence) of mtDNA (Mitochondrial DNA) and Y-chromosomes were considered for the study of human evolution, because

Option 1:
Their structure is known in greater detail

Option 2:
They can be studied from the samples of fossil remains

Option 3:
They are small and therefore, easy to study (elf they are uniparental in origin and do not take part in recombination

Option 4:
They are uniparental in origin and do not take part in recombination

Correct Answer:
They are uniparental in origin and do not take part in recombination

Solution:
Wilson and Sarich choose mitochondrial DNA (mtDNA) for the study of maternal line inheritance. While Y-chromosomes were considered for the study of human evolution, particularly male domain. It is possible because they are uniparental in origin and do not take part in recombination.Hence, the correct answer is option 4.

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Q. 2 Given below are two statement: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The first stage of gametophyte in the life cycle of moss in protonema stage.

Reason R: Protonema develops directly from spores produced in capsule. In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both A and R are correct and R is the correct explanation of A.

Option 2:
Both A and R are correct but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are correct and R is the correct explanation of A.

Solution:
The correct option is Option 1) Both A and R are correct, and R is the correct explanation of A.

Assertion A states that the first stage of the gametophyte in the life cycle of a moss is the protonema stage. This is correct. In mosses, the gametophyte generation begins with the development of a filamentous structure called a protonema. The protonema is the first stage in the life cycle and is formed from the germination of moss spores.

Reason R provides the correct explanation for Assertion A. It states that the protonema develops directly from spores produced in the capsule. This is also correct. The spores of mosses are produced within the capsule, which is part of the sporophyte generation. When the spores are released and land in a suitable environment, they germinate and give rise to the protonema.

Therefore, both Assertion A and Reason R are correct, and Reason R provides the correct explanation for Assertion A.

Q. 3 Identify the pair of heterosporous pteridophytes among the following :

Option 1:
Lycopodium and Selaginella

Option 2:
Selaginella and Salvinia

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Option 3:
Psilotum and Salvinia

Option 4:
Equisetum and Salvinia

Correct Answer:
Selaginella and Salvinia

Solution:
The correct option is Option 2) Selaginella and Salvinia.

Heterospory refers to the condition in which a plant produces two different types of spores: microspores and megaspores. These spores give rise to distinct male and female gametophytes.

Among the given options, Selaginella and Salvinia are both heterosporous pteridophytes.

Selaginella is a genus of clubmosses that exhibits heterospory. It produces two types of spores: microspores, which develop into male gametophytes, and megaspores, which develop into female gametophytes. The male gametophytes produce sperm, while the female gametophytes produce eggs.

Salvinia is a genus of aquatic ferns that is also heterosporous. It produces microspores and megaspores, which give rise to male and female gametophytes, respectively.

Therefore, Option 2) Selaginella and Salvinia is the correct pair of heterosporous pteridophytes among the given options.

Q. 4 Given below are two statement:

One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.

Reason R: Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the Correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

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Option 4:
A is false but R is true.

Correct Answer:
A is true but R is false.

Solution:
The correct answer is Option 3) A is true but R is false.

Assertion A is true. Gymnosperms are plants that produce naked seeds, and their pollen grains are typically released from the microsporangium (pollen sacs) and are dispersed by air currents. This is known as anemophily, which is a common mode of pollination in gymnosperms.

Reason R is false. Air currents do carry the pollen grains, but they are not specifically directed to the mouth of the archegonia. In gymnosperms, the male gametes are discharged from the pollen grains and are transported to the female reproductive structures called ovules. The male gametes typically travel through a pollen tube to reach the archegonia, where fertilization takes place.

Therefore, while Assertion A is true, Reason R is incorrect as it does not provide a correct explanation for Assertion A.

Q. 5 Select the correct statements with reference to chordates.

A. Presence of a mid-dorsal, solid and double nerve cord.
B. Presence of closed circulatory system.
C. Presence of paired pharyngeal gillslits.
D. Presence of dorsal heart.
E. Triploblastic pseudocoelomate animals.

Choose the **correct** answer from the option given below:

Option 1:
A, C and D only

Option 2:
B and C only

Option 3:
B, D and E only

Option 4:
C, D and E only

Correct Answer:
B and C only

Solution:
The correct answer is Option 2) B and C only.

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Statement B is correct. Chordates have a closed circulatory system, meaning that the blood is contained within vessels and circulated throughout the body.

Statement C is correct. Chordates have paired pharyngeal gill slits, which are openings in the pharynx region used for respiration or filter feeding in some species.

For incorrect options,

Statement A is incorrect. Chordates have a notochord, which is a flexible rod-like structure running along the mid-dorsal region. However, they have a hollow nerve cord, not a solid and double nerve cord.

Statement D is incorrect. Chordates have a ventral heart, not a dorsal heart. The heart is located on the ventral side of the body.

Statement E is not relevant to chordates. Chordates are triploblastic, meaning they have three germ layers (ectoderm, mesoderm, and endoderm), but they are not pseudocoelomate. Pseudocoelomate animals have a body cavity called a pseudocoelom, which is not present in chordates.

Therefore, the correct answer is Option 2) B and C only.

Q. 6 The unique mammalian characteristics are:

Option 1:
Hairs, tympanic membrane and mammary glands

Option 2:
Hairs, pinna and mammary glands

Option 3:
Hairs, pinna and indirect development

Option 4:
Pinna, monocondylic skull and mammary glands

Correct Answer:
Hairs, pinna and mammary glands

Solution:
The correct answer is Option 2) Hairs, pinna, and mammary glands.

Hairs: Mammals are characterized by the presence of hairs or fur on their bodies. Hairs serve various functions such as insulation, protection, and sensory perception.

Pinna: The pinna refers to the external, visible part of the mammalian ear. It is a characteristic feature of mammals and helps in collecting sound waves and directing them into the ear canal.

Mammary glands: Mammary glands are unique to mammals and are responsible for producing milk to nourish the young. They are specialized glands that undergo development and enlargement during pregnancy and lactation.

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For incorrect options,

Tympanic membrane: This is incorrect as the tympanic membrane, also known as the eardrum, is not unique to mammals. It is found in various animals, including reptiles, birds, and amphibians.

Indirect development: This is also incorrect as most mammals exhibit direct development, where the young are born in a relatively advanced state and undergo minimal or no metamorphosis.

Monocondylic skull: This is incorrect as the monocondylic skull, which refers to having a single occipital condyle, is not unique to mammals. It is found in various groups of vertebrates, including reptiles and amphibians.

Therefore, the correct answer is Option 2) Hairs, pinna, and mammary glands.

Q. 7 Radial symmetry is NOT found in adults of phylum

Option 1:
Ctenophora

Option 2:
Hemichordata

Option 3:
Coelenterata

Option 4:
Echinodermata

Correct Answer:
Hemichordata

Solution:
The correct answer is:

Option 2) Hemichordata

Radial symmetry is not found in adults of the phylum Hemichordata. Hemichordates, which include organisms like acorn worms, exhibit a bilateral symmetry as adults. Radial symmetry is commonly observed in organisms such as cnidarians (formerly referred to as Coelenterata) like jellyfish and corals, as well as in echinoderms such as starfish and sea urchins. Ctenophora, or comb jellies, also exhibit radial symmetry as adults.

Q. 8 Given below are two statements:

Statement I: RNA mutates at a faster rate.
Statement II: Viruses having RNA genome and shorter life span mutate and evolve faster.
In the light of the above statements, choose the correct answer from the options given below:

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Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I false but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true.

Solution:
Option 1) Both Statement I and Statement II are true.

Both statements are correct. RNA generally mutates at a faster rate compared to DNA due to the inherent properties of RNA molecules. Viruses with RNA genomes tend to have higher mutation rates, and their shorter life spans allow for faster evolution through the accumulation of mutations. These mutations contribute to the diversity and adaptability of RNA viruses.

Q. 9 Axile placentation is observed in

Option 1:
Mustard, Cucumber and Primrose

Option 2:
China rose, Beans and Lupin

Option 3:
Tomato, Dianthus and Pea

Option 4:
China rose, Petunia and Lemon

Correct Answer:
China rose, Petunia and Lemon

Solution:
The correct option is Option 4) China rose, Petunia, and Lemon.

Axile placentation is a type of arrangement of ovules within the ovary of a flower. In axile placentation, the ovules are attached to the central axis or the walls of a compound ovary. The ovary is divided into multiple chambers, and each chamber contains one or more ovules.

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Among the given options, China rose (Hibiscus rosa-sinensis), Petunia (Petunia spp.), and Lemon (Citrus limon) exhibit axile placentation. In these plants, the ovary is composed of several chambers, and the ovules are attached to the central axis or the walls of the ovary chambers.

Mustard, Cucumber, Primrose, China rose, Beans, Lupin, Tomato, Dianthus, and Pea mentioned in the other options may exhibit different types of placentation, but not specifically axile placentation.

Therefore, Option 4) China rose, Petunia, and Lemon is the correct option for plants that exhibit axile placentation.

Q. 10 Family Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens , pick out the characteristics specific to family Fabaceae but not found in Solanaceae or Liliaceae.

Option 1:
Diadelphous and Dithecous anthers

Option 2:
Polyadelphous and epipetalous stamens

Option 3:
Monoadelphous and Monothecous anthers

Option 4:
Epiphyllous and Dithecous anthers

Correct Answer:
Diadelphous and Dithecous anthers

Solution:

The correct option is Option 1) Diadelphous and Dithecous anthers.

Family Fabaceae, also known as the legume family, is characterized by several distinguishing features, including the stamen characteristics. Diadelphous and Dithecous anthers are specific to the family Fabaceae and are not found in Solanaceae or Liliaceae.

Diadelphous anthers refer to the stamens being fused into two groups, with one group containing nine stamens and the other group containing one stamen. This arrangement is commonly observed in the flowers of the Fabaceae family.

Dithecous anthers refer to the anthers having two thecae or pollen sacs. Each theca contains pollen grains. This is another characteristic feature specific to the Fabaceae family.

In contrast, Solanaceae and Liliaceae do not typically exhibit diadelphous or dithecous anthers. Solanaceae family members often have stamens that are either free or fused together in different arrangements, and the anthers may have different structures. Liliaceae family members usually have six stamens with anthers that are usually not fused in distinct groups.

Therefore, Option 1) Diadelphous and Dithecous anthers is the correct option for the stamen characteristics specific to the family Fabaceae but not found in Solanaceae or Liliaceae.

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Q. 11 Given below are two statements :

Statement I : Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

Statement II : Exarch condition is the most common feature of the root system. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false..

Option 3:
Statement I is correct but Statement II is false

Option 4:
Statement I is correct but Statement II is false

Correct Answer:
Statement I is correct but Statement II is false

Solution:
The correct option is Option 4) Statement I is correct, but Statement II is false.

Statement I is correct. Endarch and exarch are terms used to describe the position of the secondary xylem in the plant body. Endarch refers to the condition where the primary xylem is located towards the center of the stem or root, and the secondary xylem is formed towards the outside. Exarch, on the other hand, refers to the condition where the primary xylem is located towards the periphery of the stem or root, and the secondary xylem is formed towards the inside.

Statement II is false. The exarch condition is not the most common feature of the root system. In fact, the exarch condition is more commonly observed in the stem rather than the root. In the root system, the most common condition is the endarch condition, where the primary xylem is located towards the center and the secondary xylem is formed towards the outside.

Therefore, Option 4) Statement I is correct, but Statement II is false is the correct answer.

Q. 12 Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A : Late wood has fewer xylary elements with narrow vessels.

Reason R : Cambium is less active in winters. In the light of the above statements, choose the correct answer from the options given below :

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Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
The correct option is Option 1) Both A and R are true, and R is the correct explanation of A.

Assertion A states that late wood has fewer xylary elements with narrow vessels. This statement is true. Late wood, also known as summer wood, refers to the portion of a tree ring that is formed later in the growing season. It tends to have a higher density and a darker color compared to early wood. Late wood is characterized by having fewer xylary elements, such as tracheids or vessels, and narrower vessels compared to early wood.

Reason R states that cambium is less active in winters. This statement is also true. Cambium is a layer of cells in the stem or root of a plant that is responsible for secondary growth. During the winter season, cambium activity is reduced or even dormant due to environmental factors such as cold temperatures and shorter days. This reduced cambium activity during winter contributes to the formation of late wood with fewer xylary elements and narrow vessels.

Therefore, both Assertion A and Reason R are true, and Reason R provides a correct explanation for Assertion A. Hence, Option 1) is the correct answer.

Q. 13 Given below are two statement: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason R: Internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves. In the light of the above statements,

choose the Correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

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Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
Option 1) Both A and R are true and R is the correct explanation of A.

The assertion A states that a flower is defined as a modified shoot where the shoot apical meristem changes to floral meristem. This is true as flowers are indeed modified shoots.

The reason R explains that the internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves. This is also true as floral appendages such as sepals, petals, stamens, and carpels are produced in place of leaves in the floral structure.

Therefore, both the assertion and the reason are true, and the reason correctly explains the assertion. Hence, the correct answer is Option 1.

Q. 14 Identify the correct statements :

A. Lenticels are the lens-shaped openings permitting the exchange of gases.

B. Bark formed early in the season is called hard bark.

C. Bark is a technical term that refers to all tissues exterior to vascular cambium.

D. Bark refers to periderm and secondary phloem.

E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below:

Option 1:
B, C and E only

Option 2:
A and D only

Option 3:
A, B and D only

Option 4:
B and C only

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Correct Answer:
A and D only

Solution:
The correct statements are:

A. Lenticels are the lens-shaped openings permitting the exchange of gases.
D. Bark refers to periderm and secondary phloem.

Therefore, the correct answer is Option 2) A and D only.

Q. 15 Which of the following is characteristic feature of cockroach regarding sexual dimorphism?

Option 1:
Dark brown body colour and anal cerci

Option 2:
Presence of anal styles

Option 3:
Presence of sclerites

Option 4:
Presence of anal cerci

Correct Answer:
Presence of anal styles

Solution:
The correct answer is Option 2) Presence of anal styles.

Sexual dimorphism refers to the differences in physical characteristics between males and females of a species. In the case of cockroaches, one characteristic feature of sexual dimorphism is the presence of anal styles in males.

Anal styles are specialized structures located at the posterior end of the abdomen in male cockroaches. They are small, finger-like projections that are used during mating. These structures help in the transfer of sperm from the male to the female during copulation.

On the other hand, option 1) Dark brown body color and anal cerci are not specifically related to sexual dimorphism in cockroaches. Both males and females of many cockroach species can have a dark brown body color and anal cerci, which are sensory appendages located at the posterior end of the abdomen.

Option 3) Presence of sclerites refers to the hardened plates or structures that form the exoskeleton of cockroaches. While there may be slight differences in sclerites between male and female cockroaches, it is not a characteristic feature of sexual dimorphism.

Therefore, the correct answer is Option 2) Presence of anal styles.

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Q. 16 Match **List I** with **List II**.

	List I		List II
A.	Mast cells	I.	Ciliated epithelium
B.	Inner surface of bronchiole	II.	Areolar connective tissue
C.	Blood	III.	Cuboidal epithelium
D.	Tubular parts of nephrons	IV.	Specialised connective tissue

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-IV, D-III

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-II, B-I, C-IV, D-III

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
The correct matching is:

A-II (Mast cells - Connective tissue)
B-I (Inner surface of bronchiole - Ciliated epithelium)
C-IV (Blood - Specialized connective tissue)
D-III (Tubular parts of nephrons - Cuboidal epithelium)

Hence, the correct answer is Option 3) A-II, B-I, C-IV, D-III.

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Q. 17 Which of the following statements are correct ?

A. Basophils are most abundant cells of the total WBCs
B. Basophils secrete histamine, serotonin and heparin
C. Basophils are involved in inflammatory response
D. Basophils have kidney shaped nucleus
N. Basophils are agranulocytes

Choose the correct answer from the options given below:

Option 1:
D and E only

Option 2:
C and E only

Option 3:
B and C only

Option 4:
A and B only

Correct Answer:
B and C only

Solution:

Statement B is correct. Basophils secrete histamine, serotonin, and heparin. These substances are involved in inflammatory and immune responses.

Statement C is correct. Basophils are involved in the inflammatory response. They release histamine and other chemicals that cause blood vessels to dilate and increase blood flow to the site of inflammation.

For incorrect options,

Statement A is incorrect. Basophils are not the most abundant cells of the total white blood cells (WBCs). Neutrophils are the most abundant WBCs.

Statement D is incorrect. Basophils have a lobed or segmented nucleus, not a kidney-shaped nucleus.

Therefore, the correct statements are B and C.

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Q. 18 Given below are two statements:
Statement I: Ligament are dense irregular tissue.
Statement II: Cartilage is dense regular tissue.

In the light of the above statements, choose the **correct** answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is true but **Statement II** is false.

Option 4:
Statement I is false but **Statement II** is ture.

Correct Answer:
Both **Statement I** and **Statement II** are false.

Solution:
The correct answer is Option 2) Both Statement I and Statement II are false.

Explanation:
Statement I: Ligaments are not dense irregular tissue. Ligaments are dense regular connective tissue that connects bones to other bones, providing stability and support to joints.

Statement II: Cartilage is not dense regular tissue. Cartilage is a type of connective tissue that is more flexible than bone but firmer than dense regular tissue. It provides support and cushioning between bones.

Therefore, both statements are incorrect.

Q. 19 In cockroach, excretion is brought about by.
A. Phallic gland B. Urecose gland
C. Nephrocytes D. Fat body
E. Collaterial glands

Choose the correct answer from the options given below:

Option 1:
A and E only

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Option 2:
A, B and E only

Option 3:
B, C and D only

Option 4:
B and D only

Correct Answer:
B, C and D only

Solution:
he correct answer is Option 3) B, C, and D only.

B. Urecose gland: The urecose gland is involved in excretion in cockroaches. It is a specialized gland located in the posterior part of the alimentary canal, and it helps eliminate metabolic waste products, including uric acid.

C. Nephrocytes: Nephrocytes, also known as "green glands," are excretory organs found in many invertebrates. However, they are not present in cockroaches. Nephrocytes are commonly found in crustaceans, such as crabs and lobsters, and are responsible for excretion.

D. Fat body: The fat body in cockroaches is involved in various metabolic functions but is not directly responsible for excretion.

For incorrect options,

A. Phallic gland: The phallic gland in male cockroaches is not involved in excretion. It is a part of the reproductive system and is responsible for producing and transferring the spermatophore during mating.

E. Collaterial glands: Collaterial glands are not involved in excretion in cockroaches. They are associated with the reproductive system and are responsible for producing and storing spermatozoa.

Based on the information above, the correct answer is Option 3) B, C, and D only.

Q. 20 Among eukaryotes, replication of DNA takes place in–

Option 1:
$M\ \ phase$

Option 2:
$S\ phase$

Option 3:
$G_1\ phase$

Option 4:
$G_2\ phase$

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Correct Answer:
$S\ phase$

Solution:
The correct option is Option 2) S phase.

During the cell cycle of eukaryotes, DNA replication occurs during the S (synthesis) phase. The cell cycle consists of different phases, including G1 (gap 1), S phase, G2 (gap 2), and M (mitosis) phase.

In the S phase, the cell's DNA is replicated, producing two identical copies of each chromosome. This replication process ensures that each daughter cell receives a complete set of genetic information during cell division. The S phase is characterized by active DNA synthesis, with DNA polymerases synthesizing new DNA strands based on the existing template strands.

Therefore, DNA replication specifically takes place during the S phase of the cell cycle in eukaryotes.

Q. 21 Which of the following stages of meiosis involves division of centromere?

Option 1:
Metaphase I

Option 2:
Metaphase II

Option 3:
Anaphase II

Option 4:
Telophase

Correct Answer:
Anaphase II

Solution:
The correct option is Option 3) Anaphase II.

During meiosis, which is the process of cell division that produces gametes (sex cells), the division of the centromere occurs during Anaphase II.

Anaphase II is the stage in meiosis II where the sister chromatids, which are the replicated copies of the chromosomes, are separated. The centromere, which holds the sister chromatids together, divides and allows the chromatids to move towards opposite poles of the cell.

In contrast, in Metaphase I, the homologous pairs of chromosomes align at the cell's equator, and the centromeres are still intact. Metaphase II is when the individual chromosomes line up at the equator, and the centromeres are ready to divide in Anaphase II.

Therefore, it is during Anaphase II of meiosis that the division of the centromere occurs.

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Q. 22 Cellulose does not form blue colour with Iodine because

Option 1:
It is a disaccharide.

Option 2:
It is a helical molecules

Option 3:
It does not contain complex helices and hence cannot hold iodine molecules.

Option 4:
It breaks down when iodine reacts with

Correct Answer:
It does not contain complex helices and hence cannot hold iodine molecules.

Solution:
The correct option is Option 3) It does not contain complex helices and hence cannot hold iodine molecules.

Cellulose is a polysaccharide composed of glucose units linked together in a linear chain. It is a major component of plant cell walls and provides structural support to plants. When iodine reacts with certain polysaccharides, such as starch, a blue color complex is formed. However, cellulose does not form a blue color with iodine.

This is because cellulose does not contain complex helices, which are necessary for the formation of the blue color complex with iodine. Starch, on the other hand, has a helical structure that allows iodine molecules to fit into the helical spaces, forming a blue color complex.

Cellulose has a more linear and rigid structure, lacking the helical arrangement required for the interaction with iodine molecules. Therefore, when iodine reacts with cellulose, it does not form the blue color complex observed with other polysaccharides like starch.

Hence, Option 3) It does not contain complex helices and hence cannot hold iodine molecules is the correct explanation for why cellulose does not form a blue color with iodine.

Q. 23 The process of appearance of recombination nodules occurs at which sub stage of prophase I in meiosis?

Option 1:
Zygotene

Option 2:
Pachytene

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Option 3:
Diplotene

Option 4:
Diakinesis

Correct Answer:
Pachytene

Solution:
The correct option is Option 2) Pachytene.

Recombination nodules, also known as recombination sites or synaptonemal complexes, are structures that facilitate the exchange of genetic material between homologous chromosomes during meiosis. These structures are important for genetic recombination and the generation of genetic diversity.

The appearance of recombination nodules occurs during the substage of prophase I called Pachytene. Prophase I is the longest and most complex phase of meiosis, consisting of several sub-stages: leptotene, zygotene, pachytene, diplotene, and diakinesis.

During Pachytene, homologous chromosomes pair up and undergo synapsis, where they become closely associated. The synaptonemal complex, which includes the recombination nodules, forms between the paired chromosomes. This complex helps to align the homologous chromosomes and promote the exchange of genetic material through a process called crossing over.

After Pachytene, the chromosomes continue to condense and undergo further changes in the subsequent sub-stages of prophase I, namely diplotene and diakinesis. In diplotene, the synaptonemal complex becomes more visible, and the homologous chromosomes begin to separate but remain connected at specific points called chiasmata. Diakinesis is the final sub-stage of prophase I, where the chromosomes fully condense and prepare for metaphase I.

Therefore, the appearance of recombination nodules occurs at the substage of prophase I called Pachytene, as stated in Option 2.

Q. 24 How many different proteins does the ribosome consist of ?

Option 1:
80

Option 2:
60

Option 3:
40

Option 4:
20

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Correct Answer:
80

Solution:

The correct option is Option 1) 80.

The ribosome is a complex molecular machine involved in protein synthesis. It is composed of two subunits, the large subunit and the small subunit, which come together during protein synthesis and dissociate afterward. These subunits consist of a combination of proteins and RNA molecules called ribosomal RNA (rRNA).

In eukaryotes, the ribosome consists of 80 different proteins in total. The large subunit contains approximately 50 different proteins, while the small subunit contains around 30 different proteins. These proteins play essential roles in various aspects of protein synthesis, including mRNA binding, tRNA binding, and catalyzing the formation of peptide bonds.

Therefore, Option 1) 80 is the correct answer for the number of different proteins that the ribosome consists of.

Q. 25 Match List I with List II :

	List I		**List II**
A.	M phase	I.	Proteins are synthesized
B.	G_2 Phase	II.	Inactive phase
C.	Quiescent stage	III.	Interval between mitosis and initiation of DNA replication
D.	G_1 Phase	IV.	Equational division

Choose the correct answer from the options given below:

Option 1:
A-III, B-II, C-IV, D-I

Option 2:
A-IV, B-II, C-I, D-III

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-II, B-IV, C-I, D-III

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Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
A. M phase IV. Equational division
B. G_2 Phase I. Proteins are synthesized
C. Quiescent stage II. Inactive phase
D. G_1 Phase III. Interval between mitosis and initiation of DNA replication

Therefore, the correct answer is Option 3) A-IV, B-I, C-II, D-III.

Q. 26 Melonate inhibits the growth of pathogenic bacteria by inhibiting the activity of

Option 1:
Succinic dehydrogenase

Option 2:
Amylase

Option 3:
Lipase

Option 4:
Dinitrogenase

Correct Answer:
Succinic dehydrogenase

Solution:
The correct option is Option 1) Succinic dehydrogenase.

Melonate inhibits the growth of pathogenic bacteria by specifically inhibiting the activity of succinic dehydrogenase, an enzyme involved in the citric acid cycle (also known as the Krebs cycle or tricarboxylic acid cycle). By inhibiting this enzyme, melonate disrupts the energy production process in bacteria, leading to their growth inhibition.

Q. 27 Which of the following are NOT considered as the part of endomembrane system?
A. Mitochondria B. Endoplasmic Reticulum
C.Chloroplast D. Golgi complex
E. Peroxisomes

Choose the most **appropriate** answer from the options given below:

Option 1:
A and D only

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Option 2:
A, C and E only

Option 3:
A and D only

Option 4:
A, D and E only

Correct Answer:
A, C and E only

Solution:
Mitochondria, Chloroplasts, and Peroxisomes are not considered part of the endomembrane system. The endomembrane system primarily includes the endoplasmic reticulum and the Golgi complex, which are involved in the synthesis, modification, and transport of proteins and lipids within the cell.

Mitochondria are responsible for cellular respiration and energy production, and they have their own distinct membrane system separate from the endomembrane system. Chloroplasts are found in plant cells and are responsible for photosynthesis, which involves the conversion of light energy into chemical energy. Peroxisomes are involved in various metabolic processes, including the breakdown of fatty acids, detoxification of harmful substances, and synthesis of certain lipids. These organelles have their own unique functions and are not part of the endomembrane system.

Option 2 is the correct answer

Q. 28 Given below are two statements:

Statement I: Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroyes enzymatic activity because proteins are denatured by heat.
Statement II: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of above statements, choose the **correct** answer from the options given below:

Option 1:
Both **statement I** and **statement II** are true.

Option 2:
Both **statement I** and **statement II** are false.

Option 3:
statement I is true but **statement II** is false.

Option 4:
Statement I is false but **statement II** is true.

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Correct Answer:
Both **statement I** and **statement II** are true.

Solution:
The correct answer is Option 1) Both statement I and statement II are true.

Statement I is true. Low temperatures can slow down or preserve enzymatic activity by reducing the rate of reactions. However, high temperatures can denature proteins, including enzymes, leading to a loss of enzymatic activity.

Statement II is also true. Competitive inhibition occurs when an inhibitor molecule closely resembles the substrate and competes with the substrate for binding to the active site of the enzyme. The inhibitor effectively "competes" with the substrate for the enzyme's active site, thereby reducing the enzyme's activity.

Therefore, both statement I and statement II are true, and Option 1 is the correct answer.

Q. 29 Which of the following functions is carried out by cytoskeleton in a cell ?

Option 1:
Nuclear division

Option 2:
Protein synthesis

Option 3:
Motility

Option 4:
Transportation

Correct Answer:
Motility

Solution:
The correct answer is Option 3) Motility.

The cytoskeleton is a network of protein filaments found in the cytoplasm of cells. It plays a crucial role in providing structural support, maintaining cell shape, and facilitating various cellular processes. One of the key functions of the cytoskeleton is cell motility, which includes the movement of cells themselves and the movement of cellular components within the cell.

The cytoskeleton is responsible for the movement of cells through processes such as amoeboid movement, ciliary/flagellar movement, and muscle contraction. It provides the necessary structure and organization for these movements to occur.

While nuclear division (Option 1) is primarily regulated by the spindle apparatus during mitosis and meiosis and protein synthesis (Option 2) is carried out by ribosomes, the cytoskeleton is not directly involved in these specific processes.

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Transportation (Option 4) can involve the cytoskeleton indirectly, as it provides tracks or highways for intracellular transport, but it is not the primary function of the cytoskeleton. The cytoskeleton's primary role in transportation is to help maintain the cellular architecture and provide support for the transport systems.

Therefore, the correct function carried out by the cytoskeleton in a cell is Option 3) Motility.

Q. 30 Select the correct statements.
A. Tetrad formation is seen during Leptotene.
B. During Anaphase, the centromeres split and chromatids separate.
C. Terminalization takes place during Pachytene.
D. Nucleolus, Golgi complex and ER are reformed during Telophase.
E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the **correct** answer from the option given below:

Option 1:
A and C only

Option 2:
B and D only

Option 3:
A, C and E only

Option 4:
B and E only

Correct Answer:
B and D only

Solution:
The correct answer is Option 2) B and D only.

Statement A: Tetrad formation is seen during Leptotene.
This statement is incorrect. Tetrad formation, also known as synapsis, occurs during the zygotene stage of prophase I of meiosis. During leptotene, chromosomes start to condense, but they have not yet formed tetrads.

Statement B: During Anaphase, the centromeres split and chromatids separate.
This statement is correct. During anaphase of mitosis or meiosis II, the centromeres split, and the sister chromatids separate, moving toward opposite poles of the cell.

Statement C: Terminalization takes place during Pachytene.
This statement is incorrect. Terminalization occurs during diplotene, not pachytene, which is a substage of prophase I of meiosis. Terminalization refers to the movement of chiasmata (sites of crossing over) towards the ends (terminals) of the homologous chromosomes.

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Statement D: Nucleolus, Golgi complex, and ER are reformed during Telophase.
This statement is correct. During telophase of mitosis or meiosis, the nuclear envelope reforms, and the nucleolus, Golgi complex, and endoplasmic reticulum (ER) are reformed.

Statement E: Crossing over takes place between sister chromatids of homologous chromosomes.
This statement is incorrect. Crossing over, also known as recombination, occurs between non-sister chromatids of homologous chromosomes during prophase I of meiosis. It results in the exchange of genetic material between homologous chromosomes.

Therefore, the correct answer is Option 2) B and D only.

Q. 31 Given below are two statements:

Statement I: During G0 phase of cell cycle, the cell is metabolically inactive.
Statement II: The centrosome undergoes duplication during S phase of interphase.
In the light of the above statements, choose the **most appropriate** answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are correct.

Option 2:
Both **Statement I** and **Statement II** are incorrect.

Option 3:
Statement I is correct but **statement II** incorrect.

Option 4:
Statement I is incorrect but **statement II** correct.

Correct Answer:
Statement I is incorrect but **statement II** correct.

Solution:
The correct answer is Option 4) Statement I is incorrect but statement II is correct.

Statement I: During G_0 phase of the cell cycle, the cell is metabolically inactive.
This statement is incorrect. The G_0 phase of the cell cycle is a resting phase where cells exit the cell cycle and temporarily halt their division. However, cells in the G_0 phase can still be metabolically active and perform their specific functions. They may carry out normal cellular activities, such as protein synthesis and energy production, even though they are not actively dividing.

Statement II: The centrosome undergoes duplication during S phase of interphase.
This statement is correct. The centrosome, a cellular organelle involved in cell division, undergoes duplication during the S phase of interphase. The centrosome consists of two centrioles, and each centriole replicates to form a new pair of centrioles. This duplication ensures that each daughter cell produced during cell division receives a complete set of centrioles.

Therefore, the correct answer is Option 4) Statement I is incorrect but statement II is correct.

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Q. 32 Given below are two statements:

Statement I: A protein is imagined as a line, the left end represented by first amino acid (C -terminal) and the right end represented by last amino acid (N-terminal)
Statement II: Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is false but Statement II is true.

Solution:
Option 4) Statement I is false but Statement II is true.

Statement I is incorrect because in the conventional representation of a protein, the N-terminal (amino-terminus) is depicted on the left end and the C-terminal (carboxy-terminus) is depicted on the right end.

Statement II is true. Adult human hemoglobin is indeed composed of four subunits, two of which are α subunits and two of which are β subunits.

Q. 33 How many ATP and $NADPH_2$ are required for the synthesis of one molecule of Glucose during Calvin cycle?

Option 1:
$12\ ATP$ and $12\ NADPH_2$

Option 2:
$18\ ATP$ and $12\ NADPH_2$

Option 3:
$12\ ATP$ and $16\ NADPH_2$

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Option 4:
18 ATP and 16 $NADPH_2$

Correct Answer:
18 ATP and 12 $NADPH_2$

Solution:
During the Calvin cycle, also known as the light-independent reactions of photosynthesis, glucose is synthesized from carbon dioxide (CO_2) using ATP and $NADPH_2$ as energy sources. The overall reaction for glucose synthesis can be represented as follows:

6 CO_2 + 18 ATP + 12 $NADPH_2$ → $C_6H_{12}O_6$ (glucose) + 18 ADP + 18 Pi + 12 $NADP^+$

For the synthesis of one molecule of glucose, 6 molecules of carbon dioxide are required. Each carbon dioxide molecule undergoes a series of reactions involving ATP and $NADPH_2$ to form one glucose molecule.

Each carbon dioxide molecule requires 3 ATP molecules for its fixation and reduction, and 2 $NADPH_2$ molecules for its reduction. Therefore, for 6 CO_2 molecules, we need 6 × 3 = 18 ATP molecules and 6 × 2 = 12 $NADPH_2$ molecules.

Hence, the correct option is Option 2) 18 ATP and 12 $NADPH_2$.

Q. 34 Given below are two statements :

Statement I: The forces generated by transpiration can lift a xylem-sized column of water over 130 meters height.

Statement II: Transpiration cools leaf surfaces sometimes 10 to 15 degrees, by evaporative cooling. In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are correct.

Solution:

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Statement I is correct because transpiration, the process of water loss from plants through stomata, creates a pulling force that can generate enough tension to lift a column of water in the xylem to considerable heights. This phenomenon is known as the transpiration pull or cohesion-tension theory.

Statement II is also correct because transpiration, as water evaporates from the leaf surfaces, leads to evaporative cooling. This cooling effect helps regulate leaf temperature and can lower leaf surface temperatures by 10 to 15 degrees Celsius.

Option 1 is the correct answer.

Q. 35 Which micronutrient is required for splitting of water molecule during photosynthesis?

Option 1:
manganese

Option 2:
molybdenum

Option 3:
magnesium

Option 4:
copper

Correct Answer:
manganese

Solution:
Manganese is absorbed by plants in the form of Mn^{2+} ions. It serves various important functions, including the activation of enzymes involved in respiration, nitrogen metabolism, and photosynthesis. However, its primary role lies in the process of photosynthesis, where it plays a crucial part in the splitting of water molecules to release oxygen.

Option 1 is the correct answer.

Q. 36 The reaction centre in PS II has an absorption maxima at.

Option 1:
680 nm

Option 2:
700 nm

Option 3:
660 nm

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Option 4:
780 nm

Correct Answer:
680 nm

Solution:
The reaction center in Photosystem II (PS II) has an absorption maximum at approximately 680 nm. This means that the reaction center is most efficient in capturing light energy at this specific wavelength. Light energy absorbed by PS II at this wavelength is used to drive the process of photosynthesis, particularly in the initial steps of the light-dependent reactions.

Option 1 is the correct answer.

Q. 37 Given below are two statement: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: ATP is used at two steps in glycolysis.

Reason R: First ATP is used in converting glucose into glucose-6 phosphate and second ATP is used in conversion of fructose-6- phosphate into fructose-1-6-diphosphate. In the light of the above statements, choose the Correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A

Solution:
The correct answer is Option 1) Both A and R are true and R is the correct explanation of A.

Assertion A states that ATP is used in two steps in glycolysis, which is true.

Reason R provides the correct explanation for Assertion A. It states that the first ATP is used in converting glucose into glucose-6-phosphate, and the second ATP is used in the conversion of fructose-6-phosphate into fructose-1,6-diphosphate. This is an accurate description of the steps in glycolysis where ATP is utilized.

Therefore, both Assertion A and Reason R are true, and Reason R correctly explains Assertion A.

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Q. 38 Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production?

Option 1:
Indole-3-butyric acid

Option 2:
Gibberellic Acid

Option 3:
Zeatin

Option 4:
Abscisic Acid

Correct Answer:
Gibberellic Acid

Solution:
The correct option is Option 2) Gibberellic Acid.

Gibberellic Acid is a phytohormone that can be used to accelerate the maturity period of plants, including conifers. When sprayed on juvenile conifers, Gibberellic Acid promotes the elongation of stems, stimulates flowering, and induces early seed production.

Gibberellic Acid is a naturally occurring plant hormone that regulates various physiological processes in plants, including growth and development. By applying Gibberellic Acid to juvenile conifers, the plant's natural growth and reproductive processes can be influenced, resulting in an expedited maturity period and earlier seed production.

Therefore, Gibberellic Acid is the phytohormone that, when sprayed on juvenile conifers, helps in hastening the maturity period and leads to early seed production.

Q. 39 Which hormone promotes internode/petiole elongation in deep water rice?

Option 1:
GA_3

Option 2:
Kinetin

Option 3:
Ethylene

Option 4:
$2,4-D$

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Correct Answer:
Ethylene

Solution:
The correct option is Option 3) Ethylene.

Ethylene is a plant hormone that plays a crucial role in promoting internode and petiole elongation in deep water rice plants. Deep water rice, also known as floating rice, is a variety of rice that is adapted to grow in flooded or waterlogged conditions.

When deep water rice plants are submerged in water, the lack of oxygen and the presence of ethylene stimulate elongation growth in the internodes and petioles. Ethylene acts as a signal to trigger the elongation of these plant parts, allowing the plant to grow taller and keep its leaves above the water surface for efficient gas exchange.

Gibberellic acid (GA3) is a hormone that is involved in promoting stem elongation in many plant species. However, ethylene is specifically associated with promoting internode and petiole elongation in deep water rice under flooded conditions.

Kinetin is a type of cytokinin hormone that is involved in cell division and growth regulation but is not directly associated with internode or petiole elongation in deep water rice.

2,4-D (2,4-dichlorophenoxyacetic acid) is a synthetic auxin hormone that is commonly used as a herbicide. It is not directly involved in promoting internode or petiole elongation in deep water rice.

Therefore, Option 3) Ethylene is the correct option for the hormone that promotes internode/petiole elongation in deep water rice.

Q. 40 Movement and accumulation of ions across a membrane against their concentration gradient can be explained by

Option 1:
Osmosis

Option 2:
Facilitated Diffusion

Option 3:
Passive Transport

Option 4:
Active Transport

Correct Answer:
Active Transport

Solution:
The correct option is Option 4) Active Transport.

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Active transport is the process by which ions or molecules are moved across a membrane against their concentration gradient, from an area of lower concentration to an area of higher concentration. This process requires the expenditure of energy in the form of ATP (adenosine triphosphate).

During active transport, specialized transport proteins, such as pumps or carriers, are involved in the movement of ions or molecules across the membrane. These transport proteins actively move the ions or molecules, using ATP as an energy source, to create a concentration gradient that is different from the equilibrium state.

This process is in contrast to passive transport, such as osmosis or facilitated diffusion, where the movement of ions or molecules occurs along the concentration gradient and does not require energy expenditure.

Therefore, Option 4) Active Transport is the correct explanation for the movement and accumulation of ions across a membrane against their concentration gradient.

Q. 41 Match **List I** with **List II** :

	List I		**List II**
A.	Oxidative decarboxylation	I.	Citrate synthase
B.	Glycolysis	II.	. Pyruvate dehydrogenase
C.	Oxidative phosphorylation	III.	Electron transport system
D.	Tricarboxylic acid cycle	IV.	EMP pathway

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-II, D-I

Option 2:
A-II, B-IV, C-I, D-III

Option 3:
A-III, B-I, C-II, D-IV

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-IV, C-III, D-I

Solution:

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A. Oxidative decarboxylation - II. Pyruvate dehydrogenase
B. Glycolysis - IV. EMP pathway
C. Oxidative phosphorylation - III. Electron transport system
D. Tricarboxylic acid cycle - I. Citrate synthase

Therefore, the correct matching is A-II, B-IV, C-III, D-I, which corresponds to Option 4).

Q. 42 Match List I with List II :

	List I		**List II**
A.	Cohesion	I.	More attraction in liquid phase
B.	Adhesion	II.	Mutual attraction among water molecules
C.	Surface tension	III.	Water loss in liquid phase
D.	Guttation	IV.	Attraction towards polar surfaces

Choose the correct answer from the options given below:

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
A. Cohesion - II. Mutual attraction among water molecules
B. Adhesion - IV. Attraction towards polar surfaces
C. Surface tension - I. More attraction in liquid phase
D. Guttation - III. Water loss in liquid phase

Therefore, the correct matching is A-II, B-IV, C-I, D-III, which corresponds to Option 1).

Q. 43 Which of the following combinations is required for chemiosmosis?

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Option 1:
membrane, proton pump, proton gradient, ATP synthase

Option 2:
membrane, proton pump, proton gradient, NADP synthase

Option 3:
proton pump, electron gradient, ATP synthase

Option 4:
proton pump, electron gradient, NADP synthase

Correct Answer:
membrane, proton pump, proton gradient, ATP synthase

Solution:
The correct combination required for chemiosmosis is:

Option 1) membrane, proton pump, proton gradient, ATP synthase.

Chemiosmosis is the process by which ATP (adenosine triphosphate) is synthesized using the energy generated by the movement of protons (H+) across a membrane. In this process, a membrane is necessary to create a barrier for the proton movement. A proton pump actively transports protons across the membrane, creating a proton gradient. This proton gradient serves as a source of potential energy. ATP synthase is an enzyme located in the membrane that utilizes this proton gradient to generate ATP.

Option 2) (membrane, proton pump, proton gradient, NADP synthase) is incorrect because NADP synthase is not involved in chemiosmosis. NADP synthase is not a known enzyme involved in ATP synthesis.

Option 3) (proton pump, electron gradient, ATP synthase) is also incorrect because chemiosmosis primarily involves the movement of protons (H+) and not electrons.

Option 4) (proton pump, electron gradient, NADP synthase) is incorrect for the same reasons as Option 2 and Option 3.

Therefore, the correct combination required for chemiosmosis is Option 1.

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Q. 44 Match List I with List II

	List I		List II.
A.	Iron	I.	Synthesis of auxin
B.	Zinc	II.	. Component of nitrate reductase
C.	Boron	III.	Activator of catalase
D.	Molybdenum	IV.	Cell elongation and differentiation

Choose the correct answer from the options given below:

Option 1:
A–III, B–II, C–I, D–IV

Option 2:
A–II, B–III, C–IV, D–I

Option 3:
A–III, B–I, C–IV, D–II

Option 4:
A–II, B–IV, C–I, D–III

Correct Answer:
A–III, B–I, C–IV, D–II

Solution:
A. Iron - III. Activator of catalase
B. Zinc - II. Component of nitrate reductase
C. Boron - I. Synthesis of auxin
D. Molybdenum - IV. Cell elongation and differentiation

Therefore, the correct answer is Option 3) A–III, B–I, C–IV, D–II.

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Q. 45 Match **List I** with **List II** with respect to human eye.

	List I		**List II**
A.	Fovea	I.	Visible coloured portionof eye that regulats diameter of pupil.
B.	Iris	II.	External layer of eye formed of dense connective tissue.
C.	Blind spot	III.	Point of greatest visual acuity or resolution.
D.	Sclera	IV.	leaves the eyeball and photoreceptor cells are absent.

Choose the **correct** answer from the options given below :

Option 1:
A–III, B–I, C–IV, D–II

Option 2:
A–IV, B–III, C–II, D–I

Option 3:
A–I, B–IV, C–III, D–II

Option 4:
A–II, B–I, C–III, D–IV

Correct Answer:
A–III, B–I, C–IV, D–II

Solution:
A. Fovea - III. Point of greatest visual acuity or resolution
The fovea is a small depression in the retina of the eye that contains a high concentration of cone cells, which are responsible for detailed and color vision. It is the area of the retina with the highest visual acuity.

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B. Iris - I. Visible colored portion of the eye that regulates the diameter of the pupil
The iris is the colored part of the eye surrounding the pupil. It controls the size of the pupil and regulates the amount of light entering the eye.

C. Blind spot - IV. Leaves the eyeball, and photoreceptor cells are absent
The blind spot, also known as the optic disc, is the area on the retina where the optic nerve leaves the eye. It does not contain photoreceptor cells, so there is no vision in that particular area.

D. Sclera - II. External layer of the eye formed of dense connective tissue
The sclera is the tough, white, outer layer of the eye that forms the protective covering of the eyeball. It is composed of dense connective tissue.

Therefore, the correct matching is:
A–III, B–I, C–IV, D–II, which corresponds to Option 1.

Q. 46 Which of the following statement are correct regarding skeletal muscle?

A. Muscle bundles are held together by collagenous connective tissue layer called fascicle.
B. Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.
C. Striated appearance of skeletal muscle fibre is due to distribution pattern of acting and myosin proteins.
D. M line is considered as functional unit of contraction called sarcomere.

Choose the **most appropriate** answer from the options given below:

Option 1:
A, B and C only

Option 2:
B and C only

Option 3:
A, C and D only

Option 4:
C and D only

Correct Answer:
B and C only

Solution:
The correct answer is Option 2) B and C only.

Statement B is correct. The sarcoplasmic reticulum of muscle fibers is a specialized type of endoplasmic reticulum that stores and releases calcium ions (Ca2+) during muscle contraction.

Statement C is correct. The striated appearance of skeletal muscle fibers is due to the arrangement and distribution pattern of actin and myosin proteins, which form the contractile units called sarcomeres.

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For incorrect options,

Statement A is incorrect. Muscle bundles in skeletal muscle are held together by a collagenous connective tissue layer called perimysium, not fascicle.

Statement D is incorrect. The M line is a structural component of the sarcomere, but the functional unit of contraction in skeletal muscle is the sarcomere itself, not specifically the M line.

Therefore, the correct answer is Option 2) B and C only.

Q. 47 Match **List I** with **List II**.

	List I (Type of Joint)		**List II (Found between)**
A.	Cartilaginous Joint	I.	Between flat skull bones
B.	Ball and Socket Joint	II.	Between adjacent vertebrae in vertebral column
C.	Fibrous Joint	III.	Between carpal and metacarpal of thumb
D.	Saddle Joint	IV.	Between Humerus and Pectoral girdle

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-II, B-IV, C-I, D-III

Option 3:
A-I, B-IV, C-III, D-II

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-IV, C-I, D-III

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Solution:
The correct answer is Option 2) A-II, B-IV, C-I, D-III.

Explanation:
A. Cartilaginous Joint is found between flat skull bones.
B. Ball and Socket Joint is found between the Humerus and Pectoral girdle.
C. Fibrous Joint is found between the carpal and metacarpal of the thumb.
D. Saddle Joint is found between adjacent vertebrae in the vertebral column.

Therefore, the correct match is:
A-II, B-IV, C-I, D-III.

Option 2) A-II, B-IV, C-I, D-III is the correct answer.

Q. 48 Once the undigested and unabsorbed substances enter the caecum, their backflow is prevented by-

Option 1:
Sphincter of Oddi

Option 2:
Ileo - caecal valve

Option 3:
Gastro - oesophageal sphincter

Option 4:
Pyloric sphincter

Correct Answer:
Ileo - caecal valve

Solution:
The correct answer is: Option 2) Ileo-caecal valve

The ileo-caecal valve is a one-way valve located between the ileum (the last part of the small intestine) and the caecum (the first part of the large intestine). It prevents the backflow of undigested and unabsorbed substances from the caecum back into the ileum, ensuring that the materials move in the correct direction through the digestive system.

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Q. 49 Match List I with List II.

	List I		List II
A.	P-wave	I.	Beginning of systole
B.	Q -wave	II.	Repolarisation of ventricles
C.	QRS complex	III.	Depolarisation of atria
D.	T-wave	IV.	Depolarisation of ventricles

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
A. P-wave: The P-wave represents the depolarization (contraction) of the atria. It corresponds to the beginning of systole, which is the contraction phase of the cardiac cycle. Therefore, the correct pairing is A-III.

B. Q-wave: The Q-wave is a small downward deflection that appears before the R-wave in the QRS complex. It doesn't have a specific physiological event associated with it. Therefore, the correct pairing is B-I.

C. QRS complex: The QRS complex represents the depolarization (contraction) of the ventricles. It is the largest and most prominent wave complex on the ECG. It occurs during the ventricular systole phase, which is the main pumping phase of the cardiac cycle. Therefore, the correct pairing is C-IV.

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D. T-wave: The T-wave represents the repolarization (relaxation) of the ventricles. It occurs during the diastole phase, which is the relaxation phase of the cardiac cycle. Therefore, the correct pairing is D-II.

Based on the explanations above, we can conclude that Option 1) A-III, B-I, C-IV, D-II is the correct answer.

Q. 50 Which of the following are NOT under the control of thyroid hormone?

A. Maintenance of water and electrolyte balance
B. Regulation of basal metabolic rate
C. Normal rhythm of sleep-wake cycle
D. Development of immune system
E. Support the process of R.B.Cs formation Choose the correct answer from the options given below:

Option 1:
A and D only

Option 2:
B and C only

Option 3:
C and D only

Option 4:
D and E only

Correct Answer:
C and D only

Solution:
The correct answer is Option 3) C and D only.

Thyroid hormone primarily influences the regulation of metabolism and growth in the body. It affects various physiological processes, but not all processes are directly controlled by thyroid hormone. Let's go through each option:

C. Normal rhythm of sleep-wake cycle: The sleep-wake cycle is primarily regulated by the hormone melatonin, which is produced by the pineal gland. Thyroid hormone does not directly control the sleep-wake cycle.

D. Development of immune system: The development of the immune system is not directly controlled by thyroid hormone. It is regulated by various other factors and immune cells.

For incorrect options,

A. Maintenance of water and electrolyte balance: This process is not directly regulated by thyroid hormone. It is primarily controlled by hormones such as antidiuretic hormone (ADH) and aldosterone.

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B. Regulation of basal metabolic rate: Thyroid hormone plays a crucial role in regulating the basal metabolic rate (BMR). It affects the rate at which the body utilizes energy.

E. Support the process of R.B.Cs formation: The production of red blood cells (RBCs) is primarily controlled by erythropoietin, a hormone produced by the kidneys. Thyroid hormone does not directly support the process of RBC formation.

Based on the explanations above, we can conclude that Option 3) C and D only is the correct answer.

Q. 51 Which of the following statements are correct?
A. An excessive loss of body fluid from the body switches off osmoreceptors.
B. ADH facilitates water reabsorption to prevent diuresis.
C. ANF causes vasodilation.
D. ADH causes increase in blood pressure. E. ADH is responsible for decrease in GFR.

Choose the correct answer from the options given below:
Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
B, C and D only

Option 3:
A, B and-E only

Option 4:
C, D and E only

Correct Answer:
B, C and D only

Solution:
The correct answer is Option 2) B, C, and D only.

B. ADH facilitates water reabsorption to prevent diuresis: This statement is correct. Antidiuretic hormone (ADH), also known as vasopressin, acts on the kidneys to increase water reabsorption. It helps prevent excessive water loss and promotes water retention in the body.

C. ANF causes vasodilation: This statement is correct. Atrial natriuretic peptide (ANF) is a hormone released by the heart in response to increased blood volume and pressure. It promotes vasodilation, which leads to the relaxation of blood vessels and increased excretion of sodium and water.

D. ADH causes an increase in blood pressure: This statement is correct. ADH acts on the blood vessels, causing vasoconstriction. This constriction increases peripheral vascular resistance, which can contribute to an increase in blood pressure.

For incorrect options,

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A. An excessive loss of body fluid from the body switches off osmoreceptors: This statement is incorrect. Osmoreceptors are specialized cells that detect changes in osmotic pressure and regulate the release of antidiuretic hormone (ADH). Excessive loss of body fluid would lead to increased osmotic pressure and activate osmoreceptors, not switch them off.

E. ADH is responsible for a decrease in GFR (glomerular filtration rate): This statement is incorrect. ADH does not directly influence the glomerular filtration rate. It primarily affects water reabsorption in the renal tubules.

Based on the evaluations, the correct answer is Option 2) B, C, and D only.

Q. 52 The parts of human brain that helps in regulation of sexual behaviour, expressian of excitement, pleasure, rage, fear etc are.

Option 1:
Limbic system & hypothalamus

Option 2:
Corpora quadrigemina & hippocampus

Option 3:
Brain stem & epithalamus

Option 4:
Corpus callosum and thalamus

Correct Answer:
Limbic system & hypothalamus

Solution:
The correct answer is Option 1) Limbic system and hypothalamus.

The limbic system and hypothalamus are the parts of the human brain that play a crucial role in regulating sexual behavior and expressing emotions such as excitement, pleasure, rage, fear, and more.

The limbic system is a collection of structures located in the cerebral hemispheres of the brain. It includes the hippocampus, amygdala, and parts of the thalamus and hypothalamus. The limbic system is involved in various functions, including emotion, behavior, motivation, and memory.

The hypothalamus is a small but important region located below the thalamus. It serves as a control center for many autonomic functions and behaviors, including regulation of body temperature, hunger, thirst, and sexual behavior. The hypothalamus plays a key role in the release of hormones involved in sexual arousal and reproductive functions.

Therefore, Option 1) Limbic system and hypothalamus is the correct answer.

Q. 53 Vital capacity of lung is____.

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Option 1:
ERV + ERV

Option 2:
IRV + ERV + TV + RV

Option 3:
ERV + ERV + TV + RV

Option 4:
IRV + ERV + TV

Correct Answer:
IRV + ERV + TV

Solution:
The vital capacity of the lungs is the maximum amount of air that can be exhaled after a maximum inhalation. It is calculated by adding the following lung volumes:

IRV (Inspiratory Reserve Volume): the additional air that can be forcefully inhaled after a normal inhalation.
ERV (Expiratory Reserve Volume): the additional air that can be forcefully exhaled after a normal exhalation.
TV (Tidal Volume): the amount of air inhaled or exhaled during normal breathing.
The residual volume (RV), which is the amount of air that remains in the lungs after maximum exhalation, is not included in the vital capacity calculation.

Therefore, the correct option for the vital capacity of the lungs is indeed Option 4) IRV + ERV + TV.

Option 4 is the correct answer.

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Q. 54 Match List I with List II.

	List I (cells)		**List II (Secretion)**
A.	Peptic cells	I.	Mucus
B.	Goblet cells	II.	Bile juice
C.	Oxyntic cells	III.	Proenzyme pepsinogen
D.	Hepatic cells	IV.	HCl and intrinsic factor for absorption of vitamin B12

Choose the **correct** answer from the options given below:

Option 1:
A–IV, B–III, C–II, D–I

Option 2:
A–II, B–I, C–III, D–IV

Option 3:
A–III, B–I, C–IV, D–II

Option 4:
A–II, B–IV, C–I, D–III

Correct Answer:
A–III, B–I, C–IV, D–II

Solution:
A. Peptic cells - III. Proenzyme pepsinogen
Peptic cells secrete the proenzyme pepsinogen, which is later activated to the enzyme pepsin and helps in the digestion of proteins.

B. Goblet cells - I. Mucus
Goblet cells secrete mucus, which helps in lubricating and protecting the epithelial lining of various organs.

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C. Oxyntic cells - IV. HCl and intrinsic factor for absorption of vitamin B12
Oxyntic cells, also known as parietal cells, secrete hydrochloric acid (HCl) to aid in the digestion of food and also produce intrinsic factor, which is essential for the absorption of vitamin B12 in the small intestine.

D. Hepatic cells - II. Bile juice
Hepatic cells, found in the liver, secrete bile juice, which is stored in the gallbladder and aids in the digestion and absorption of fats.

Therefore, the correct matching is:
A–III, B–I, C–IV, D–II, which corresponds to Option 3.

Q. 55 Match **List I** with **List II**.

	List I		**List II**
A.	CCK	I.	Kidney
B.	GIP	II.	Heart
C.	ANF	III.	Gastric gland
D.	ADH	IV.	Pancreas

Choose the **correct** answer from the options given below:

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-IV, B-II, C-III, D-I

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:

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A. CCK - IV. Pancreas
CCK (cholecystokinin) is a hormone that is released by the small intestine and stimulates the pancreas to release digestive enzymes.

B. GIP - III. Gastric gland
GIP (gastric inhibitory peptide) is a hormone that is released by the small intestine and inhibits gastric acid secretion.

C. ANF - II. Heart
ANF (atrial natriuretic peptide) is a hormone that is produced by the heart and is involved in the regulation of blood pressure and fluid balance.

D. ADH - I. Kidney
ADH (antidiuretic hormone), also known as vasopressin, is produced by the hypothalamus and released by the pituitary gland. It acts on the kidneys to regulate water balance and urine concentration.

Therefore, the correct matching is:
A-IV, B-III, C-II, D-I, which corresponds to Option 1.

Q. 56 Match List I with List II .

	List I		**List II**
A.	Taenia	I.	Nephridia
B.	Paramoecium	II.	Contractile vacuole
C.	Periplaneta	III.	Flame cells
D.	Pheretima	IV.	Urecose gland

Choose the correct answer from the options give below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-I, B-II, C-IV, D-III

Option 3:
A-III, B-II, C-IV, D-I

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Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-II, C-IV, D-I

Solution:
A. Taenia - III. Flame cells
B. Paramoecium - II. Contractile vacuole
C. Periplaneta - IV. Urecose gland
D. Pheretima - I. Nephridia

Therefore, the correct answer is:

Option 3) A-III, B-II, C-IV, D-I

Q. 57 **Assertion (A):** Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in the cortex and medulla.

Reason (R): Juxta medullary nephrons have a short loop of Henle whereas, cortical nephrons have a longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
A is true but R is false.

Solution:
Assertion A is true, stating that nephrons are of two types: cortical and juxtamedullary, based on their relative position in the cortex and medulla.

However, Reason R is false. Juxtamedullary nephrons have a longer loop of Henle that extends deep into the medulla, while cortical nephrons have a shorter loop of Henle that does not extend as far into the medulla. The statement in Reason R is the opposite of the actual arrangement of the loop of Henle in cortical and juxtamedullary nephrons.

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Therefore, while Assertion A is true, Reason R is false. The correct answer is Option 3 which is A is true but R is false.

Q. 58 Large, colourful, fragrant flowers with nectar are seen in:

Option 1:
insect pollinated plants

Option 2:
bird pollinated plants

Option 3:
bat pollinated plants

Option 4:
wind pollinated plants

Correct Answer:
insect pollinated plants

Solution:
- Entomophily is the scientific term used to describe insect pollination.
- Flowers that rely on insect pollination tend to be large, vibrant in color, emit a pleasant fragrance, and contain abundant nectar.
- To enhance visibility, multiple flowers are often arranged in clusters known as inflorescences.
- These flowers possess nectar glands and emit strong fragrances to attract insects.
- Option 1 is the correct answer.

Q. 59 Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly?

Option 1:
Genital herpes

Option 2:
Gonorrhoea

Option 3:
Hepatitis-B

Option 4:
HIV Infection

Correct Answer:
Gonorrhoea

Solution:
The correct option is Option 2) Gonorrhoea.

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Gonorrhoea is a common sexually transmitted disease caused by the bacterium Neisseria gonorrhoeae. When detected early and treated properly with appropriate antibiotics, gonorrhoea can be completely cured. It is important to seek medical attention and adhere to the prescribed treatment to ensure effective eradication of the infection. However, it is worth noting that the emergence of antibiotic-resistant strains of Neisseria gonorrhoeae is a growing concern, emphasizing the importance of early detection and appropriate treatment.

Q. 60 Which of the following statements are correct regarding female reproductive cycle?

A. In non-primate mammals cyclical changes during reproduction are called oestrus cycle.
B. First menstrual cycle begins at puberty and is called menopause.
C. Lack of menstruation may be indicative of pregnancy.
D. Cyclic menstruation extends between menarche and menopause.

Choose the **most appropriate** answer from the options given below:-

Option 1:
A and D only

Option 2:
A and B only

Option 3:
A, B and C only

Option 4:
A, C and D only

Correct Answer:
A, C and D only

Solution:
The correct option is Option 4) A, C, and D only.

A. In non-primate mammals, cyclical changes during reproduction are indeed called the estrus cycle, not the menstrual cycle. The estrus cycle is characterized by the presence of an estrus phase, during which the female is sexually receptive.

B. The first menstrual cycle in females begins at puberty, marking the onset of reproductive capability. Menopause, on the other hand, refers to the cessation of menstrual cycles and reproductive function in women, typically occurring around the age of 45-55.

C. Lack of menstruation can be indicative of pregnancy. During pregnancy, menstruation ceases due to the hormonal changes that sustain the pregnancy.

D. Cyclic menstruation extends between menarche (the onset of the first menstrual cycle) and menopause (the cessation of menstrual cycles).

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Therefore, statements A, C, and D are correct, while statement B is incorrect.

Q. 61 Given below are two statements: one is labelled as **Assertion** A and the other is labelled as **Reason R**.

Assertion A: Amniocentesis for sex determination is one of the strategies of Reproductive and Child Health Care Programme.
Reason R: Ban ion amniocentesis checks increasing menace of female foeticide.

In the light of the above statements, choose the **correct** answer from the options given below:

Option 1:
Both **A** and R are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true and **R** is NOT the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false but **R** is true.

Correct Answer:
A is false but **R** is true.

Solution:
The correct option is Option 4) A is false but R is true.

Amniocentesis is a prenatal diagnostic procedure that involves the collection of amniotic fluid from the amniotic sac surrounding the fetus. It is primarily used for genetic testing and detecting chromosomal abnormalities, not specifically for sex determination. Therefore, Assertion A is false.

On the other hand, Reason R is true. The reason for banning amniocentesis for sex determination is to address the increasing problem of female foeticide, which is the selective abortion of female fetuses due to a preference for male children. The ban is aimed at preventing the misuse of prenatal diagnostic techniques for gender-based discrimination.

Since Assertion A is false and Reason R is true, the correct option is 4) A is false but R is true.

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Q. 62 Given below are two statements:

Statement I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct:
Statement II: The cavity of the cervix is called cervical canal which along with vagina' forms birth canal.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but Statement II is false .

Option 4:
Statement I incorrect but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true.

Solution:
The correct answer is Option 1) Both Statement I and Statement II are true.

Explanation:
Statement I: The vas deferens is a tube that receives a duct from the seminal vesicle and carries sperm from the epididymis to the urethra. It is a part of the male reproductive system and is involved in the transportation of sperm during ejaculation.

Statement II: The cervix is the lower part of the uterus that connects the uterus to the vagina. It contains a cavity called the cervical canal, which is a passageway for sperm to enter the uterus and for menstrual blood to exit the uterus during menstruation. The cervix, along with the vagina, forms the birth canal through which a baby passes during childbirth.

Both statements accurately describe anatomical structures and their functions in the male and female reproductive systems, respectively. Therefore, Option 1) Both Statement I and Statement II are true.

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Q. 63 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Endometrium is necessary for implantation of blastocyst.
Reason R: In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.
In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A .**

Option 2:
Both **A** and **R** are true and **R** is NOT the correct explanation of **A .**

Option 3:
A is true but **R** is false.

Option 4:
A is true but **R** is true.

Correct Answer:
Both **A** and **R** are true and **R** is NOT the correct explanation of **A .**

Solution:
Option 2) Both A and R are true, but R is NOT the correct explanation of A.

The assertion (A) that the endometrium is necessary for implantation of the blastocyst is true. The endometrium provides the necessary environment for the embryo to attach and establish a connection with the maternal blood supply.

The reason (R) provided, stating that the degeneration of the corpus luteum causes the disintegration of the endometrium, is not the correct explanation for the necessity of the endometrium. The degeneration of the corpus luteum and the subsequent decrease in hormone levels contribute to the shedding of the endometrium during menstruation but do not directly explain the requirement of the endometrium for blastocyst implantation.

Therefore, option 2 is the correct answer. Both statements are true, but the reason does not explain the assertion.

Q. 64 What is the function of tassels in the corn cob?

Option 1:
To attract insects

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Option 2:
To trap pollen grains

Option 3:
To disperse pollen grains

Option 4:
To protect seeds

Correct Answer:
To disperse pollen grains

Solution:
The correct option is Option 3) To disperse pollen grains

Tassels are the male reproductive structures found at the top of the corn cob in corn plants. Their primary function is to produce and release pollen grains, which are the male gametes. The tassels are composed of numerous long and slender filaments, each bearing anthers at their tips.

The anthers within the tassels contain the pollen grains. When the corn plant is mature and ready for pollination, the tassels release large amounts of pollen into the air. The purpose of releasing pollen is to ensure successful fertilization of the female reproductive structures called silk.

The silks, found in the ear of the corn cob, are the female reproductive structures. The pollen released from the tassels is carried by wind or other pollinators to the silks, where the process of fertilization occurs.

The tassels, with their filaments and anthers, effectively trap and release the pollen grains, allowing them to be dispersed and transported to the female flowers for pollination and fertilization. This is essential for successful seed production in corn plants.

Therefore, the main function of tassels in the corn cob is to trap but majorly it is to release pollen grains for the purpose of pollination and fertilization.

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Q. 65 Match **List I** with **List II**.

	List I		List II
A.	Vasectomy	I.	Oral method
B.	Coitus interruptus	II.	Barrier method
C.	Cervical caps	III.	Surgical method
D.	Saheli	IV.	Natural method

Choose the **correct** answer from the options given below:-

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-IV, B-II, C-I, D-III

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:
The correct option is Option 2) A-III, B-IV, C-II, D-I.

A. Vasectomy - III. Surgical method
B. Coitus interruptus - IV. Natural method
C. Cervical caps - II. Barrier method
D. Saheli - I. Oral method

Q. 66 The phenomenon of pleiotropism refers to

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Option 1:
presence of several alleles of a single gene controlling a single crossover.

Option 2:
presence of two alleles, each of the two genes controlling a single trait.

Option 3:
a single gene affecting multiple phenotypic expression

Option 4:
more than two genes affecting a single character.

Correct Answer:
a single gene affecting multiple phenotypic expression

Solution:
Pleiotropy refers to a phenomenon in which a single gene can manifest multiple phenotypic expressions. In most cases, this occurs due to the gene's impact on various metabolic pathways that contribute to different observable traits.

Option 3 is the correct answer

Q. 67 Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by

Option 1:
Thomas Hunt Morgan

Option 2:
Sutton and Boveri

Option 3:
Alfred Sturtevant

Option 4:
Henking

Correct Answer:
Alfred Sturtevant

Solution:
Alfred Sturtevant was a geneticist who played a significant role in the field of genetics, particularly in the mapping of genes on chromosomes. He was the first to propose the use of recombination frequency as a measure of the distance between genes on the same chromosome. Sturtevant's work was based on the observations made by Thomas Hunt Morgan, who studied the patterns of inheritance in fruit flies. However, it was Sturtevant who developed the concept of genetic mapping using recombination

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frequencies, which later became known as Sturtevant's mapping function. His groundbreaking work paved the way for the understanding of gene linkage and the construction of genetic maps.
Option 3 is the correct answer.

Q. 68 Unequivocal proof that DNA is the genetic material was first proposed by

Option 1:
Frederic Griffith

Option 2:
Alfred Hershey and Martha Chase

Option 3:
Avery, Macleoid and McCarthy

Option 4:
Wilkins and Franklin

Correct Answer:
Alfred Hershey and Martha Chase

Solution:
The correct option is Option 2) Alfred Hershey and Martha Chase.

Alfred Hershey and Martha Chase conducted the Hershey-Chase experiment in 1952, which provided unequivocal proof that DNA is the genetic material. In their experiment, they used bacteriophages, which are viruses that infect bacteria. They labeled the DNA of the bacteriophage with radioactive phosphorus-32 and the protein coat with radioactive sulfur-35.

By infecting bacteria with these labeled bacteriophages and then separating the viral DNA and proteins from the infected bacteria, Hershey and Chase showed that only the radioactive DNA, not the protein, was transferred to the next generation of phages. This demonstrated that DNA, not protein, carried the genetic information and served as the hereditary material.

Their experiment provided strong evidence supporting the idea that DNA, rather than protein, is the genetic material responsible for the transmission of inherited traits. Therefore, Hershey and Chase were the ones who first proposed the unequivocal proof that DNA is the genetic material.

Q. 69 What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

Option 1:
Transcription of tRNAs $(28S, 18S\ and\ 5.8S)$

Option 2:
Transcription of tRNA, 5 srRNA and snRNA

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Option 3:
Transcription of precursor of mRNA

Option 4:
Transcription of only snRNAs

Correct Answer:
Transcription of tRNA, 5 srRNA and snRNA

Solution:
The correct option is Option 2) Transcription of tRNA, 5S rRNA, and snRNA.

RNA polymerase III is an enzyme responsible for transcribing specific types of RNA molecules in eukaryotes. Its primary role is the transcription of three major types of RNA:

tRNA (transfer RNA): tRNA molecules play a crucial role in protein synthesis by carrying specific amino acids to the ribosomes. RNA polymerase III transcribes the genes encoding tRNA molecules.

5S rRNA (5S ribosomal RNA): 5S rRNA is a component of the large ribosomal subunit involved in protein synthesis. RNA polymerase III transcribes the genes encoding 5S rRNA.

snRNA (small nuclear RNA): snRNA molecules are involved in the processing of pre-mRNA and in the assembly of spliceosomes, which are responsible for removing introns from pre-mRNA. RNA polymerase III transcribes the genes encoding snRNA.

Therefore, RNA polymerase III plays a critical role in transcribing tRNA, 5S rRNA, and snRNA, contributing to important cellular functions such as protein synthesis and RNA processing.

Q. 70 Expressed Sequence Tags (ESTs) refers to

Option 1:
All genes that are expressed as RNA.

Option 2:
All genes that are expressed as proteins.

Option 3:
All genes whether expressed or unexpressed.

Option 4:
Certain important expressed genes.

Correct Answer:
All genes that are expressed as RNA.

Solution:

The correct option is Option 1) All genes that are expressed as RNA.

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Expressed Sequence Tags (ESTs) are short DNA sequences that are generated from complementary DNA (cDNA) libraries. These sequences represent fragments of messenger RNA (mRNA) molecules that have been transcribed from expressed genes in a specific tissue or organism.

ESTs are obtained by converting mRNA into cDNA using the enzyme reverse transcriptase. The cDNA is then cloned into a vector and sequenced. These sequences provide information about the genes that are actively transcribed and expressed in the tissue or organism from which the mRNA was isolated.

ESTs are particularly useful in gene discovery and gene expression studies. They provide valuable insights into the identity and expression patterns of genes, allowing researchers to identify and study genes that may be involved in specific biological processes or diseases.

Therefore, ESTs refer to all genes that are expressed as RNA, as stated in Option 1.

Q. 71 Which of the following statements are correct about Klinefelter's Syndrome?

A. This disorder was first described by Langdon Down (1866).

B. Such an individual has overall masculine development. However, the feminine development is also expressed.

C. The affected individual is short statured.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below.

Option 1:
A and B only

Option 2:
C and D only

Option 3:
B and E only

Option 4:
A and E only

Correct Answer:
B and E only

Solution:
The correct statements about Klinefelter's Syndrome are:

B. Such an individual has overall masculine development. However, the feminine development is also expressed.
E. Such individuals are sterile.

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Therefore, the correct answer is Option 3) B and E only.

Q. 72 Select the correct group/set of Australian Marsupials exhibiting adaptive radiation

Option 1:
Tasmanian wolf, Bobcat, Marsupial mole

Option 2:
Numbat, Spottted cuscus, Flying phalanger

Option 3:
Mole, Flying squirrel, Tasmanian tiger cat

Option 4:
Lemur, Anteater, Wolf

Correct Answer:
Numbat, Spottted cuscus, Flying phalanger

Solution:
The correct option is Option 2) Numbat, Spotted cuscus, Flying phalanger.

Adaptive radiation refers to the diversification of a group of organisms into various ecological niches, resulting in the development of different species with specialized adaptations. In the case of Australian marsupials, there are several examples of adaptive radiation.

Numbat, Spotted cuscus, Flying phalanger: This option includes marsupials that have undergone adaptive radiation. The numbat is a small marsupial that feeds mainly on termites and has specialized adaptations for this diet. The spotted cuscus is a tree-dwelling marsupial that has adapted to life in the rainforest. The flying phalanger, also known as the sugar glider, is a marsupial that has evolved adaptations for gliding through the air.

Option 1) is incorrect because Tasmanian wolf, Bobcat, Marsupial mole: This option includes species that are not examples of Australian marsupials. The Tasmanian wolf, also known as the thylacine, is extinct. The bobcat is not a marsupial but a small wild cat found in North America. The marsupial mole is native to Australia but is not considered an example of adaptive radiation.

Option 3) is incorrect because Mole, Flying squirrel, Tasmanian tiger cat: This option includes species that are not Australian marsupials. The mole is not a marsupial but a type of mammal found in various parts of the world. The flying squirrel is not native to Australia but is found in other regions. The Tasmanian tiger cat is not a recognized marsupial species.

Option 4) is incorrect because Lemur, Anteater, Wolf: This option includes species that are not Australian marsupials. Lemurs are primates found in Madagascar, not Australia. Anteaters are found in Central and South America, and wolves are found in various parts of the world.

Therefore, the correct option is Option 2) Numbat, Spotted cuscus, Flying phalanger, as these species are examples of Australian marsupials that have undergone adaptive radiation.

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Q. 73 Given below are two statements:

Statements I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.
Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the **correct** answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but statement II is false.

Option 4:
Statement I incorrect but statement II is true.

Correct Answer:
Statement I incorrect but statement II is true.

Solution:
The correct answer is Option 4) Statement I is incorrect, but statement II is true.

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

This statement is incorrect. In prokaryotes, the DNA is not associated with proteins like histones to form nucleosomes as seen in eukaryotes. Instead, prokaryotic DNA is typically found in a region called the nucleoid, where it is condensed and organized to some extent, but without the association of histones or other proteins.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

This statement is true. In eukaryotes, DNA is wrapped around histone proteins to form nucleosomes. The DNA molecule is negatively charged due to its phosphate backbone, while the histone proteins are positively charged. The interaction between the negatively charged DNA and the positively charged histones allows for the compaction and organization of DNA into a higher-order structure.

Therefore, the correct answer is Option 4) Statement I is incorrect, but statement II is true.

Q. 74 Broad plam with single palm crease is visible in a person suffering from –

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Option 1:
Down's syndrome

Option 2:
Turner's syndrome

Option 3:
Klinefelter's syndrome

Option 4:
Thalassemia

Correct Answer:
Down's syndrome

Solution:
The correct answer is Option 1) Down's syndrome.

A broad palm with a single palm crease, also known as a simian crease or a single transverse palmar crease, is commonly seen in individuals with Down's syndrome. Down's syndrome is a genetic disorder caused by the presence of an extra copy of chromosome 21. This condition is characterized by various physical and intellectual disabilities, and certain physical features are often associated with it.

One of the common physical features observed in individuals with Down's syndrome is a single transverse palmar crease, where the normal two creases in the palm fuse into one. This can give the appearance of a broad palm with a single crease extending across the palm.

Therefore, the correct answer is Option 1) Down's syndrome.

Q. 75 Which one of the following is sequence on corresponding coding strand, if the sequence on mRNA formed is as follows
5'AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3'?

Option 1:
5'UAGCUAGCUAGCUAGCUAGCUAGC UAGC 3'

Option 2:
3'UAGCUAGCUAGCUAGCUAGCUAGCUAGC 5'

Option 3:
5'ATCGATCGATCGATCGATCGATCGATCG 3'

Option 4:
3'ATCGATCGATCGATCGATCGATCGATCG 5'

Correct Answer:
5'ATCGATCGATCGATCGATCGATCGATCG 3'

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Solution:
To find the corresponding coding strand, we replace each base according to the base pairing rules:

A (mRNA) pairs with U (coding DNA)
U (mRNA) pairs with A (coding DNA)
C (mRNA) pairs with G (coding DNA)
G (mRNA) pairs with C (coding DNA)

Applying these base pairing rules to the given mRNA sequence, we get:

5'AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3'
3'TAGCTAGCTAGCTAGCTAGCTAGC TAGC 5'

Therefore, the correct corresponding coding strand sequence is 3'TAGCTAGCTAGCTAGCTAGCTAGC TAGC 5'.

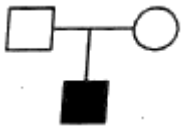
Option 3 is the correct answer.

Q. 76 Which one of the following symbols represents mating between relatives in human pedigree analysis?

Option 1:

Option 2:

Option 3:



Option 4:

Correct Answer:

Solution:
In human pedigree analysis, mating between relatives is represented by a double horizontal line connecting a male and a female. This symbol indicates consanguineous mating, which means that the individuals involved share a common ancestor. The double horizontal line represents a marriage or mating relationship, while the vertical lines extending downward from the double line represent their offspring.

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Option 2 is the correct answer.

Q. 77 Match **List I** with **List II**.

	List I		List II
A.	Gene 'a'	I.	β-galactosidase
B.	Gene 'y'	II.	Treansacetylase
C.	Gene 'I'	III.	Permease
D.	Gene 'z'	IV.	Repressor protein

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-I, C-IV, D-III

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
A. Gene 'a' - II. Transacetylase
Transacetylase is an enzyme involved in the metabolism of lactose.

B. Gene 'y' - III. Permease
Permease is a protein that facilitates the movement of molecules across the cell membrane.

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C. Gene 'I' - IV. Repressor protein
Repressor protein regulates the expression of genes by binding to the operator region of DNA and preventing transcription.

D. Gene 'z' - I. β-galactosidase
β-galactosidase is an enzyme that hydrolyzes lactose into glucose and galactose.

Therefore, the correct matching is:
A-II, B-III, C-IV, D-I, which corresponds to Option 2.

Q. 78 In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as :

Option 1:
Differentiation

Option 2:
Dedifferentiation

Option 3:
Development

Option 4:
Senescence

Correct Answer:
Dedifferentiation

Solution:
The correct option is Option 2) Dedifferentiation. In tissue culture experiments, when leaf mesophyll cells are placed in a culture medium and start to divide and proliferate, they undergo dedifferentiation. Dedifferentiation refers to the process where specialized cells revert to a less specialized or undifferentiated state. In the case of tissue culture, dedifferentiation allows the cells to lose their specialized characteristics and regain their ability to divide and form callus, which is an undifferentiated mass of cells. This process provides an opportunity for researchers to manipulate and regenerate new plants from these dedifferentiated cells.

Q. 79 Which of the following is NOT an advantage of inbreeding?

Option 1:
It decreases homozygosity.

Option 2:
It exposes harmful recessive genes that are eliminated by selection.

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Option 3:
Elimination of less desirable genes and accumulation of superior genes takes place due to it.

Option 4:
It decreases the productivity of inbred population, after continuous inbreeding.

Correct Answer:
It decreases the productivity of inbred population, after continuous inbreeding.

Solution:
The correct answer is Option 4) It decreases the productivity of the inbred population after continuous inbreeding.

Inbreeding refers to the mating of individuals who are closely related, such as siblings or close relatives. While inbreeding can have certain advantages in specific situations, such as in selective breeding programs, it also has several disadvantages.

Advantages of inbreeding include:

- Increased uniformity: Inbreeding can increase the uniformity of a population by concentrating desirable traits and reducing genetic variation.
- Identification of recessive traits: Inbreeding can expose harmful recessive genes that are otherwise masked in outbred populations. This allows for the identification and elimination of these harmful traits through selective breeding.
- Fixation of desired traits: Inbreeding can help fix or stabilize desirable traits in a population by repeatedly mating individuals with those traits. This is particularly useful in breeding programs aimed at developing specific breeds or lines.

Option 4 states that inbreeding decreases the productivity of the inbred population after continuous inbreeding. This is a correct statement as continuous inbreeding can lead to inbreeding depression. Inbreeding depression is a phenomenon where reduced fitness, lower fertility, and decreased productivity are observed in inbred populations due to the accumulation of deleterious recessive traits.

Therefore, Option 4 is the correct answer as it describes a disadvantage of inbreeding.

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Q. 80 Match **List I** with **List II**

	List I		**List II**
A.	Heroin	I.	Effect on cardiovascular system
B.	Marijuana	II.	Slow down body function
C.	Cocaine	III.	Painkiller
D.	Morphine	IV.	Interfere with transport of dopamine

Choose the **correct** answer from the options given below:

Option 1:
A–II, B–I, C–IV D–III

Option 2:
A–I, B–II, C–III, D–IV

Option 3:
A–IV, B–III, C–II, D–I

Option 4:
A–III, B–IV, C–I, D–II

Correct Answer:
A–II, B–I, C–IV D–III

Solution:
The correct answer is Option 2) A–I, B–II, C–III, D–IV.

Explanation:
A. Heroin is a narcotic drug that acts as a painkiller.
B. Marijuana is known to slow down body functions.
C. Cocaine is a stimulant drug that interferes with the transport of dopamine, a neurotransmitter.
D. Morphine is a strong painkiller, often used in medical settings.

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Therefore, the correct match is:
A–I (Heroin - Painkiller)
B–II (Marijuana - Slow down body function)
C–III (Cocaine - Interfere with transport of dopamine)
D–IV (Morphine - Effect on cardiovascular system)

Q. 81 In which blood corpuscles, the HIV undergoes replication and produces progeny viruses ?

Option 1:
T_H Cells

Option 2:
B–lymphocytes

Option 3:
Basophils

Option 4:
Eosinophils

Correct Answer:
T_H Cells

Solution:
The correct answer is Option 1) TH Cells (T-helper cells).

HIV (Human Immunodeficiency Virus) primarily infects and replicates in T-helper cells, which are a type of white blood cell involved in the immune response. HIV specifically targets CD4 receptors on the surface of T-helper cells. Once inside the T-helper cells, the virus replicates and produces progeny viruses, leading to the destruction of T-helper cells and the weakening of the immune system. This is why HIV infection can result in immunodeficiency and increased susceptibility to various infections and diseases.

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Q. 82 Match **List I** with **List II.**

	List I		List II
A.	Ringworm	I.	Haemophilus influenzae
B.	Filariasis	II.	Trichophyton
C.	Malaria	III.	Wuchereria bancrofti
D.	Pneumonia	IV.	Plasmodium vivax

Choose the **correct** answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

A. Ringworm is caused by the fungus Trichophyton.
B. Filariasis is caused by the parasite Wuchereria bancrofti.
C. Malaria is caused by the parasite Plasmodium vivax.
D. Pneumonia is caused by the bacterium Haemophilus influenzae.

Therefore, the correct match is:
A-II, B-III, C-IV, D-I.

Option 1) A-II, B-III, C-IV, D-I is the correct answer.

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Q. 83 In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.

Option 1:
Copper

Option 2:
Zinc

Option 3:
Tungsten or gold

Option 4:
Silver

Correct Answer:
Tungsten or gold

Solution:
The correct option is Option 3) Tungsten or gold.

In the gene gun method, which is a technique used to introduce foreign DNA into host cells, microparticles of either tungsten or gold are commonly used.

The gene gun method, also known as biolistic particle delivery system, involves coating tiny microparticles with foreign DNA and propelling them into the target cells. The microparticles act as carriers for the foreign DNA and are propelled into the host cells using a high-pressure gas or a mechanical force.

Tungsten and gold are preferred materials for the microparticles because they are dense and can effectively penetrate the cell walls and membranes of the target cells. The microparticles are typically coated with the desired DNA and then shot into the host cells, where they can integrate into the cell's genome and express the foreign genes.

Therefore, in the gene gun method, microparticles of either tungsten or gold are commonly used as carriers for the introduction of alien DNA into host cells.

Q. 84 Upon exposure to UV radiation, DNA stained with ethidium bromide will show

Option 1:
Bright red colour

Option 2:
Bright blue colour

Option 3:
Bright yellow colour

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Option 4:
Bright orange colour

Correct Answer:
Bright orange colour

Solution:
The correct option is Option 4) Bright orange colour.

Ethidium bromide is a fluorescent dye commonly used to stain DNA and visualize it under ultraviolet (UV) light. When ethidium bromide binds to DNA, it intercalates between the base pairs of the DNA molecule.

When exposed to UV radiation, ethidium bromide emits a fluorescent signal that can be visualized. Under UV light, DNA stained with ethidium bromide typically appears as bright orange bands or spots.

The orange color is a result of the interaction between ethidium bromide and the DNA molecule. The intercalation of ethidium bromide into the DNA helix causes a shift in the absorbance and emission wavelengths, resulting in the characteristic orange fluorescence when exposed to UV light.

Therefore, Option 4) Bright orange colour is the correct option for how DNA stained with ethidium bromide appears when exposed to UV radiation.

Q. 85 During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

Option 1:
RNA

Option 2:
DNA

Option 3:
Histones

Option 4:
Polysaccharides

Correct Answer:
DNA

Solution:
The correct option is Option 2) DNA.

During the purification process in recombinant DNA technology, chilled ethanol is often used to precipitate out DNA. This technique is based on the principle that DNA is less soluble in cold ethanol than in aqueous solutions.

When chilled ethanol is added to the DNA solution, it causes the DNA molecules to come out of solution and form a visible precipitate. This is because the ethanol disrupts the hydrogen bonding and hydrophobic interactions that hold the DNA molecules together, leading to their aggregation and

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precipitation.

By centrifuging the mixture, the precipitated DNA can be collected as a pellet, while other contaminants remain in the supernatant. The DNA pellet can then be further purified and processed for various applications in recombinant DNA technology.

Therefore, Option 2) DNA is the correct answer for the component that precipitates out when chilled ethanol is added during the purification process for recombinant DNA technology.

Q. 86 Main steps in the formation of Recombinant DNA are given below.

Arrange these steps in a correct sequence.

A. Insertion of recombinant DNA into the host cell.

B. Cutting of DNA at specific location by restriction enzyme.

C. Isolation of desired DNA fragment.

D. Amplification of gene of interest using PCR.

Choose the correct answer from the options given below:

Option 1:
B, C, D, A

Option 2:
C, A, B, D

Option 3:
C, B, D, A

Option 4:
B, D, A, C

Correct Answer:
B, C, D, A

Solution:
The correct sequence of steps in the formation of Recombinant DNA is:

Option 1) B, C, D, A

The correct sequence is as follows:

Step 1: Cutting of DNA at specific locations by a restriction enzyme (B)
Step 2: Isolation of the desired DNA fragment (C)
Step 3: Amplification of the gene of interest using PCR (D)
Step 4: Insertion of the recombinant DNA into the host cell (A)

Therefore, the correct answer is Option 1.

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Q. 87 Which one of the following techiques does not serve the purpose of early diagnosis of a disease for its early treatment ?

Option 1:
Recombinant DNA Technology

Option 2:
Serum and Urine analysis

Option 3:
Polymerase Chain Reaction (PCR) technique

Option 4:
Enzyme Linked Immuno-Sorbent Assay (ELISA) technique

Correct Answer:
Serum and Urine analysis

Solution:
The correct answer is Option 2) Serum and Urine analysis.

Serum and urine analysis are useful diagnostic techniques, but they do not specifically serve the purpose of early diagnosis of a disease for its early treatment. Serum analysis involves testing the blood serum for various markers, such as antibodies or specific substances, to detect the presence of a disease or assess its progression. Urine analysis involves examining urine for abnormalities that may indicate certain diseases or conditions.

On the other hand, the other options mentioned do serve the purpose of early diagnosis for early treatment:

Option 1) Recombinant DNA Technology: This technology involves the manipulation of DNA to produce specific molecules or substances for diagnostic purposes, such as the production of recombinant proteins or DNA probes for detecting disease markers.

Option 3) Polymerase Chain Reaction (PCR) technique: PCR is a powerful molecular biology technique used to amplify specific DNA sequences. It is widely used in disease diagnosis, including the early detection of infections or genetic disorders.

Option 4) Enzyme-Linked Immuno-Sorbent Assay (ELISA) technique: ELISA is a technique used to detect and quantify specific substances, such as antibodies or antigens, in samples. It is commonly used in disease diagnosis, including the early detection of infections or the presence of specific molecules associated with diseases.

Therefore, the correct answer is Option 2) Serum and Urine analysis.

Q. 88 Which of the following is not a cloning vector?

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Option 1:
BAC

Option 2:
YAC

Option 3:
pBR322

Option 4:
Probe

Correct Answer:
Probe

Solution:

The correct answer is:

Option 4) Probe

A probe is not a cloning vector. It is a short, single-stranded DNA or RNA molecule that is used to detect a specific sequence in a sample. It is not used for the replication and cloning of DNA fragments like the other options (BAC, YAC, pBR322) which are commonly used cloning vectors.

Q. 89 Identify the correct statements :

(A) Detrivores perform fragmentation.

(B) The humus is further degraded by some microbes during mineralization.

(C) Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.

(D) The detritus food chain begins with living organisms.

(E) Earthworms break down detritus into smaller particles by a process called catabolism.

Option 1:
A, B, C only

Option 2:
B, C, D only

Option 3:
C, D, E only

Option 4:
D, E, A only

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Correct Answer:
A, B, C only

Solution:
The correct option is Option 1) A, B, C only.

(A) Detrivores perform fragmentation. This statement is correct. Detrivores are organisms that break down organic matter into smaller particles through fragmentation, facilitating the decomposition process.

(B) The humus is further degraded by some microbes during mineralization. This statement is correct. Humus, the partially decomposed organic matter, undergoes further degradation by microbes during the process of mineralization, where organic compounds are converted into inorganic forms.

(C) Water-soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching. This statement is correct. Leaching is the process by which water-soluble inorganic nutrients are washed down through the soil profile and can get precipitated or accumulate in deeper layers, affecting nutrient availability.

(D) The detritus food chain begins with living organisms. This statement is incorrect. The detritus food chain begins with dead organic matter, such as fallen leaves or dead animals. It involves the decomposition and consumption of detritus by organisms.

(E) Earthworms break down detritus into smaller particles by a process called catabolism. This statement is incorrect. Earthworms break down detritus into smaller particles through a process called physical fragmentation or grinding, not catabolism.

Therefore, Option 1) A, B, C only is the correct choice.

Q. 90 The thickness of ozone in a column of air in the atmosphere is measured in terms of :

Option 1:
Dobson units

Option 2:
Decibels

Option 3:
Decameter

Option 4:
Kilobase

Correct Answer:
Dobson units

Solution:
The correct option is Option 1) Dobson units. The thickness of ozone in a column of air in the atmosphere is commonly measured in Dobson units (DU). Dobson units are a unit of measurement used to quantify the concentration of ozone in the Earth's atmosphere. They are named after G.M.B. Dobson, a pioneer in atmospheric ozone research.

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Q. 91 In the equation

GPP – R = NPP

GPP is Gross Primary Productivity

NPP is Net Primary Productivity

R here is _____.

Option 1:
Photosynthetically active radiation

Option 2:
Respiratory quotient

Option 3:
Respiratory loss

Option 4:
Reproductive allocation

Correct Answer:
Respiratory loss

Solution:
The correct option is Option 3) Respiratory loss.

In the given equation, GPP (Gross Primary Productivity) represents the total amount of energy fixed by plants through photosynthesis. R represents the respiratory loss, which is the energy consumed by the plant through respiration. Respiration is the process by which organisms, including plants, convert stored energy (such as glucose) into usable energy (ATP) for cellular activities.

NPP (Net Primary Productivity) represents the net amount of energy available for plant growth and biomass production after accounting for the energy lost through respiration. Therefore, in the equation GPP - R = NPP, R represents the respiratory loss, indicating the energy consumed by plants during respiration.

Q. 92 Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species?

Option 1:
Habitat loss and fragmentation

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Option 2:
Over exploitation for economic gain

Option 3:
Alien species invasions

Option 4:
Co-extinctions

Correct Answer:
Habitat loss and fragmentation

Solution:
The correct option is Option 1) Habitat loss and fragmentation.

"The Evil Quartet" refers to four major drivers of species extinction, as identified by the renowned ecologist Paul Ehrlich. These drivers are habitat loss and fragmentation, over exploitation for economic gain, alien species invasions, and co-extinctions.

Among these, habitat loss and fragmentation are widely considered the most significant cause driving the extinction of species. As human activities continue to modify and destroy natural habitats, species are losing their homes and the resources they need to survive. Fragmentation refers to the breaking up of habitats into smaller, isolated patches, which further exacerbates the negative impacts on species by limiting their mobility, access to resources, and gene flow.

While the other factors listed in the options (over exploitation for economic gain, alien species invasions, and co-extinctions) also contribute to species loss, habitat loss and fragmentation are recognized as the primary drivers with the most far-reaching and detrimental consequences for biodiversity.

Q. 93 The historic Convention on Biological Diversity, 'The Earth Summit' was held in Rio de Janeiro in the year :

Option 1:
1985

Option 2:
1992

Option 3:
1986

Option 4:
2002

Correct Answer:
1992

Solution:
The correct option is Option 2) 1992.

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The historic Convention on Biological Diversity, also known as "The Earth Summit," took place in Rio de Janeiro in the year 1992. The summit was officially called the United Nations Conference on Environment and Development (UNCED) and was held from June 3 to June 14, 1992.

During this summit, the Convention on Biological Diversity (CBD) was adopted, which is an international treaty aimed at promoting the conservation and sustainable use of biodiversity and the fair and equitable sharing of its benefits. The CBD has been ratified by numerous countries and is considered a key global framework for addressing biodiversity loss and environmental sustainability.

Therefore, the Earth Summit, where the Convention on Biological Diversity was established, occurred in Rio de Janeiro in 1992.

Q. 94 Given below are two statements :

Statement I : Gause's Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II : In general, carnivores are more adversely affected by competition than herbivores. In the light of the above statements,

choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false..

Option 3:
Statement I is correct but Statement II is false.

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Statement I is correct but Statement II is false.

Solution:
The correct option is Option 3) Statement I is correct but Statement II is false.

Statement I, known as Gause's Competitive Exclusion Principle, is true. It states that two closely related species competing for the same resources cannot coexist indefinitely, and the competitively inferior species will be eliminated eventually.

However, Statement II is false. It is not true that carnivores are more adversely affected by competition than herbivores in general. The impact of competition on different species can vary depending on various factors such as resource availability, ecological niche, and adaptive traits.

Therefore, the correct answer is Option 3) Statement I is correct but Statement II is false.

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Q. 95 Which one of the following statements is NOT correct?

Option 1:
The micro-organisms involved in biodegradation of organic matter in a sewage polluted water body consume a lot of oxygen causing the death of aquatic organisms.

Option 2:
Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries

Option 3:
Water hyacinth grows abundantly in eutrophic water bodies and leads to an imbalance in the ecosystem dynamics of the water body

Option 4:
The amount of some toxic substances of industrial waste water increases in the organisms at successive trophic levels.

Correct Answer:
Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries

Solution:
The correct answer is Option 2) Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries.

This statement is not correct. Algal blooms caused by excess organic matter, particularly from sources like agricultural runoff or sewage pollution, can have negative impacts on water quality. The excessive growth of algae can deplete oxygen levels in the water, leading to oxygen-deprived zones harmful to aquatic organisms. It can also block sunlight from reaching other aquatic plants, affecting their growth. Additionally, algal blooms can lead to the release of toxins harmful to fish and other organisms.

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Q. 96 Match List I with List II :

	List I (Interaction)		**List II (Species A and B)**
A.	Mutualism	I.	+(A), O(B)
B.	Commensalism	II.	–(A), O(B)
C.	Amensalism	III.	+(A), –(B)
D.	Parasitism	IV.	+(A), +(B)

Choose the correct answer from the options given below:

Option 1:
A-IV, B-II, C-I, D-III

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
A. Mutualism - IV. +(A), +(B)
B. Commensalism - II. –(A), O(B)
C. Amensalism - I. +(A), O(B)
D. Parasitism - III. +(A), –(B)

Therefore, the correct answer is Option 2) A-IV, B-I, C-II, D-III.

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Q. 97 Match **List I** with **List II**.

	List I (Interacting species)		**List II** (Name of interaction)
A.	A Leopard and a Lion in a forest/grassland	I.	Competition
B.	A Cuckoo laying egg in a Crow's nest	II.	Brood parasitism
C.	Fungi and root of a higher plant in Mycorrtizae	III.	Gastric gland
D.	A cattle egret and a Cattle in a field	IV.	Commensalism

Choose the **correct** answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-I, B-II, C-IV, D-III

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-II, B-III, C-I, D-IV

Correct Answer:
A-I, B-II, C-III, D-IV

Solution:
A. A Leopard and a Lion in a forest/grassland - I. Competition
Leopard and Lion compete for resources, such as prey, within their shared habitat.

B. A Cuckoo laying an egg in a Crow's nest - II. Brood parasitism
Cuckoos lay their eggs in the nests of other bird species, such as Crows, for them to raise the cuckoo offspring.

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C. Fungi and root of a higher plant in Mycorrhizae - III. Mutualism
Mycorrhizae is a mutually beneficial association between fungi and the roots of higher plants, where the fungi provide nutrients to the plant, and the plant provides carbohydrates to the fungi.

D. A cattle egret and a Cattle in a field - IV. Commensalism
Cattle egrets follow cattle and feed on insects stirred up by the grazing activity of the cattle. The cattle are unaffected by the presence of the egret.

Therefore, the correct matching is:
A-I, B-II, C-III, D-IV, which corresponds to Option 1.

Q. 98 Given below are two statements:

Statement I: Electrostatic precipitator is most widely used in thermal power plant.
Statement II: Electrostatic precipitator in thermal power plant removes ionising radiations
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Boh Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I incorrect but Statement II is correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
The correct answer is Option 3) Statement I is correct but Statement II is incorrect.

Statement I: Electrostatic precipitators are indeed widely used in thermal power plants. They are used to control and reduce air pollution by removing particulate matter (such as dust and ash) from the flue gases emitted during the combustion of fossil fuels.

Statement II: However, electrostatic precipitators are not used to remove ionizing radiations in thermal power plants. Ionizing radiation refers to radiation with enough energy to remove tightly bound electrons from atoms, causing them to become charged ions. Electrostatic precipitators are not designed to capture or control ionizing radiation; their main function is the removal of particulate matter from the exhaust gases.

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Q. 99 Which of the following statements is correct?

Option 1:
Eutrophication' refers to increase in domestic sewage and waste water in lakes.

Option 2:
Biomagnification refers to increase in concentration of the toxicant at successive trophic levels.

Option 3:
Presence of large amount of nutrients in water restricts 'Algal Bloom'

Option 4:
Algal Bloom decreases fish mortality

Correct Answer:
Biomagnification refers to increase in concentration of the toxicant at successive trophic levels.

Solution:
The correct statement is:

Option 2) Biomagnification refers to an increase in the concentration of the toxicant at successive trophic levels.

Biomagnification is the process in which the concentration of certain substances, such as pollutants or toxins, increases as they move up the food chain. This phenomenon occurs because organisms at higher trophic levels consume a larger number of organisms from lower trophic levels, leading to the accumulation of the substances in their bodies.

Eutrophication (Option 1) refers to the excessive enrichment of water bodies with nutrients, typically from sources such as agricultural runoff or excessive sewage, which can lead to increased algal growth and subsequent ecosystem imbalances.

Option 3 states that the presence of large amounts of nutrients in water restricts algal bloom, which is incorrect. Algal blooms are often associated with an abundance of nutrients in the water, particularly phosphorus and nitrogen.

Option 4 suggests that algal blooms decrease fish mortality, which is also incorrect. Algal blooms can have detrimental effects on fish and other aquatic organisms due to factors such as oxygen depletion and the release of harmful algal toxins.

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Q. 100 Match **List I** with **List II.**

	List I		List II
A.	Logistic growth	I.	Unlimited resource availability condition
B.	Exponential growth	II.	The limited resource availability condition
C.	Expanding age pyramid	III.	The percent individuals of pre-reproductive age is the largest followed by reproductive and post-reproductive age groups
D.	Stable age pyramid	IV.	The percent individuals of pre-reproductives and reproductive age groups are same

Choose the **correct** from the option given below:

Option 1:
A-II, B-I, C-III, D-IV

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-I, C-III, D-IV

Solution:
A. Logistic growth corresponds to II. Limited resource availability condition. Logistic growth occurs when a population's growth rate slows down as it approaches the carrying capacity of its environment, where resources become limited.

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B. Exponential growth corresponds to I. Unlimited resource availability condition. Exponential growth occurs when a population's growth rate is not limited by resources and continues to increase exponentially over time.

C. Expanding age pyramid corresponds to III. The percent individuals of pre-reproductive age is largest followed by reproductive and post-reproductive age groups. An expanding age pyramid indicates a population with a high proportion of individuals in the younger age groups, suggesting a high birth rate and potential for future population growth.

D. Stable age pyramid corresponds to IV. The percent of individuals of pre-reproductives and reproductive age groups are the same. A stable age pyramid indicates a population with a relatively balanced distribution across different age groups, indicating a stable population size over time.

Therefore, the correct matching is A-II, B-I, C-III, D-IV.Hence, the correct answer is option 1.

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Physics

Q. 1 Given below are two statements:
Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.
Statement II: Atoms of each element are stable and emit their characteristic spectrum.
In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:
Both Statement I and Statement II are correct

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II is incorrect

Option 4:
Statement I is incorrect but Statement II is correct

Correct Answer:
Statement I is correct but Statement II is incorrect

Solution:
Statement I is true as atoms are electrically neutral because they contain equal number of positive and negative charges.
Statement II is wrong as atom of most of the elements are stable and emit characteristic spectrum. But this statement is not true for every atom.

Q. 2 If $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)\mathrm{m}$ represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are

Option 1:
5 cm, 2 s

Option 2:
5 m, 2 s

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Option 3:
5 cm, 1 s

Option 4:
5 m, 1 s

Correct Answer:
5 m, 2 s

Solution:
$x = 5\sin\left(\pi t + \frac{\pi}{3}\right)\mathrm{m}$
Amplitude = 5 m

$$\omega = \pi = \frac{2\pi}{T}$$
$$T = \frac{2\pi}{\pi} = 2\ \mathrm{s}$$

Hence, the answer is the option (2).

Q. 3 A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2 ω while keeping the same radius, the tension in the string becomes:

Option 1:
T

Option 2:
4T

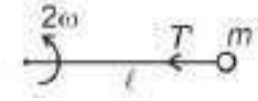
Option 3:
T/4

Option 4:
$\sqrt{2}T$

Correct Answer:
4T

Solution:

$T = m\ell\omega^2$



$T' = m\ell(2\omega)^2$
$T' = 4T$

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Q. 4 In an ideal transformer, the turns ratio is $\frac{N_P}{N_S} = \frac{1}{2}$.The ratio $V_S : V_P$ is equal to (the symbols carry their usual meaning) :

Option 1:
1:2

Option 2:
2:1

Option 3:
1:1

Option 4:
1:4

Correct Answer:
2:1

Solution:
We have

According to transformer ratio,
$\frac{V_S}{V_P} = \frac{N_S}{N_P} = 2 : 1$

Hence, the correct option is the (2).

Q. 5 A logic circuit provides the output Y as per the following truth table :

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is :

Option 1:
$A \cdot B + \bar{A}$

Option 2:
$A \cdot \bar{B} + \bar{A}$

Option 3:
$\bar{B}$

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Option 4:
B

Correct Answer:
$\bar{B}$

Solution:

A	B	Y
0 0 1 1	0 1 0 1	1 0 1 0

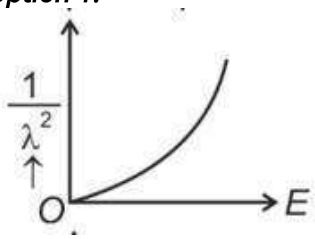
According to given truth table, output is independent on value of A

$\therefore$ Output $Y = \bar{B}$

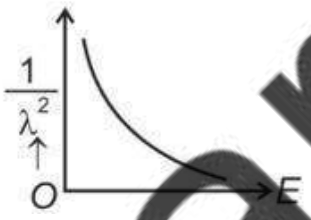
Hence, the answer is the option (3).

Q. 6 The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$

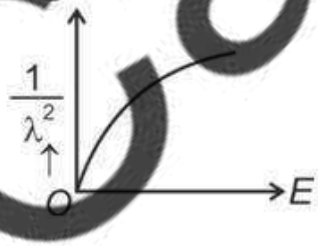
Option 1:



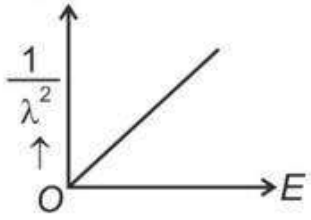
Option 2:



Option 3:



Option 4:



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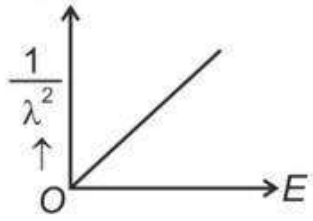
Correct Answer:



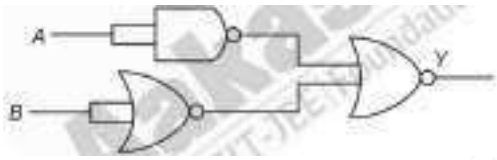
Solution:
de-Broglie wavelength $\lambda = \frac{h}{P} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$ where $E = \frac{1}{2}mv^2$

Squaring both sides,

$$\lambda^2 = \frac{h^2}{4m^2E}$$
$$\Rightarrow \frac{1}{\lambda^2} = (\text{ constant })E$$

Graph passes through origin with constant slope.

Q. 7 The output (Y) of the given logic gate is similar to the output of an/a



Option 1:
NAND gate

Option 2:
NOR gate

Option 3:
OR gate

Option 4:
AND gate

Correct Answer:
AND gate

Solution:
$Y_1 = \overline{A \cdot A}$
$= \bar{A}$
$Y_2 = \overline{B + B}$

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$$= \bar{B}$$

$$Y = \overline{Y_1 + Y_2}$$

$$= \overline{\bar{A} + \bar{B}}$$

$$= \overline{\bar{A}} \cdot \overline{\bar{B}}$$

= A.B is similar to the output of AND Gate.

Hence, the answer is the option (4).

Q. 8 In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 seconds. The moment of inertia of the needle is $9.8 \times 10^{-6}\ \mathrm{kg\ m^2}$. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}\mathrm{Am}^2$, then the value of 'x' is :

Option 1:
$5\pi^2$

Option 2:
$128\pi^2$

Option 3:
$50\pi^2$

Option 4:
$1280\pi^2$

Correct Answer:
$1280\pi^2$

Solution:

Time period of Oscillation, $T = 2\pi\sqrt{\frac{I}{MB}}$

$$\Rightarrow \frac{1}{4} = 2\pi\sqrt{\frac{9.8\times10^{-6}}{M\times0.049}}$$

$$\Rightarrow \frac{1}{16} = 4\pi^2 \times \frac{9.8\times10^{-6}}{M\times49\times10^{-3}}$$

$$\Rightarrow \quad M = \frac{4\pi^2 \times 9.8 \times 10^{-6}}{49 \times 10^{-3}} \times 16$$

$$= \frac{4\pi^2 \times 9.8 \times 16 \times 10^{-3}}{49}$$

$$= 12.8\pi^2 \times 10^{-3} \times 10^{-2} \times 10^2$$

$$= 1280\pi^2 \times 10^{-5}\mathrm{Am}^2$$

Hence, the answer is the option (4).

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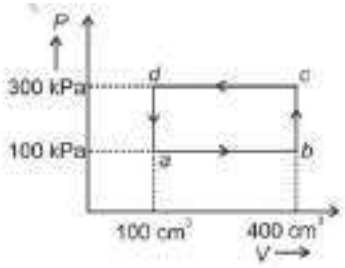
Q. 9 A thermodynamic system is taken through the cycle abcda. The work done by the gas along the path bc is:



Option 1:
Zero

Option 2:
30 J

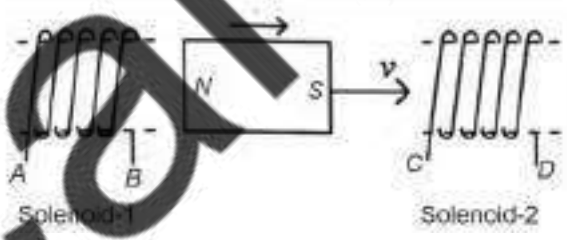
Option 3:
–90 J

Option 4:
–60 J

Correct Answer:
Zero

Solution:
Path bc is an isochoric process.
∴ Work done by gas along path bc is zero.

Q. 10



In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

Option 1:
AB and DC

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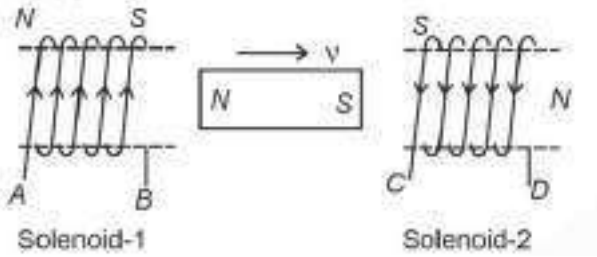
Option 2:
BA and CD

Option 3:
AB and CD

Option 4:
BA and DC

Correct Answer:
AB and DC

Solution:



North of magnet is moving away from solenoid 1 so end B of solenoid 1 is South and as south of magnet is approaching solenoid 2 so end C of solenoid 2 is South.

Q. 11 An unpolarised light beam strikes a glass surface at Brewster's angle. Then -

Option 1:
The reflected light will be partially polarised.

Option 2:
The refracted light will be completely polarised.

Option 3:
Both the reflected and refracted light will be completely polarised.

Option 4:
The reflected light will be completely polarised but the refracted light will be partially polarised.

Correct Answer:
The reflected light will be completely polarised but the refracted light will be partially polarised.

Solution:
According to Brewster's law, reflected rays are completely polarized and refracted rays are partially polarized. An unpolarized light beam is incident on a surface at an angle of incidence equal to Brewster's angle. Then, the reflected beam is polarized completely and the refracted beam is polarized partially. Also, both these beams used to be at a right angle to each other.

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Hence, the answer is the option (4).

Q. 12 A wire of length 'l' and resistance 100 Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

Option 1:
26 Ω

Option 2:
52 Ω

Option 3:
55 Ω

Option 4:
60 Ω

Correct Answer:
52 Ω

Solution:

$$R = \frac{\rho l}{A}$$

$$R' = \frac{\rho l}{10A} = \frac{R}{10}$$

When divided into 10 parts

Series resistance:

$$R_S = 5 \times \frac{R}{10} \quad [\text{series}]$$

$$R_S = 50$$

Parallel resistance:

$$R_P = \frac{R}{50} \quad [\text{parallel}]$$

$$R_{\text{eq}} = R_S + R_P = 52\Omega$$

Hence, the answer is the option (2).

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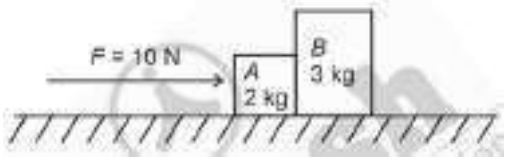
Q. 13 A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is :



Option 1:
zero

Option 2:
4N

Option 3:
6N

Option 4:
10 N

Correct Answer:
6N

Solution:
$F = (M_1 + M_2)a$

$a = \dfrac{10}{2+3} = 2\ \text{ms}^{-2}$
$F' = M_2(2) = 3 \times 2\ \text{N} = 6\ \text{N}$

Q. 14 Two bodies A and B of the same mass undergo a completely inelastic one-dimensional collision. Body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is

Option 1:
1:2

Option 2:
2:1

Option 3:
4:1

Option 4:
1:4

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Correct Answer:
2:1

Solution:
As per the question, the system undergoes a completely inelastic collision
Using conservation of linear momentum
Initial momentum = Final momentum

$$\Rightarrow mv_1 = mv_2 + mv_2$$
$$\Rightarrow mv_1 = 2mv_2$$
$$\Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$$

Hence, the answer is the option (2).

Q. 15 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**. **Assertion** A: The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude, $4 \times 10^{-6}\text{Cm}$, is $\pm 9 \times 10^3$ V.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{SI}$ units)

Reason R: $V = \pm\frac{2P}{4\pi\epsilon_0 r^2}$ where r is the distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true and R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
A is true but R is false.

Solution:

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The potential V at any point, at distance r from centre of dipole $= \frac{KP\cos\theta}{r^2}$

At axial point where $\theta = 0^\circ, V = \frac{KP}{r^2} = \frac{9\times10^9\times4\times10^{-6}}{2^2} = 9\times10^3\ \text{V}$

At axial point where $\theta = 180^\circ, V = \frac{-KP}{r^2} = -9\times10^3\ \text{V}$

Hence, the answer is the option (3).

Q. 16 Match List I with List II.

	List I **(Spectral Lines of Hydrogen for transitions from)**		**List II** **(Wavelengths (nm))**
A	$n_2 = 3$ to $n_1 = 2$	i	410.2
B	$n_2 = 4$ to $n_1 = 2$	ii	434.1
C	$n_2 = 5$ to $n_1 = 2$	iii	656.3
D	$n_2 = 6$ to $n_1 = 2$	iv	486.1

Choose the correct answer from the options given below:

Option 1:
A-II, B-I, C-IV, D-III

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:
Energy difference $\Delta E = \frac{hc}{\lambda}$

$\therefore \lambda \propto \frac{1}{\Delta E}$

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$(\Delta E)_{6-2} > (\Delta E)_{5-2} > (\Delta E)_{4-2} > (\Delta E)_{3-2}$
$\lambda_{6-2} < \lambda_{5-2} < \lambda_{4-2} < \lambda_{3-2}$

A-III, B-IV, C-II, D-I

Hence, the answer is the option (2).

Q. 17 In a vernier callipers, (N + 1) divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:

Option 1:
$\frac{1}{10N}$

Option 2:
$\frac{1}{100(N+1)}$

Option 3:
100 N

Option 4:
10(N + 1)

Correct Answer:
$\frac{1}{100(N+1)}$

Solution:
Sol. V.C = MSD – VSD .....(1)

given : (N + 1) VSD = N MSD

$\text{VSD} = \left(\frac{N}{N+1}\right)\text{MSD}$.....(2)

From (1) and (2)

$$V.C = (MSD) - \frac{N}{N+1}(MSD)$$
$$= MSD\left(1 - \frac{N}{N+1}\right) = \frac{MSD}{N+1}$$
$$= \frac{0.01}{N+1} = \frac{1}{100(N+1)}$$

Hence, the answer is the option (2).

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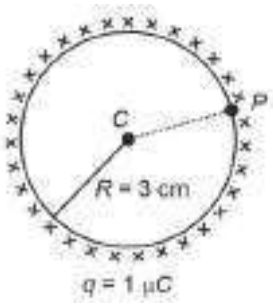
Q. 18 A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is:

(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ SI units)



Option 1:
3×10^5

Option 2:
1×10^5

Option 3:
0.5×10^5

Option 4:
zero

Correct Answer:
zero

Solution:
For uniformly charged spherical shell,

$$V = \frac{kq}{R} \quad (\text{For } r \leq R)$$
$$\therefore \quad V_C = V_P$$
$$V_C - V_P = \text{Zero}$$

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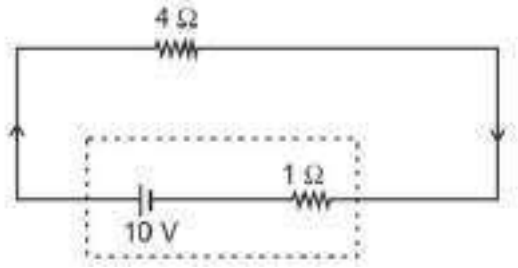
Q. 19 The terminal voltage of the battery, whose emf is 10 V and internal resistance 1Ω , when connected through an external resistance of 4 Ω as shown in the figure is:



Option 1:
4 V

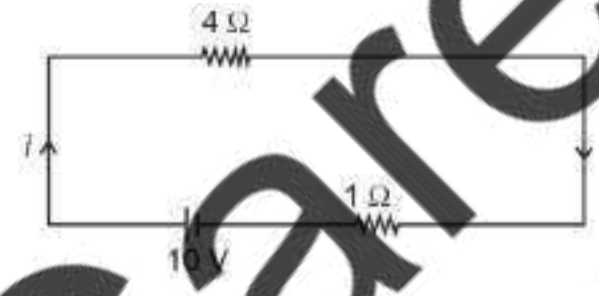
Option 2:
6 V

Option 3:
8 V

Option 4:
10 V

Correct Answer:
8 V

Solution:



Current in circuit $i = \frac{10}{4+1} = 2\ \mathrm{A}$

Terminal voltage = E – iR

= 10 – 2 × 1 = 8 V

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Q. 20 If c is the velocity of light in free space, the correct statements about photons among the following are:

A. The energy of a photon is $E = h\nu$
B. The velocity of a photon is c.
C. The momentum of a photon, $p = \frac{h\nu}{c}$
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses a positive charge.

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
A, B, C and D only

Option 3:
A, C and D only

Option 4:
A, B, D and E only

Correct Answer:
A, B, C and D only

Solution:
(A) If c is the velocity of light
so, $E = hv$ (Energy of photon)
(B) Velocity of a photon is equal to the velocity of light i.e. c.

(C) $\lambda = \frac{h}{p}$

$$p = \frac{h}{\lambda}$$
$$p = \frac{hv}{c}$$

(D) In photon-electron collision both total energy and total momentum are conserved.

Hence, the answer is the option (2).

Q. 21 A particle moving with uniform speed in a circular path maintains:

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Option 1:
Constant velocity

Option 2:
Constant acceleration

Option 3:
Constant velocity but varying acceleration

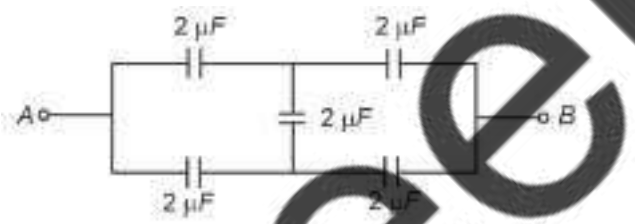
Option 4:
Varying velocity and varying acceleration

Correct Answer:
Varying velocity and varying acceleration

Solution:
A particle moving with uniform speed in a circular path maintains varying velocity and varying acceleration. It is because the direction of both velocity as well as acceleration will change continuously.

Hence, the answer is the option (4).

Q. 22 In the following circuit, the equivalent capacitance between terminal A and terminal B is :



Option 1:
$2\mu F$

Option 2:
$1\mu F$

Option 3:
$0.5\mu F$

Option 4:
$4\mu F$

Correct Answer:
$2\mu F$

Solution:

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Q 22 Image

Given circuit is balanced Wheatstone bridge

Image

$$C_{AB} = 1 + 1$$
$$= 2\mu F$$

Q. 23 A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If the surface tension of water is 0.07 N m^{-1}, then the excess force required to take it away from the surface is

Option 1:
19.8 mN

Option 2:
198 mN

Option 3:
1.98 mN

Option 4:
99 mN

Correct Answer:
19.8 mN

Solution:
Excess force $= T \times 2\pi R$

$= \frac{7}{100} \times 2 \times 3.14 \times \frac{4.5}{100}$

$= 197.82 \times 10^{-4}$
$= 19.8 \times 10^{-3}\ \mathrm{N}$
$= 19.8\mathrm{mN}$

Hence, the answer is the option (1).

Q. 24 The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus,respectively, are $8 \times 10^{8}\ \mathrm{N\ m^{-2}}$ and $2 \times 10^{11}\ \mathrm{N\ m^{-2}}$, is :

Option 1:
4 mm

Option 2:
0.4 mm

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Option 3:
40 mm

Option 4:
8 mm

Correct Answer:
4 mm

Solution:
In the case for maximum elongation,
Stress = Elastic limit

$$\Delta l_{max} = \frac{stress_{\text{elastic}} \times L}{\text{Young's modulus}} = \frac{8 \times 10^8 \times 1}{2 \times 10^{11}} = 4 \times 10^{-3} = 4 \text{ mm}$$

i.e. maximum elongation is 4 mm

Hence, the answer is the option (1).

Q. 25 $^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$

In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are

Option 1:
280, 81

Option 2:
286, 80

Option 3:
288, 82

Option 4:
286, 81

Correct Answer:
286, 81

Solution:
image

$A \rightarrow 286$
$Z = 81$

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Q. 26 At any instant of time t, the displacement of any particle is given by 2t – 1 (SI unit) under the influence of the force of 5 N. The value of instantaneous power is (in SI units):

Option 1:
10

Option 2:
5

Option 3:
7

Option 4:
6

Correct Answer:
10

Solution:
$x = 2t - 1$

$v = \frac{dx}{dt} = 2\ \mathrm{m\ s^{-1}}$

P = F. v = 2 × 5 = 10 W

Hence, the answer is the option (1).

Q. 27 The quantities which have the same dimensions as those of solid angle are:

Option 1:
strain and angle

Option 2:
stress and angle

Option 3:
strain and arc

Option 4:
angular speed and stress

Correct Answer:
strain and angle

Solution:

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Solid angle $d\Omega = \frac{dA}{r^2}$ has dimensions $\left[\mathrm{M}^0\ \mathrm{L}^0\ \mathrm{T}^0\right]$

Strain $= \frac{\Delta l}{l}$ has dimensions $\left[\mathrm{M}^0\ \mathrm{L}^0\ \mathrm{T}^0\right]$

Angle measured in radians is also dimensionless $\left[\mathrm{M}^0\ \mathrm{L}^0\ \mathrm{T}^0\right]$

$\theta = \frac{l}{r}$

Hence, the answer is the option (1).

Q. 28 The moment of inertia of a thin rod about an axis passing through its midpoint and perpendicular to the rod is 2400 g cm^2. The length of the 400 g rod is nearly:

Option 1:
8.5 cm

Option 2:
17.5 cm

Option 3:
20.7 cm

Option 4:
72.0 cm

Correct Answer:
8.5 cm

Solution:
Moment of inertia of rod $= I = \frac{m\ell^2}{12}$

$$\Rightarrow \quad 2400 = 400\frac{\ell^2}{12}$$
$$\Rightarrow \quad 72 = \ell^2$$
$$\Rightarrow \quad \ell = \sqrt{72} = 8.48\ \mathrm{cm} \simeq 8.5\ \mathrm{cm}$$

Hence, the answer is the option (1).

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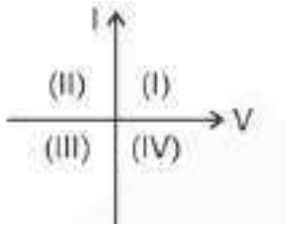
Q. 29 Consider the following statements A and B and identify the correct answer:



A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.
B. In a reverse biased pn junction diode, the current measured in (μA), is due to majority charge carriers.

Option 1:
A is correct but B is incorrect

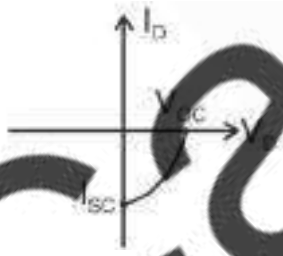
Option 2:
A is incorrect but B is correct

Option 3:
Both A and B are correct

Option 4:
Both A and B are incorrect

Correct Answer:
A is correct but B is incorrect

Solution:
A: Solar cell characteristics



B: In reverse biased pn junction diode, the current measured in (μA), is due to minority charge carrier.

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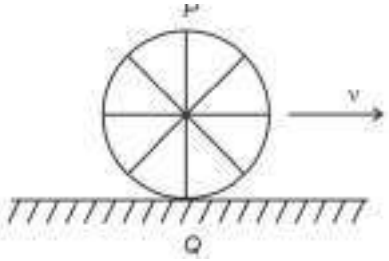
Q. 30 A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?



Option 1:
Point P moves slower than point Q

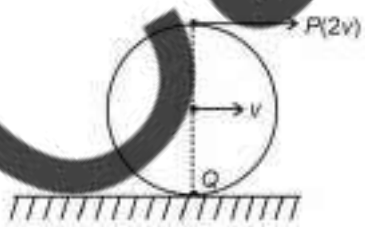
Option 2:
Point P moves faster than point Q

Option 3:
Both the points P and Q move with equal speed

Option 4:
Point P has zero speed

Correct Answer:
Point P moves faster than point Q

Solution:
In the case of pure rolling,



The topmost point will have velocity 2v while point Q i.e. lowest point will have zero velocity.
Hence point P moves faster than point Q.

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Q. 31 A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

Option 1:
44 mT

Option 2:
4.4 T

Option 3:
4.4 mT

Option 4:
44 T

Correct Answer:
4.4 mT

Solution:
The magnitude of magnetic field due to circular coil of N turns is given by

$$B_C = \frac{\mu_0 i N}{2R}$$

$$= \frac{4\pi \times 10^{-7} \times 7 \times 100}{2 \times 0.1}$$
$$= 4.4 \times 10^{-3}\ \mathrm{T}$$
$$= 4.4\mathrm{mT}$$

Hence, the answer is the option (3).

Q. 32 If the monochromatic source in Young's double slit experiment is replaced by white light, then

Option 1:
Interference pattern will disappear

Option 2:
There will be a central dark fringe surrounded by a few coloured fringes

Option 3:
There will be a central bright white fringe surrounded by a few coloured fringes

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Option 4:
All bright fringes will be of equal width

Correct Answer:
There will be a central bright white fringe surrounded by a few coloured fringes

Solution:
At central point on screen, path difference is zero for all wavelength. So, central bright fringe is white and other fringes depend on wavelength as $\beta = \frac{\lambda D}{d}$.

Therefore, other fringes will be coloured.

Q. 33 **Match List-I with List-II.**

	List-I (Material)		List-I (Susceptibility (χ))
A	Diamagnetic	**I**	$\chi = 0$
B	Ferromagnetic	**II**	$0 > \chi \geq -1$
C	Paramagnetic	**III**	$\chi >> 1$
D	Non-magnetic	**IV**	$0 < \chi < \varepsilon$(a small positive number)

Choose the correct answer from the options given below

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

(Material)		(Susceptibility (χ))
Diamagnetic	**II**	$0 > \chi \geq -1$

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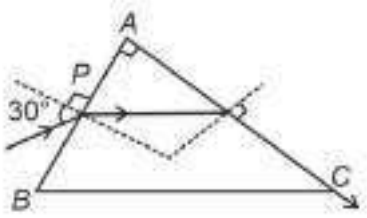
Ferromagnetic	**IIi**	$\chi >> 1$
Paramagnetic	**IV**	$0 < \chi < \varepsilon$
Non-magnetic	**I**	$\chi = 0$

Hence, the answer is the option (1).

Q. 34 A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is:



Option 1:
$\frac{\sqrt{5}}{4}$

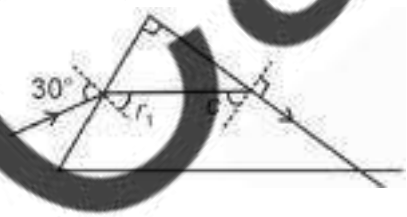
Option 2:
$\frac{\sqrt{5}}{2}$

Option 3:
$\frac{\sqrt{3}}{4}$

Option 4:
$\frac{\sqrt{3}}{2}$

Correct Answer:
$\frac{\sqrt{5}}{2}$

Solution:



A = 90°

In prism,
$r_1 + c = A$
$r_1 = 90^\circ - c$ .....(1)

$\sin c = \frac{1}{\mu} \Rightarrow \cos c = \frac{\sqrt{\mu^2 - 1}}{\mu}$

$\Rightarrow$ Apply Snell's law, on incidence surface

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$1 \cdot \sin 30^\circ = \mu \sin(r_1) \Rightarrow 1 \times \frac{1}{2} = \mu \times \sin(90^\circ - c)$

$\frac{1}{2} = \mu \times \frac{\sqrt{\mu^2 - 1}}{\mu}$

On squaring $\frac{1}{4} = \mu^2 - 1$

$\Rightarrow \mu^2 = \frac{5}{4} \Rightarrow \mu = \frac{\sqrt{5}}{2}$

Q. 35 The mass of a planet is $\frac{1}{10}$ th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is:

Option 1:
$19.6\ \mathrm{m\ s^{-2}}$

Option 2:
$9.8\ \mathrm{m\ s^{-2}}$

Option 3:
$4.9\ \mathrm{m\ s^{-2}}$

Option 4:
$3.92\ \mathrm{m\ s^{-2}}$

Correct Answer:
$3.92\ \mathrm{m\ s^{-2}}$

Solution:
$g' = \frac{GM'}{R'^2} = \frac{GM}{10\left(\frac{R}{2}\right)^2}$

$= \frac{4}{10}\frac{GM}{R^2} = 0.4 \times 9.8$

= 3.92 m s^{-2}

Hence, the answer is the option (4).

Q. 36 The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of 2R from the surface of the earth is:

Option 1:
$\frac{5GmM}{6R}$

Option 2:
$\frac{2GmM}{3R}$

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Option 3:
$\frac{GmM}{2R}$

Option 4:
$\frac{GmM}{2R}$

Correct Answer:
$\frac{2GmM}{3R}$

Solution:
Apply energy conservation,

$U_i + K_i = U_f + K_f$

$\Rightarrow \quad -\frac{GMm}{R} + K_i = -\frac{GMm}{3R} + \frac{1}{2}mv^2$

$\Rightarrow \quad -\frac{GMm}{R} + K_i = -\frac{GMm}{3R} + \frac{1}{2} \times m \times \frac{GM}{3R}$

$\Rightarrow \quad K_i = -\frac{1}{6}\frac{GMm}{R} + \frac{GMm}{R}$

$K_i = \frac{5}{6}\frac{GMm}{R}$

Q. 37 A metallic bar of Young's modulus, $0.5 \times 10^{11}\ \mathrm{N\,m^{-2}}$ and coefficient of linear thermal expansion $10^{-5}{}^{\circ}\mathrm{C}^{-1}$, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is :

Option 1:
5 × 10^3 N

Option 2:
50 × 10^3 N

Option 3:
100 × 10^3 N

Option 4:
2 × 10^3 N

Correct Answer:
50 × 10^3 N

Solution:
Thermal strain = Longitudinal strain $= \alpha\Delta T$

$\Rightarrow$ Longitudinal strain, $\delta = 10^{-5} \times 10^2 = 10^{-3}$
$\Rightarrow$ Compressive stress $= \delta \times$ Young's Modulus

$$= 10^{-3} \times 0.5 \times 10^{11}$$
$$= 0.5 \times 10^8$$

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\Rightarrow \quad \text { Compressive force }=0.5 \times 10^8 \times 10^{-3}=0.5 \times 10^5

$= 5 \times 10^4 \times \frac{10}{10}$

= 50 × 10^3 N

Q. 38 A small telescope has an objective of focal length 140 cm and an eyepiece of focal length 5.0 cm. The magnifying power of a telescope for viewing a distant object is:

Option 1:
34

Option 2:
28

Option 3:
17

Option 4:
32

Correct Answer:
28

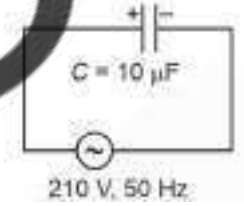
Solution:
$f_0 = 140 \text{ cm}$ and $f_e = 5 \text{ cm}$

For distant object,

$m = \frac{f_0}{f_e} = \frac{140}{5} = 28$

Hence, the answer is the option (2).

Q. 39 A $10\mu\text{F}$ capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is
nearly $(\pi = 3.14)$:



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Option 1:
0.58 A

Option 2:
0.93 A

Option 3:
1.20 A

Option 4:
0.35 A

Correct Answer:
0.93 A

Solution:
Capacitive Reactance $x_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\times 3.14\times 50\times 10\times 10^{-6}}$

$= \frac{1000}{3.14}$

$V_{\text{rms}} = 210\ \text{V}$

$i_{\text{rms}} = \dfrac{V_{\text{rms}}}{X_C} = \dfrac{210}{X_C}$

Peak current $= \sqrt{2} i_{\text{rms}} = \sqrt{2} \times \frac{210}{1000} \times 3.14 = 0.932$

$\simeq 0.93\ \text{A}$

Q. 40 If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

Option 1:
$\sqrt{3}$

Option 2:
$\sqrt{2}$

Option 3:
$2\sqrt{3}$

Option 4:
4

Correct Answer:
$\sqrt{2}$

Solution:

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$T' = 2\pi\sqrt{\frac{\ell'}{g}}$ where $\ell' = \frac{\ell}{2}$

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T' = \frac{x}{2}T$$

$$2\pi\sqrt{\frac{\ell}{2g}} = \frac{x}{2}2\pi\sqrt{\frac{\ell}{g}}$$

$$\frac{1}{\sqrt{2}} = \frac{x}{2} \Rightarrow x = \sqrt{2}$$

Hence, the answer is the option (2).

Q. 41 The property which is not of an electromagnetic wave travelling in free space is that:

Option 1:
They are transverse in nature

Option 2:
The energy density in electric field is equal to energy density in magnetic field

Option 3:
They travel with a speed equal to $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$

Option 4:
They originate from charges moving with uniform speed

Correct Answer:
They originate from charges moving with uniform speed

Solution:
The EM waves originate from an accelerating charge. The charge moving with uniform velocity produces a steady-state magnetic field.

Hence, the answer is the option (4).

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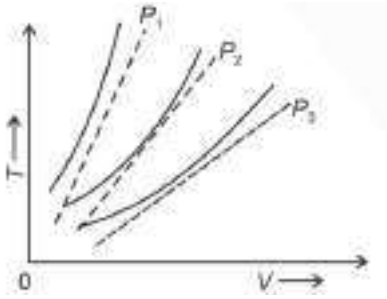
Q. 42 The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.



Then the correct relation is:

Option 1:
$P_3 > P_2 > P_1$

Option 2:
$P_1 > P_3 > P_2$

Option 3:
$P_2 > P_1 > P_3$

Option 4:
P1 > P2 > P3

Correct Answer:
P1 > P2 > P3

Solution:
At same temperature, curve with higher volume corresponds to lower pressure.

$$V_3 > V_2 > V_1$$
$$\Rightarrow P_1 > P_2 > P_3$$

(We draw a straight line parallel to volume axis to get this)

Q. 43 A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is:

Option 1:
$\beta t/\alpha$

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Option 2:
$\alpha t/\beta$

Option 3:
$\alpha\beta t$

Option 4:
$\alpha\beta/t$

Correct Answer:
$\alpha t/\beta$

Solution:
From principle of homogeneity

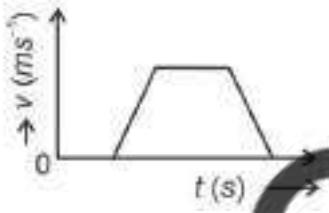
$[F] = \left[\alpha t^2\right] = [\beta t]$

$[\alpha] = \frac{[F]}{[t^2]}$ and $[\beta] = \frac{[F]}{[t]}$

$\therefore \quad [\alpha][t] = [\beta]$

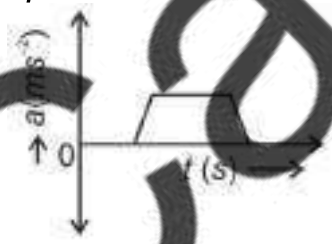
$\therefore \quad \frac{\alpha t}{\beta} =$ dimensionless

Q. 44 The velocity (v) – time (t) plot of the motion of a body is shown below:

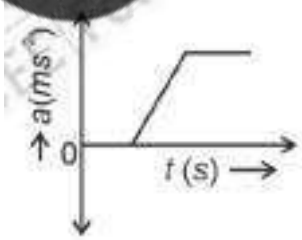


The acceleration (a) – time (t) graph that best suits this motion is :

Option 1:



Option 2:



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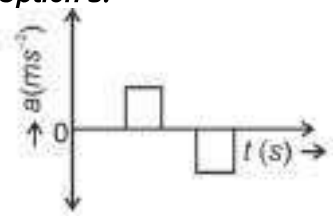
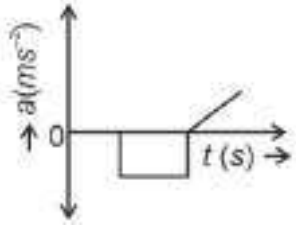
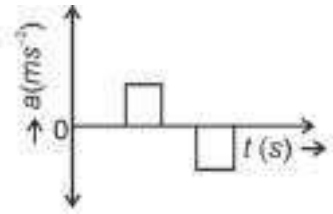
Option 3:



Option 4:

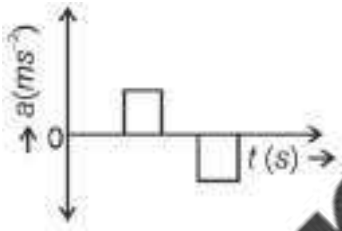


Correct Answer:



Solution:

Sol. Initially, the body has zero velocity and zero slope. Hence the acceleration would be zero initially.
After that, the slope of v-t curve is constant and positive.
After some time, velocity becomes constant and acceleration is zero.
After that, the slope of v-t curve is constant and negative.



Q. 45 Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

Option 1:

1:1

Option 2:

2:9

Option 3:

1:2

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Option 4:
2:3

Correct Answer:
2:9

Solution:
Power Consumed $= P = \frac{V^2}{R}$

$$\frac{P_A}{P_B} = \frac{R_B}{R_A}$$
$$R_A = 2R_B$$

For Series Combination

$$P_S = \frac{V^2}{3R_B}$$

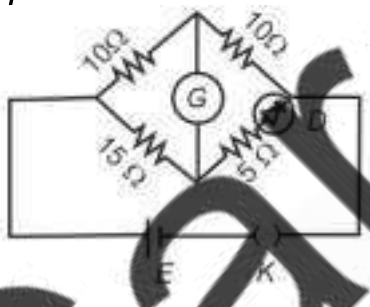
For Parallel Combination

$$P_P = \frac{3V^2}{2R_B}$$
$$\frac{P_S}{P_P} = \frac{2}{9}$$

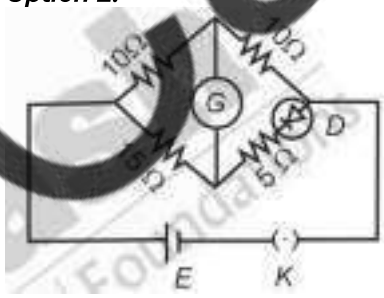
Hence, the answer is option (2).

Q. 46 Choose the correct circuit which can achieve the bridge balance.

Option 1:



Option 2:



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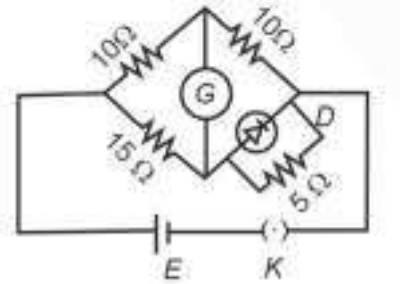
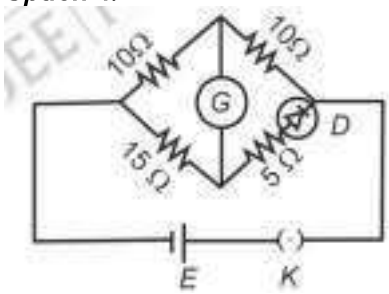
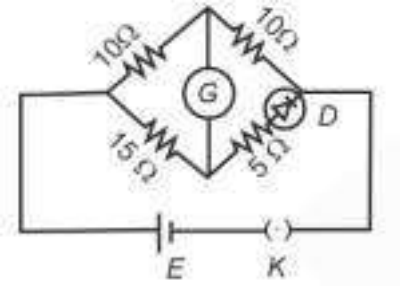
Option 3:



Option 4:



Correct Answer:



Solution:
In option (1),

$\frac{10}{15} = \frac{10}{5+R_D}$

The diode can conduct and have resistance $R_D = 10\Omega$ because diode have dynamic resistance. In that case bridge will be balanced.

Q. 47 A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to:

A. hold the sheet there if it is magnetic.
B. hold the sheet there if it is non-magnetic.
C. move the sheet away from the pole with uniform velocity if it is conducting.
D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.
Choose the correct statement(s) from the options given below:

Option 1:
B and D only

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Option 2:
A and C only

Option 3:
A, C and D only

Option 4:
C only

Correct Answer:
A and C only

Solution:
A. A magnetic pole will repel or attract magnetic sheet so force is need.
B. If sheet is non-magnetic, no force needed.
C. If it is conducting, then there will be addy current in sheet, which opposes the motion. So forces is needed move sheet with uniform speed.
D. The non-conducting and non-polar sheet do not interact with magnetic field of magnet.

Q. 48 If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then
A. the charge stored in it, increases.
B. the energy stored in it, decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.

Choose the most appropriate answer from the options given below:

Option 1:
A, B and E only

Option 2:
A, C and E only

Option 3:
B, D and E only

Option 4:
A, B and C only

Correct Answer:
A, C and E only

Solution:
Given $V' = V$ = Constant

(i) $C' = \frac{\varepsilon_0 A}{d'}, C = \frac{\varepsilon_0 A}{d}$

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$d' < d$
$C' > C$

Hence, final capacitance greater than initial capacitance,

(ii) $U' = \frac{1}{2}C'V^2$

(ii) $U' = \frac{1}{2}CV^2$

$U' > U$
Hence final energy is greater than initial energy

(iii) $\frac{Q'}{V'} = C'$ and $\frac{Q}{V} = C$

$\frac{Q'}{V'} \neq \frac{Q}{V}$
(iv) Product of charge and voltage

$X' = Q'V = C'V^2$
$X = QV = CV^2$
$X' > X$

Q. 49 An iron bar of length L has magnetic moment M. It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is :

Option 1:
M

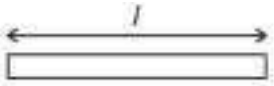
Option 2:
$\frac{M}{2}$

Option 3:
2M

Option 4:
$\frac{M}{\sqrt{3}}$

Correct Answer:
$\frac{M}{2}$

Solution:



$M = ml$

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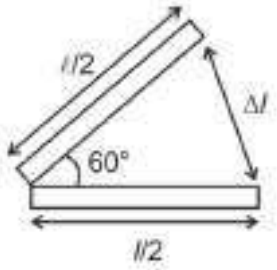

$\Delta I = 2\frac{I}{2}\sin 30^\circ$

$= \frac{1}{2}$

$M' = m//2$
$= M/2$

Q. 50 A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates:

Option 1:
There is no current

Option 2:
Displacement current of magnitude equal to I flows in the same direction as I

Option 3:
Displacement current of magnitude equal to I flows in a direction opposite to that of I

Option 4:
Displacement current of magnitude greater than I flows but can be in any direction

Correct Answer:
Displacement current of magnitude equal to I flows in the same direction as I

Solution:

According to modified Ampere's law

$\oint B.dI = \mu_0 (I_C + I_D)$

For Loop L_1 $I_C \neq 0$ and $I_D = 0$

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For Loop L_2 $I_C = 0$ and $I_D \neq 0$

Due to KCL $I_C = I_D$

Chemistry

Q. 1 On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as

Option 1:
Crystallization

Option 2:
Sublimation

Option 3:
Distillation

Option 4:
Chromatography

Correct Answer:
Sublimation

Solution:
(1) **Crystallization :** It is based on difference in the solubilities of the compound and impurities in a suitable solvent.
(2) **Sublimation :** It is the purification technique based on principle that on heating, some solid substances change from solid to vapour state without passing through liquid state.
(3) **Distillation :** It is used to separate volatile liquids from non-volatile impurities and the liquids having sufficient difference in their boiling point.
(4) **Chromatography :** It is based on separation by using stationary and mobile phase.

Hence, the answer is the option (2).

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Q. 2 Match List I with List II.

List-I (Process)	List -II (Conditions)
A.Isothermal process	I. No heat exchange
B.Isochoric process	II.Carried out at constant temperature
C.Isobaric process	III.Carried out at constant volume
D.Adiabatic process	IV.Carried out at constant pressure

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-IV, B-II, C-III, D-I

Option 3:
A-I, B-II, C-III, D-IV

Option 4:
A-II, B-III, C-IV, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
(A) Isothermal process $\Rightarrow$ Temperature is constant throughout the process
(B) Isochoric process $\Rightarrow$ Volume is constant throughout the process
(C) Isobaric process $\Rightarrow$ Pressure is constant throughout the process
(D) Adiabatic process $\Rightarrow$ No exchange of heat (q) between system and surrounding

Hence, the answer is the option (4).

Q. 3 In which of the following equilibria, $\mathrm{K_p}$ and $\mathrm{K_c}$ are NOT equal?

Option 1:
$\mathrm{PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}}$

Option 2:
$\mathrm{H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}}$

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Option 3:

$\mathrm{CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}}$

Option 4:

$\mathrm{2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}}$

Correct Answer:

$\mathrm{PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}}$

Solution:

$\mathrm{K_p = K_c(RT)^{\Delta n_g}}$

$\mathrm{for\ K_p \neq K_c}$

$\mathrm{\Delta n_g \neq 0}$

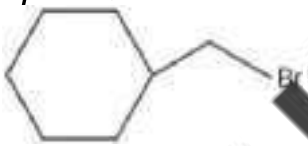
$\mathrm{\Delta n_g = n_p - n_r}$

(1) $\mathrm{\Delta n_g = 2 - 1 = 1}$
(2) $\mathrm{\Delta n_g = 2 - 2 = 0}$
(3) $\mathrm{\Delta n_g = 2 - 2 = 0}$
(4) $\Delta n_g = 2 - 2 = 0$

Hence, the answer is the option (1).

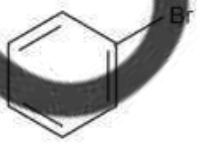
Q. 4 The compound that will undergo S_N1 reaction with the fastest rate is

Option 1:



Option 2:

Option 3:



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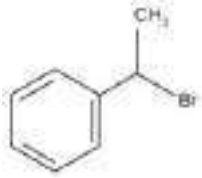
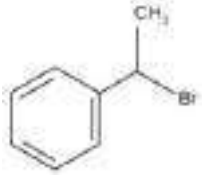
Option 4:

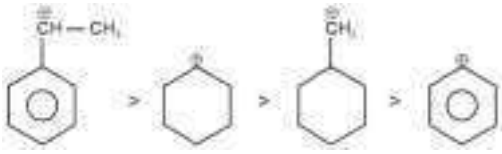


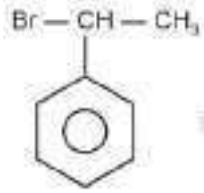
Correct Answer:



Solution:
Reactivity towards S_N1 depends upon stability of carbocation.

Order of stability is



Hence  is most reactive.

Q. 5 A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is :

Option 1:
n-hexane

Option 2:
2-methylpentane

Option 3:
2,3-dimethylbutane

Option 4:
2,2-dimethylbutane

Correct Answer:
2,3-dimethylbutane

Solution:
$CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - CH_3$ has no tertiary carbon

(n-Hexane)

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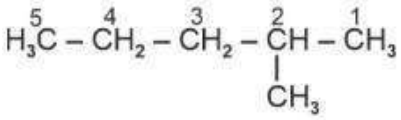
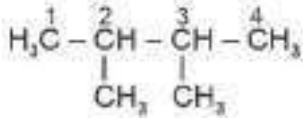
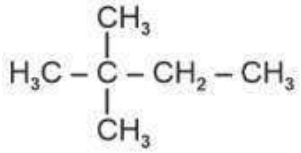
 has only one tertiary carbon

(2,3-Methylpentane)

 has two tertiary carbon.

(2,3-Dimethylbutane)

 has no tertiary carbon

(2,2-Dimethylbutane)

Q. 6 Given below are two statements:

Statement I: The boiling point of hydrides of Group 16 elements follow the order

$\mathrm{H_2O > H_2Te > H_2Se > H_2\ S}$.

Statement II: On the basis of molecular mass, $\mathrm{H_2O}$ is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in $\mathrm{H_2O}$, it has higher boiling point.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:

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Statement I is correct, because boiling point of hydrides of group 16 follows the order

$\mathrm{H_2O > H_2Te > H_2Se > H_2\ S.}$

Statement II due to intermolecular H-bonding $\mathrm{H_2O}$ shows higher boiling point than respective hydrides of group 16.
(Both Statement are true)
Order from $\mathrm{H_2Te}$ to $\mathrm{H_2\ S}$ is due to decreasing molar mass.

Hence, the answer is the option (1).

Q. 7 For the reaction $\mathrm{2\ A \rightleftharpoons B + C, K_C = 4 \times 10^{-3}}$. At a given time, the composition of reaction mixture is:

$[A] = [B] = [C] = 2 \times 10^{-3}\mathrm{M}.$

Then, which of the following is correct?

Option 1:
Reaction is at equilibrium.

Option 2:
Reaction has a tendency to go in forward direction.

Option 3:
Reaction has a tendency to go in backward direction.

Option 4:
Reaction has gone to completion in forward direction.

Correct Answer:
Reaction has a tendency to go in backward direction.

Solution:
$\mathrm{2\ A \rightleftharpoons B + C, K_C = 4 \times 10^{-3}}$

At a given time t, $\mathrm{Q_c}$ is to be calculated and been compared with $\mathrm{K_c}$.

$$Q_C = \frac{[B][C]}{[A]^2} = \frac{\left(2 \times 10^{-3}\right)\left(2 \times 10^{-3}\right)}{\left(2 \times 10^{-3}\right)^2}$$

$$Q_C = 1$$

As $Q_c > K_c$, so reaction has a tendency to move backward.

Hence, the answer is the option (3).

Q. 8 Activation energy of any chemical reaction can be calculated if one knows the value of

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Option 1:
rate constant at standard temperature

Option 2:
probability of collision

Option 3:
orientation of reactant molecules during collision

Option 4:
rate constant at two different temperatures

Correct Answer:
rate constant at two different temperatures

Solution:
To calculate value of $\mathrm{E_a}$
Equation used is

$$\log\left(\frac{k_2}{k_1}\right)=\frac{E_a}{2.303R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

Hence E_a can be calculated if value of rate constant k is known at two different temperatures T_1 and T_2

Hence, the answer is the option (4).

Q. 9 Given below are two statements :

Statement I: Both $[\mathrm{Co(NH_3)_6}]^{3+}$ and $[\mathrm{CoF_6}]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.
Statement II: $[\mathrm{Co(NH_3)_6}]^{3+}$ is diamagnetic whereas $[\mathrm{CoF_6}]^{3-}$ is paramagnetic.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

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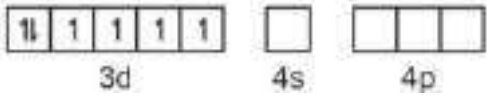
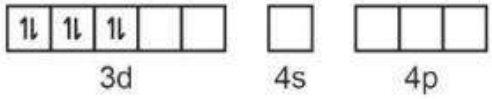
Solution:
In $[Co(NH_3)_6]^{3+}$, Co^{3+} ion is having $3\ d^6$ configuration.

Electronic configuration of Co^{3+} :



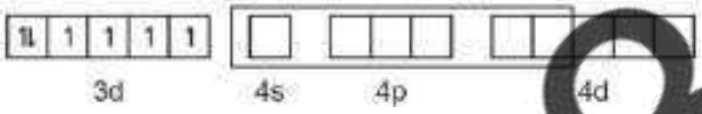
In presence of NH_3 ligand, pairing of electrons takes place and it becomes diamagnetic complex ion.

In presence of NH_3 ligand:



$\therefore \quad [Co(NH_3)_6]^{3+}$ is octahedral with d^2sp^3 hybridisation and it is diamagnetic in nature.
In case of $[CoF_6]^{3-}$, Co is in +3 oxidation state and it is having $3\ d^6$ configuration.
In presence of weak field F^-ligand, pairing does not take place.

In presence of F– ligands :



$\therefore \quad In\ [CoF_6]^{3-}, Co^{3+}$ is $sp^3\ d^2$ hybridised with four unpaired electrons, so it is paramagnetic in nature.

Q. 10 The highest number of helium atoms is in

Option 1:
4 mol of helium

Option 2:
4 u of helium

Option 3:
4 g of helium

Option 4:
2.271098 L of helium at STP

Correct Answer:
4 mol of helium

Solution:
(1) $4\ mol\ of\ He = 4NA_AHe$ atoms

(2) $4u\ of\ He = \frac{4u}{4u} = 1He$ atom

(3) $4\ g$ of Helium $= \frac{4\ g}{4\ g}$ mole $= 1$ mole $= N_A$ He atom

(4)

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$$
\begin{aligned}
2.2710982 \text{ of He at STP } &= \frac{2.271}{22.710982} \text{ mole} \\
&= 0.1 \text{ mole} \\
&= 0.1 \text{ } N_A \text{He atom}
\end{aligned}
$$

Hence, the answer is the option (1).

Q. 11 Arrange the following elements in increasing order of first ionization enthalpy:

Li, Be, B, C, N

Choose the correct answer from the options given below:

Option 1:
Li < Be < B < C < N

Option 2:
Li < B < Be < C < N

Option 3:
Li < Be < C < B < N

Option 4:
Li < Be < N < B < C

Correct Answer:
Li < B < Be < C < N

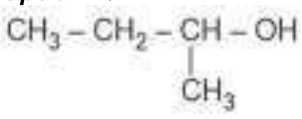
Solution:
The increasing order of the first ionization enthalpy is Li < B < Be < C < N

Element	First ionization enthalpy $\left(\Delta_i H / kJ mol^{-1}\right)$
Li	520
Be	899
B	801
C	1086
N	1402

Hence, the answer is the option (2).

Q. 12 Which one of the following alcohols reacts instantaneously with Lucas reagent?

Option 1:
$CH_3 - CH_2 - CH_2 - CH_2OH$

Option 2:


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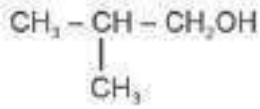
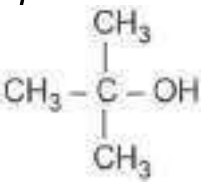
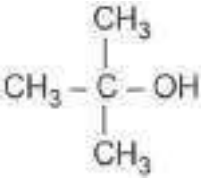
Option 3:



Option 4:



Correct Answer:



Solution:

Tertiary alcohols react instantaneously with Lucas reagent and gives immediate turbidity. In case of tertiary alcohols, they form halides easily with Lucas reagent (conc. HCl and $ZnCl_2$)

Q. 13 'Spin only' magnetic moment is same for which of the following ions?

A. Ti^{3+}

B. Cr^{2+}

C. $\mathrm{Mn}^{2+}

D. Fe^{2+}

E. Sc^{3+}

Choose the most appropriate answer from the options given below.

Option 1:

B and D only

Option 2:

A and E only

Option 3:

B and C only

Option 4:

A and D only

Correct Answer:

B and D only

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Solution:

Ions	No.of unpaired electrons	Configuration
Ti^{3+}	1	$3d^1$
Cr^{2+}	4	$3d^4$
Mn^{2+}	5	$3d^5$
Fe^{2+}	4	$3d^6$
Sc^{3+}	0	$3d^0$

Spin only magnetic moment is given by $\sqrt{n(n+2)}BM$

$\therefore$ Cr^{2+} and Fe^{2+} will have same spin only magnetic moment.

Hence, the answer is the option (1).

Q. 14 The reagents with which glucose does not react to give the corresponding tests/products are

A. Tollen's reagen

B. Schiff's reagent

C. HCN

D. NH_2OH

E. $NaHSO_3$

Choose the correct options from the given below:

Option 1:
B and C

Option 2:
A and D

Option 3:
B and E

Option 4:
E and D

Correct Answer:
B and E

Solution:
Despite having the aldehyde group glucose does not give Schiff's test and it does not form the hydrogen sulphite addition product with $NaHSO_3$.

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Q. 15 Given below are two statements:

Statement I : Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is correct but Statement II is false

Option 4:
Statement I is incorrect but Statement II is true

Correct Answer:
Both statement I and Statement II are true

Solution:
• Aniline does not undergo Friedel-Crafts alkylation reaction due to salt formation with aluminium chloride, the Lewis acid, which is used as a catalyst.

• Aniline (aromatic primary amine) cannot be prepared by Gabriel phthalimide synthesis because aryl halides do not undergo nucleophilic substitution with anion formed by phthalimide.

Hence, the answer is the option (1).

Q. 16 The energy of an electron in the ground state $(n = 1)$ for He^{+}ion is $-x$ J, then that for an electron in $n = 2$ state for Be^{3+} ion in J is

Option 1:
$-\mathrm{X}$

Option 2:
$-\frac{x}{9}$

Option 3:
$-4\mathrm{x}$

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Option 4:
$-\frac{4}{9}x$

Correct Answer:
$-\mathrm{X}$

Solution:
$E_n = -R_H\left(\frac{Z^2}{n^2}\right)J$

For $\mathrm{He}^+(n=1)$,

$$E_n = -x = -R_H\left(\frac{2^2}{1^2}\right) = -4R_H$$

$$\therefore \quad R_H = \frac{x}{4}$$

For $B^{3+}(n=2)$

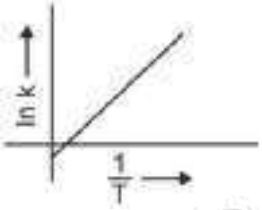
$$E_n = -R_H\left(\frac{Z^2}{n^2}\right)J$$

$$= -\frac{x}{4} \times \left(\frac{4\times 4}{2\times 2}\right) = - \times \mathrm{J}$$

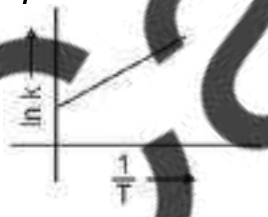
Hence, the answer is the option (1).

Q. 17 Which plot of ln k vs $\frac{1}{\mathrm{T}}$ is consistent with Arrhenius equation?

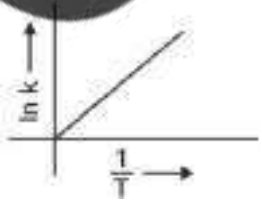
Option 1:



Option 2:



Option 3:



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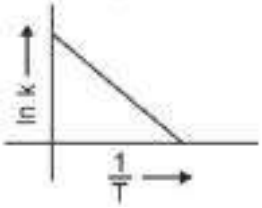
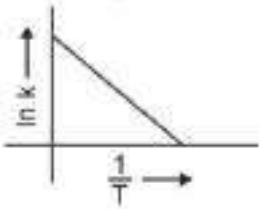
Option 4:



Correct Answer:

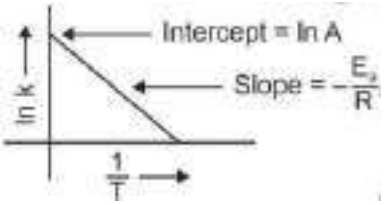


Solution:
The Arrhenius equation is given as

$$\mathrm{k} = \mathrm{Ae} - \frac{\mathrm{E_a}}{\mathrm{RT}}$$

$$\therefore \ln \mathrm{k} = \ln \mathrm{A} - \frac{\mathrm{E_a}}{\mathrm{RT}}$$

In $kv/s\,\frac{1}{T}$ gives a straight line graph with slope $= -\frac{E_a}{R}$ and intercept $= \ln A$



Q. 18 Given below are two statements:

Statement I : The boiling point of three isomeric pentanes follows the order n-pentane > isopentane > neopentane

Statement II : When branching increases, the molecule attains a shape of sphere. This results in smaller surface area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are correct

Option 2:
Both Statement I and Statement II are incorrect

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Option 3:
Statement I is correct but Statement II is incorrect

Option 4:
Statement I is incorrect but Statement II is correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:
Both statement I and statement II are correct.
Boiling point of n-pentane = 309 K
isopentane = 301 K
neopentane = 282.5

As branching increases molecules attain the shape of a sphere results in smaller area of contact thus weak intermolecular forces between spherical molecules, which are overcome at relatively lower temperature. Leading to decrease in boiling point.

Q. 19 The E° value for the $\mathrm{Mn}^{3+}/\mathrm{Mn}^{2+}$ couple is more positive than that of $\mathrm{Cr}^{3+}/\mathrm{Cr}^{2+}$ or $\mathrm{Fe}^{3+}/\mathrm{Fe}^{2+}$ due to change of

Option 1:
d^5 to d^4 configuration

Option 2:
d^5 to d^2 configuration

Option 3:
d^4 to d^5 configuration

Option 4:
d^3 to d^5 configuration

Correct Answer:
d^4 to d^5 configuration

Solution:
$\mathrm{E}^{0}_{\mathrm{Mn}^{3+}/\mathrm{Mn}^{2+}} > \mathrm{E}^{0}_{\mathrm{Cr}^{3+}/\mathrm{Cr}^{2+}}$ or $\mathrm{E}^{\circ}_{\mathrm{Fe}^{3+}//\mathrm{Fe}^{2+}}$

Electronic configuration of $\mathrm{Mn}^{3+} = [Ar]3d^4$
Electronic configuration of $\mathrm{Mn}^{2+} = [\mathrm{Ar}]3d^5$
Electronic configuration of $\mathrm{Cr}^{3+} = [\mathrm{Ar}]3d^{\beta}$
Electronic configuration of $\mathrm{Cr}^{2+} = [\mathrm{Ar}]3d^4$

As Mn^{3+} from d^4 configuration goes to more stable d^5 configuration (Half filled), due to more exchange energy in d^5 configuration.

Hence, the answer is the option (3).

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Q. 20 In which of the following processes entropy increases?

A. A liquid evaporates to vapour.

B. Temperature of a crystalline solid lowered from $130\ \mathrm{K}$ to $0\ \mathrm{K}$.

C. $2\mathrm{NaHCO}_{3(\mathrm{s})} \rightarrow \mathrm{Na}_2\mathrm{CO}_{3(\mathrm{s})} + \mathrm{CO}_{2(\mathrm{g})} + \mathrm{H}_2\mathrm{O}_{(\mathrm{g})}$

D. $\mathrm{Cl}_{2(\mathrm{g})} \rightarrow 2\mathrm{Cl}_{(\mathrm{g})}$

Choose the correct answer from the options given below:

Option 1:
A and C

Option 2:
A, B and D

Option 3:
A, C and D

Option 4:
C and D

Correct Answer:
A, C and D

Solution:
When a liquid evaporates to vapour entropy increases.

$2\mathrm{NaHCO}_{3(\mathrm{s})} \rightarrow \mathrm{Na}_2\mathrm{CO}_{3(\mathrm{s})} + \mathrm{CO}_{2(\mathrm{g})} + \mathrm{H}_2\mathrm{O}_{(\mathrm{g})}$

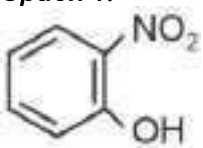
A number of gaseous product molecules increase so entropy increases.
$\mathrm{Cl}_{2(g)} \rightarrow 2\mathrm{Cl}_{(g)}$

$1\ \mathrm{mole}\mathrm{Cl}_{2(\mathrm{g})}$ form $2\ \mathrm{mol}\mathrm{Cl}_{(\mathrm{gg})}$. So entropy increases.

Hence, the answer is the option (3).

Q. 21 Intramolecular hydrogen bonding is present in

Option 1:



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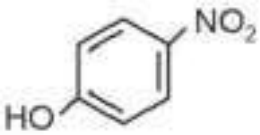
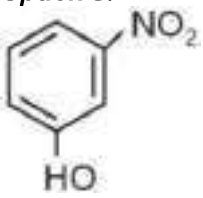
Option 2:



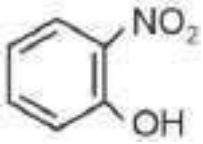
Option 3:



Option 4:

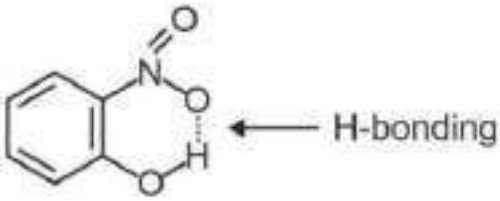
HF

Correct Answer:



Solution:

In o-nitrophenol intramolecular H-bonding is present.



Q. 22 Arrange the following elements in increasing order of electronegativity:

N, O, F, C, Si

Choose the correct answer from the options given below:

Option 1:

Si < C < N < O < F

Option 2:

Si < C < O < N < F

Option 3:

O < F < N < C < Si

Option 4:

F < O < N < C < Si

Correct Answer:

Si < C < N < O < F

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Solution:
Electronegativity increases across the period when moving left to right. It decreases on moving down the group.

The correct option is Si < C < N < O < F

Hence, the answer is the option (1).

Q. 23 Which reaction is NOT a redox reaction?

Option 1:
$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

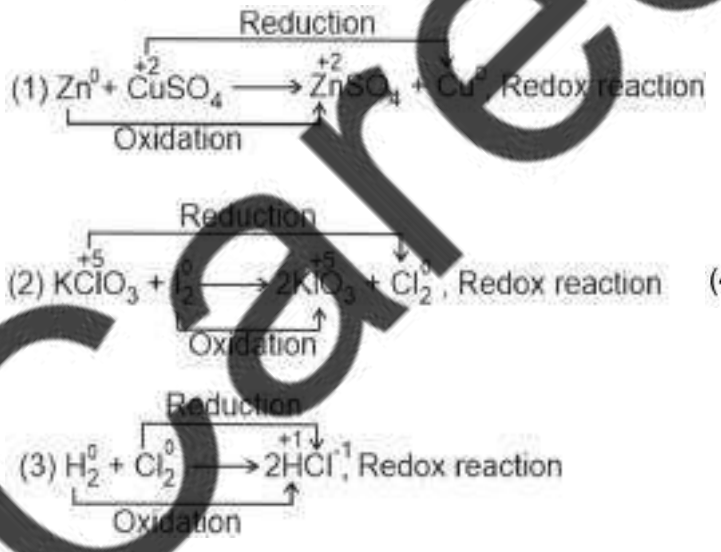
Option 2:
$2KClO_3 + I_2 \rightarrow 2KIO_3 + Cl_2$

Option 3:
$H_2 + Cl_2 \rightarrow 2HCl$

Option 4:
$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

Correct Answer:
$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$

Solution:



(4) $\overset{+2}{Ba}\overset{-1}{Cl_2} + \overset{+1}{Na_2}\overset{-2}{SO_4} \longrightarrow \overset{+2}{Ba}SO_4 + 2\overset{+1}{Na}\overset{-1}{Cl}$

This is not a redox reaction as there is no change in the oxidation state.

Hence, the answer is the option (4).

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Q. 24 Match List I with List II.

List I (Conversion)	List II (Number of Faraday required)
A. 1 mol of H_2O to O_2	I. 3F
B. 1 mol^{of} MnO_4^- to Mn^{2+}	II. 2F
C. 1.5 mol of Ca from molten $CaCl_2$	III. 1F
D. 1 mol of FeO to Fe_2O_3	IV. 5F

Choose the correct answer from the options given below:

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-III, B-IV, C-I, D-II

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:

$4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

for 2 mole of $H_2O = 4\ F$ charge is required for 1 mole of $H_2O = \frac{4\ F}{2} = 2\ F$ required

$\overset{+7}{Mn}\ O_4^- \rightarrow \overset{+2}{Mn}^{2+}$

for 1 mole$\frac{5\ F}{MnO_4^-}$ charge is required

$Ca^{2+} \xrightarrow{+2e^-} Ca$

For 1 mole Ca^{2+} ion required $= 2\ F$

1.5 mole Ca^{2+} ion required $= \frac{2}{1} \times 1.5 = 3\ F$

$\overset{+2}{FeO} \rightarrow \overset{+3}{Fe_2}O_3$

for 1 mole$FeO, 1\ F$ charge is required.

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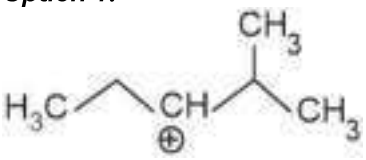
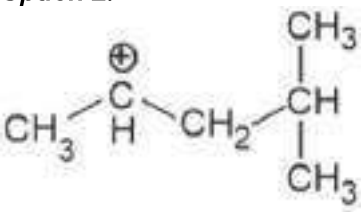
Hence, the answer is the option (1).

Q. 25 The most stable carbocation among the following is :

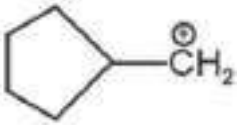
Option 1:



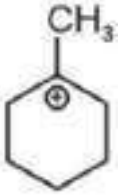
Option 2:



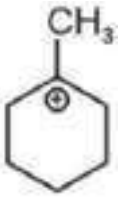
Option 3:



Option 4:



Correct Answer:



Solution:

The stability of carbocation can be described by the hyperconjugation. Greater the extent of hyperconjugation, more is the stability of carbocation.

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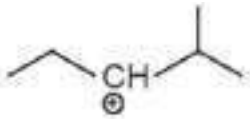
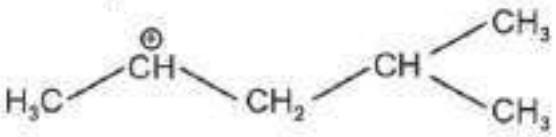
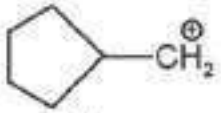
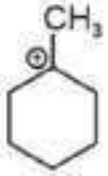
(1)  → 3 α-H

(2)  → 5 α-H

(3)  → 1 α-H

(4)  → 7 α-H

Stability order of carbocations = (4) > (2) > (1) > (3)

Q. 26 Match List I with List II.

List I (Molecule)	List II (Number and types of bond/s between two carbon atoms)
A. ethane	I. one σ-bond and two π-bonds
B. ethene	II.two π-bonds
C. Carbon molecule, C_2	III. one σ-bond
D. ethyne	IV. one σ-bond and one π-bond

Choose the correct answer from the options given below:

Option 1:
A-I, B-IV, C-II, D-III

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-IV, C-II, D-I

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:

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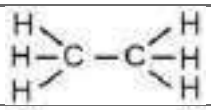
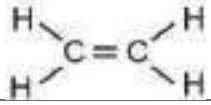
A. Ethane 	one $(C-C)\sigma$ bond
B. Ethene 	one $(C-C)\sigma$ and one $(C-C)\pi$ bond
C. C_2	two $(C-C)\pi$ bonds
D.Ethyne $H-C\equiv C-H$	two $(C-C)\pi$ bonds and one $(C-C)\sigma$ bond

Hence, the answer is the option (3).

Q. 27 Among Group 16 elements, which one does NOT show –2 oxidation state?

Option 1:
O

Option 2:
Se

Option 3:
Te

Option 4:
Po

Correct Answer:
Po

Solution:
Oxygen shows –2, –1, +1 and +2 oxidation states
Selenium shows –2, +2, +4 and +6 oxidation states
Tellurium shows –2, +2, +4 and +6 oxidation states
Polonium shows +2 and +4 oxidation states

Hence, the answer is the option (4).

Q. 28 The Henry's law constant (K_H) values of three gases (A, B, C) in water are $145, 2\times10^{-5}$ and $35 kbar$, respectively. The solubility of these gases in water follow the order:

Option 1:
B>A>C

Option 2:
B > C > A

Option 3:
A > C > B

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Option 4:
A $>$ B $>$ C

Correct Answer:
B $>$ C $>$ A

Solution:
Value of Henry's law constant $\propto \frac{1}{\text{Solubility of gas}}$

The higher the value of $\mathrm{K_H}$ at a given pressure, the lower the solubility of the gas in the liquid.

K_H value of gases (given) : $A > C > B$

$\therefore$ Order of solubility of gases in water : $\mathrm{B > C > A}$

Hence, the answer is the option (2).

Q. 29 Fehling's solution 'A' is

Option 1:
aqueous copper sulphate

Option 2:
alkaline copper sulphate

Option 3:
alkaline solution of sodium potassium tartrate (Rochelle's salt)

Option 4:
aqueous sodium citrate

Correct Answer:
aqueous copper sulphate

Solution:
Fehling solution 'A' = Aqueous copper sulphate

Fehling solution 'B' = Alkaline sodium potassium tartrate (Rochelle salt)

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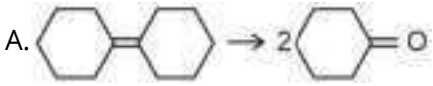
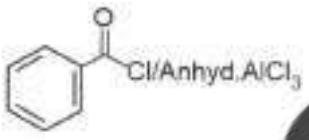
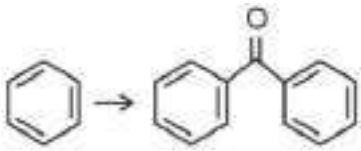
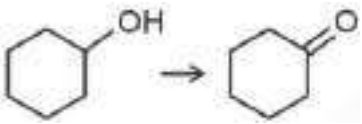
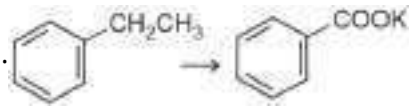
Q. 30 Match List I with List II.

List I (Reaction)	List II (Reagents/Condition)
A. 	
B. 	II. CrO_3
C. 	III. $KMnO_4/KOH, \Delta$
D. 	IV. (i) O_3 (ii) $Zn - H_2O$

Choose the correct answer from the options given below:

Option 1:
A-IV, B-I, C-III, D-II

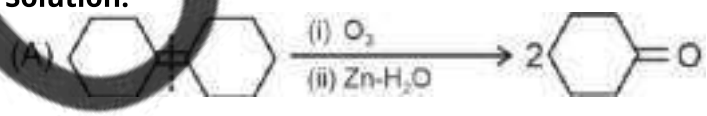
Option 2:
A-III, B-I, C-II, D-IV

Option 3:
A-IV, B-I, C-II, D-III

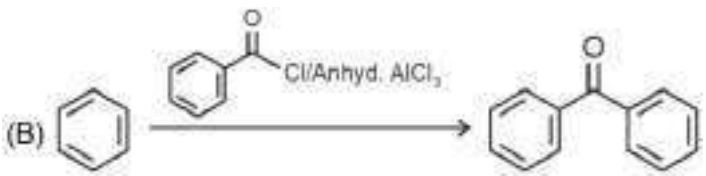
Option 4:
A-I, B-IV, C-II, D-III

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:



It is reductive ozonolysis



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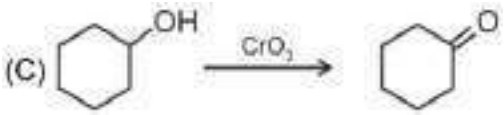
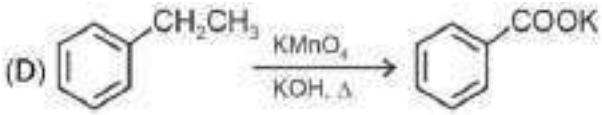
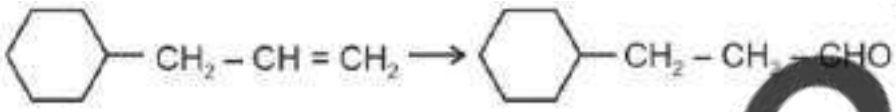
It is Friedel-Crafts acylation reaction.



Secondary alcohols are oxidised to ketones by CrO_3



Q. 31 Identify the correct reagents that would bring about the following transformation.



Option 1:
(i) H_2O/H^+
(ii) CrO_3

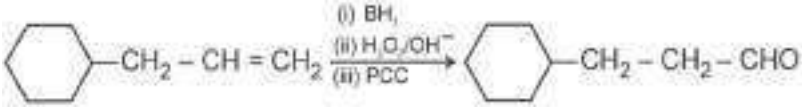
Option 2:
(i) BH_3
(ii) $H_2O_2/\overset{\ominus}{O}H$
(iii) PCC

Option 3:
(i) BH_3
(ii) $H_2O_2/\overset{\ominus}{O}H$
(iii) alk. $KMnO_4$
(iv) $H_3O^{\oplus}$

Option 4:
(i) H_2O/H^+
(ii) PCC

Correct Answer:
(i) BH_3
(ii) $H_2O_2/\overset{\ominus}{O}H$
(iii) PCC

Solution:



Mechanism :

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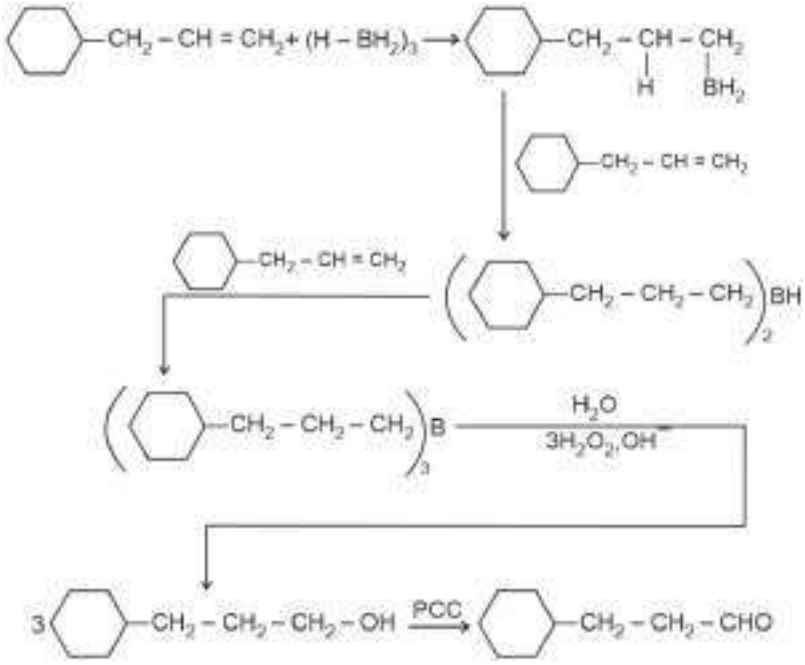

Q. 32 Match List I with List II

List I (Quantum Number)	List II (Information provided)
A.m_I	I. Shape of orbital
B. m_s	II.Size of orbital
C.l	III. Orientation of orbital
D. n	IV. Orientation of spin of electron

Choose the correct answer from the options given below :

Option 1:
A-I, B-III, C-II, D-IV

Option 2:
A-III, B-IV, C-I, D-II

Option 3:
A-III, B-IV, C-II, D-I

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:

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• Magnetic quantum number ml informs about the orientation of the orbital.
• Spin quantum number ms informs about the orientation of the spin of the electron.
• Azimuthal quantum number (l) informs about the shape of the orbital
• Principal quantum number (n) informs about the size of the orbital

Hence, the answer is the option (2).

Q. 33 Match List I with List II.

List I (Compound)	List II (Shape/geometry)
A. $\mathrm{NH_3}$	I. Trigonal Pyramidal
B. $\mathrm{BrF_5}$	II. Square Planar
C. $\mathrm{XeF_4}$	III. Octahedral
D. $\mathrm{SF_6}$	IV. Square Pyramidal

Choose the correct answer from the options given below:

Option 1:
A-I, B-IV, C-II, D-III

Option 2:
A-II, B-IV, C-III, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-II, B-III, C-IV, D-I

Correct Answer:
A-I, B-IV, C-II, D-III

Solution:
$\mathrm{NH_3} \Rightarrow sp^3$ hybridised with 1 lone pair.
Structure will be Trigonal Pyramidal.

$\mathrm{BrF_5} \Rightarrow sp^3d^2$ hybridised with 1 lone pair.
Structure will be Square Pyramidal.

$\mathrm{XeF_4} \Rightarrow sp^3d^2$ with two lone pairs.
Structure will be Square Planar.

$\mathrm{SF_6} \Rightarrow sp^3d^2$ with no lone pair.
Structure will be Octahedral.

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A-I, B-IV, C-II, D-III

Hence, the answer is the option (1).

Q. 34 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to

Option 1:
750 mg

Option 2:
250 mg

Option 3:
Zero mg

Option 4:
200 mg

Correct Answer:
250 mg

Solution:

$$\mathrm{M}=\frac{\mathrm{W} \times 1000}{\mathrm{M}_2 \times \mathrm{V}(\text{ in mL})}$$

$$W=\frac{M \times M_2 \times V(\text{ in } mL)}{1000}=\frac{0.75 \times 36.5 \times 25}{1000}$$

$$=0.684 \text{ g}(\text{ Mass of HCl})$$

$$\underset{36.5 \text{ g}}{\mathrm{HCl}}+\underset{40 \text{ g}}{\mathrm{NaOH}} \longrightarrow \mathrm{HCl}+\mathrm{NaOH}$$

$$36.5 \text{ g} \mathrm{HCl} \text{ reacts with } \mathrm{NaOH}=40 \text{ g}$$

$$0.684 \text{ g} \mathrm{HCl} \text{ reacts with } \mathrm{NaOH}=\frac{40}{36.5} \times 0.684 \simeq 0.750 \text{ g}$$

$$\text{Amount of } \mathrm{NaOH} \text{ left } =1 \text{ g}-0.750 \text{ g}=0.250 \text{ g}=250 \mathrm{mg}$$

Hence, the answer is the option (2).

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Q. 35 Match List I with List II.

List I (Complex)	List II (Type of isomerism)
A. $[Co(NH_3)_5(NO_2)]Cl_2$	I. Solvate isomerism
B. $[Co(NH_3)_5(SO_4)]Br$	II.Linkage isomerism
C. $[Co(NH_3)_6][Cr(CN)_6]$	III.Ionization isomerism
D. $[Co(H_2O)_6]Cl_3$	IV. Coordination isomerism

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-I, B-III, C-IV, D-II

Option 3:
A-I, B-IV, C-III, D-II

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

A. $[Co(NH_3)_5(NO_2)]Cl_2$	II.Linkage isomerism due to ' N ' and ' O ' linkage by NO_2
B. $[Co(NH_3)_5(SO_4)]Br$	III. Ionization isomerism
C. $[Co(NH_3)_6][Cr(CN)_6]$	IV. Coordination isomerism
D. $[Co(H_2O)_6]Cl_3$	I. Solvate isomerism

Hence, the answer is the option (1).

Q. 36 During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

Option 1:
dilute hydrochloric acid

Option 2:
concentrated sulphuric acid

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Option 3:
dilute nitric acid

Option 4:
dilute sulphuric acid

Correct Answer:
dilute sulphuric acid

Solution:
During the preparation of Mohr's salt, dilute sulphuric acid is added to prevent the hydrolysis of Fe^{2+} ion.

Hence, the answer is the option (4).

Q. 37 Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.

A. Al^{3+}

B. Cu^{2+}

C. Ba^{2+}

D. Co^{2+}

E. Mg^{2+}

Choose the correct answer from the options given below.

Option 1:
B, A, D, C, E

Option 2:
B, C, A, D, E

Option 3:
E, C, D, B, A

Option 4:
E, A, B, C, D

Correct Answer:
B, A, D, C, E

Solution:

Group	Cations
Group - II	Cu^{2+}
Group - III	Al^{3+}
Group - IV	CO^{2+}
Group - V	Ba^{2+}

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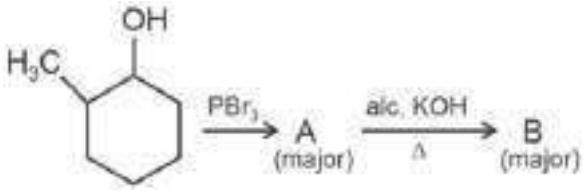
Group - VI	Mg^{2+}

The correct order of group number of ions is $\underset{(B)}{Cu^{2+}} < \underset{(A)}{Al^{3+}} < \underset{(D)}{Co^{2+}} < \underset{(C)}{Ba^{2+}} < \underset{(E)}{Mg^{2+}}$

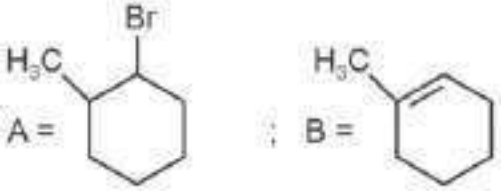
$\therefore$ The correct order is B, A, D, C, E

Hence, the answer is the option (1).

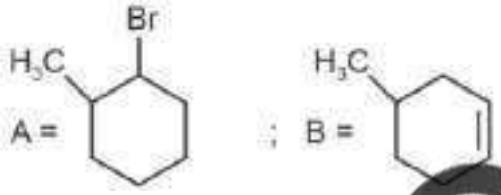
Q. 38 Major products A and B formed in the following reaction sequence, are



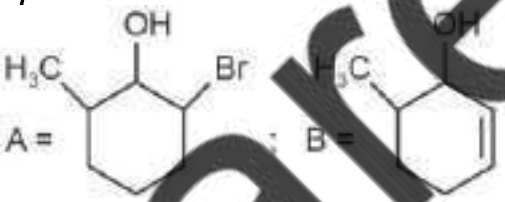
Option 1:



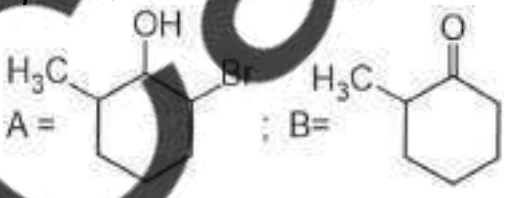
Option 2:



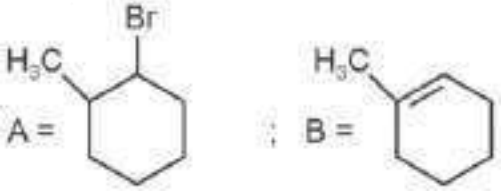
Option 3:



Option 4:



Correct Answer:



Solution:

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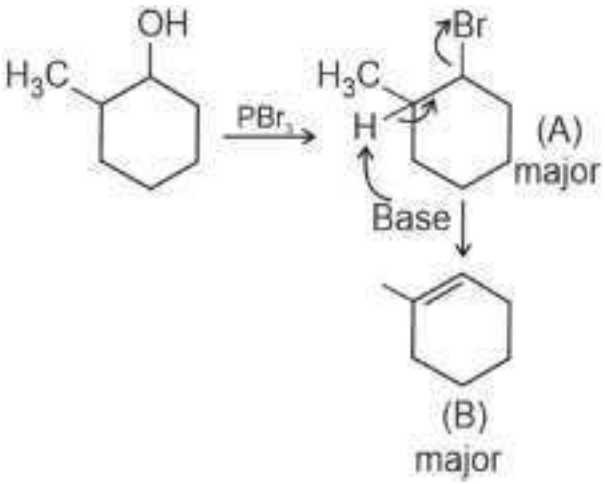


Q. 39 The pair of lanthanoid ions which are diamagnetic is

Option 1:
Ce^{4+} and Yb^{2+}

Option 2:
Ce^{3+} and Eu^{2+}

Option 3:
Gd^{3+} and Eu^{3+}

Option 4:
Pm^{3+} and Sm^{3+}

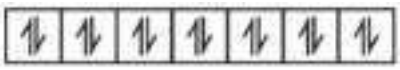
Correct Answer:
Ce^{4+} and Yb^{2+}

Solution:
Magnetic moment $\mu = \sqrt{n(n+2)}$

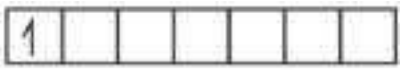
$n \rightarrow$ number of unpaired electron

$Ce^{4+} \Rightarrow (Xe)4f^0$

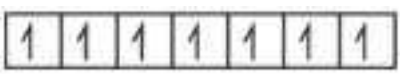
$\mu = 0$ Diamagnetic

$Yb^{2+} \Rightarrow (Xe)4f^{14}$ 

$\mu = 0$ Diamagnetic

$Ce^{3+} \Rightarrow (Xe)4f^1$ 

$\mu = \sqrt{3}$ Paramagnetic

$Eu^{2+} \Rightarrow (Xe)4f^7$ 

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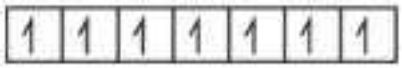
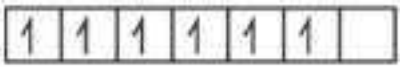
$\mu = \sqrt{63}$ Paramagnetic

$Gd^{3+} \Rightarrow (Xe)4f^7$ 

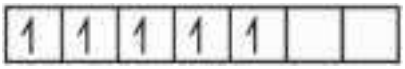
$\mu = \sqrt{63}$ Paramagnetic

$Eu^{3+} \Rightarrow (Xe)4f^6$ 

$\mu = \sqrt{48}$ Paramagnetic

$Pm^{3+} \Rightarrow (Xe)4f^4$

$\mu = \sqrt{24}$ Paramagnetic

$Sm^{3+} \Rightarrow (Xe)4f^5$ 

$\mu = \sqrt{35}$ Paramagnetic

Hence Ce^{4+} and Yb^{2+} are only diamagnetic.

Hence, the answer is the option (1).

Q. 40 Identify the correct answer.

Option 1:
Three resonance structures can be drawn for ozone

Option 2:
BF_3 has non-zero dipole moment

Option 3:
Dipole moment of NF_3 is greater than that of NH_3

Option 4:
Three canonical forms can be drawn for CO_3^{2-} ion

Correct Answer:
Three canonical forms can be drawn for CO_3^{2-} ion

Solution:
(1) In ozone; there are two resonating structures.

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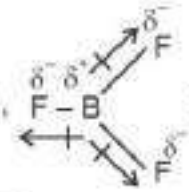
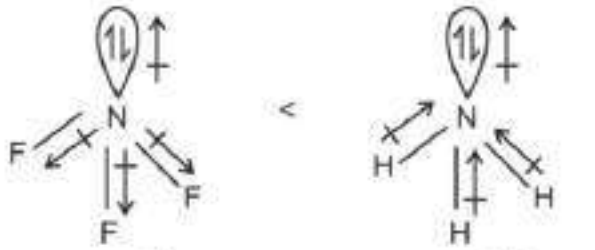
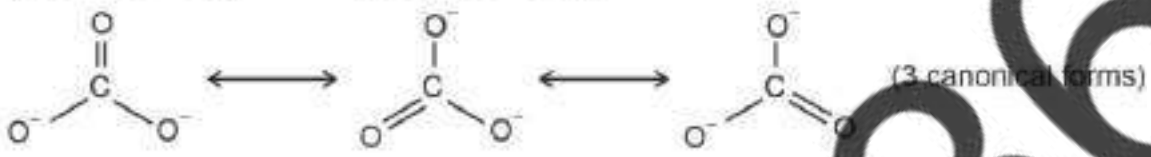
(2) BF_3 i.e.,  ; Dipole moment = 0

(3) 

(0.80×10^{-30} Cm) < (4.90×10^{-30} Cm)

(4) 

(3 canonical forms)

Q. 41 A compound X contains 32% of A, 20 $\%$ of B and the remaining percentage of C. Then, the empirical formula of X is :

(Given atomic masses of A=64 ; B=40 ; C=32 u)

Option 1:
A_2BC_2

Option 2:
ABC_3

Option 3:
AB_2C_2

Option 4:
ABC_4

Correct Answer:
ABC_3

Solution:

Element	Mass percentage %	No. of moles	No.of moles/Smallest number	Simplest whole number
A	32%	$\frac{32}{64}=\frac{1}{2}$	$\frac{1}{2}\times 2$	=1
B	20%	$\frac{20}{40}=\frac{1}{2}$	$\frac{1}{2}\times 2$	=1
C	48%	$\frac{48}{32}=\frac{3}{2}$	$\frac{3}{2}\times 2$	=3

So, the empirical formula of X = A: B: C

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1: 1 : 3

$\therefore$ The correct empirical formula of the compound X is $\mathrm{ABC_3}$

Hence, the answer is the option (2).

Q. 42 The work was done during the reversible isothermal expansion of one mole of hydrogen gas at $25^{\circ}\mathrm{C}$ from a pressure of 20 atmospheres to 10 atmospheres.

(Given R=2.0 $\mathrm{calK^{-1}\ mol^{-1}}$)

Option 1:
0 calorie

Option 2:
-413.14 calories

Option 3:
413.14 calories

Option 4:
100 calories

Correct Answer:
-413.14 calories

Solution:

$$W_{\text{rev, iso}} = -2.303nRT\log\frac{P_i}{P_f}$$

$$= -2.303 \times 1 \times 2 \times 298 \times \log 2$$

$$= -2.303 \times 1 \times 2 \times 298 \times 0.3$$

$$= -413.14 \text{ calories}$$

Hence, the answer is the option (2).

Q. 43 Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulfate solution for 100 seconds is

(Given: Molar mass ofCu: 63 $\mathrm{g\ mol^{-1}}$,1 F=96487 C

Option 1:
$3.15\ \mathrm{g}$

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Option 2:
0.315 g

Option 3:
31.5 g

Option 4:
0.0315 g

Correct Answer:
0.315 g

Solution:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

$$\text{Mass of Cu deposited (w)} = \frac{M \times i \times t}{nF}$$

$$= \frac{63 \times 9.6487 \times 100}{2 \times 96487}$$

$$= 0.315 \text{ g}$$

Hence, the answer is the option (2).

Q. 44 Consider the following reaction in a sealed vessel at equilibrium with concentrations of

$N_2 = 3.0 \times 10^{-3}M$, $O_2 = 4.2 \times 10^{-3}M$ and $NO = 2.8 \times 10^{-3}M$

$$2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$$

If 0.1 mol L^{-1} of $NO_{(g)}$ is taken in a closed vessel, what will be the degree of dissociation (α) of $NO_{(g)}$ at equilibrium?

Option 1:
0.00889

Option 2:
0.0889

Option 3:
0.8889

Option 4:
0.717

Correct Answer:
0.717

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Solution:

$$2NO_{(g)} \rightleftharpoons N_{2(\ g)} + O_{2(\ g)}$$

$$K_c = \frac{[N_2]\,[O_2]}{[NO]^2}$$

$$= \frac{3 \times 10^{-3} \times 4.2 \times 10^{-3}}{2.8 \times 10^{-3} \times 2.8 \times 10^{-3}}$$

$$= 1.607$$

$$2NO_{(g)} \rightleftharpoons N_{2(\ g)} + O_{2(\ g)}$$

$t = 0$	0.1	0	0
$0.1 - 0.1\alpha$	0.05α	0.05α	

$$K_c = \frac{0.05\alpha \times 0.05\alpha}{(0.1-0.1\alpha)^2}$$

$$K_c = \frac{0.05\alpha \times 0.05\alpha}{0.01(1-\alpha)^2}$$

$$1.607 = \frac{(0.05)^2\alpha^2}{0.01(1-\alpha)^2}$$

$$\frac{\alpha}{1-\alpha} = 2.54$$

$$\frac{\alpha^2}{(1-\alpha)^2} = \frac{1.607 \times (0.1)^2}{(0.05)^2}$$

$$\alpha = 2.54 - 2.54\alpha$$

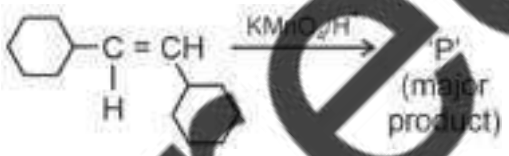
$$3.54\alpha = 2.54$$

$$\alpha = \frac{2.54}{3.54} = 0.717$$

$$\frac{\alpha}{1-\alpha} = \frac{1.27 \times 0.1}{0.05}$$

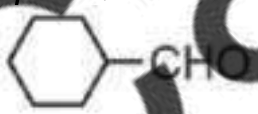
Hence, the answer is the option (4).

Q. 45 For the given reaction:

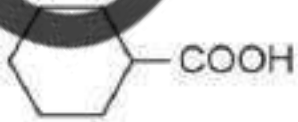


'P' is

Option 1:



Option 2:



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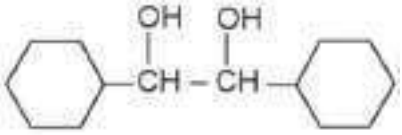
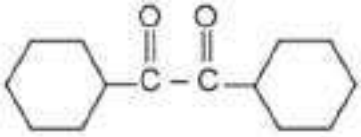
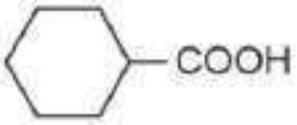
Option 3:



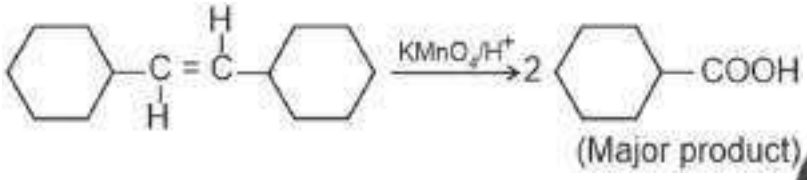
Option 4:



Correct Answer:



Solution:



Hence, the answer is the option (2).

Q. 46 The rate of a reaction quadruples when the temperature changes from $27^{\circ}\mathrm{C}$ to $57^{\circ}\mathrm{C}$. Calculate the energy of activation.

Given R=$8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}$, $\log 4$=0.6021

Option 1:
$38.04\ \mathrm{kJ/mol}$

Option 2:
$380.4\ \mathrm{kJ/mol}$

Option 3:
$3.80\ \mathrm{kJ/mol}$

Option 4:
$3804\ \mathrm{kJ/mol}$

Correct Answer:
$38.04\ \mathrm{kJ/mol}$

Solution:

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$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log\left(\frac{4}{1}\right) = \frac{E_a}{2.303R}\left(\frac{1}{300} - \frac{1}{330}\right)$$

$$E_a = \frac{(\log(4)) \times 2.303 \times 8.314 \times 300 \times 330}{30}$$

$$= 3.804 \times 10^4 \text{ J/mol}$$

$$= 38.04 \text{ kJ/mol}$$

Hence, the answer is the option (1).

Q. 47 Identify the major product C formed in the following reaction sequence:

$$CH_3 - CH_2 - CH_2 - I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} \underset{\text{(major)}}{C}$$

Option 1:
propylamine

Option 2:
butylamine

Option 3:
butanamide

Option 4:
α-bromobutanoic acid

Correct Answer:
propylamine

Solution:

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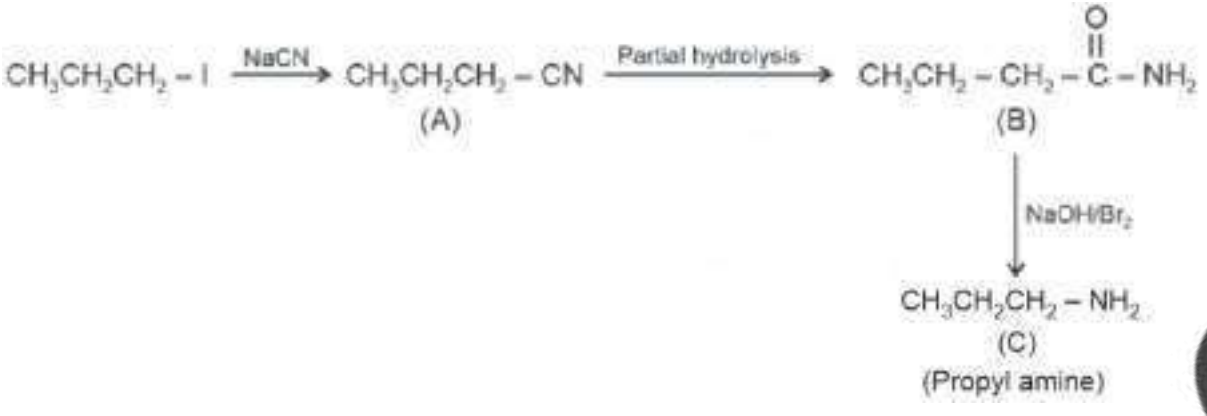

- Step I is S_N reaction with $\overset{\ominus}{C}\ N$ nucleophile.
- Step-II will give amide.
- Step-III is Hoffmann bromamide degradation reaction.

Hence, the answer is the option (1).

Q. 48 The products A and B obtained in the following reactions, respectively, are

$$3ROH + PCl_3 \rightarrow 3RCl + A$$

$$ROH + PCl_5 \rightarrow RCl + HCl + B$$

Option 1:
$POCl_3$ and H_3PO_3

Option 2:
$POCl_3$ and H_3PO_4

Option 3:
H_3PO_4 and $POCl_3$

Option 4:
H_3PO_3 and $POCl_3$

Correct Answer:
H_3PO_3 and $POCl_3$

Solution:
These reactions are preparation of haloalkanes from alcohols.

$$3ROH + PCl_3 \longrightarrow 3RCl + H_3PO_3$$
(A)

$$ROH + PCl_5 \longrightarrow RCl + HCl + POCl_3$$
(B)

A and B are H_3PO_3 and $POCl_3$ respectively.

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Q. 49 The plot of osmotic pressure () vs concentration $\left(\mathrm{molL}^{-1}\right)$ for a solution gives a straight line with slope $25.73\ \mathrm{Lbarmol}^{-1}$. The temperature at which the osmotic pressure measurement is done is

(Use $\mathrm{R}=0.083\ \mathrm{Lbar}^{\mathrm{mol}}\mathrm{m}^{-1}\ \mathrm{K}^{-1}$)

Option 1:
$37^{\circ}\mathrm{C}$

Option 2:
$310^{\circ}\mathrm{C}$

Option 3:
$25.73^{\circ}\mathrm{C}$

Option 4:
$12.05^{\circ}\mathrm{C}$

Correct Answer:
$37^{\circ}\mathrm{C}$

Solution:
$\Pi=\mathrm{CRT}$

$\text{Slope }=\mathrm{RT}$

$25.73=0.083\times\mathrm{T}$

$\mathrm{T}=\dfrac{25.73}{0.083}=309.47\approx 310\ \mathrm{K}$

$\therefore\quad \text{Temperature in }^{\circ}\mathrm{C}=310-273$

$=37^{\circ}\mathrm{C}$

Q. 50 Given below are two statements :

Statement I: $\left[\mathrm{Co}(\mathrm{NH}_3)_6\right]^{3+}$ is a homoleptic complex whereas $\left[\mathrm{Co}(\mathrm{NH}_3)_4\mathrm{Cl}_2\right]^{+}$is a heteroleptic complex.

Statement II : Complex $\left[\mathrm{Co}(\mathrm{NH}_3)_6\right]^{3+}$ has only one kind of ligands but $\left[\mathrm{Co}(\mathrm{NH}_3)_4\mathrm{Cl}_2\right]^{+}$ has more than one kind of ligands.

In the light of the above statements, choose the correct answer from the options given below.

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Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
$[Co(NH_3)_6]^{3+}$ is a homoleptic complex as only one type of ligands (NH_3) is coordinated with Co^{3+} ion. While $[Co(NH_3)_4Cl_2]^{+}$is a heteroleptic complex in which Co^{3+} ion is ligated with more than one type of ligands, i.e., NH_3 and Cl^{-}.

Biology

Q. 1 Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of:

Option 1:
Cofactor inhibition

Option 2:
Feedback inhibition

Option 3:
Competitive inhibition

Option 4:
Enzyme activation

Correct Answer:
Competitive inhibition

Solution:
Correct answer is option (3) because malonate shows close structural similarity with the substrate and it competes with the substrate for the substrate binding site of the enzyme succinic dehydrogenase. Option (1), (2) and (4) are incorrect as enzyme activation, co-factor inhibition are not showing structural similarity with substrate.

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Q. 2 Given below are two statements:

Statement I : Bt toxins are insect group specific and coded by a gene cry IAc.

Statement II : Bt toxin exists as inactive protoxin in B. thuringiensis. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is true but Statement II is false

Solution:
The correct answer is option (3) as specific Bt toxin genes were isolated from Bacillus thuringiensis and incorporated into the several crop plants such as cotton. The choice of genes depends upon the crop and the targeted pest as most Bt toxins are insect-group specific. The toxin is coded by a gene named cry. There are a number of them, for example, the proteins encoded by the genes cry IAc and cry IIAb control the cotton bollworms, that of cry IAb controls corn borer.

Q. 3 Match List I with List II

	List-I		List-II
A.	Rhizopus	I.	Mushroom
B	Ustilago	II	Smut fungus
C	Puccinia	III	Bread mould
D	Agaricus	IV	Rust fungus

Choose the correct answer from the options given below:

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Option 1:
A-III, B-II, C-IV, D-I

Option 2:
A-I, B-III, C-II, D-IV

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-III, B-II, C-IV, D-I

Solution:
Rhizopus is a bread mould fungus. Ustilago is a smut fungi. Puccinia is known as rust fungi. Agaricus is commonly called mushroom.

A-III
B-II
C-IV
D-I

Q. 4 The capacity to generate a whole plant from any cell of the plant is called:

Option 1:
Totipotency

Option 2:
Micropropagation

Option 3:
Differentiation

Option 4:
Somatic hybridization

Correct Answer:
Totipotency

Solution:
Totipotency is defined as the capacity to generate a whole plant from any cell of the plant.

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Q. 5 The equation of Verhulst-Pearl logistic growth is $\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$.

From this equation, K indicates:

Option 1:
Intrinsic rate of natural increase

Option 2:
Biotic potential

Option 3:
Carrying capacity

Option 4:
Population density

Correct Answer:
Carrying capacity

Solution:
In the equation $\frac{\mathrm{dN}}{\mathrm{dt}} = \mathrm{rN}\left(\frac{\mathrm{K}-\mathrm{N}}{\mathrm{K}}\right), \mathrm{K}$ represents carrying capacity.

Q. 6 Identify the set of correct statements:

A. The flowers of Vallisneria are colourful and produce nectar.

B. The flowers of water lily are not pollinated by water.

C. In most of water-pollinated species, the pollen grains are protected from wetting.

D. Pollen grains of some hydrophytes are long and ribbon like.

E. In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below.

Option 1:
C, D and E only

Option 2:
A, B, C and D only

Option 3:
A, C, D and E only

Option 4:
B, C, D and E only

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Correct Answer:
B, C, D and E only

Solution:
Flowers of Vallisneria are not colourful and do not produce nectar. Waterlily is pollinated by insect or wind. In water-pollinated species, pollen grains are protected from wetting by a mucilaginous covering. In some hydrophytes such as Vallisneria pollen grains are carried passively by water current.

Q. 7 Match List I with List II

List I	List II
A. Two or more alternative forms of a gene	I. Back cross
B. Cross of F_1 progeny with homozygous recessive parent	II.Ploidy
C.Cross of F_1 progeny with any of the parents	III.Allele
D.Number of chromosome sets in plant	IV.Test cross

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
A. Two or more alternative forms of gene are called alleles.
B. Cross of F_1 progeny with homozygous recessive parent is a test cross.
C. Cross of F_1 progeny with any of the parents is a back cross.
D. Number of chromosome sets in plant is called ploidy.

Q. 8 A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?

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Option 1:
Only red flowered plants

Option 2:
Red flowered as well as pink flowered plants

Option 3:
Only pink flowered plants

Option 4:
Red, Pink as well as white flowered plants

Correct Answer:
Red flowered as well as pink flowered plants

Solution:
Pink colour flower in snapdragon have genotype **Rr**
Red flowered snapdragon have genotype **RR** when they both are crossed

♂ / ♀	R	R
R	RR	RR
r	Rr	Rr

Phenotype

Red : Pink : White

2 2 0

So the progeny that we get are red and pink flowered plants only

---

Q. 9 Given below are two statements:

Statement I : Chromosomes become gradually visible under light microscope during leptotene stage.

Statement II : The beginning of diplotene stage is recognized by dissolution of synaptonemal complex.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

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Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
• During leptotene stage the chromosomes become gradually visible under the light microscope.
• The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and the tendency of the recombined homologous chromosomes of the bivalents to separate from each other except at the site of crossover.

Thus both statement I and II are correct.

Q. 10 The lactose present in the growth medium of bacteria is transported to the cell by the action of

Option 1:
Beta-galactosidase

Option 2:
Acetylase

Option 3:
Permease

Option 4:
Polymerase

Correct Answer:
Permease

Solution:
The y gene lac operon codes for permease enzyme, which increase the permeability of cell to β-galactosides.

So, the lactose present in the growth medium of bacteria is transported into the cell by the action of permease.

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Q. 11 These are regarded as major causes of biodiversity loss:

A. Over exploitation

B. Co-extinction

C. Mutation

D. Habitat loss and fragmentation

E. Migration

Choose the correct option:

Option 1:
A, C and D only

Option 2:
A, B, C and D only

Option 3:
A, B and E only

Option 4:
A, B and D only

Correct Answer:
A, B and D only

Solution:
Major causes of biodiversity losses are

(1) Habitat loss and fragmentation

(2) Over-exploitation

(3) Alien species invasions

(4) Co-extinctions

Hence correct option is A, B and D only.

Q. 12 Bulliform cells are responsible for

Option 1:
Inward curling of leaves in monocots.

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Option 2:
Protecting the plant from salt stress.

Option 3:
Increased photosynthesis in monocots.

Option 4:
Providing large spaces for storage of sugars.

Correct Answer:
Inward curling of leaves in monocots.

Solution:
In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface is exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss.

Q. 13 Which of the following is an example of actinomorphic flower?

Option 1:
Datura

Option 2:
Cassia

Option 3:
Pisum

Option 4:
Sesbania

Correct Answer:
Datura

Solution:
Datura shows actinomorphic flower. In Cassia, Pisum and Sesbania, zygomorphic flowers are seen.

Q. 14 In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

Option 1:
BB

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Option 2:
bb

Option 3:
Bb

Option 4:
BB/Bb

Correct Answer:
bb

Solution:
To determine the genotype of a black seed colour at F_2, the black seed from F_2 is crossed with the white seed colour. This is called a test cross.

$\therefore$ To determine the genotype of $(\mathrm{BB}/\mathrm{Bb})$ black seed we need to cross them with white seed $i.e.\ \mathrm{bb}$.

Q. 15 Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles do not show any expression and both the characters appear as such in F_2generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called factor.

E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

Option 1:
A, B and C only

Option 2:
A, C, D and E only

Option 3:
B, C and D only

Option 4:
A, B, C, D and E

Correct Answer:
A, C, D and E only

Solution:
According to Law of Dominance

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(1) Characters are controlled by discrete units called factors
(2) Factors occur in pairs
(3) In a dissimilar pair of factors one member of the pair dominates (dominant) the other recessive

The law of dominance is used to explain the expression of only one of the parental characters in a monohybrid cross.

Law of segregation is based on the fact that the alleles do not show any expression and both the characters are recovered as such in F_2 generation

---

Q. 16 Formation of interfascicular cambium from fully developed parenchyma cells is an example for

Option 1:
Differentiation

Option 2:
Redifferentiation

Option 3:
Dedifferentiation

Option 4:
Maturation

Correct Answer:
Dedifferentiation

Solution:
The phenomenon of formation of interfascicular cambium from fully differentiated parenchyma cells is called dedifferentiation.

Q. 17 The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called

Option 1:
in-situ conservation

Option 2:
Biodiversity conservation

Option 3:
Semi-conservative method

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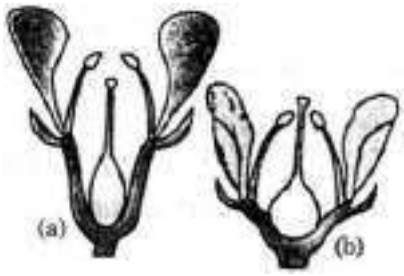
Option 4:
Sustainable development

Correct Answer:
Biodiversity conservation

Solution:
The type of conservation in which threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called ex-situ conservation which is a type of biodiversity conservation.

Q. 18 Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (a) and (b)



Option 1:
(a) Epigynous; (b) Hypogynous

Option 2:
(a) Hypogynous; (b) Epigynous

Option 3:
(a) Perigynous; (b) Epigynous

Option 4:
(a) Perigynous; (b) Perigynous

Correct Answer:
(a) Perigynous; (b) Perigynous

Solution:
If gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level, it is called perigynous.

Both diagram shows perigynous condition.

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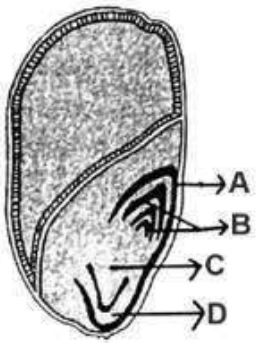
Q. 19 Identify the part of the seed from the given figure which is destined to form root when the seed germinates.



Option 1:
A

Option 2:
B

Option 3:
C

Option 4:
D

Correct Answer:
C

Solution:
Radicle is destined to form root.

In the given diagram 'C' represent radicle

Q. 20 Auxin is used by gardeners to prepare weed-free lawns. But no damage is caused to grass as auxin

Option 1:
promotes apical dominance.

Option 2:
promotes abscission of mature leaves only.

Option 3:
does not affect mature monocotyledonous plants.

Option 4:
can help in cell division in grasses, to produce growth.

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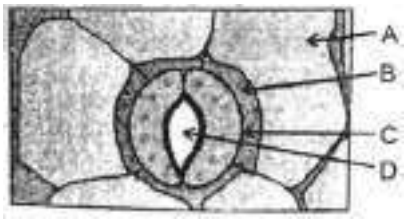
Correct Answer:
does not affect mature monocotyledonous plants.

Solution:
Auxin does not affect mature monocot plants. In monocots, especially grasses show limited translocation and cause rapid degradation of external auxin.

Q. 21 In the given figure, which component has thin outer walls and highly thickened inner walls?



Option 1:
A

Option 2:
B

Option 3:
C

Option 4:
D

Correct Answer:
A

Solution:
Guard cells of stomata have thin outer wall and highly thickened inner walls.

Q. 22 How many molecules of ATP and NADPH are required for every molecule of CO_2 fixed in the Calvin cycle?

Option 1:
2 molecules of ATP and 3 molecules of NADPH

Option 2:
2 molecules of ATP and 2 molecules of NADPH

Option 3:
3 molecules of ATP and 3 molecules of NADPH

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Option 4:
3 molecules of ATP and 2 molecules of NADPH

Correct Answer:
3 molecules of ATP and 2 molecules of NADPH

Solution:
For fixation of 1 molecule of CO_2 in Calvin cycle 3 ATP molecules and 2 NADPH molecules are required.

Q. 23 Tropical regions show greatest level of species richness because

A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.

B. Tropical environments are more seasonal.

C. More solar energy is available in tropics.

D. Constant environments promote niche specialization.

E. Tropical environments are constant and predictable.

Choose the correct answer from the options given below.

Option 1:
A, C, D and E only

Option 2:
A and B only

Option 3:
A, B and E only

Option 4:
A, B and D only

Correct Answer:
A, C, D and E only

Solution:
Only statement B is incorrect because tropical environments unlike temperate ones, are less seasonal, relatively more constant and predictable.

Thus statements A, C, D and E are correct.

Q. 24 The cofactor of the enzyme carboxypeptidase is:

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Option 1:
Zinc

Option 2:
Niacin

Option 3:
Flavin

Option 4:
Haem

Correct Answer:
Zinc

Solution:
The correct answer is option (1) as the cofactor of the enzyme carboxypeptidase is zinc.

Niacin is associated with coenzyme NAD and NADP.

Option (4) is incorrect as haem is the prosthetic group in peroxidase and catalase.

Q. 25 Which of the following are required for the dark reaction of photosynthesis?

A. Light

B. Chlorophyll
C. CO_2
D. ATP
E. NADPH

Choose the correct answer from the options given below:

Option 1:
A, B and C only

Option 2:
B, C and D only

Option 3:
C, D and E only

Option 4:
D and E only

Correct Answer:
C, D and E only

Solution:

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For dark reaction of photosynthesis there are the requirement of

$$\left.\begin{array}{l} CO_2 \\ ATP \\ NADPH \end{array}\right]$$

Q. 26 A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;

Option 1:
Repressor, Operator gene, Structural gene

Option 2:
Structural gene, Transposons, Operator gene

Option 3:
Inducer, Repressor, Structural gene

Option 4:
Promotor, Structural gene, Terminator

Correct Answer:
Promotor, Structural gene, Terminator

Solution:
A transcription unit of DNA is defined primarily by the three regions in the DNA:

(i) A promoter
(ii) The structural gene
(iii) A terminator

The promoter is said to be located towards 5□-end (upstream) of the structural gene (the reference is made with respect to the polarity of coding strand)
The terminator is located towards 3□-end (downstream) of the coding strand.

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Q. 27 Match List I with List II

List I	List II
A. Clostridium butylicum	I. Ethanol
B. Saccharomyces cerevisiae	II. Streptokinase
C. Trichoderma polysporum	III. Butyric acid
D.Streptococcus sp.	IV.Cyclosporin

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-II, B-IV, C-III, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-IV, B-I, C-III, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
A. Clostridium butylicum — Butyric acid

B. Saccharomyces cerevisiae — Ethanol

C. Trichoderma polysporum — Cyclosporin-A

D. Streptococcus sp. — Streptokinase

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Q. 28 Match List I with List II

	List - I		List - II
A.	Nucleous	I.	Site of formation of glycolipid
B.	Centriole	II.	Organization like the cartwheel
C.	Leucoplasts	III.	Site for active ribosomal RNA synthesis
D.	Golgi apparatus	IV.	For storing nutrients

Choose the correct answer from the options given below:

Option 1:
A-III, B-II, C-IV, D-I

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-III, B-IV, C-II, D-I

Option 4:
A-I, B-II, C-III, D-IV

Correct Answer:
A-III, B-II, C-IV, D-I

Solution:
• Nucleolus is a site for active ribosomal RNA synthesis
• Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the cartwheel.
• Leucoplasts are the colourless plastids of varied shapes and sizes with stored nutrients.
• Golgi apparatus is the important site for formation of glycoproteins and glycolipids.

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Q. 29 What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?

A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.

B. It may get integrated into the genome of the recipient.

C. It may multiply and be inherited along with the host DNA.

D. The alien piece of DNA is not an integral part of chromosome.

E. It shows ability to replicate.

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
D and E only

Option 3:
B and C only

Option 4:
A and E only

Correct Answer:
B and C only

Solution:
Correct answer is option (3) because

The fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism are:
(B) It may get integrated into the genome of the recipient
(C) It may multiply and be inherited along with the host DNA

$\Rightarrow$ This piece of DNA would not be able to multiply itself in the progeny cells of the organism but when gets integrated into the genome of the recipient, it may multiply and be inherited along with the host DNA.

Q. 30 Spindle fibers attach to kinetochores of chromosomes during

Option 1:
Prophase

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Option 2:
Metaphase

Option 3:
Anaphase

Option 4:
Telophase

Correct Answer:
Metaphase

Solution:
Spindle fibers attach to kinetochores of chromosome in metaphase stage.

Q. 31 Lecithin, a small molecular weight organic compound found in living tissues, is an example of:

Option 1:
Amino acids

Option 2:
Phospholipids

Option 3:
Glycerides

Option 4:
Carbohydrates

Correct Answer:
Phospholipids

Solution:
The correct answer is option (2).

Some lipids have phosphorous and a phosphorylated organic compound in them. These are phospholipids. They are found in cell membrane. Lecithin is one example.

Option (3) is incorrect as glycerides are another group of lipids in which both glycerol and fatty acids are present.

Option (1) and (4) are incorrect as amino acids and carbohydrates are separate groups of biomolecules.

Q. 32 Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of:

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Option 1:
8 bp

Option 2:
6 bp

Option 3:
4 bp

Option 4:
10 bp

Correct Answer:
6 bp

Solution:
The correct answer is option (2).

The first restriction endonuclease – Hind II, whose functioning depends on a specific DNA nucleotide sequence was isolated. It was found that Hind II always cut DNA molecules at a particular point by recognising sequence of six base pairs.

Option (1), (3) and (4) are incorrect because they have either more than 6 or less than 6 bp.

Q. 33 Given below are two statements:

Statement I : Parenchyma is living but collenchyma is dead tissue.

Statement II : Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperms.

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

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Solution:
Collenchyma is also living tissue.

Gymnosperm lack xylem vessels but presence of xylem vessels is the characteristic of angiosperm.

Q. 34 List of endangered species was released by

Option 1:
GEAC

Option 2:
WWF

Option 3:
FOAM

Option 4:
IUCN

Correct Answer:
IUCN

Solution:
List of endangered species was released by – IUCN.

Q. 35 Which one of the following is not a criterion for classification of fungi?

Option 1:
Morphology of mycelium

Option 2:
Mode of nutrition

Option 3:
Mode of spore formation

Option 4:
Fruiting body

Correct Answer:
Mode of nutrition

Solution:
The morphology of the mycelium, mode of spore formation and fruiting bodies form the basis for the division of the kingdom fungi into various classes.

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Q. 36 The DNA present in chloroplast is:

Option 1:
Linear, double stranded

Option 2:
Circular, double stranded

Option 3:
Linear, single stranded

Option 4:
Circular, single stranded

Correct Answer:
Circular, double stranded

Solution:
The DNA present in chloroplast is circular double stranded.

Q. 37 Match List I with List II

	List - I		List - II
A.	Citric acid cycle	I.	Cytoplasm
B.	Glycolysis	II.	Mitochondrial matrix
C.	Electron transport system	III.	Intermembrane space of mitochondria
D.	Proton gradient	IV.	Inner mitochondrial membrane

Choose the correct answer from the options given below:

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-II, B-I, C-IV, D-III

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-III, C-II, D-I

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Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
Citric acid cycle occurs in mitochondrial matrix.

Glycolysis occurs in cytosol in most of the organism.

Electron transport system is present in the inner mitochondrial membrane.

Proton gradient is formed across the intermembrane space of mitochondria.

Q. 38 Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.

Option 1:
Malic acid $\rightarrow$ Oxaloacetic acid

Option 2:
Succinic acid $\rightarrow$ Malic acid

Option 3:
Succinyl-CoA $\rightarrow$ Succinic acid

Option 4:
Isocitrate $\rightarrow$ α-ketoglutaric acid

Correct Answer:
Succinyl-CoA $\rightarrow$ Succinic acid

Solution:
Oxidation involves the loss of electrons (often as part of hydrogen) from a molecule, leaving to an increase in its oxidation state. This process is typically associated with the transfer of electrons to an electron acceptor which is reduced in the process.

The conversion of succinyl CoA to succinic acid does not involve oxidation of substrate.

Q. 39 Identify the correct description about the given figure:

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Option 1:
Wind pollinated plant inflorescence showing flowers with well exposed stamens.

Option 2:
Water pollinated flowers showing stamens with mucilaginous covering.

Option 3:
Cleistogamous flowers showing autogamy.

Option 4:
Compact inflorescence showing complete autogamy

Correct Answer:
Wind pollinated plant inflorescence showing flowers with well exposed stamens.

Solution:
The given diagram shows a wind pollinated plant showing compact inflorescence and well exposed stamens.

Stamens are exposed so complete autogamy does not occur.

Q. 40 Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield?

Option 1:
Auxin

Option 2:
Gibberellin

Option 3:
Cytokinin

Option 4:
Abscisic acid

Correct Answer:
Gibberellin

Solution:
Sugarcanes store carbohydrate as sugar in their stems. Spraying sugarcane crop with gibberellins increases the length of the stem, thus increasing the yield.

Q. 41 In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is $100x\left(\mathrm{kcalm}^{-2}\right)\mathrm{yr}^{-1}$, what would be the GPP (Gross Primary Productivity) of the third trophic level of the same ecosystem?

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Option 1:
$\frac{x}{10}\left(\mathrm{kcalm}^{-2}\right)\mathrm{yr}^{-1}$

Option 2:
$x\left(\mathrm{kcalm}^{-2}\right)\mathrm{yr}^{-1}$

Option 3:
$10x\left(\mathrm{kcalm}^{-2}\right)\mathrm{yr}^{-1}$

Option 4:
$\frac{100x}{3x}\left(\mathrm{kcalm}^{-2}\right)\mathrm{yr}^{-1}$

Correct Answer:
$10x\left(\mathrm{kcalm}^{-2}\right)\mathrm{yr}^{-1}$

Solution:
NPP at first trophic level would be the GPP for second trophic level. NPP at second trophic level would be GPP for third trophic level. Therefore, $100x\left(\mathrm{kcal}/\mathrm{m}^2/\mathrm{yr}\right)$ would be GPP at second trophic level and $100x \times 10\%\left(\mathrm{kcal}/\mathrm{m}^2/\mathrm{yr}\right)$ i.e., $10x\left(\mathrm{kcal}/\mathrm{m}^2/\mathrm{yr}\right)$ energy would be GPP at third trophic level.

Q. 42 Which of the following statement is correct regarding the process of replication in E.coli?

Option 1:
The DNA dependent DNA polymerase catalyses polymerization in one direction that is $3' \rightarrow 5'$

Option 2:
The DNA dependent RNA polymerase catalyses polymerization in one direction, that is $5' \rightarrow 3$ '

Option 3:
The DNA dependent DNA polymerase catalyses polymerization in $5' \rightarrow 3'$ as well as $3' \rightarrow 5$ ' direction

Option 4:
The DNA dependent DNA polymerase catalyses polymerization in 5' $\rightarrow 3'$ direction

Correct Answer:
The DNA dependent DNA polymerase catalyses polymerization in 5' $\rightarrow 3'$ direction

Solution:
In Prokaryotes, like E.coli during replication, the DNA dependent DNA polymerase catalyse polymerization only in one direction, that is $5' \rightarrow 3'$

Q. 43 Which of the following are fused in somatic hybridization involving two varieties of plants?

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Option 1:
Callus

Option 2:
Somatic embryos

Option 3:
Protoplasts

Option 4:
Pollens

Correct Answer:
Protoplasts

Solution:
Protoplast of two varieties of plants are fused in somatic hybridization.

Q. 44 Match List I with List II

List - I	List - II
A. Rose	I. Twisted aestivation
B. Pea	II. Perigynous flower
C. Cotton	III. Drupe
D. Mango	IV. Marginal placentation

Choose the correct answer from the options given below :

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-I, B-II, C-III, D-IV

Option 3:
A-IV, B-III, C-II, D-I

Option 4:
A-II, B-III, C-IV, D-I

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:

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Rose have half-inferior ovary, thus it is known as Perigynous flower.

In Pea, the placenta form a ridge along the ventral suture of the ovary and ovules are borne on this ridge forming two rows.

In Cotton, twisted aestivation is present.

In Mango, fruit is drupe.

Q. 45 Match List I with List II

List - I	List -II
A. Frederick Griffth	I. Genetic code
B. Francois Jacob & Jacque Monod	II. Semi-conservative mode of DNA replication
C. Har Gobind Khorana	III. Transformation
D. Meselson & Stahl	IV.Lac operon

Choose the correct answer from the options given below:

Option 1:
A-III, B-II, C-I, D-IV

Option 2:
A-III, B-IV, C-I, D-II

Option 3:
A-II, B-III, C-IV, D-I

Option 4:
A-IV, B-I, C-II, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
Frederick Griffith series of experiment witness miraculous transformation in the bacteria.

The elucidation of Lac operon was a result of a close association between geneticist, Francois Jacob and a biochemist, Jacques Monod.

Meselson and Stahl gave semi-conservative mode of DNA replication.

Har Gobind Khorana developed chemical method to define combination of bases in genetic code.

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Q. 46 Match List-I with List-II

List - I	List - II
A. GLUT- 4	I. Hormone
B. Insulin	II. Enzyme
C. Trypsin	III. Intercellular ground substance
D. Collagen	IV. Enables glucose transport into cells

Choose the correct answer from the options given below.

Option 1:
A-IV, B-I, C-II, D-III

Option 2:
A-I, B-II, C-III, D-IV

Option 3:
A-II, B-III, C-IV, D-I

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
Correct answer is option (1)

List - I	List - II
A. GLUT- 4	IV. Enables glucose transport into cells
B. Insulin	I. Hormone
C. Trypsin	II. Enzyme
D. Collagen	III. Intercellular ground substance

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Q. 47 Match List I with List II

List I	List II
A. Robert May	I.Species-Area relationship
B.Alexander von Humboldt	II.Long term ecosystem experiment using out door plots
C.Paul Ehrlich	III.Global species diversity at about 7 million
D.David Tilman	IV.Rivet popper hypothesis

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-I, B-III, C-II, D-IV

Option 4:
A-III, B-IV, C-II, D-I

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
Robert May places the global species diversity at about 7 million.

Alexander von Humboldt gave species-area relationship.

Paul Ehrlich used an analogy "Rivet popper hypothesis" to explain the role of species in the ecosystem.

David Tilman performed long term ecosystem experiments using out door plots.

Q. 48 Given below are two statements:

Statement I: In C_3 plants, some O_2 binds to RuBisCO, hence CO_2 fixation is decreased.

Statement II: In C_4 plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the correct answer from the options given below:

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Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is true but Statement II is false

Solution:
In C_3 plant, some O_2 bind to RuBisCO, and hence CO_2 fixation is decreased. Statement II is incorrect, photorespiration does not occur in C_4 plants as they lack RuBisCO in mesophyll. Hence statement I is the only correct option.

Q. 49 Match List I with List II

List - I (Types of stamens)	List -II (Example)
A. Monoadelphous	I. Citrus
B. Diadelphous	II. Pea
C. Polyadelphous	III. Lily
D. Epiphyllous	IV. China-rose

Choose the correct answer from the options given below:

Option 1:
A-IV, B-II, C-I, D-III

Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-I, B-II, C-IV, D-III

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-IV, B-II, C-I, D-III

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Solution:
In China rose monoadelphous androecium is present.

Diadelphous androecium is found in pea plant.

Polyadelphous androecium is found in citrus.

Epiphyllous androecium is found in lily.

Q. 50 Read the following statements and choose the set of correct statements:

In the members of Phaeophyceae,

A. Asexual reproduction occurs usually by biflagellate zoospores.
B. Sexual reproduction is by oogamous method only.
C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.
D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.
E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

Option 1:
A, B, C and D only

Option 2:
B, C, D and E only

Option 3:
A, C, D and E only

Option 4:
A, B, C and E only

Correct Answer:
A, C, D and E only

Solution:
In members of Phaeophyceae sexual reproduction is by oogamous, isogamous or anisogamous methods.

Therefore correct set of statements are A, C, D and E.

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Q. 51 Match List I with List II :

List I	List II
A. Cocaine	I. Effective sedative in surgery
B. Heroin	II. Cannabis sativa
C. Morphine	III. Erythroxylum
D. Marijuana	IV. Papaver somniferum

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C-I, D-II

Option 2:
A-I, B-III, C-II, D-IV

Option 3:
A-II, B-I, C-III, D-IV

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
The correct option is (4) as
A. Cocaine – Obtained from plant Erythroxylum coca, stimulating action on CNS.
B. Heroin – Formed by the acetylation of morphine which is obtained from plant Papaver somniferum.
C. Morphine – Obtained from Papaver somniferum, is an effective sedative in surgery.
D. Marijuana – Obtained from Cannabis sativa, produces hallucinogenic effect and affects cardiovascular system of the body.

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Q. 52 Match List I with List II :

	List I		List II
A	Down's syndrome	I	11^{th} chromosome
B	α-Thalassemia	II	'X' chromosome
C	β -Thalassemia	III	21^{st} chromosome
D	Klinefelter's syndrome	IV	16^{th} chromosome

Choose the correct answer from the options given below :

Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-I, C-II, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
Sol. Down's syndrome is due to presence of an additional copy of chromosome number 21. Klinefelter's syndrome is caused due to presence of an additional copy of X-chromosome. α -Thalassemia is controlled by two closely linked genes on chromosome 16 of each parent. β-Thalassemia is controlled by a single gene HBB on chromosome 11 of each parent.

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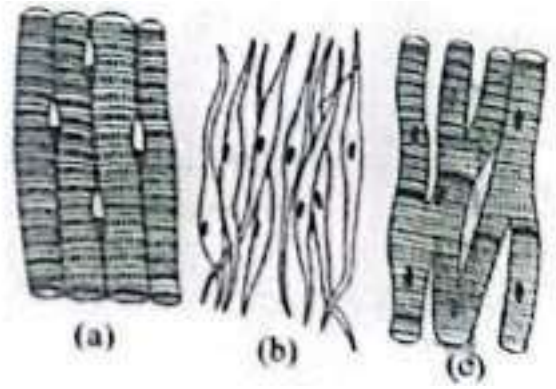
Q. 53 Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body:



Name of muscle/location

Option 1:
(a) Smooth - Toes
(b) Skeletal – Legs
(c) Cardiac – Heart

Option 2:
(a) Skeletal - Triceps
(b) Smooth – Stomach
(c) Cardiac – Heart

Option 3:
(a) Skeletal - Biceps
(b) Involuntary – Intestine
(c) Smooth – Heart

Option 4:
(a) Involuntary – Nose tip
(b) Skeletal – Bone
(c) Cardiac – Heart

Correct Answer:
(a) Skeletal - Triceps
(b) Smooth – Stomach
(c) Cardiac – Heart

Solution:
The correct answer is option (2) as
Figure (a) represents skeletal muscle fibres which are closely attached to skeletal bones. In a typical muscle such as triceps and biceps, striated muscle fibres are bundled together in a parallel fashion.

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Figure (b) represents smooth muscle fibres which are present in the wall of internal organs such as the blood vessels, stomach and intestine.
Figure (c) represents cardiac muscle fibres which are exclusively present in the heart.

Q. 54 Match List I with List II :

List I	List II
A. Pterophyllum	I. Hag fish
B. Myxine	II. Saw fish
C. Pristis	III. Angel fish
D. Exocoetus	IV. Flying fish

Choose the correct answer from the options given below :

Option 1:
A-II, B-I, C-III, D-IV

Option 2:
A-III, B-I, C-II, D-IV

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-III, B-II, C-I, D-IV

Correct Answer:
A-III, B-I, C-II, D-IV

Solution:
The correct option is option no. (2) as
Pterophyllum is the scientific name for Angel fish.
Myxine is the scientific name for Hag fish.
Pristis is the scientific name for Saw fish.
Exocoetus is the scientific name for Flying fish.

Q. 55 Which of the following is not a component of Fallopian tube?

Option 1:
Uterine fundus

Option 2:
Isthmus

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Option 3:
Infundibulum

Option 4:
Ampulla

Correct Answer:
Uterine fundus

Solution:
The correct answer is option (1) as uterine fundus is the upper, dome-shaped part of the uterus, above the opening of fallopian tubes.
• Option (2) is incorrect as isthmus is the last and narrow part of the oviduct that links to the uterus.
• Option (3) is incorrect as infundibulum is the part of oviduct which is closer to the ovary.
• Option (4) is incorrect as ampulla is the wider part of the oviduct.

Q. 56 Match List I with List II :

	List I		List II
A	Pleurobrachia	I	Mollusca
B	Radula	II	Ctenophora
C	Stomochord	III	Osteichthyes
D	Air bladder	IV	Hemichordata

Choose the correct answer from the options given below :

Option 1:
A-IV, B-II, C-III, D-I

Option 2:
A-II, B-I, C-IV, D-III

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
The correct answer is option (2) as
A. Pleurobrachia – is a member of phylum Ctenophora.
B. Radula – is a rasping feeding organ present in phylum Mollusca.

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C. Stomochord – Rudimentary structure similar to notochord found in the collar region of members of phylum Hemichordata.
D. Air bladder – is found in Osteichthyes which provides them buoyancy.

Q. 57 Which of the following are Autoimmune disorders?
A. Myasthenia gravis
B. Rheumatoid arthritis
C. Gout
D. Muscular dystrophy
E. Systemic Lupus Erythematosus (SLE)

Choose the most appropriate answer from the options given below:

Option 1:
A, B & D only

Option 2:
A, B & E only

Option 3:
B, C & E only

Option 4:
C, D & E only

Correct Answer:
A, B & E only

Solution:
The correct answer is option (2) as Myasthenia gravis, Rheumatoid arthritis and Systemic Lupus Erythematosus (SLE) are autoimmune disorders.
Muscular dystrophy is a genetic disorder which progressively affects the skeletal muscles.
Gout is the inflammation of joints due to deposition of uric acid crystals.
Option (1), (3) and (4) are not the correct answer because all of them are not autoimmune disorders.

Q. 58 Given below are two statements :
Statement I : In the nephron, the descending limb of loop of Henle is impermeable to water and permeable
to electrolytes.
Statement II : The proximal convoluted tubule is lined by simple columnar brush border epithelium and
increases the surface area for reabsorption.
In the light of the above statements, choose the correct answer from the option given below :

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Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are false

Solution:
Correct answer is option (2) because
Statement I is false as the descending limb of loop of Henle is permeable to water and almost impermeable to electrolytes.
Statement II is false as proximal convoluted tubule is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption.

Q. 59 Which of the following is not a steroid hormone?

Option 1:
Cortisol

Option 2:
Testosterone

Option 3:
Progesterone

Option 4:
Glucagon

Correct Answer:
Glucagon

Solution:

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The correct answer is option (4) as glucagon is a proteinaceous hormone secreted from pancreas. Options (1), (2) and (3) are not correct as they are steroid in nature.

Q. 60 Given below are two statements:
Statement I: The presence or absence of hymen is not a reliable indicator of virginity.
Statement II: The hymen is torn during the first coitus only.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is true but Statement II is false

Solution:
The correct answer is option no. (3) because the presence or absence of hymen is not a reliable indicator of virginity because hymen can also be broken by a sudden jolt, insertion of a vaginal tampon, active participation in some sports and in some women the hymen persists even after coitus.

Q. 61 Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

Option 1:
High pO_2 and High pCO_2temperature

Option 2:
High pO_2 and Lesser H^+ concentration

Option 3:
Low pCO_2 and High H^+ concentration

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Option 4:
Low pCO_2 and High

Correct Answer:
High pO_2 and Lesser H^+ concentration

Solution:
The correct answer is option (2) as
Conditions favourable for formation of oxyhaemoglobin in alveoli are high pO_2, less H^+ concentration low pCO_2 and low temperature.
Option (1), (3) and (4) are not correct as they do not favour the formation of oxyhaemoglobin.

Q. 62 Match List I with List II :

List I	List II
A. Axoneme	I. Centriole
B. Cartwheel pattern	II. Cilia and flagella
C. Crista	III. Chromosome
D. Satellite	IV. Mitochondria

Choose the correct answer from the options given below :

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-IV, B-II, C-III, D-I

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
• Axoneme is seen in cilia and flagella
• Centriole shows cartwheel appearance
• Crista is found in mitochondria
• Satellite is present in chromosomes

Q. 63 The flippers of the Penguins and Dolphins are the example of the

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Option 1:
Adaptive radiation

Option 2:
Natural selection

Option 3:
Convergent evolution

Option 4:
Divergent evolution

Correct Answer:
Convergent evolution

Solution:
The correct answer is option (3), because the flippers of the Penguins and Dolphins perform similar function but they are not anatomically similar structures. This is example of analogous structures.
• Option (1) is incorrect as adaptive radiation is the process of evolution of different species in a given geographical area starting from a point and literally radiating to the other areas of geography.
• Option (2) is incorrect as natural selection is a key mechanism of evolution.
• Option (4) is incorrect as divergent evolution results in the formation of homologous structures.

Q. 64 Given below are some stages of human evolution.
Arrange them in correct sequence. (Past to Recent)

A. Homo habilis
B. Homo sapiens
C. Homo neanderthalensis
D. Homo erectus

Choose the correct sequence of human evolution from the options given below:

Option 1:
D-A-C-B

Option 2:
B-A-D-C

Option 3:
C-B-D-A

Option 4:
A-D-C-B

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Correct Answer:
A-D-C-B

Solution:
Correct answer is option (4) because the correct sequence of stages of human evolution from past to recent is
Homo habilis → Homo erectus → Homo neanderthalensis → Homo sapiens

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Q. 65 In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on

Option 1:
5^{th} segment

Option 2:
10^{th} segment

Option 3:
8^{th} and 9^{th} segment

Option 4:
11^{th} segment

Correct Answer:
10^{th} segment

Solution:
Correct answer is option (2), because in both sexes of cockroach, 10th segment bears a pair of jointed filamentous structures called anal cerci.
Options (1), (3) and (4) are incorrect because 5^{th}, 8^{th} and 9^{th} segments do not bear such structures. In adult cockroaches only 10^{th} segments are present in abdomen. 11^{th} abdominal segment is absent.

Q. 66 Which of the following statements is incorrect?

Option 1:
A bio-reactor provides optimal growth conditions for achieving the desired product

Option 2:
Most commonly used bio-reactors are of stirring type

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Option 3:
Bio-reactors are used to produce small scale bacterial cultures

Option 4:
Bio-reactors have an agitator system, an oxygen delivery system and foam control system

Correct Answer:
Bio-reactors are used to produce small scale bacterial cultures

Solution:
Correct answer is option (3)
The statement (3) is incorrect because bioreactors are used for processing of large volumes (100 –1000 litres) of culture.
Small volume cultures cannot yield appreciable quantities of products. To produce in large quantities the development of bioreactors is required.

Q. 67 Match List I with List II :

	List I		List II
A	Typhoid	I	Fungus
B	Leishmaniasis	II	Nematode
C	Ringworm	III	Protozoa
D	Filariasis	IV	Bacteria

Choose the correct answer from the options given below:

Option 1:
A-I, B-III, C-II, D-IV

Option 2:
A-IV, B-III, C-I, D-II

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-IV, B-III, C-I, D-II

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Solution:
The correct answer is option (2) as
Typhoid – Caused by Salmonella typhimurium (Bacteria)
Leishmaniasis – Caused by protozoan i.e., Leishmania donovani
Ringworm – Caused by fungi belonging to the genera Microsporum, Trichophyton and Epidermophyton
Filariasis – Caused by Wuchereria bancrofti and Wuchereria malayi (Nematode)

Q. 68 Consider the following statements :
A. Annelids are true coelomates
B. Poriferans are pseudocoelomates
C. Aschelminthes are acoelomates
D. Platyhelminthes are pseudocoelomates
Choose the correct answer from the options given below :

Option 1:
B only

Option 2:
A only

Option 3:
C only

Option 4:
D only

Correct Answer:
A only

Solution:
The correct answer is option no. (2), because annelids are true coelomate animals. Options (1), (3) and (4) are incorrect because poriferans are acoelomates, aschelminths are pseudocoelomates and platyhelminthes are acoelomates.

Q. 69 Which of the following is not a natural/traditional contraceptive method?

Option 1:
Coitus interruptus

Option 2:
Periodic abstinence

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Option 3:
Lactational amenorrhea

Option 4:
Vaults

Correct Answer:
Vaults

Solution:
The correct answer is option (4) as
Vault is a barrier method of contraception which is made of rubber that is inserted into the female reproductive tract to cover the cervix during the coitus.
• Option (1) is incorrect as coitus interruptus is a natural method of contraception in which male partner withdraws his penis from the vagina just before ejaculation so as to avoid insemination.
• Option (2) is incorrect as periodic abstinence is also a natural method of contraception in which couples avoid coitus during the fertile period.
Option (3) is incorrect as lactational amenorrhea is also a natural method of contraception which is based on the fact that ovulation and therefore the cycle do not occur during the period of intense lactational following parturition.

Q. 70 Match List I with List II :

	List I		List II
A	Pons	I	Provides additional space for Neurons, regulates posture and balance.
B	Hypothalamus	II	Controls respiration and gastric secretions.
C	Medulla	III	Connects different regions of the brain.
D	Cerebellum	IV	Neuro secretory cells

Choose the correct answer from the options given below :

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-I, B-III, C-II, D-IV

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Option 4:
A-II, B-I, C-III, D-IV

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:
The correct answer is option (2) as

A	Pons	-	Part of hindbrain, it connects different regions of the brain.
B	Hypothalamus	-	Also have neuro secretory cells which secrete hormones.
C	Medulla oblongata	-	Part of hindbrain which controls respiration and gastricsecretions.
D	Cerebellum	-	Part of hindbrain with convoluted surface which provides additional space for neurons, also regulates posture and balance.

Q. 71 Which one of the following factors will not affect the Hardy-Weinberg equilibrium?

Option 1:
Genetic recombination

Option 2:
Genetic drift

Option 3:
Gene migration

Option 4:
Constant gene pool

Correct Answer:
Constant gene pool

Solution:
The correct answer is option (4) as a constant gene pool will not disturb the Hardy-Weinberg equilibrium.
Option (1), (2) & (3) will affect the equilibrium leading to evolution.

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Q. 72 Match List I with List II :

List I	List II
A. α-1 antitrypsin	I. Cotton bollworm
B. Cry IAb	II. ADA deficiency
C. Cry IAc	III. Emphysema
D Enzyme replacement therapy	IV Corn borer

Choose the correct answer form the options given below:

Option 1:
A-II, B-I, C-IV, D-III

Option 2:
A-III, B-I, C-II, D-IV

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-II, B-IV, C-I, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
The correct answer is option (3) as
\alph– I antitrypsin → Is used for treatment of Emphysema
Cry I Ab gene → Controls corn borer
Cry I Ac gene → Controls cotton bollworms
Enzyme replacement therapy → Can be used as treatment option in ADA deficiency.

Q. 73 Following are the stages of pathway for conduction of an action potential through the heart
A. AV bundle
B. Purkinje fibres
C. AV node
D. Bundle branches
E. SA node
Choose the correct sequence of pathway from the options given below

Option 1:
E-C-A-D-B

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Option 2:
A-E-C-B-D

Option 3:
B-D-E-C-A

Option 4:
E-A-D-B-C

Correct Answer:
E-C-A-D-B

Solution:
Correct answer is option (1) because the correct pathway of conduction of action potential is
SA → AV node → AV bundle → Bundle branches → Purkinje fibres

Q. 74 The "Ti plasmid" of Agrobacterium tumefaciens stands for

Option 1:
Tumour inhibiting plasmid

Option 2:
Tumor independent plasmid

Option 3:
Tumor inducing plasmid

Option 4:
Temperature independent plasmid

Correct Answer:
Tumor inducing plasmid

Solution:
The correct answer is option (3) as Ti plasmid of Agrobacterium tumefaciens is tumor inducing plasmid, containing T-DNA which causes tumor in several dicot plants.
Options (1), (2) and (4) are not correct.

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Q. 75 Match List I with List II :

	List I		List II
A	Expiratory capacity	I	Expiratory reserve volume + Tidal volume +Inspiratory reserve volume
B	Functional residual capacity	II	Tidal volume + Expiratory reserve volume
C	Vital capacity	III	Tidal volume + Inspiratory reserve volume
D	Inspiratory capacity	IV	Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-II, B-I, C-IV, D-III

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
Expiratory capacity = Tidal volume + Expiratory reserve volume
Functional residual capacity = Expiratory reserve volume + Residual volume
Vital capacity = Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
Inspiratory capacity = Tidal volume + Inspiratory reserve volume

Q. 76 Following are the stages of cell division :
A. Gap 2 phase
B. Cytokinesis
C. Synthesis phase
D. Karyokinesis
E. Gap 1 phase
Choose the correct sequence of stages from the options given below :

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Option 1:
C-E-D-A-B

Option 2:
E-B-D-A-C

Option 3:
B-D-E-A-C

Option 4:
E-C-A-D-B

Correct Answer:
E-C-A-D-B

Solution:
The correct sequence of stages of cell division is

Gap 1 phase → Synthesis phase → Gap 2 phase
(E) (C) (A)

→ Karyokinesis → Cytokinesis
(D) (B)

The correct sequence will be → E → C → A → D → B

Q. 77 Match List I with List II :

	List I		List II
A	Fibrous joints	I	Adjacent vertebrae, limited movement
B	Cartilaginous joints	II	Humerus and Pectoral girdle, rotational movement
C	Hinge joints	III	Skull, don't allow any movement
D	Ball and socket joints	IV	Knee, help in locomotion

Choose the correct answer from the options given below :

Option 1:
A-IV, B-II, C-III, D-I

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Option 2:
A-I, B-III, C-II, D-IV

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
The correct answer is option no. (4) as
• Fibrous joints do not allow any movement. This type of joint is shown by the flat skull bones which fuse end-to-end with the help of dense fibrous connective tissues in the form of sutures.
• Cartilaginous joint is present between the adjacent vertebrae in the vertebral column and this permits limited movements.
• Hinge joint is a type of synovial joint present in knee, help in locomotion
• Ball and socket joint is also a type of synovial joint present between humerus and pectoral girdle and allows rotational movement.

Q. 78 Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :
Assertion A : FSH acts upon ovarian follicles in female and Leydig cells in male.
Reason R : Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in
male human being.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

Option 4:
A is false but R is true

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Correct Answer:
A is false but R is true

Solution:
The correct answer is option (4) as FSH is a gonadotropin affects ovarian follicles in females and causes their growth but in males LH affects Leydig cells leading to secretion of androgens.
Growing ovarian follicles secrete estrogen in females while interstitial cells secrete androgen in male human being.
Hence, Assertion is false and Reason is true.

Q. 79 Match List I with List II :

	List I		List II
A	Diakinesis	I	Synaptonemal complex formation
B	Pachytene	II	Completion of terminalisation of chiasmata
C	Zygotene	III	Chromosomes look like thin threads
D	Leptotene	IV	Appearance of recombination nodules

Choose the correct answer from the options given below

Option 1:
A-IV, B-II, C-III, D-I

Option 2:
A-I, B-II, C-IV, D-III

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
(A) Diakinesis – Completion of terminalisation of chiasmata
(B) Pachytene – Appearance of recombination nodules
(C) Zygotene – Synaptonemal complex formation
(D) Leptotene – Chromosomes look like thin threads

A-II, B-IV, C-I, D-III

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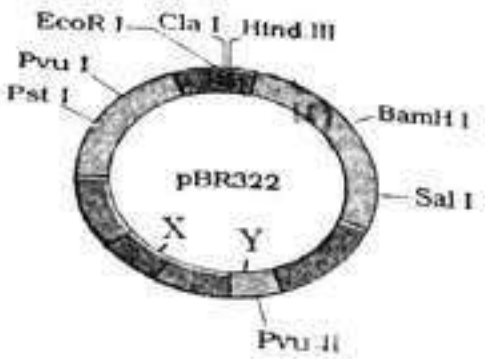
Q. 80 The following diagram showing restriction sites in E. coli cloning vector pBR322. Find the role of 'X' and 'Y'genes :



Option 1:
The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.

Option 2:
The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved
in the replication of Plasmid.

Option 3:
The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.

Option 4:
Gene 'X' is responsible for recognitions sites and 'Y' is responsible for antibiotic resistance.

Correct Answer:
The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved
in the replication of Plasmid.

Solution:
Correct answer is option (2), because
'X' in the given diagram is ori while 'Y' is rop.
'X' which is ori is responsible for controlling the copy number of the linked DNA and 'Y' which is rop codes for protein involved in the replication of plasmid.
Options (1), (3) and (4) are incorrect as 'X' and 'Y' are not related to these functions.

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Q. 81 Match List I with List II :

	List I		List II
A	Common cold	**I**	Plasmodium
B	Haemozoin	**II**	Typhoid
C	Widal test	**III**	Rhinoviruses
D	Allergy	**IV**	Dust mites

Choose the correct answer from the options given below :

Option 1:
A-II, B-IV, C-III, D-I

Option 2:
A-I, B-III, C-II, D-IV

Option 3:
A-III, B-I, C-II, D-IV

Option 4:
A-IV, B-II, C-III, D-I

Correct Answer:
A-III, B-I, C-II, D-IV

Solution:
Correct answer is option (3) because
• Common cold is caused by Rhinoviruses
• Haemozoin is released in blood due to ruptured RBCs after Plasmodium infection.
• Widal test is used to confirm the typhoid fever.
• Allergy is caused due to dust mites.

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Q. 82 Match List I with List II :

	List I		List II
A	Non-medicated IUD	**I**	Multiload 375
B	Copper releasing IUD	**II**	Progestogens
C	Hormone releasing IUD	**III**	Lippes loop
D	Implants	**IV**	LNG-20

Choose the correct answer from the option given below:

Option 1:
A-III, B-I, C-II, D-IV

Option 2:
A-I, B-III, C-IV, D-II

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
Correct answer is option (4) because
• Lippes loop is a non-medicated IUD.
• Multiload 375 is a copper releasing IUD.
• LNG -20 is a hormone releasing IUD.
• Progestogens are used as implants.

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Q. 83 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A : Breast-feeding during initial period of infant growth is recommended by doctors for bringing a
healthy baby.

Reason R : Colostrum contains several antibodies absolutely essential to develop resistance for the new
born baby.

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both A and R are correct and R is the correct explanation of A

Option 2:
Both A and R are correct but R is NOT the correct explanation of A

Option 3:
A is correct but R is not correct

Option 4:
A is not correct but R is correct

Correct Answer:
Both A and R are correct and R is the correct explanation of A

Solution:
Correct answer is option (1)
Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby as colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.

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Q. 84 Match List I with List II :

	List I		Lst II
A	Lipase	**I**	Peptide bond
B	Nuclease	**II**	Ester bond
C	Protease	**III**	Glycosidic bond
D	Amylase	**IV**	Phosphodiester bond

Choose the correct answer from the options given below :

Option 1:
A-IV, B-II, C-III, D-I

Option 2:
A-III, B-II, C-I, D-IV

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-IV, B-I, C-III, D-II

Correct Answer:
A-II, B-IV, C-I, D-III

Solution:
The correct answer is option (3) as

A	Lipase	-	Digests ester bond found in lipids.
B	Nuclease	-	Helps in digestion of phosphodiester bonds found in nucleic acids.
C	Protease	-	Helps in digestion of peptide bond found in proteins.
D	Amylase	-	Digests/breaks the glycosidic bonds found in carbohydrates i.e., digest starch into smaller molecules, ultimately yielding maltose, which in turn is cleaved into two glucose molecules by maltase.

Q. 85 Which one is the correct product of DNA dependent RNA polymerase to the given template?
3'TACATGGCAAATATCCATTCA5'

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Option 1:
5'AUGUACCGUUUAUAGGUAAGU3'

Option 2:
5'AUGUAAAGUUUAUAGGUAAGU3'

Option 3:
5'AUGUACCGUUUAUAGGGAAGU3'

Option 4:
5'ATGTACCGTTTATAGGTAAGT3'

Correct Answer:
5'AUGUACCGUUUAUAGGUAAGU3'

Solution:
Template DNA is :
3'TACATGGCAAATATCCATTCA5'
5'AUGUACCGUUUAUAGGUAAGU3' m-RNA

Q. 86 Match List I with List II related to digestive system of cockroach.

	List I		Lst II
A	The structures used for storing of food	**I**	Gizzard
B	Ring of 6-8 blind tubules at junction of foregut and midgut.	**II**	Gastric Caeca
C	Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	**III**	Malpighian tubules
D	The structures used for grinding the food.	**IV**	Crop

Choose the correct answer from the options given below :

Option 1:
A-IV, B-II, C-III, D-I

Option 2:
A-I, B-II, C-III, D-IV

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Option 3:
A-IV, B-III, C-II, D-I

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-IV, B-II, C-III, D-I

Solution:
The correct answer is option no. (1) as

The structure used for griding the food particles	-	Gizzard
The structure used for storing of food	-	Crop
Ring of 6-8 blind tubules at junction of foregut and midgut which assists in secretion of digestive juices	-	Gastric Caeca
Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut which assists in elimination of nitrogenous wastes	-	Malpighian tubules

Q. 87 Match List I with List II :

	List I		**Lst II**
A	RNA polymerase III	**I**	snRNPs
B	Termination of transcription	**II**	Promotor
C	Splicing of Exons	**III**	Rho factor
D	TATA box	**IV**	SnRNAs, tRNA

Choose the correct answer from the options given below :

Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-III, C-I, D-II

Correct Answer:
A-IV, B-III, C-I, D-II

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Solution:
• In eukaryotes, RNA polymerase III codes for snRNAs, tRNA and 5s rRNA.
• Splicing of exons is performed by snRNPs.
• TATA box is present in promoter region of transcription unit.
• Rho factor is responsible for termination of transcription.

Q. 88 The following are the statements about non-chordates:
A. Pharynx is perforated by gill slits.
B. Notochord is absent.
C. Central nervous system is dorsal.
D. Heart is dorsal if present.
E. Post anal tail is absent.
Choose the most appropriate answer from the options given below:

Option 1:
A & C only

Option 2:
A, B & D only

Option 3:
B, D & E only

Option 4:
B, C & D only

Correct Answer:
B, D & E only

Solution:
The correct answer is option no. (3) as the features of non-chordates among the given statements are:
B. Notochord is absent.
D. Heart is dorsal if present.
E. Post anal tail is absent.
Statements A and C are features of chordates.
Hence, option (3) is correct and options (1), (2) and (4) are incorrect.

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Q. 89 Match List I with List II :

	List I		Lst II
A	Exophthalmic goiter	**I**	Excess secretion of cortisol, moon face & hypergylcemia.
B	Acromegaly	**II**	Hypo-secretion of thyroidhormone and stunted growth.
C	Cushing's syndrome	**III**	Hyper secretion of thyroidhormone & protruding eye balls.
D	Cretinism	**I**	Excessive secretion of growth hormone.

Choose the correct answer from the options given below :

Option 1:
A-I, B-III, C-II, D-IV

Option 2:
A-IV, B-II, C-I, D-III

Option 3:
A-III, B-IV, C-II, D-I

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
The correct answer is option no. (4) as

A	Exophthalmic goiter	I	Hyper secretion of thyroid hormone and characterized by protruding eye balls
B	Acromegaly	II	Excessive secretion of growth hormone
C	Cushing's syndrome	III	Excess secretion of cortisol, moon face and hyperglycaemia
D	Cretinism	IV	Hypo-secretion of thyroid hormone and characterized by stunted growth

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Q. 90 Given below are two statements:
Statement I: Mitochondria and chloroplasts both double membranes bound organelles.
Statement II: Inner membrane of mitochondria is relatively less permeable, as compared chloroplast.
In the light of the above statements, choose the mis appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
Both mitochondria and chloroplasts are double membrane bound cell organelles.
Transport of ions occurs across the inner membrane of mitochondria. The inner membrane of chloroplast is impermeable to ions and metabolites. Therefore, it is said that inner membrane of mitochondria is relatively more permeable to that of chloroplast.

Q. 91 Given below are two statements:
Statement I: The cerebral hemispheres are connected by nerve tract known as corpus callosum.
Statement II: The brain stem consists of the medulla oblongata, pons and cerebrum.
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1:
Both Statement I and Statement II are correct.

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Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Statement I is correct but Statement II is incorrect.

Solution:
The correct answer is option (3) as statement I is correct but statement II is incorrect.
In human brain, a deep cleft divides the cerebrum longitudinally into two halves, which are termed as the left and right cerebral hemispheres. The cerebral hemispheres are connected by a tract of nerve fibres called corpus callosum.
Three major regions make up the brain stem i.e. mid brain, pons and medulla oblongata.
Cerebrum is a part of forebrain which does not form brain stem.
Options (1), (2) and (4) are incorrect.

Q. 92 Regarding catalytic cycle of an enzyme action, select the correct sequential steps :
A. Substrate enzyme complex formation.
B. Free enzyme ready to bind with another substrate.
C. Release of products.
D. Chemical bonds of the substrate broken.
E. Substrate binding to active site.
Choose the correct answer from the options given below :

Option 1:
E, A, D, C, B

Option 2:
A, E, B, D, C

Option 3:
B, A, C, D, E

Option 4:
E, D, C, B, A

Correct Answer:
E, A, D, C, B

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Solution:
The correct answer is option (1) which is E, A, D, C, B.
The catalytic cycle of an enzyme action can be described in the following steps.
(1) First, the substrate binds to the active site of the enzyme, fitting into the active site.
(2) The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate.
(3) The active site of the enzyme, now in close proximity of the substrate breaks the chemical bonds of the
substrate and the new enzyme-product complex is formed.
(4) The enzyme releases the products of the reaction and the free enzyme is ready to bind to another molecule of the substrate and run through the catalytic cycle once again.
Options (2), (3) and (4) are incorrect as the steps mentioned are in the wrong sequence.

Q. 93 Given below are two statements:
Statement I: Gause's competitive exclusion principle states that two closely related species competing for
different resources cannot exist indefinitely.
Statement II: According to Gause's principle, during competition, the inferior will be eliminated. This may be
true if resources are limiting.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is false but Statement II is true.

Solution:
Gause's competitive exclusion principle states that two closely related species competing for the same resources cannot exist indefinitely and the competitively inferior one will be eliminated eventually. This may be true if resources are limiting.

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Q. 94 Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.

Image

Option 1:
FSH, Leydig cells, Sertoli cells, spermiogenesis.

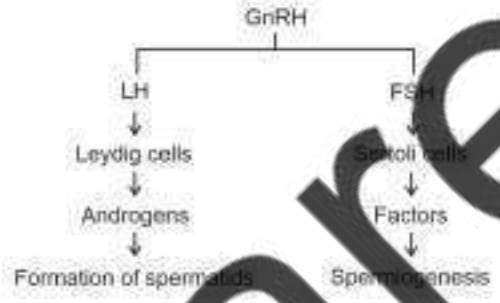
Option 2:
ICSH, Interstitial cells, Leydig cells, spermiogenesis.

Option 3:
FSH, Sertoli cells, Leydig cells, spermatogenesis.

Option 4:
ICSH, Leydig cells, Sertoli cells, spermatogenesis.

Correct Answer:
FSH, Leydig cells, Sertoli cells, spermiogenesis.

Solution:
The correct answer is option no. (1) as



(A) is FSH which is a pituitary hormone.
(B) is Leydig cells which are found in the interstitial space outside of the seminiferous tubules.
(C) is Sertoli cells are found inside the seminiferous tubules.
(D) is Spermiogenesis which is a process that helps in transformation of spermatids into spermatozoa.

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Q. 95 Match List I with List II :

	List I		Lst II
A	P wave	**I**	Heart muscles are electrically silent
B	QRS complex	**II**	Depolarisation of ventricles.
C	T wave	**III**	Depolarisation of atria.
D	T-P gap	**IV**	Repolarisation of ventricles.

Choose the correct answer from the options given below :

Option 1:
A-I, B-III, C-IV, D-II

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-II, B-III, C-I, D-IV

Option 4:
A-IV, B-II, C-I, D-III

Correct Answer:
A-III, B-II, C-IV, D-I

Solution:
The correct answer is option no. (2) as
A. P wave - III. Depolarisation of atria.
B. QRS complex - II. Depolarisation of ventricles.
C. T wave - IV. Repolarisation of ventricles.
D. T-P gap - I. Heart muscles are electrically silent.

Q. 96 Choose the correct statement given below regarding juxta medullary nephron.

Option 1:
Juxta medullary nephrons are located in the columns of Bertini.

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Option 2:
Renal corpuscle of juxta medullary nephron lies in the outer portion of the renal medulla.

Option 3:
Loop of Henle of juxta medullary nephron runs deep into medulla.

Option 4:
Juxta medullary nephrons outnumber the cortical nephrons.

Correct Answer:
Loop of Henle of juxta medullary nephron runs deep into medulla.

Solution:
The correct answer is option no, (3) because the length of loop of Henle of juxta medullary nephron is longer than the length of loop of Henle of cortical nephron and runs deep into medulla.
Option (1) is incorrect as juxta medullary nephron are not present in columns of Bertini.
Option (2) is incorrect because renal corpuscle of juxta medullary nephron lies in inner cortical region.
Option (4) is incorrect as juxta medullary nephrons are lesser in number than cortical nephrons.

Q. 97 Given below are two statements :
Statement I : Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are
produced.
Statement II : Both bone marrow and thymus provide micro environments for the development and
maturation of T-lymphocytes.
In the light of above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

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Correct Answer:
Both Statement I and Statement II are correct.

Solution:
The correct answer is option no. (1) as both statements I and II are correct.
In humans, the bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.

Both bone-marrow and thymus provide micro-environments for the development and maturation of T-lymphocytes.

Options (2), (3) and (4) are incorrect.

Q. 98 Match List I with List II :

	List I		Lst II
A	Unicellular glandular epithelium	**I**	Salivary glands
B	Compound epithelium	**II**	Pancreas
C	Multicellular glandular epithelium	**III**	Goblet cells of alimentary canal
D	Endocrine glandular epithelium	**IV**	Moist surface of buccal cavity

Choose the correct answer from the options given below :

Option 1:
A-II, B-I, C-III, D-IV

Option 2:
A-IV, B-III, C-I, D-II

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
The correct answer is option no. (3) as
A. Unicellular glandular epithelium (III) Goblet cells of alimentary canal
B. Compound epithelium (IV) Lines moist surface of buccal cavity

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C. Multicellular glandular epithelium (I) Salivary glands
D. Endocrine glandular epithelium (II) Pancreas

Q. 99 Match List I with List II :

	List I		Lst II
A	Mesozoic Era	**I**	Lower invertebrates
B	Proterozoic Era	**II**	Fish & Amphibia
C	Cenozoic Era	**III**	Birds & Reptiles
D	Paleozoic Era	**IV**	Mammals

Choose the correct answer from the options given below :

Option 1:
A-II, B-I, C-III, D-IV

Option 2:
A-III, B-I, C-II, D-IV

Option 3:
A-I, B-II, C-IV, D-III

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
The correct answer is option no. (4)
(A) Mesozoic Era – (III) Birds & Reptiles
(B) Proterozoic Era – (I) Lower invertebrates
(C) Cenozoic Era – (IV) Mammals
(D) Paleozoic Era – (II) Fish & Amphibia

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Q. 100 As per ABO blood grouping system, the blood group of father is B^+, mother is A^+ and child is O^+ Their respective genotype can be

A .$|^{B}i/I^{A}i/ii$

B. $|^{B\ B}/|^{A\ A}/ii$

C. $|A|^{B}/i|^{A}/|^{B}j$

D. $|A_i/|B_i/\ |\ A_i$

E. $il^{B}/i|^{A}/|^{A\ B}$

Choose the most appropriate answer from the options given below :

Option 1:
A only

Option 2:
B only

Option 3:
C & B only

Option 4:
D & E only

Correct Answer:
A only

Solution:
Genotype of father with blood group $B^+ = |B//i|^{B}$
Genotype of mother with blood group $A^+ = |A//i|^{A}$
Genotype of child with blood group $O^+ = ii$
Hence only 'A' is correct.

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